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Maintenance Management in Network Utilities

Framework and Practical Implementation

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Dedicated to Antonio Sola

Foreword

I am pleased and honored by the opportunity to write a short foreword to *Maintenance Management in Service Distribution Networks*. The book's authors, Juan F. Gomez Fernández and Adolfo Crespo Márquez, bring a wealth of experience to bear on their topic, maintenance management strategies for network utilities, such as gas, water, electricity and telecommunications.

Juan F. Gomez Fernández is part of the Spanish Research and Development Group in Industrial Management of the University of Seville. He has managed network maintenance and deployment departments in various national distribution network companies in both the private and public sector. Adolfo Crespo Márquez is full professor at the School of Engineering of the University of Seville, in the Department of Industrial Management. He has published extensively in the area of maintenance management and has been involved in many maintenance projects, especially in the mining sector.

Network utilities, a type of public utility, distribute essential services in a number of sectors. Commonly called Distribution Network Services Providers (DNSP), these companies rely on network infrastructures whereby services are directly provided by physical interconnection, fixed or mobile, to customers in their homes, dwellings or places of business. They require the development of special maintenance management capabilities to satisfy the needs of their customers, who are "the core of the business activities". In short, in this sector, good maintenance management information systems allow companies to maintain control, acquire essential information and improve decision-making.

The first section "Introduction to Maintenance Management in Network Utilities" defines network utilities and reviews the basic concepts and models needed to design, develop, and implement the network utility framework.

In part two, "Processes and Tools to Manage Maintenance in Network Utilities", the authors go on to develop the basic concepts of maintenance management framework for DNSPs, providing readers with a useful overview of practical modeling tools and systems for network maintenance management. Chapters deal with such important issues as the management of maintenance strategy and monitoring, managing, and preventing incidents.

Having established the basis of their argument and defined their terms, the authors turn to more pragmatic matters. In part three, “Management Framework Evaluation,” they offer a practical view of the network utility maintenance management evaluation process, focusing on the reliability concept and maturity models. Of special interest is their development of a Maturity Model for maintenance management and a Performability Evaluation Model to measure maintenance contribution. These practical models should prove helpful to those in DNSP management and reliability management and control.

Through its interesting and valuable case studies, this work provides a detailed description of framework implementation. It illustrates the validity of its arguments by using real company situations, backed up by a wealth of scholarly information. This book considers specific characteristics of maintenance in network utilities, discussing existing standards and best management practices. Its practical approach will certainly prove useful to those working in the area. At the same time, its solid scholarly research is evidenced in the extensive bibliography sections at the end of each chapter.

Andrew K. S. Jardine
University of Toronto

Preface

Network utilities are companies dealing with network infrastructure, (such as distribution of gas, water, electricity or telecommunications), and they require the development of special maintenance management capabilities in order to satisfy the needs of their customers. In this sector, maintenance management information systems are essential to ensure control, gain knowledge and improve decision-making.

The aim of this book is the study of specific characteristics of maintenance in these types of companies. We discuss existing standards and the best management practices with the scope of defining a suitable ad hoc framework for the implementation of maintenance management.

The work offers a very practical approach to the problem, defining a series of stages to follow in order to manage maintenance framework properly.

The work is divided into three major parts that cover different contents with the following intentions:

- Part I. Introduction to Maintenance Management in Network utilities: Reviews the basic concepts and models needed for the design, development and implementation of the framework in network utilities companies.
- Part II. Processes and Tools to Manage Maintenance in Network Utilities: Develops the basic pillars of the maintenance management framework for these companies, providing the reader with a consistent background in practical modeling tools and systems for network maintenance management.
- Part III. Management Framework Evaluation: Offers a practical view of the network utility maintenance management evaluation process, focusing on the reliability concept and maturity models. Probably the most relevant aspect of this part of the book is the definition of a maturity model for maintenance management and, a performability evaluation model to measure the maintenance contribution from different points of views that can be found in the relevant literature. Through different case studies, this work provides a detailed description about framework implementation and illustrates the experiences in real company situations.

This book tries to show that maintenance management optimization in network utilities is a relevant issue. Due to their huge infrastructure and organization, maintenance management needs to be based on proper processes, decision-making tools and systems (technologies) in order to achieve suitable levels of maintenance organization and service.

In order to do so, we have characterized suitable processes, tools and systems, evaluating their application to service provider companies, but in order to obtain a comprehensive framework covering all maintenance points of view: organizational, functionality, economic, quality, safety and improvement.

Searching this management integration we have established a processes framework, where the maintenance functions have been characterized in order to fulfill the mission and objectives of the department, but in line with the overall objectives of the organization.

At the same time we have explored current issues related to systems integration and connection and how they relate to the sector of activity of the network utilities. We pay special attention to the support provided to management activities at different levels and to the standards for interoperability.

Developments in information systems and processes have followed parallel paths in industrial companies and in IT companies. The present framework seeks convergence between functions and systems in order to optimize maintenance by properly managing its knowledge. This convergence is justified mainly through the implementation of emerging e-maintenance strategies, which due to the spread of the Internet and modern communications, allow remote control, which was previously impossible for infrastructures of such dimensions.

Furthermore, another issue of vital importance in maintenance is the assessment of its real contribution to the organization, giving maintenance tasks the responsibility and the importance corresponding to their achievements.

The assessment of maintenance in network utilities companies is presented using different levels of maturity and excellence. We pursue and increase in the collective intelligence of the maintenance organization by expanding, improving and disseminating knowledge in a standardized manner and with the proper information technology support.

We seek constant maintenance innovation developing and valuing the knowledge of the maintenance organization using use criteria of intellectual capital for this continuous improvement.

In summary, the framework for network utilities has been developed with the aims of:

- Unifying concepts and terminology to facilitate communication and to guide sustainable development of maintenance.
- Defining the common bases of knowledge and the appropriate technologies for optimization.
- Facilitating staff motivation, consensus in decision-making and the involvement of management with an integrative approach.

- Allowing maintenance management to be auditable and comparable with those of other organizations or reference standards from different fields and different levels of detail.
- Identifying improvements and risks arising from its application by the performance, achievement of objectives and degree of standardization.
- Quantifying in terms of cost the maintenance implications in different fields.
- Ensuring service quality from customers, business and society perspectives.

The conclusion of the work supports the proposal of a reference framework, which we do believe is useful as a recommendation for the distribution networks sector. This framework consists of a set of standardized processes and proven technologies and systems, integrated for continuous improvement of maintenance activities. The maturity level of the framework can be measured as well as the intellectual capital developed by the organization in accordance with its contribution in performability and with the excellence in management.

This framework will ensure suitable life cycle of resources and customers' loyalty in network utilities companies, through a sustainable and stable evolution which progressively reduces uncertainty and reactivity through knowledge generation and dissemination.

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Materials and knowledge gained during the development of these projects also serve as part of subjects that we are currently teaching: “Production and Maintenance Engineering” and “Maintenance” in the School of Engineering of Seville; “Operations Management” in the University of Bergamo; “Advanced Models for Maintenance Management” in the School of Engineering of the Technical University “Federico Santa María de Chile” and many different

Modules that we teach in collaboration with INGEMAN (Asociación para el Desarrollo de la Ingeniería de Mantenimiento) and other institutions in Spain and in other countries.

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Professor Benoit Iung, from the Nancy Research Centre for Automatic Control (CRAN) of the Nancy University, offered a very important support reviewing our framework definition and development, especially during the generation of other common publications included as parts of this book.

Carlos Parra (from IngeCon, Venezuela) provided the data in the case study concerning RCFA in [Chap. 11](#). Carlos is a leading international expert in the field of Operational Reliability.

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maintenance management models and the practical implementation of the e-maintenance concept, respectively (in [Chaps. 1 and 2](#)). In addition, Roberto Serna and Francisco Álvarez provided their collaboration and their experience in the development of information and communication technologies projects and change management projects, respectively (in [Chaps. 2 and 13](#)).

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Part I
Introduction to Maintenance Management
in Network Utilities

Chapter 1

A Background on Utilities and Maintenance Management

1.1 Utility Definition and Characteristics

The expression “public utility” or “utility”, first used in 1903, is defined by Merriam-Webster’s Collegiate® Dictionary, Eleventh Edition (<http://www.merriam-webster.com>), as “*a business organization performing a public service and subject to special governmental regulation*”. In the same manner, the Encyclopædia Britannica (<http://www.britannica.com>) explains “public utility” as “*an enterprise that provides certain classes of services to the public, including common carrier transportation; telephone and telegraph; power, heat, and light; and community facilities for water, sanitation, and similar services*”. Therefore, “public utility” could be defined as a business organization that provides, by means of an infrastructure, specific services to the public, subject to special governmental regulation to facilitate coverage according to social criteria. Within this definition, we may find, amongst other types of businesses, airlines, railroads, buses, trucking, canals, ferries, electricity, light, oil, natural gas, heat, water, sanitation, waste collection, or telecommunications companies.

Public utilities may be characterized from different perspectives. For instance, they can be initially studied from a *strategic* or from an *operational* viewpoint.

From a *strategic* perspective, a key issue is the statement of the market. In order to understand this, we have to distinguish three main aspects:

- *Ownership of the utility.* Depending on the property owner, the utility organization could be publicly-owned if the property is of a state or a public administration, or privately-owned if the owner is a private company. In some countries a mixture or composition of both forms of ownership is designed to ensure the control of the business or the infrastructure, and is quite frequently used as starting point for a process of market liberalization.
- *Market structure.* Taking into consideration the market structure, there are two criteria that must be defined. The first criteria is dominance of the market to control access to the services. Dominance could be monopolistic, duopolistic or

oligopolistic, depending on the domination by a single company, by two companies, or by a small number of companies. The second criteria is that of legal regulation of the market, in which the regulation mechanisms are imposed by government regulation according to state requirements or by industry self-regulation following private association requirements, or in a form of industry market such as the laws of supply and demand. A special case is that of perfect competition, which describes the presence of several non-dominant providers or suppliers in the market with similar services, with accessible information of the market, and with no restrictions upon the free entry or exit of providers in the market. The main motivations for totally or partially regulating utility services, are those of security, exploitation of resources, reinvestment of company profits, the use of the public domain to deploy the infrastructure, equity control of the market and prices, the utilization of a specific technology, employment and production norms, the guarantee of human and environmental conservation, and universal access to the service [84].

- *Ownership of the infrastructure.* As a consequence of liberalization, the infrastructure could be owned by a company in whole or in parts to facilitate competition. Two common forms of infrastructure division are by geographical territories or by hierarchical dependence, that is to say, horizontal and vertical disaggregation. In addition, each part of the disaggregated infrastructure could be shared in equal conditions by a variety of utilities instead of developing their independent infrastructure; what is known as the “unbundling” concept. Even though unbundling disaggregation implies profitable economic synergies and knowledge specialization, regulatory rules will be more complex than in the case of individual providers or suppliers with independent infrastructures.

From an *operational* perspective, the location in which the services are provided permits the classification of public utilities into the following two groups:

- Social utilities based on social infrastructures where services are supplied to users in permanent sites of public access, for example; transportation services, airlines, railroads, buses or ferries.
- Network utilities based on network infrastructures where the services are directly provided by physical interconnection, fixed or mobile, to customers in their homes, dwellings or places of business, for example; water, electricity or telecommunications services.

Note the intentional consideration that is made between users and customers, since ‘users’ are considered here as momentary clients of the available services in the social infrastructures, while ‘customers’ refers to prolonged clients of services through a network infrastructure linked to them or their residence.

According to the 1993 System of National Accounts (SNA) and the International Standard Industrial Classification (ISIC), public utilities are not absolutely considered inside the service sector; basically those with social infrastructure are included in the service sector whilst others with network infrastructure are contained within the industry sector. Note that this classification is not in line with

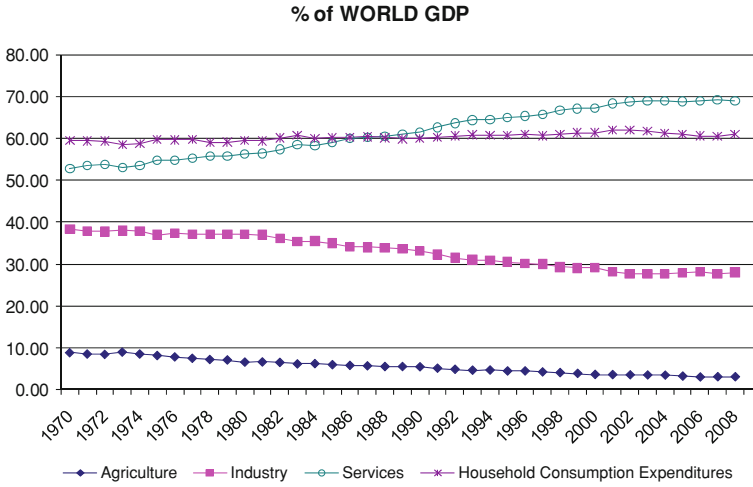


Fig. 1.1 Sector trends in the World GDP [94]

Levitt’s [56] ideas, “*There is no such thing as service industries. There are only industries whose service components are greater or lesser than those other industries. Everybody is in service*”.

The service sector occupies the highest percentage in the economy and is the fastest growing sector in recent decades, from a share of 52.88% of world GDP in 1970 to 69% in 2008; whereas the industry and agriculture sectors have seen their share of world GDP decrease from 38.28 and 8.85% in 1970 to 28 and 3% in 2008, respectively (the World Bank Development Indicators [94], see Fig. 1.1). Focusing on the two utilities, social and network, in the case of U.S., from 1985 to 2008, both have maintained their level around 3% for social utilities (considering transportation and warehousing) and 5% for network utilities (including utilities, broadcasting and telecommunications) of the U.S. GDP [89].

In addition to assessing the importance of both utilities in terms of percentage of GDP, we must also consider their weight on the average annual consumer expenditure, with network utilities at about 7% and social utilities at around 1.1% ([90], see Table 1.1), or the stock market value of shares for network utilities, which stood at a figure of 15% of U.K. GDP in 1995 [67].

Many authors have studied the progression of utilities by referring to the different circumstances that can influence their development: marketing demand and competition, standard of living, technological innovations, policy regulation, environmental and safety worries, resources mishandling, etc. These circumstances focus the studies on the infrastructure life cycle, determining their relationships as a dynamic evolution of the long-life infrastructure, following the S-shaped pattern divided into three clearly distinguished periods in order to guide the growth [24]:

Table 1.1 Consumer expenditures in 2007 (U.S. Bureau of Labor Statistic)

Type of expenditure	1985 (%)	1990 (%)	1995 (%)	2000 (%)	2005 (%)	2007 (%)
Average annual expenditure (dollars)	100	100	100	100	100	100
Food, alcoholic beverages and tobacco	17.0	17.1	15.7	15.4	14.4	13.9
Housing, apparel, entertainment and reading	34.8	35.3	36.4	36.0	35.3	36.5
Health & Personal care products and services	6.0	6.5	6.6	6.9	6.9	6.9
Personal insurance, pensions and cash contributions	12.0	12.0	12.1	12.0	14.8	14.4
Education	1.4	1.4	1.5	1.7	2.0	1.9
Miscellaneous	2.3	3.0	2.4	2.0	1.7	1.6
Customer-operated Transportation	18.4	17.0	17.5	18.4	17.0	16.6
Public Transportation	1.1	1.1	1.1	1.1	1.0	1.1
Utilities, fuels, and public services	7.0	6.7	6.8	6.5	6.9	7.0
Natural gas	1.2	0.9	0.8	0.8	1.0	1.0
Electricity	2.8	2.7	2.7	2.4	2.5	2.6
Fuel oil and other fuels	0.5	0.4	0.3	0.3	0.3	0.3
Telephone	1.9	2.1	2.2	2.3	2.3	2.2
Water and other public services	0.6	0.7	0.8	0.8	0.8	0.9

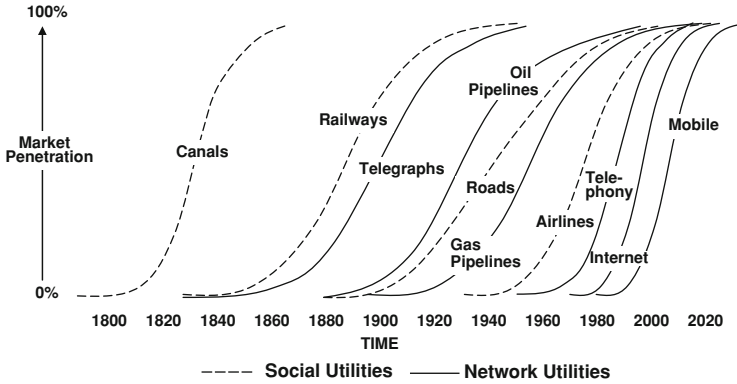


Fig. 1.2 Dynamic evolution of utilities (adapted from [30])

- First period of Development, in which the technologies are competitively advantageous and the market position is being started and formed.
- Second period of Deployment, in which the technologies are efficiently exploited and the market captured.
- Third period of Maturity, where technologies are at their limits and the market is saturated.

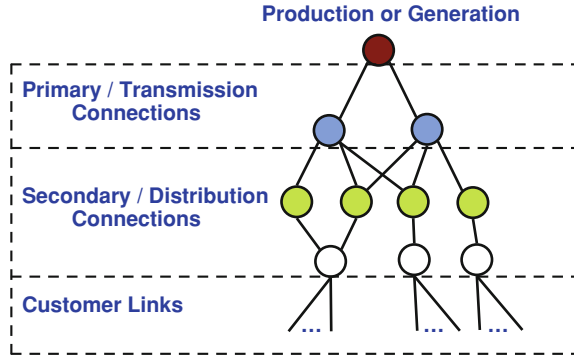
This dynamic behaviour has to be considered in order to decide future investments in the network and its management, because other technologies or type of utilities could occupy the market niche, fulfilling the Kondratiev [54] economy cycles (see Fig. 1.2).

Initially maintenance research activities were focused primarily on industrial environments, although in the last two decades these research activities have been growing in the utility industries, due to liberalization initiatives and technological improvements [91]. These issues have fostered the competence breaking down with the monopolistic market structure of past decades. In our study, from an operational perspective, we will focus on public utilities based on a network infrastructure, network utilities also called Distribution Network Services Providers “DNSP” [84].

Therefore, in general words, we can consider a DNSP as a utility company providing certain services to clients that are supported and distributed by a network infrastructure linked directly to the customers or their residences (households or companies). This infrastructure is often organized and composed of elements arranged and interrelated in a hierarchical form, consisting of four main parts (see Fig. 1.3) that could be segmented and operated by different owners:

- Production or generation, where the services are generated.
- Transmission or primary network, functional locations to transform the services to be transported to remote areas by connections with huge capacity.

Fig. 1.3 Infrastructure of network utilities



- Distribution or secondary network, technical sites and connections to disperse the services within each area.
- Customer links or final network, end elements and connections to supply the services to the customers.

For example, in developed countries, the most common situation implies a combination of multi-privately-owned generation, distribution and customer links with area segmentation of limited jurisdiction, plus a state-owned transmission.

As these companies are supported by a network infrastructure, they have special necessities in management and control, implying that models, systems and tools are not employed in the same manner as in other industries. We have found that these companies have the following characteristics:

- Produced services are generated and consumed simultaneously, as a result they are distributed extensively.
- Services have to be delivered to customers, and some services are considered universal by states determining a baseline level of provision.
- Network infrastructure has a hierarchical form replicated by distribution areas or jurisdictions.
- High number and type of elements compose the network, which are geographically dispersed and in non-optimal environmental conditions.
- Elements are interrelated among themselves.
- There are high numbers and types of customers classified in geographical and market segmentation.
- Physical relationship exists between the network elements and the customer service by levels of aggregation.
- High social interaction, particularly with the customer. At least, one communication with customers is established monthly or bimonthly through consumer bills.
- Services are consumable with a tangible component based on resource consumption and an intangible component based on service experiences.
- The network is dynamic and suffers configuration and operational changes.
- Extremely demanding in human resources, network capacity and spare parts.
- Capital-intensive with a long-term return on investment (ROI).

That is to say, in the core of the network the resources (water, gas electricity, information, etc.) are manipulated to be transferred to the customers (households, commercial or industrial companies with different service necessities). A portfolio of services is provided to satisfy distinct customers demand profiles (seasonal and/or peak demand) and to provide a reliable service supply. Due to the wide distance between generation and customer, numerous equipment and connections are deployed in each area to conduct the services. To harmonize the distribution of elements according to the customer demanded capacity, equipment and connections are designed in proportion to levels of capacity aggregation as a hierarchy network. In order to design the network, we must not only consider demand, but also the environment and geographical characteristic of the territory, which has a great influence in the equipment operating conditions. Furthermore, the capacity has to be regulated and the dispersed equipment operated to adapt to customers demand. This normally requires very demanding technical activities to sustain the contracted service level agreements. As a consequence, the utilities are capital-intensive [67] with decades for pay-back on investments.

Additionally, there is an intense and long lasting relationship with customers and consequently, a diverse range of interactions with them. This can be appreciated in terms of service provisioning, service level agreements, concerns about the technologies employed, safety and environmental preservation, etc. At the end of the day, customer opinion is essential in competitive markets and extremely decisive for the consideration of future investments [91]. Customer requirements, attitudes and behaviour are not always the same, even among similar groups or at different times. Subsequently they have to be analyzed considering the positive and negative feelings of customers concerning service issues and the supplying company, defining customer segmentation to correctly see the needs of actual or potential customers [27]. Consequently, in a competitive market, these companies try to increase the market share and the “customer life-cycle value” by:

- retaining their actual customers;
- building loyalty to their customers and;
- capturing new potential customers in geographical territories.

In summary, network utilities are customer oriented organizations in which the supply chain is focused on the customer who emerges as the core of the business activities. These network characteristics and the narrow relationship with the customer condition the very high level of operation and maintenance requirements in order to properly sustain the service level agreements over time.

1.2 Current Maintenance Management in Network Utilities

Maintenance is defined in EN [40] as “*the combination of all technical, administrative and managerial actions during the life cycle of an item intended to retain it in, or restore it to a state in which it can perform the required function*”. In the

Table 1.2 Profits before improvements due to maintenance [66]

1% of improvement in each area	Produced profit (%)
Productivity	3
Availability	3
Cost reduction	0.5–3.5
Rates reduction	0.7–1.2
Price increase	0.5–0.9

same vein, Maintenance Management is defined as “*all the activities of management that determine the maintenance objectives or priorities, strategies and responsibilities and implement them by means such as maintenance planning, maintenance control and supervision, and several improving methods including economical aspects*”. So, maintenance has to take into account all the necessary functions to provide a given service. Maintenance and maintenance management are nowadays considered as highly complex fields involving many disciplines such as: operations, human resources, analytical accounting and economics, safety and environmental management, etc. [13]. Due to this variety, the maintenance decision making process requires proper tools to pursue:

- Economics, lower expenditure.
- Increase in productivity, minimizing the impact of any faults, thus resulting in a lower production cost and service pricing.
- Prioritization, quick decisions distinguishing critical resources and activities for the business.
- Quality, the maximum achievable for a better service, which results in major customer satisfaction, the capture of new customers and improves the company image.
- Forecasting, to prevent or minimize damage.
- Dealing with high social pressure, in cases when an activity could possibly only be noticed due to its failures. A situation that when it occurs, it must be remembered, can lead to a lot of pressure.
- Management of suitable human resources adequate for maintenance with its restrictions.

As the reader may guess, the pursuit of some of these objectives may result in the deterioration of others. For example, maintenance activities are always under pressure to cut costs, underestimating the implications that cost cutting has in terms of risk increase or global impact on the entire company [9, 60].

One figure that illustrates the importance of maintenance activities is their weight in GDP (9.4% in Spain, [2]). Other international studies show that maintenance activities contribution to the value added by the companies can vary from 15 to 40% depending on the sector [61, 63]. For example, Murthy et al. [66], show that any improvement in maintenance activities implies larger improvements in the organization (see Table 1.2). That is to say, “*it involves an adequate human*

capacity with a high professional level of ethics and deontology in the face of the liability and social impact of their decisions” [15].

Focusing on the customer oriented network utility industry, we can affirm, based on different contributions [66], that maintenance has become a key department, contributing to customer satisfaction and business profitability. Many authors remark how maintenance contributes to the service quality, but also enriches the entire company experience surrounding the service provisioning [55, 99]:

- *Customer relationship.* More reliable services improve customer satisfaction and loyalty, and enhance the business’ image and ability to capture new customers.
- *Increase productivity and reduction of cycle times.* Good maintenance preserves equipment from deterioration, minimizing production disruptions and overhaul stops.
- *Reduction of costs.* As a result of efficient maintenance, the cost of production is reduced and therefore lower price of service is obtained.
- *Quality and features.* If maintenance keeps the performance of services inside the contracted levels throughout a period of time, the availability and the quality of the services are improved.
- *Personnel motivation.* If operations activities are carried out within standards, staffs are motivated, thus avoiding personnel frustration, conflicts and pressure on corrective activities. Good maintenance empowers human reliability and reduces errors.
- *Image and reputation.* having the same effect as the first point above. These not only influence customer satisfaction but also personnel motivation—that of being part of a highly valued company.

A base network utilities maintenance management scenario consists of managing multiple and dispersed functional locations, equipment, connections, services and customers. An operational policy with an imposition of services continuity even under non-optimal operation conditions would illustrate some typical faults such as the following:

- Tough management of a high number of elements/assets supplied by a large number of companies with remote access.
- High dedication to the resolution of incidents and problems, decision-making with scarce knowledge of the adequate movement and solutions along the network.
- Non-automated operations, manual dedication to repetitive tasks.
- Reactive management, with no real-time response.
- Tendency to distribute self-organizations per areas, without a central authority and rules, thus obstructing integration and standardization.
- Network documentation in paper format and with complications in correlating, decision-making and obtaining maintenance knowledge.

- Absence of a unique and updated inventory or network configuration, which supports the operational activities, changes and determination of the exact position of problems.
- Isolated transmission of information and knowledge into islands of specialties or departments, which hampers teamwork and improvements in intergroup activities.

This scenario can be drawn as an ill-process, where most of the time is dedicated to unproductive tasks [11]. This situation tends to become more complex, and increasingly difficult to manage, especially because systems depend on time, network growth and geographical dispersion.

Of course this base scenario reflects a non-appropriate situation and the fact that network utilities companies require a well structured maintenance management system to control maintenance and to improve from this base operational scenario.

In our literature review, we have found references to technical, administrative and managerial actions conforming current maintenance strategies applied in network utilities: corrective, preventive, predictive, proactive and perfective maintenance. Let us look at some of the definitions found for each type of these maintenance strategies:

- *Corrective maintenance* [92]. Necessary actions, carried out on the network, in order to remedy or alleviate incidents producing degradation of services being delivered through said network.
- *Preventive maintenance* [33, 45]. A set of planned routines carried out on network elements in order to maintain them in an optimum level of performance and thus reduce the reoccurrence of incidents. Therefore this would also include the next type of strategies.
- *Predictive maintenance, based on condition* [61, 62]. A set of analyses aimed at estimating the occurrence and behaviour of an incident.
- *Proactive maintenance* [70]. A set of activities designed to detect and correct an incident before it occurs avoiding its effects within the network and within the service.
- *Perfective maintenance*. In the spirit of continuous improvement [35]) this type of activities are included within a set of projects that are normally designed after the start of the operational phase of a distribution network. Their target is to improve the network performance and/or maintainability as well as the services provided through it. These activities are also called “Design-out Maintenance (DOM)” by other authors [26].

Maintenance, in such companies, is often a department or function located in the final process of the main supply chain. Due to this fact, maintenance assumes not only its specific mission, but also may improve existing deviations in other departments’ missions to meet the global supply chain results.

Human capital has been found to be crucial for a proper maintenance [21], and in Network Utilities, according to our point of view and experience this is especially true in this particular type of companies where maintenance departments

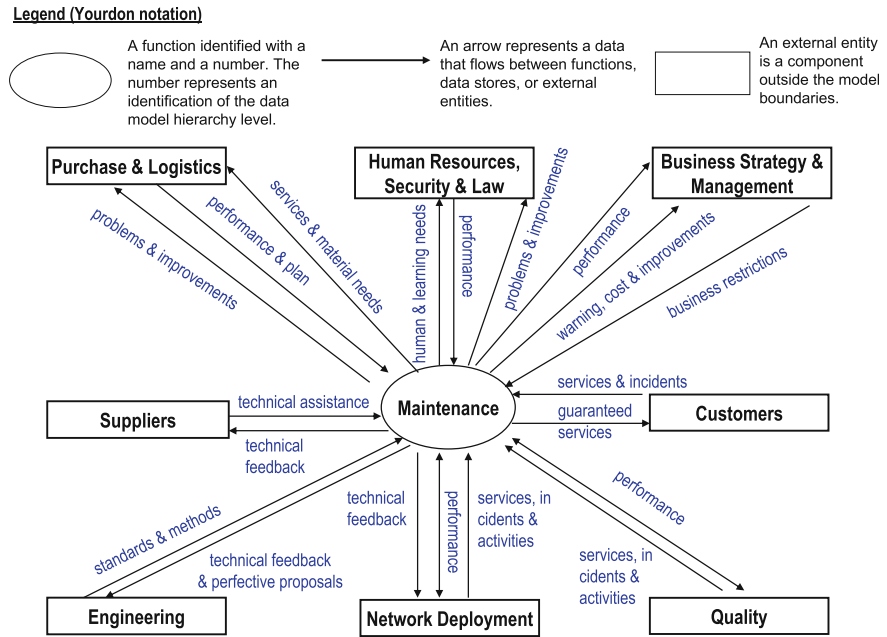


Fig. 1.4 Information flow of maintenance with other entities

hold an important number of operational relationships, having intense internal and external information exchange. Maintenance staff have to deal with complex technical aspects, flexibility, changes, adaptation, and they must have creativity, learning capacity, a willingness to work in teams with equipment, decision-making ability and the facility to work in multiple and interrelated operations. We may say the same as Prusak [80] that “the main source to create competitive advantages lies primarily in their knowledge, in what it knows, in how it uses what it knows and in its capacity to do new things”.

More precisely, in Fig. 1.4 we have modelled, using Yourdon notation, the information exchange for maintenance departments in network utilities. Maintenance not only has to interrelate with operational departments but also with other strategic or supporting departments:

- Engineering has to develop the instructions and methods to operate the network and services from an operational point of view, whilst maintenance feed-back is important to improve them.
- Network deployment has to build the infrastructure assuring that the present services will not be affected and offering all the necessary conditions and information to maintain it. Again, maintenance experiences are useful for network implementation.
- Suppliers support maintenance with technical assistance according to its requests and recommendations.

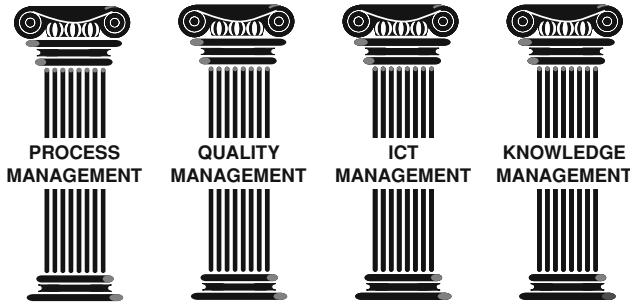


Fig. 1.5 Maintenance management supporting pillars for network utilities

- Some of the customer complaints about service level agreements have to be solved and reported by the maintenance department.
- Purchase and logistics have to (if we consider logistics within the central services) provide tools, vehicles, spare parts, and other services in adequate quantity and sufficient time fulfilling the maintenance needs.
- Human resources, security and law, have to take care of labour and environmental conditions, health and safety, and the corresponding quantity and performance of human resources.
- With regard to business strategy and management, maintenance, just like any other department, has to follow the strategy of the company, in line with the business goals and restrictions.

Maintenance management has to take into account all these circumstances for effective and efficient network, services and customers administration.

In this book we try to develop an integrated maintenance management framework to improve all maintenance decision-making processes in the case of Network Utilities. We try to overcome the simple use of typical management techniques, applying knowledge across the business and offering a comprehensive approach for the management of maintenance. Therefore, we need to focus on strong supporting pillars design for better maintenance management.

1.3 Supporting Pillars of Maintenance Management

A myriad of considerations, data, policies, techniques and tools affect the effective execution of maintenance, particularly in modern technologically endowed utilities. In such instances, an integrated, rather than the conventional “silo” style approach to maintenance management would play a pivotal role. However, much difficulty in the practice of maintenance management arises from the mix-up between the actions and the tools designed to enable them. This issue often remains unresolved by practitioners and unaddressed by researchers. To help

resolve this, we describe the essentials supporting pillars to enable the maintenance management process to yield the desired results in network utilities.

Due to our previous characterization of network utilities, we consider that the following four pillars or disciplines should have very good foundations for proper maintenance management in these type of companies (see Fig. 1.5): process, customers, knowledge, and information and communication technologies (ICT) management.

1.3.1 Orientation to Process Management

“In the first place, service is a process” [7], *“service is an act, a performance, an effort”* [82]. Through processes it is possible to orientate improvement to the efficiency in activities and resources management. When the maintenance function is focused on services, its activities are linked more easily with the company objectives, because the change in processes and organizations is facilitated [44].

According to Davenport and Short [18], a process is *“a set of tasks logically related to run for attaining a certain result of business”*. And according to [69], a process is *“a logic organization of persons, materials, energy, equipment and procedures in designed activities to produce a final specified result”*. There is, consequently, a double view of each process, a strategic view for its contribution to business and an operational view. Maintenance management should be carried out as a redesign using an integrated methodology. *“Radical business processes redesign to achieve a dramatic improvement of criticism measures about processes performance, such as cost, quality, service and speed”* [31].

During our research on network utilities companies, we have found that, independently of the type of service provided, (water, gas, etc.), the maintenance functions and management problems are similar. Then a maintenance management based on processes management, applying analogous abstractions for all these companies, seems to be a reasonable strategy. With that purpose, we intend to derive a unique framework considering similar processes to regulate maintenance performance in these companies.

In order to do so, and to develop different business applications for maintenance, it is necessary to have a methodology which allows a clear definition of the processes, their execution and data requirements.

Indeed, maintenance projects tend to deal with an important quantity of data and information: requirements, objects, relations, restrictions, functions, etc. An interesting opportunity here is to obtain and manage all that information structure using the Business Process Modelling approach.

We will propose the use of the Business Process Management (BPM) methodology in order to characterize maintenance management processes, with the aim to improve the efficiency through the management of business process that are modelled, automatized, integrated, controlled and continuously optimized [68]. This involves managing change in a complete processes life cycle.

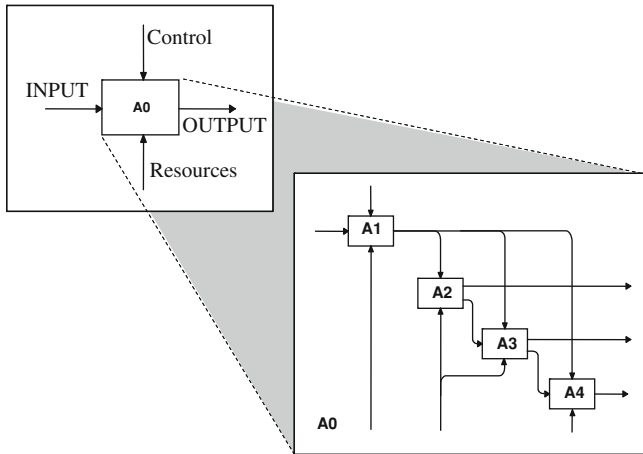


Fig. 1.6 IDEF, integration definition for function modelling

Using BPM methodology it will be possible to model a particular maintenance management process and afterwards “connect” this model with a general information system. In this way, a flexible management process will be created, that if required, could be quickly and even automatically modified into the enterprise information system [23].

Business process modelling (BPM) has frequently been used in industry since the 1990s, for example:

- To obtain a global vision of processes by means of support, control and monitoring activities [85],
- To facilitate the comprehension of the business key mechanisms,
- To be a base for the creation of appropriate information systems,
- To improve the business structure and operation,
- To show the structure of changes made in business,
- To identify outsourcing opportunities,
- To facilitate the alignment of the information and communication technologies with the business needs and strategies [6], and
- For other activities such as automatic documents processing [50].

Nevertheless, during recent years, the research in this field has increased enormously and with it the applications of technological advances, business process modelling has been used many diverse fields such as: planning of managerial resources (ERP), integration of managerial applications (EAI), management of the client relations (CRM), management of work flows (WFM) and communication among users to facilitate the management requirements [81, 85].

Some of the benefits reported from the adoption of business process modelling are: improvement of the accomplishment speed of business processes, increase of

client satisfaction, optimization and elimination of unnecessary tasks and the incorporation of clients and partners in the business processes [74].

In our case, we will use IDEF standard (Integrated Definition Methods) because [22, 28]: it is easy to employ and understand, allows for modelling processes oriented towards customers and indicators, systems and resources can be described. Besides this, there is a lot of software packages to implement these models, for instance: AXXIS (ARIS Toolset. August Wilhelm Scheer. Germany), BDF (Texas Instruments. Enterprise Integration Laboratory. Department of Industrial Engineering, University of Toronto), BPWin (Logic Work, Inc. Princeton) or Meta Workflow Analyzer (Metasoftware Corporation. Business Process Management Group—BPMG, Cambridge), amongst others.

This standard represents each phase/process of maintenance management as a box with inputs, outputs, used resources and control signals [36, 87] (see Fig. 1.6). In addition to these representations, the mission and vision of the organization, the activator and receiver of the process, and the set of indicators to control and guide the process towards the goals also have to be defined.

Software processes and Business processes present certain similarities. The most common similarity is the fact that both try to capture the principal characteristics of a group of partially tidy activities that are carried out to achieve a specific goal. The aim of a software process is to obtain a software product [1], while the aim of a business process is to obtain beneficial results (generally a product or service) for clients or others affected by the process [86].

In fact, the origins of the different business process modelling languages are inspired in the software modelling languages. A computer approach has defined this modelling as the “designing of software applications before coding” [68], this approach has allowed the development of several languages and applications for code generation and processes automation, which have notoriously increased in quantity and diversity especially during the last two decades.

There is a large number of business process modelling languages. An initial classification of these modelling languages can separate graphical from textual languages [96]. Unified Modelling Language (UML) is a well known graphical language used for specifying, visualizing, constructing and documenting systems. The UML has proven to be successful in the modelling of many large and complex systems [95].

Another possibility is to classify languages according to how they facilitate the understanding of business processes, although other graphical representation can be frequently employed to simulate or represent the information flows and its transformations inside the internal steps of processes. A good example of languages facilitating the process representation is the Data Flow Diagrams, DFD [52]. DFD allows the representation of:

- The sequence of steps in a process through procedures with standardized guidelines.
- The decision-making steps defining necessary information and tools.
- The functioning of information systems outlining them visually.

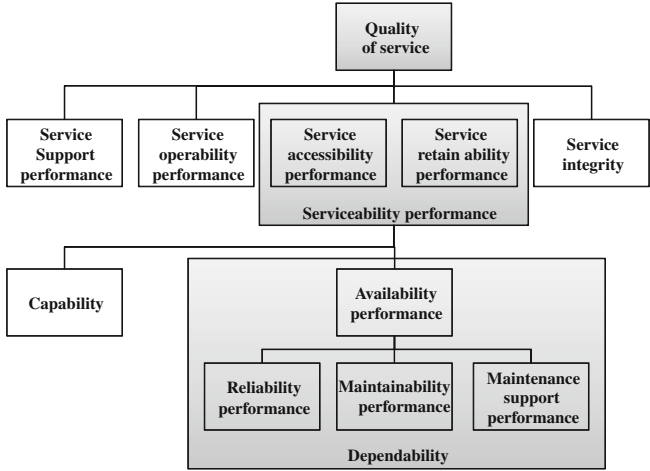


Fig. 1.7 The quality of a service [38]

As example of this, [43] emphasizes the use of DFD, as the previous employed DFD Yourdon Notation [98], for the design of information systems representing the exchange of data between activities. Although, for representing flowcharts, the most commonly used graphic symbols are those defined by the ISO5807 [42] standard or the IBM Flowcharting Templates.

1.3.2 Orientation to Customer Management

We know that ISO defines “quality” as the ability of a product or service to satisfy user’s needs. In the Network Utilities sector we are concerned with the quality of services provided through the network. In order to present a group of well structured technical concepts dealing with the quality of a service we have searched for those terms in Electropedia.org. Electropedia is produced by the International Electrotechnical Commission (IEC) [38], which is the world’s leading organization that prepares and publishes International standards for all electrical, electronic and related technologies—collectively known as “electro-technology”. Electropedia contains all the terms and definitions in the International Electrotechnical Vocabulary or IECV which is also published as a set of publications in the IEC 60050 series.

Within IEC 60050 series, area 191 is about dependability and quality of service. In this standard the quality of a service is defined (IEC 60050-191-19-01) as the collective effect of service performance which determines the degree of satisfaction of a user of the service. The quality of service is characterized by the combined aspects of service support performance, service operability

performance, serveability performance, service integrity and other factors specific to each service (Fig. 1.7). Let us review these definitions and concepts:

- *Service support performance* (IEC 60050-191-19-05) is the ability of an organization to provide a service and assist in its utilization. An example of service support performance in telecommunications is the ability to provide assistance in commissioning a basic service, or a supplementary service such as a call waiting service or a directory enquiry service.
- *Service operability performance* is (IEC 60050-191-19-05) the ability of a service to be successfully and easily operated by a user.
- *Serveability performance or serviceability performance* is (IEC 60050-191-19-05) the ability of a service to be obtained within specified tolerances and other given conditions when requested by the user and continue to be provided for a requested duration. Serveability performance may be subdivided into the service accessibility performance and the service retainability performance.
- Finally *service integrity* (IEC 60050-191-19-07) is the ability of a service, once obtained, to be provided without excessive impairments. For a telecommunication service, for instance, this is characterized by the transmission performance of the system used.

So far we have talked about the abilities of the service or of the organization, when we start to study the terms related to serviceability performance we realize that we also define item (assets) related concepts (IEC 60050-191-02), as follows:

- *Capability* is the ability of an item to meet a service demand of given quantitative characteristics under given internal conditions. Where internal conditions refer, for example, to any combination of faulty and non-faulty sub-items (for telecommunication services this is called trafficability performance).
- *Dependability* is the collective term used to describe the availability performance and its influencing factors: reliability performance, maintainability performance and maintenance support performance. Dependability is used only for general descriptions in non-quantitative terms.
- *Availability performance* is defined as the ability of an item to be in a state to perform a required function under given conditions at a given instant of time or over a given time interval, assuming that the required external resources are provided. This ability depends on the combined aspects of the reliability performance, the maintainability performance and the maintenance support performance (Required external resources, other than maintenance resources do not affect the availability performance of the item) defined as follows:
 - *Reliability performance* is the ability of an item to perform a required function under given conditions for a given time interval. It is normally assumed that the item is in a state to perform this required function at the beginning of the time interval. Generally, reliability performance is quantified using appropriate measures. In some applications, these measures include an expression of reliability performance as a probability, which is also called reliability.

- *Maintainability performance* is the ability of an item under given conditions of use, to be retained in, or restored to, a state in which it can perform a required function, when maintenance is performed under given conditions and using stated procedures and resources.
- *Maintenance support performance* is the ability of a maintenance organization, under given conditions, to provide upon demand, the resources required to maintain an item, under a given maintenance policy. The given conditions are related to the item itself and to the conditions under which the item is used and maintained.

We have described how serviceability performance of a system depends on its capacity and dependability performance. This is equivalent to saying that its serveability performance (that we could also define as service effectiveness) will be determined by its design, current configuration, operational status or condition, proper operation and—most importantly—its proper maintenance. A level of dependability performance to ensure proper service serveability performance is achieved by activating a series of processes of continuous improvement that systematically incorporates diagnostic tools, analytical methodologies and new technologies, to optimize the design, management, planning, implementation and control associated with the operation and maintenance of those assets configuring our network. It is therefore necessary to act in a comprehensive manner on those assets, from their design to their operation, as well as on those aspects related to organizational processes and people.

Once reviewed *technical concepts* dealing with the service quality, let us now look at the service quality concept from a *business perspective*.

In a given business context different customers may pay different importance to the distinct features determining the quality of a service. We have to translate in a practical manner, these customer expectations into service features. Quality will then be inspected and measured through attributes or characteristics determining the perceived service quality by the customer. For that purpose, surveys of consumer groups may derive the main quality attributes providing value for customer satisfaction.

We will now present a generic example (see Table 1.3), built with information from different authors, where different product/service attributes are listed. For instance, Juran [48] defines five groups of quality characteristics in relation to the customer perspective: time-oriented (such as reliability or availability), contract (such as security or warranty), structural (that of shape, length or width), sensory (for example, taste or appeal) and ethical (such as sensitivity and courtesy). On the other hand Garvin [25] describes eight dimensions of quality: performance (basic functions required by users), features, reliability, conformance (as standards), durability (service life), serviceability, aesthetics (odor, taste, touch, etc.) and perceived quality. According to Parasuraman et al. [71] ten are the demanded attributes of service quality: reliability, security, competence (of the organization to develop the service), responsiveness (of the organization), credibility, tangibles, accessibility (for

Table 1.3 Attributes of service quality from a customer point of view (survey)

Juran	Parasuraman et al.	Garvin
Time-oriented (reliability, availability)	Reliability	Reliability
Contract (security, warrantee)	Security	Durability
Structural (shape, length)	Tangibles (of the service)	Features
Sensory (taste, appeal)	Understanding the customer	Conformance
Ethical (courtesy)	Responsiveness (of the organization)	Serviceability
	Competence to develop the service	Performance
	Credibility	Perceived quality
	Courtesy	Aesthetic
	Communication (suitable and updated)	
	Access (for the customer)	

customers), communication (adequate and updated to the client), courtesy, understanding and knowledge about your customer. Consistent with the above attributes, we can conclude that it is harder to measure quality in services than in products [65] because of the greater weight of the intangible characteristics that we can find for services. Finally, we have seen that many market research surveys about service quality simplify these features, due to correlations amongst them, into five attributes: tangibles (of the service), reliability, responsiveness capability, assurance and empathy. Nevertheless, the reader must remember that “*every customer perceives service quality differently*” [76], in his/her own terms, and by a series of conditions that he/she determines. These interactions are called “critical incidents” [8, 98]. Therefore, through a study of critical incidents for customers, we can also evaluate the perception about maintenance. These critical incidents could be considered, for instance in a network utilities maintenance context, as the complaints of customers due to failures in the service.

In addition, from this last business perspective, and especially for network utilities companies, we find maintenance to be crucial because of its contribution to the improvement of the quality of services. Maintenance improves the service tangible features through prevention, reaching best possible network and service durability and reliability, improving responsiveness or the capacity to face contingencies, ensuring feedback to the design phase with experience, and improving other intangible aspects such as credibility.

In previous paragraphs we have explored the service quality concept from a technical perspective and also from a business viewpoint. In both cases we have identified the importance of maintenance for reaching service quality. Therefore, we have to consider quality management as a pillar of our maintenance management in Network Utilities, or in other words, we have to understand the service quality management process in order to develop our business maintenance management process.

Table 1.4 Opportunities of ICT in reengineering of business process (BPR)

Technologies	Advantages	Opportunities
Computation	Reduce production cost	Automatic tasks Decrease faces in information process Eliminate tasks
Communication	Reduce coordination cost	Reduce times and distances Tasks and process integration Collection and distribution of information
Storage and Systems	Reduce information cost	Process and tasks monitoring Information Analysis and Making decisions Archive and development of skills and experience Modelling and display of processes

[Source: 20]

1.3.3 Orientation to ICT Management

Information and Communication Technologies (ICT) can greatly optimize maintenance management, due to the proper exchange of updated information and the coordination through automatic procedures (see Table 1.4) [19]. In addition, the information systems should facilitate and improve the business processes quality and productivity [75].

Information systems help us to optimize maintenance management, its knowledge and decision-making processes [15, 19, 34, 49, 53]. This factor is even more important in network utilities because of their specific infrastructure characteristics [29].

In order to illustrate this point we can refer to several studies that show how the complete implementation of Computerised Maintenance Management System (CMMS) could reduce the annual maintenance budget by 10 to 30% [12, 78] ensuring at the same time a much better maintenance tasks control.

There are several key benefits of ICT systems [3,19, 34, 70, 78, 79] that can be specifically highlighted for maintenance in Network Utilities:

- Real time information about infrastructure configuration, service history and customer relationships.
- Automation and efficiency by eliminating duplication and/or non-value tasks, including automatic notifications, thus improving productivity.
- Analysis of possible interdependence among infrastructure, its variables, services, processes and existing customers.
- Provision of higher levels of intelligence and modelling layers, allowing automatic and fast root cause and weak point analysis.
- Visibility of business operations and their impact on performance, availability and safety of services. They allow a better vision (end-to-end) and control (decision making) toward the goals or objectives.
- Assessment of critical risks and priority assignment.

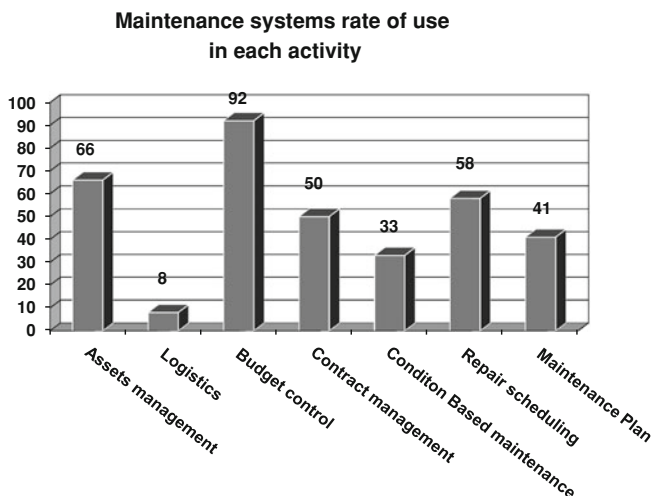


Fig. 1.8 Maintenance systems ratio of use (source [47])

- Alignment of maintenance and other departments with the business objectives.
- Integration through powerful applications going beyond the companies boundaries.
- Effective activities implementation control. Effective control of other partners and improvement in the use of assets and staff. Reductions in spending and investments, improving the ROI (Return On Investment).
- Knowledge management, standardization and unique source of information (timing), reducing problems related to data quality.

Nowadays, in Network Utilities, we can rely on information systems to generate automatic procedures and to facilitate decision-making with minor dependence [10] on: network size; technology; people; cost of human resources and time. However, a survey from a wide range of industries including network utilities, published by the Plant Maintenance Resource Center—PMRC [78], has revealed that although almost 82% of interviewees have implemented a CMMS (the most adopted are SAPTM, MAXIMOTM and MP2TM), only 77% of them (often managers and maintenance supervisors) do know the reason why their company implemented it. Furthermore, CMMS use has been found to be limited to the cost accounting, spare part management and scheduling activity at least in a medium-large sized company. Small-medium companies often do not adopt CMMS. The following figure shows the percentage of system utilization in each maintenance activity [47] Fig. 1.8.

As a consequence of the above mentioned points, the proper exchange of information and the coordination for maintenance is a core activity [72] and many authors confirm the requirement of an updated maintenance information and documentation system as compulsory when trying to create company competitive

advantage through excellent maintenance management. There is still room for improvement in this area [4]. Here, an important point is the interoperability among different systems, this should be ensured for the exchange and coordination of information.

Focusing on network utilities, there are four special issues concerning ICTs in these type of companies:

- Remote access to different sites is complicated due to huge geographical distances.
- Complexity, due to the different types of elements from different vendors, needs to be managed.
- Dynamic and automatically configured work that has to be released, considering these hierarchical multiservice networks, according to the different required capacities; and
- There is a need for scalability in relation to the network growth.

Therefore, before we start the development of any CMMS, the scope of the project (considering minimum requirements) should be determined as well as the execution time and the cost available. This has to be done objectively and realistically (going beyond the logic needs can lead to an “*analysis paralysis*”, see [46]).

We do recommend that the CMMS, as with any other enterprise system, be developed according to the following principles:

- Flexibility, to allow changes in the organization and processes without affection.
- Modularity, for easy management and improvement.
- Scalability in functionality and cost, with the aim of adapting to the requirements of each time in a stable manner with the growth.
- Practical, facilitating understanding and avoiding complexity, the system should eliminate redundant tasks and reduce the administrative burden.
- Interoperability with the rest of the company systems, an open system architecture is recommended in order not to jeopardize potential improvements.
- Confidentiality and integrity, the access and the use of the systems have to be risk-free.

At present, taking into account previous considerations and due to the rapid implementation of new communication and information technologies in maintenance, we can talk about “e-maintenance” [83, 37]. This is a new emerging concept which defines the “*capacity to monitor assets of infrastructure, linking production and maintenance systems, gathering customer feedback from remote sites, and integrating that with top level business applications*”. A more general definition is that e-maintenance is a “*maintenance management concept whereby assets are monitored and managed over the Internet*” [14].

Although the e-maintenance term has been in use since 2000 as a component of e-manufacturing, there is still no standardized definition of e-maintenance, given by any official institution. Different engineers and scientists have considered e-maintenance in different manners: as a philosophy; as a strategy to obtain real

time data; as a support for decision-making; as a tool to make predictions or prognosis analysis, and other heterogeneous definitions [57]. In 2006, Muller et al. [64] stated that the term “e-maintenance” was not yet consistently defined in maintenance theory and practice. Engineers or scientists might consider e-maintenance as a concept, or as a philosophy, or as a phenomenon, etc.

From a pragmatic point of view, and with the aim of summarizing, we may say that e-maintenance is “*the set of maintenance processes that uses the e-technologies to enable proactive decisions in a particular organization*” (definition partially derived from [57]). Those maintenance processes and the methods required to operate them (technologies, procedures, standards, etc.) are specific for each type of organization, according to its particular characteristics, as in the case of network utilities.

This means that, for similar organizations, e-maintenance could involve a similar sphere of activities. Frequently these activities are defined as: e-monitoring, e-prognosis, e-diagnosis, e-management, e-service, remote maintenance, collaborative maintenance... and many other names. Such activities are supported by means of a variety of hardware and software technologies for example, the wireless and mobile devices, embedded systems, web based applications, P2P networks, multi-agent applications, specific software architectures, communication protocols, amongst the others.

In literature, there is also a great amount of contributions analyzing and proposing different e-maintenance processes and their applications using a myriad of e-technologies. From this variety of diversified information, some authors have made the effort of classifying the e-maintenance processes, thus contributing towards the structuring of this new discipline. For example, Levrat et al [57] determine four classes of possible e-maintenance applications: for strategy, for diagnosis and prognosis, for maintenance policy assessment and scheduling, and for maintenance deployment and implementation.

Karim et al. [51] summarise and classify the multiple industrial/academic artefacts that can be used for the information exchange in e-maintenance. Indeed, the information exchange standards are classified as maintenance specific contributions (like S1000D [59], PROTEUS, etc.) and generic contributions (like XML, [39] and [41]).

1.3.4 Orientation to Knowledge Management

The maintenance function requires up-to-date data, information and ultimately knowledge about the assets. This is a must for proper planning, scheduling, execution and improvement of their maintenance activities. The development, management and distribution of assets maintenance knowledge is considered as a foundation for the continuous improvement in maintenance management [77, 88, 93].

Davenport and Prusak [17] define knowledge as “*a fluid mix of framed experience, values, contextual information, and expert insight that provides a*

framework for evaluating and incorporating new experiences and information". Maintenance knowledge can therefore be managed through practices and experiences, and represented in a series of procedures and techniques.

According Dandois and Ponte [16], Knowledge Management (KM) can be addressed from four distinct perspectives:

- Strategic, where the KM has to be guided in line with the business strategy requirements or in response to external changes (market, competitors or reputation).
- Structural, where the KM is supported by an internal structure based on processes, searching its effective implementation.
- Technological, where the KM takes advantages of the technological process possibilities for its improvement in terms of quality and speed management.
- Human, where under the KM principles, the human capital is transformed focusing on promoting positive and collaborative attitudes towards the innovation and the exchange of knowledge.

The reader may realise how we have been considering these four perspectives when defining our maintenance management approach and the corresponding supporting pillars. The pillars refer to each of the four following KM perspectives: our strategic approach is service and customer-oriented; our structural approach is based on processes; our technology-oriented approach is focused on ICT possibilities for improvement; and our human approach, valuing intellectual capital, pivots or depends on knowledge.

In utilities, we have experienced that there is a large amount of dispersed knowledge in the organizations. This knowledge is frequently unprofitable, unknown or inaccessible, and therefore cannot help to improve what we have mentioned as the *base operational scenario*, where we may find:

- High dedication to the resolution of incidents and problems, deciding with scarce knowledge the adequate movement and solution along the network.
- Non-automated operations, manual dedication to repetitive tasks.
- Reactive management, with no real-time response.
- Isolated transmission of information and knowledge into islands of specialties or departments, which hampers teamwork and improvements in intergroup activities.

In network utilities maintenance the speed as well as the rigor of decisions are key issues, KM tools can therefore become crucial tools. These tools may help in reducing the time and resources devoted to decisions to solve repetitive problems and can also facilitate the maintenance organization strategic alignment, avoiding silos of knowledge among different areas.

Maintenance knowledge is frequently tacit, empirical and dispersed among technicians working in different shifts, 24 h a day, 7 days a week. In addition, to make explicit knowledge out of tacit knowledge requires e-learning (remote), "training on the job" [73, 77, 88], coaching, mentoring and producing reference books for each maintenance activity [32]. These activities are difficult to

accomplish, due to the fact that many procedures and technical instructions refer normally to equipment under “laboratory conditions” and not to real equipment operating conditions. Obviously, when equipment is distributed in the field, it suffers from a wide range of diverse environmental effects. Consequently, as we have mentioned above, the coordination and the exchange of information and documentation is a core activity for maintenance [72], guiding activities results to match or exceed the desired results.

Thanks to KM, the maintenance organization can be more versatile and with a multipurpose staff. Flexible maintenance organizations should be enhanced to facilitate the exchange of knowledge and the teamwork in a confident and motivational environment, avoiding obsolescence and focusing on continuous improvement. It has also been proven [32] that investments in ICT and human capital have to be adequate to enhance knowledge management:

- IT investments in order to reuse and share knowledge, not only purchasing any Knowledge Management Software on the market, but also to promote the generation of new ideas and best-practices.
- Capturing and creating good professionals [5] according to their competences (know-how, abilities, attitudes and motivations) and their willingness and interest to contribute to the company strategy and performance [58].

Consequently, human resources have to be evaluated not only by their operational performance, but also by their behaviour concerning issues such as commitment, teamwork, trusting relationships, multi-tasking environment, creativity, initiative, knowledge sharing, flexibility, learning, decision-making and the use of tools and ICT.

There is a close relationship between ICT and KM, due to the fact that information and communication technologies impulse the KM because of their advantages in the administration and transference of information:

- Managing the activities and documenting the transactions, in short generating accessible knowledge automatically or immediately.
- Presenting the information necessary for the management and control of the company.
- Providing support for decision-making through visual and symbolic representations, simulations and analysis.

From a market point of view, the value of an organization is increasingly associated with intangibles. Knowledge of human resources is difficult to assimilate and/or reproduce by competitors. This knowledge, the company collective experience, creates value throughout the organization with maximum effectiveness for present and future times, or even to respond to unexpected environmental changes.

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Chapter 2

International Standards, Best Practices and Maintenance Management Models as Reference

2.1 Introduction

This chapter will introduce the reader to a set of standards, best practices and management models that have been considered for the elaboration of the following sections of this book. Obviously we cannot build our framework from zero, there are very relevant references in previous works, good practices and successful stories that we can consider when designing a framework for network utilities maintenance.

For the identification of these tools we have reviewed different types of standards and for different sectors where they have proven to be effective. The backgrounds of the authors, in addition to their experience gained in different projects, has led this chapter to an interesting discussion regarding the convergence of different standards from the telecom and the industrial sectors at present. This has been found to be extremely relevant and something that will facilitate the process of systems interoperability.

As we will discuss later, interoperability is a must for the integration of industrial automation systems with operation and maintenance systems (O&M systems). The definition of the requirements in order to improve the collaboration and interoperability among these systems along the products/assets life cycle is a key topic for the future of service distribution networks maintenance. Nowadays, this convergence requires special attention due to the huge amount of development that can be appreciated in private and open standards.

Besides systems interoperability standards development, a review of new and advanced maintenance management models is also presented in this chapter. The idea is to offer the reader different clues and perspectives for the future design of network utilities operation and maintenance management systems.

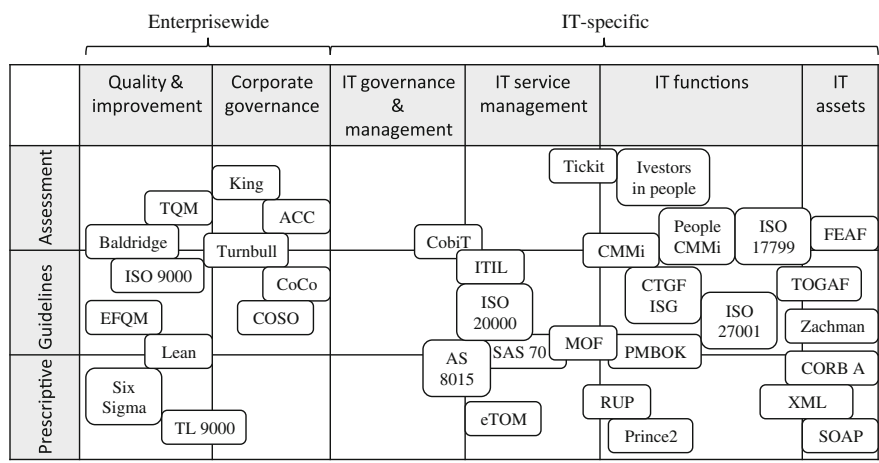


Fig. 2.1 International standard frameworks (Source Gartner, Inc., www.gartner.com)

2.2 Process-Oriented Standards and Best Practices

Concerning business process models, we have found that in network utilities there are many standards capturing best industry practices for the whole sphere of the company and others that are specific to Information and Communication Technology (ICT). We can therefore use them and learn from them to seek competitive advantages in network maintenance management (see Fig. 2.1).

Some of these frameworks are the most common in use as foundations of maintenance management schemes and are presented in Table 2.1.

Unfortunately, “no single model fits all conditions”. Standards and information systems should facilitate and improve business processes productivity [60], although processes cannot be at the “strong mercy” of standards. An interesting thing to note is that a rigorous application of any standard could expose the objective of its implementation to danger. This could even put at risk the proper operation of the company [34].

We now present different models and frameworks, defining the main reasons why we believe they offer good support to our network utility maintenance management process and framework:

- EFQM [21]. The European model of excellence from the European Foundation for Quality Management (E.F.Q.M.) is a model used as reference to maximize the management efficiency continuous improvement, coordinating activities and resources of the company.
- ISO9001 [39]. The family of ISO 9000 standards is a list of procedures concerning quality management used to develop management reference frameworks, as a guide for quality and processes orientation.

Table 2.1 Employed standards as foundation

Model	Basic description	Reference
EFQM	Management by processes and quality	EFQM [21]
ISO9001	Management by processes and quality	ISO 9001 [39]
TPM	Maintenance and quality	Wireman [86]
ITIL	e-business and processes	ITSMF [41]
eTOM	Network management and processes	eTOM [23]
CMMI	System management and processes, to evaluate maturity of companies	CMMI [16]
COBIT	As a model for audit and optimisation of processes	Cobit [18]

- TPM Total productive maintenance, as presented by the Japanese Institute of Plant Maintenance (JIPM), is a maintenance program centered on machinery to reduce quality failures, breakdowns and accidents and, to improve the productivity and motivation without quality reduction [55, 86]. TPM is used as a guide in maintenance, paying special attention to the knowledge of personnel.
- ITIL (ITSMF) [41]. Information technology infrastructure library, from the Office of Government Commerce (OGC) UK, is a framework about the best practices related to the delivery of IT services, searching quality and effectiveness.
- eTOM [23] Enhanced telecommunication operations map, from the Telemanagement Forum (TMF), is composed of a group of enterprises supplying services or telecommunication applications. This is a process reference framework for telecommunication organizations with the aim of guaranteeing the interoperability in network management, business systems and operating systems. eTOM describes the necessary processes to automate and interconnect systems or elements.
- CMMI [16]. Capability maturity model[®] integration (CMMI), belongs to the Software Engineering Institute (SEI) research center, sponsored by the USA Department of Defense and managed by the Carnegie-Mellon University. This is a framework to improve processes-oriented toward service delivery. It provides a list of the essential elements of management to reach a determined level of maturity in management to make efficient processes [15, 54]. Among its versions is the remarkable CMMI-SVC V1.2 [17], which is a processes guide for service providers to establish, manage and deliver services integrating knowledge. The CMMI is now proposed to evaluate the involved process areas implicated in maintenance. Its main interest is to show the contribution of the management in the improvement of the maintenance activities through an evolutionary path, which may spread from an unstable management toward a more mature and disciplined management. The maturity is quantified in five progressive levels: this helps to take into account the evolutionary path that can be followed during a maturity growth. In the following Table 2.2 the significance of each score is explained.
- COBIT [18]. Control Objectives for Information and related Technology, from Information Systems Audit and Control Association (ISACA) and IT Governance Institute (ITGI), is a best practices framework for the IT management, using a set of generally accepted control objectives. This reference could be utilized to define the objectives and practices in any framework concerning ICT facilities.

Table 2.2 Significance of the five progressive levels of CMMI

Level	Description of the level
Level 1 Initial	This represents a process with unpredictable result. The process is unstable and unorganised. It is defined by who performs it without explicit procedures
Level 2 Repeatable and reactive	This represents a process characterized by repeatable performance. The process is planned, implemented, monitored and checked according to pre-defined objectives
Level 3 Defined and proactive	This represents a process characterized by a proper implementation program in the company. The process is based on well-defined methodologies, techniques and supporting technologies. Proper procedures are established to drive this process
Level 4 Managed	The process is controlled, adopting quantitative techniques and, if this is the case, statistical analysis. The business objectives are checked by the comprehension of the results of the quantitative analysis
Level 5 Optimized	Focused on continuous improvement aligned with business objectives. A corporate policy is established to manage the quality of the process, based on quantitative data and feedbacks about the processes. If this is the case, also new methodologies, techniques and technologies are tested

Once we have looked over the basic international standards relating to business process models, we will also analyse the state of art of others specific standards in the area of the ICT. More precisely, we will look for standards considering the interoperability principle as a rule, a regulation, an assessorship, a norm or a requirement.

2.3 ICT Standards for Systems Interoperability, Integrity and Scalability for Maintenance Management

Different recommendations for ICT systems and their interoperability have been analyzed from European, USA and worldwide organizations. See the next Figure for the encompassed organizations (Fig. 2.2):

- *International Standard Organization (ISO)*. Within the ISO there are specific technical committees dedicated to defining standards on specific issues, except those related to telecommunications engineering (which is responsibility of the International Telecommunications Union—ITU) and electrical engineering (a responsibility of the International Electrotechnical Commission—IEC). There we find two committees for the development of software:
 - JTC1 concerning information technologies
 - TC184 concerning industrial automation systems & integration.

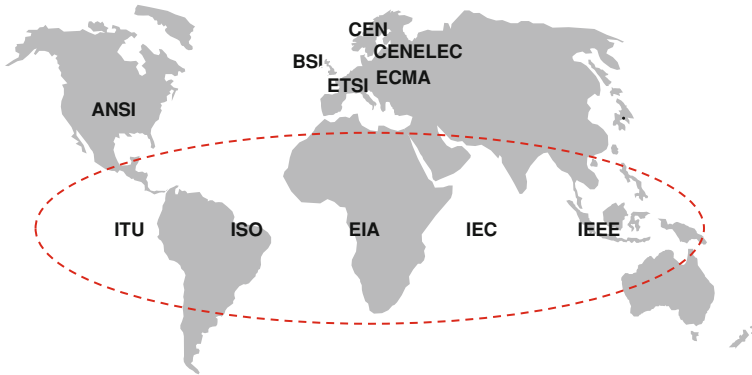


Fig. 2.2 ICT international organizations for standardization

- *International Telecommunications Union (ITU)*. Establishes the telecommunications standards.
- *International Electrotechnical Commission (IEC)*. Focuses on the standardization of electrical and electronic technologies. It is centered around the committee TC56 concerning Dependability.
- *Institute of electrical and electronics engineers (IEEE)*. Focused on electrical and electronic technologies. One of the most important bodies is the committee Computer Society Software Engineering Standards Committee (SESC).
- *American National Standards Institute (ANSI) and National Institute of Standardisation and Technology (NIST) from USA*. The Information Systems Conference Committee (ISCC) focuses on the development of IT standards and the ANSI ISO Council (AIC) develops the relationship with ISO.
- *The European Conference of Posts and Telecommunications Administrations (CEPT)* is an association of telecommunications companies, in which in 1988 the European Telecommunications Standards Institute (ETSI) was created.
- *Comité Européen de Normalisation (CEN)*. Standardizes ICT is concerned mainly in security of customers and the environment.
- *Comité Européen de Normalisation Electrotechnique (CENELEC)*. Is centered on the definition of electro-techniques standards.
- *British Standards Institution (BSI)*. Its PAS 99 (publicly available specification) is concerned with management systems, PASS 55 (optimal management of physical assets) about lifecycle management of capital investments minimizing risks and their integration according to ISO standards.
- *Electronic Industries Association (EIA)*. There are specific organizations in relation to:
 - G-33 for data management and configuration.
 - G-34 concerning the software.
 - G-47 centered on systems engineering.

- *European Computer Manufacturers Association (ECMA)*. Is an association of suppliers that in cooperation with ISO, IEC, CEN and CENELEC, ETSI and ITU, develops standards about ICT and electronics of consumption.

For the last two decades, these organizations have been working together to define standards on information technology and its applicability in all company space, seeking a global system reference framework that integrates operations and knowledge in all company functions. There are an important number of standards that have been developed to communicate systems, facilitating knowledge transfer [29]. Such standards consider companies operating in different sectors, systems of different hierarchical functional levels and/or they apply to different moments along the enterprise life cycle. At the same time, and specifically in network utilities, one can find different types of networks that may also require our attention in terms of communication issues. Therefore, to support this, they mainly focus on the internal structure of the systems and on the external communications with other systems:

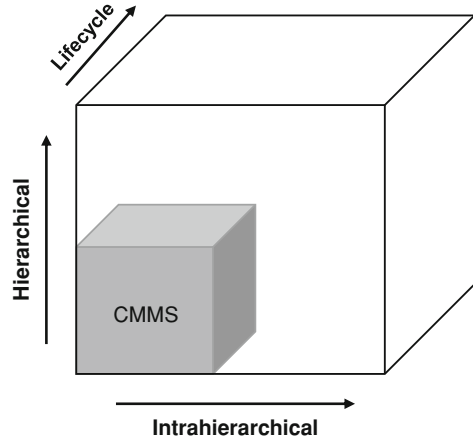
- *Software engineering or the establishment and use of sound engineering principles to economically develop software that is reliable and works on real machines efficiently* [24]. Mainly based on the principles of integrity and scalability, relating to applications and how information is processed in the business or assets.
- *Interoperability or ability of two or more systems or components to exchange and use information* [33]. Relative to communications and how information is transferred or stored.

The design of the interconnection among systems requires a special study; there are many reengineering projects that have failed due to poor system definition or their interactions [12]. The implementation of systems is a specific and distinguishing feature among companies, even within the same sector. For example, liabilities and departments are not defined in the same way, moreover within the same company there are different levels of information (information of control systems or manufacturing systems), different information flows and, communication among diverse assets with dissimilar languages, so it is valuable to optimize the quality and applicability of the existing information in order to improve the life cycle of the company.

Given the large number of suppliers of existing systems, it seems reasonable the use of the interconnection standards in order to facilitate operations, avoiding supplier dependency and not limiting future developments of the systems. A large number of standards are developed to interconnect systems depending on:

- The functional hierarchy level: strategic, tactical or operational.
- The application field: engineering, industry or IT [76].
- The application throughout the life cycle (or value chain): for customer relations, production, operation and maintenance, planning, economic/financial, human resources, design and engineering, etc.
- Communication between computers or humans.
- Property: public (open) or private (proprietary).

Fig. 2.3 Three levels of system integration



Modern enterprise interoperability is characterized by [68] in three types of integration (see Fig. 2.3):

1. Vertical integration among the company hierarchy.
2. Horizontal integration within each hierarchical level of the company.
3. Temporal and longitudinal integration along the life cycle of the product or service.

The means of transportation for information also influences the design of interconnection, that is, you can use different types of communications infrastructure: public or private, fixed or mobile, local or global, owned or shared, etc. The interconnection must be done in a controlled and limited way for greater operability and security. Internet is the most used means of transportation in e-Business to interconnect systems using the TCP/IP protocol.

In the case of enterprise operating systems, the development and the standardization have followed different paths depending on the application field, one focused on the industrial sector, and the other focused on the telecommunications sector. In these two sectors, one can highlight international efforts undertaken by large companies and public organizations to establish open interoperability solutions.

Let us review emerging standardization efforts about interoperability in each sector, focusing on the most widespread and accepted for operations and maintenance.

2.3.1 Interoperability Standards in the Industrial Sector

A representative case of collaboration in the industrial sector of electric, water and gas utilities is lead by the not-for-profit corporation UCA[®] International Users

Group (see <http://www.ucaiug.org>), consisting of utility user and supplier companies that is dedicated to promoting the integration and interoperability through the use of international standards-based technology for real-time applications, focusing mainly in open and public standards. It is a User Group for IEC 61850, the Common Information Model—Generic Interface Definition (CIM/GID as per IEC 61970/61968), advanced metering and demand response via OpenDR. The Users Group does not write standards and shall, where appropriate, work closely with those bodies that have primary responsibility for the completion of standards (notably IEC TC 57: power systems management and associated information exchange) in areas of interest where standards bodies may not yet be active or where the interests of users goes beyond the purview of the presently identified standards (such as the completion of users guides, industry education, transfer of technology, marketing support, identification of users needs and industry demonstrations to prove concepts).

On the other hand, from a broader perspective than the utilities, the technical committee ISO TC184 is dedicated to the industrial automation systems and their integration for operation and maintenance. This committee leads the definition of requirements to improve the collaboration and interoperability among O&M systems along a product life cycle. Inside TC184, there are several subcommittees and working groups. For instance, subcommittee SC5 focuses on architecture, communications and integration frameworks. This subcommittee deals with the definition of standards in collaboration with other organizations such as MIMOSA “Machinery Information Management Open Systems Alliance [50], ISA-SP95” International Society for Automation, and OPC “OLE for Process Control [58].

Considering applications within the industrial sector, we can find a great variety of standards such as: “OLE for Process Control” [58], “Condition monitoring and diagnostics of machines” ISO 13374 [40], “Industrial automation systems and integration—Diagnostics, capability assessment, and maintenance applications integration” ISO 18435 [37], “Machinery Information Management Open Systems Alliance” [50], “Enterprise-control system integration” [38], etc.

More recently, a nonprofit and coordinated approach to exchange operation and maintenance data using the open standards has emerged as a global trend pursuing systems interoperability, this standard is OpenO&M “Open Operations & Maintenance”. The initiative involves the collaboration of different organizations concerning industrial standards like ISO, OPC and MIMOSA [59].

As a set of harmonized standards, OpenO&M mainly includes the following references:

- ISO 18435 [37] “Industrial automation systems and integration. Diagnostics, capability assessment and maintenance applications integration”. This standard defines a set of models and interfaces for the vertical, horizontal and temporal integration of information about production, diagnostics and maintenance of industrial systems. It takes into account the entire life cycle of the systems and is based on knowledge generated about the state/condition of the assets. These assets may suffer reconfiguration according to certain circumstances and

therefore information systems are required to seek for efficiency and effectiveness between their operation and maintenance.

- In conjunction with the ISO 18435, ISO 15745 [36] (Application Integration Framework) sets the standard framework for the integration of applications based on UML models and XML schemes. Four hierarchical levels of information are identified:
 - Level IV: business planning and supply chain management plan, Corresponding to the enterprise or site decisions, supervision and scheduling in a short-time, months or weeks, about production, operation, logistics, etc.
 - Level III: manufacturing operations management and control information. Related to local area decisions, supervision and scheduling in a day to day basis. Decisions are concerned with production, capability, maintenance, etc.
 - Level II: batch, continuous and discrete control. Concerning sub-areas (or production units) control, prognosis, supervision and nearly real-time monitoring of physical processes.
 - Level I: sensing and manipulating the production process. Regarding units or asset utilization, configuration and data acquisition.
 - Level 0: resource identification and location. Dealing with human resources, material resources (hardware and software), facilities, documents, consumption services and assets. Valuable to classify and allocate them, not only to production, but also to monitoring, storage and communication.
- ISO/IEC 62264 “enterprise-control system integration” [38] (also known as ISA-95) defines an integration model of the different company control systems. It is a standard of automated communication among control and business systems with the ability to be applied in any manufacturing environment, in all industries and in all types of processes. It provides a unified terminology for activity models, defining the transitions among them with their models of objects and attributes of information. This standard has been chosen by major system suppliers of Manufacturing Execution System (MES) and Enterprise Resource Planning (ERP) systems such as SAP. This model is used to exchange data among the operation and maintenance activities at various levels, defining the information flows among management activities through the use of generic activities for modeling as seen in the Fig. 2.4.
- MIMOSA Information Standards “operations and maintenance information open systems alliance”. Defines the exchange of information among the operation and maintenance of industrial assets. Offers architecture to structure the operation and maintenance information in open systems. MIMOSA identifies three fundamental parts (see Fig. 2.5):
 - a. Business management applications (OSA-EAITM) “open systems architecture for enterprise application integration”;
 - b. Condition-based management (OSA-CBMTM) “open systems architecture for condition-based maintenance”; and
 - c. Reliability centered management.

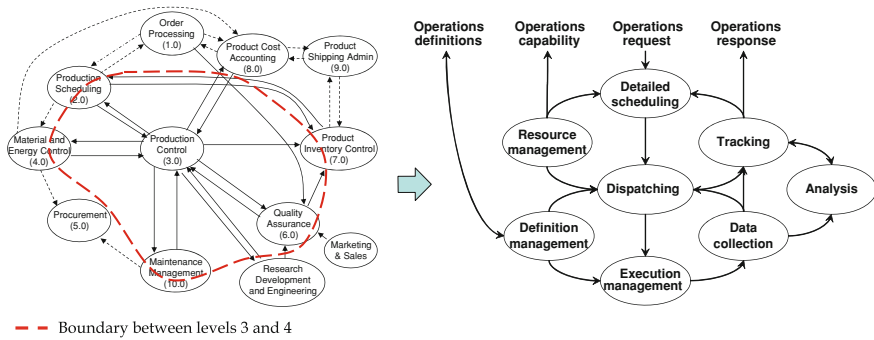


Fig. 2.4 Generic activity model and processes per ANSI/ISA S95

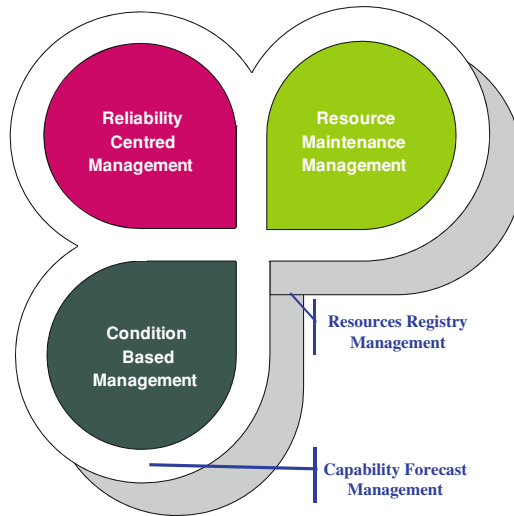


Fig. 2.5 MIMOSA architecture

- By integration of these previous parts MIMOSA ensures proper “Capability Forecast and Resources Registry Management” producing appropriate predictions and a consistent inventory of system resources.
- Recent standards related to condition monitoring, such as: ISO 13374 “Condition monitoring and diagnostics of machines” [40] describe the assets data acquisition, data manipulation and diagnosis.
- OPC Foundation standards. These standards are focussed on data acquisition and communication among assets based on Microsoft’s technologies OLE COM (component object model) and DCOM (distributed component object model). Within these assets we can include sensors, instrumentation, PLCs, RTUs, DCSs, HMIs and historical alarm systems, etc.

- OAGi Standards “open applications group”. Standards concentrated on the integration of enterprise-level applications. The standard OAGIS (open applications group integration specification) defines the requirements to exchange data among business systems (B2B, A2A), including operation and maintenance systems. This standard is integrated inside OpenO&M by a XML world-class solution B2MML “Business To Manufacturing Markup Language”, developed by WBF “World Batch Forum” in collaboration with ISA95 and OAGIS (XML Working Group).

2.3.2 *Interoperability Standards in the Telecommunications Sector*

Looking at the telecommunications sector, we can also find reference frameworks to establish interconnectivity and communication incorporating different elements into a single network management and control [6]:

- Open System Interconnection OSI;
- ISO 10040 [35]; and
- Telecommunication Network Management, TNM (ITU-T) [77].

More precisely, M.3100 recommendation from the International Telecommunications Union (ITU) describes another hierarchy management model for networks that can be considered similar to OpenO&M. This standard defines a unified management of assets and services, integrated into a common platform of levels and functional areas [66]. It describes four hierarchical levels of management:

- *Element management*. The model’s lowest level concerns control of single elements. It manages a subset of network elements, maintaining statistical records and other information about the elements.
- *Network management*. As an overview of the network, it operates network capabilities to support customer services, using statistics, records and other information about the capabilities of the network.
- *Service management*. Related to services customer, it is an interface with the customers, controlling contractual aspects of the service, performance, usage, etc., and maintaining statistical data about QoS “Quality of Service”.
- *Business management*. The upper level focuses on the overall management of the company from a business point of view.

Across these four levels of TNM, five functional areas are developed, with some similarities to the recommended operational processes of ITIL, but characterized by different levels of abstraction and features (see Fig. 2.6):

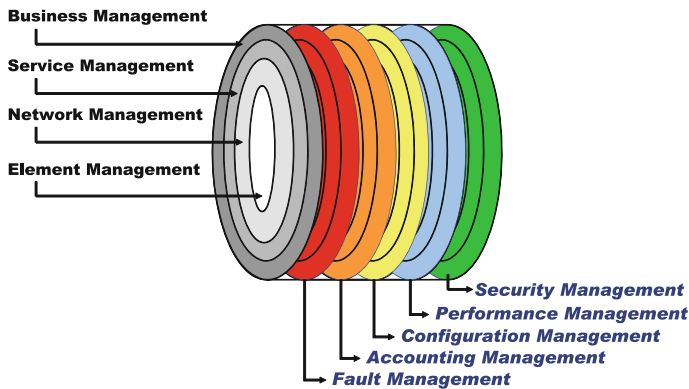


Fig. 2.6 TNM, network management model

- *Fault Management*. Focuses on identifying, examining and correcting faults, with performance indicators as reliability or survivability.
- *Accounting Management*. Centered on the control and monitoring of consumed costs and resources (billing, pricing, contracts, etc.).
- *Configuration Management*. This area plans and operates the configuration, monitoring the status and the installation (provisioning).
- *Performance Management*. The performance is evaluated and addressed to an optimum (traffic, QoS, etc.).
- *Security Management*. Possible management risks are considered through prevention, detection, continuity and recovery.

To support the management of these levels and areas, two system domains are employed, sometimes separately, and on other occasions, together:

- *Operation Support System (OSS)*. Focuses on the operational support processes uniforming and centralizing the remote systems and functions of the network; and
- *Business Support System (BSS)*. Concerning the contractual activities of the service and the customer relationships such as billing, QoS, traffic, payments, etc.

Thanks to the wide diffusion of Internet, IP-based networking protocols are the most widely used. In this sense, Ethernet and the protocol TCP/IP are the accepted communications standard in telecommunications sector and in all company levels. Based on this, and within the TNM element management level, the protocol SNMP [72] is the most commonly employed to manage IT infrastructure (routers, switches, firewalls, UPS, air conditioning, etc.). SNMP is an open protocol designed to control and monitoring network equipments. It is based on two entities: manager and agent where you perform the internal functions of control and administration of the controlled equipments respectively.

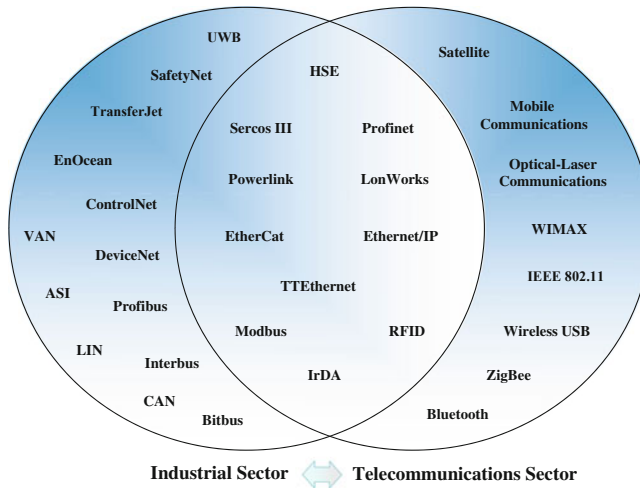


Fig. 2.7 Communications networks and protocols

2.3.3 Convergence Between Industrial and Telecommunications Systems

As the reader may realize, in both sectors the aim is to standardize the information processing to optimize the operation and maintenance. This trend, thanks to the evolution of the *knowledge society*, is carried out through the design of powerful applications going beyond the organization boundaries, strengthening the convergence between sectors and systems.

The development of systems between the two sectors has been carried out parallelly in the hierarchical levels of businesses and networks. At the same time, the convergence between both sectors standards has been searched (mainly in ISO 18435 levels 0, I and II) with the intention of taking advantage of each standard, empowering the integration with the others. This is in line with the international recommendations of the European project *Networked Control Systems Tolerant to Faults* [56].

Protocols and communication networks require special attention due to the huge amount of developed private and open standards [26]. As an example we list the following here (see Fig. 2.7):

- Ethernet-based searching Internet transmission such us EtherCat, EthernetIP, TTEthernet, Ethernet Powerlink, Modbus TCP, Profinet, SafetyNet, SERCOS III, or High Speed Ethernet (HSE).
- Fieldbus-based focus on industrial sector such us Bitbus, Interbus, local interconnect network (LIN), controller area network (CAN), vehicle area network (VAN), Profibus, Profibus DP, DeviceNet, ControlNet, ASI or LonWorks.

- Wireless-based from telecommunications sector such as IrDA, RFID (Radio Frequency Identification), bluetooth, IEEE 802.11, Wireless USB, EnOcean, TransferJet, Ultra-wideband (UWB from WiMedia Alliance), ZigBee, WIMAX, Mobile communications, Optical Laser communications or Satellite communications.

The nexus in this convergence is the aforementioned extensive use of Internet. Internet protocol TCP/IP is used by the systems of both sectors to transport information. As a result, both the industrial and telecommunication systems employ Internet to control and monitorize the remote network elements. For instance, OPC from the industrial sector and Simple Network Management Protocol [72] from the telecommunications sector have evolved approximating their developments with new devices and equipment and improving characteristics for integration in subsequent new releases such as security and network growth. A comparative description between the OPC and SNMP characteristics can be seen in detail in [46].

Nowadays the tendency is to converge both by gateways or servers to monitor and manage the two networks in a homogeneous way, increasing the knowledge about incidents (e.g. mistakes in communications instead of absence of alarms). Industrial and telecommunications servers and clients could coexist at the same time, but companies could also elect one and integrate the elements of the other by interfaces (i.e. in the case of OPC) or “ping” consultations (i.e. in the case of SNMP). Therefore, the network management system has to allow the convergence among the industrial and telecommunications open standards, due to the amount of information to manage in millions of dispersed elements, in different environmental conditions and several times per second, indeed in compliance with e-maintenance recommendations.

Summarizing, the following Fig. 2.8 reflects the main interoperability standards among systems in the industrial and telecommunications sectors, related to life cycle and at the management level. In the figure the standards within OpenO&M are represented with a dark background.

In this way, inside the energy utility sector, the GridWise[®] Architecture Council (GWAC) (www.gridwiseac.org), which was formed by the U.S. Department of Energy to promote and enable interoperability among the many entities that interact with the nation’s electric power system, considers the interoperability among standards of both sectors in different technical layers, in which the communications networking and syntax issues are information technology oriented. The GWAC encompasses much of the concepts of the Open Systems Interconnection (OSI) 7-layer communication model, including Ethernet, OPC or SNMP communications among systems and overlapping international information models such as the common information model (CIM), OpenO&M or Object models based on XML schema definition. The GWAC members are recognized and respected practitioners and leaders with broad-based knowledge and expertise in power systems, information technology, telecommunications, markets and financial systems, buildings, industrial controls, security and other related sectors.

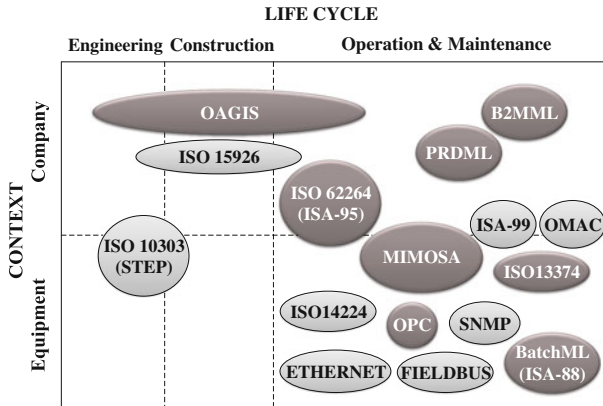


Fig. 2.8 Interoperability standards in network utilities

2.4 Maintenance Management Models

Maintenance has been experiencing a slow but constant evolution across the years, from the former concept of “necessary evil”, up to being considered an integral function of the company and a way of competitive advantage.

For approximately three decades, companies realized that if they wanted to adequately manage maintenance they would have to include it within the general scheme of the organization, and to manage it in interaction with other functions [61]. The initial challenge was therefore to integrate maintenance within the management scheme of the company.

Some benefits of having maintenance management models integrated within modern organizations are, amongst others, the following Vanneste and Wassenhove [80]; Cholasuke et al. [14]; Prasad et al. [65]; López and Crespo [47]:

- Maintenance leadership and support;
- More comprehensible organizational scheme;
- Achievement of high productivity;
- Overall equipment emergencies reduction;
- Improvement in production efficiency;
- Accident reduction;
- Verification of the investment profit;
- Development of a flexible;
- Multi-skilled organization.

Nowadays, designing the ideal model to drive maintenance activities has become a research topic and a fundamental question to accomplish. This design will condition the maintenance effectiveness and efficiency and will importantly contribute to fulfil the enterprise objectives [65].

Different authors have proposed models, frames or systems seeking to manage maintenance in the best way. Using the most advanced techniques and proposing innovative concepts; every model put forward has strengths and weaknesses, which are a matter of study in the following sections based on López and Crespo [47]

publication, included as a main part of this Section.

In subsequent paragraphs, we will follow a chronological tour through some representative maintenance management models. These models will be described in a general way and then classified according to their initial introduction as declarative models or as process-oriented models. Later proposed innovations of each model will be highlighted and model elements will be compared with ISO 9001 [39] as a reference management standard, and as according to other criteria that will be mentioned and that could be considered suitable for this study. An intensive search and compilation of maintenance management models that can found in literature, from 1990 up to the present day, will be presented for this analysis.

Finally, we will try to derive some conclusions about desirable characteristics of a modern, effective and efficient maintenance management model. Potential applications of these models supporting industrial needs as well as their future challenges will also be discussed.

2.4.1 The Literature Review

The bibliographical search was carried out using the following electronic databases:

- Blackwell synergy;
- Business source premier—EBSCOhost;
- Compendex (engineering village)—Elsevier engineering information;
- Current contents connect—ISI;
- ISI web of knowledge—ISI;
- NTIS—Ovid (SilverPlatter);
- Scopus—Elsevier;
- Springer link; and
- Wiley InterScience.

From this exploration, completed on Feb 18th 2008, a series of 14 articles were selected, these articles are: Pintelon and Van Wassenhove [62], Pintelon and Gelders [61], Vanneste and Wassenhove [80], Riis et al. [67], Hassanain et al. [28], Tsang [79], Waeyenbergh and Pintelon [83], Murthy et al. [53], Cholasuke et al. [14], Abudayyeh et al. [2], Pramod et al. [64], Prasad et al. [65], Tam et al. [75], and Soderholm et al. [73].

The criteria for the selection of the aforementioned 14 articles were:

1. The article has to propose a global maintenance management model and it does not have to be only focused on a particular management area or maintenance tool.
2. The model proposed in the article does not have to be a computer model or Computerized Maintenance Management Systems (CMMS).
3. The article had to be published only in indexed scientific journals.
4. The article had to present not only a review or an application, but a new model proposal.
5. The model in the article had to be represented preferably using a graphical diagram.

Besides the aforementioned articles, a bibliographical search was carried out in which the following books were found and selected, on the premise that the models proposed in them fulfil the criteria mentioned previously: Campbell [10, 11], Kelly and Harris [44], Wireman [85], Duffuaa et al. [20], Kelly [43], Crespo [19].

In this way 20 contributions were selected, presenting the same number of maintenance management models, that will be compared chronologically in different steps or with a specified criteria, and then some results and conclusions were identified. To synthesize the content of each and every one of these models we used a table to concentrate the information gathered.

Based on this synthesis, an initial classification is proposed, dividing the models into two types: declarative models (referenced from the concept “declarative language” that we have found in the Encyclopaedia Britannica), and process-oriented models (from “business process orientation”, a concept based upon the work of Porter—[63], among others). What is the difference between these two types of models?

- *Declarative models* mention the management maintenance components, although they do not refer to the intercommunication/link between those components in an explicit form. In this type of models a clear information flow among the components is not distinguished, and therefore, some functional, interrelational and synchronization aspects cannot be clearly appreciated. However, some of these models are very complete, including a great variety of aspects and tools related to maintenance.
- *Process-oriented models* normally offer a clear information flow among their components. In some of these models, inputs and outputs of the maintenance management model are identified. In others, a closed loop sequence of steps is clearly represented. Though in many cases we may suppose that these models seem to be of easier application in organizations than declarative models, they require proper definition concerning the coordination among their elements in order to be effective, and this definition is sometimes missing.

We can observe that a process-oriented model seems to impose a more organized scheme; certainly the complexity degree for its implementation process is greater than in a declarative model, where it is possible to take only the elements that are suitable to add to the already operating organization, and thus to obtain

Table 2.3 Model classification

Declarative models	Process- oriented models
Pintelon and Van Wassenhove [62]	Vanneste and Wassenhove [80]
Pintelon and Gelders [61]	Campbell [10]
Cholasuke et al. [14]	Kelly and Harris [44]
Prasad et al. [65]	Riis et al. [67]
Tam et al. [75]	Wireman [85]
	Duffuaa et al. [20]
	Hassanain et al. [28]
	Tsang [79]
	Waeyenbergh and Pintelon [83]
	Murthy et al. [53]
	Abydayyeh et al. ([2]
	Pramod et al. [64]
	Kelly [43]
	Soderholm et al. [73]
	Crespo [19]

fast innovations and benefits in maintenance management [63]. It is undeniable that every type of model has its own pros and cons; therefore it is convenient to study and to analyze all of them to be able of distinguish which one may be better applied to certain kinds of scenarios and conditions.

In Table 2.3 we can appreciate how the majority of the models found are process-oriented models; however, some of the declarative models, such as [63], are especially wide, and can most definitely serve as an “implementation and operations guide” for any maintenance management model.

Some important aspects of this study are related with the chronological analysis of the different author’s contributions; Fig. 2.9 represents the twenty models studied in this work arranged through a time line. In this figure we can observe that the interest in generating new proposals has remained constant during almost the last two decades.

In many books and articles about maintenance, the existence of different generations or stages of maintenance management models is commented, but that evolution is not explained in an explicit form, describing the integration of each new element and/or technique into the models.

Since history lessons can be of great interest for us, we have summarized in Table 2.4 the innovations that we have identified in selected maintenance management models (without trying to be extremely precise), according to a chronological order.

It is necessary to mention that the indicated innovations correspond to those subjects appearing for the first time inside a maintenance management model; it does not mean that these elements are new concepts (out of our maintenance management context).

In Table 2.4 we can see how maintenance management models have been acquiring new elements and trends through the years, such as: approach to processes; innovating proposals in technical aspects; use of standard languages for

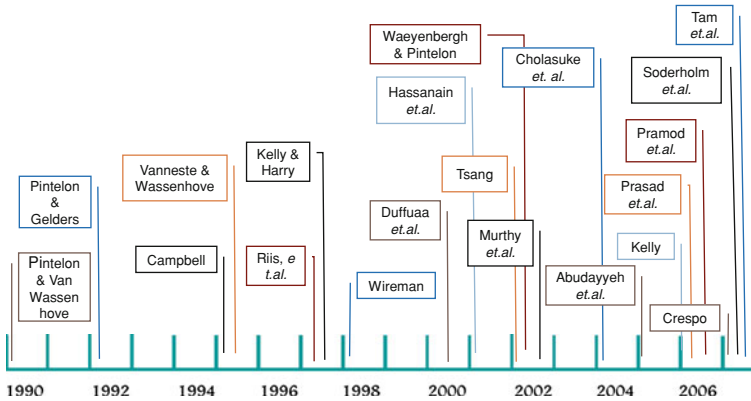


Fig. 2.9 Time line for the maintenance management models

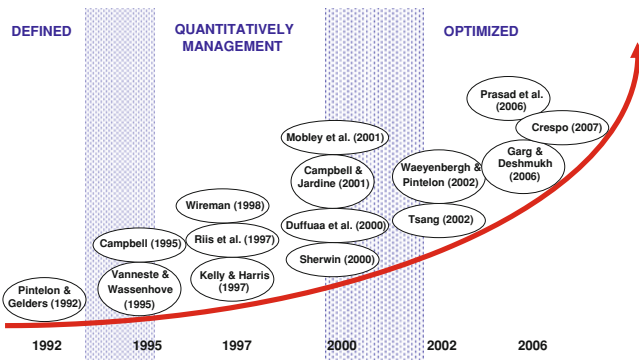


Fig. 2.10 Overall view of maintenance frameworks and models based on processes

information exchange (in order to be used subsequently in CMMS and other computer applications); successive incorporation of quantitative techniques and computer tools (due to the increasing amount of maintenance, operational and financial data generated); evaluation and constant improving of maintenance operations (for instance, using automated tools); analysis of the assets life cycle besides the evaluation of the maintenance function; integration of the assets strategy with the maintenance strategy, etc.

2.4.2 Comparative Analysis and Evolution of the Maintenance Management Models

In order to compare and to analyze the previously mentioned models, we have designed a check list which tries to capture different important elements to appear in an *advanced maintenance management model*.

Table 2.4 Innovations of maintenance management models in chronological order

Year	Innovations	Author(s)
1990	Proposes a complete scheme of maintenance indicators	Pintelon and Van Wassenhove [62]
1992	Exposes the need for a proper link between maintenance and other organizational functions Highlights the importance of using quantitative techniques for maintenance management decision making Glimpses the utilization of expert systems Mentions total productive maintenance (TPM) and reliability centered maintenance (RCM)	Pintelon and Gelders [61]
1995	Proposes an analysis focused on effectiveness and efficiency of maintenance Emphasizes the importance of the managerial leadership in maintenance management Introduces the concept of “maintenance reengineering”	Vanneste and Wassenhove [80] Campbell [10]
1997	Proposes an integrated modeling approach based on the concepts of situational management theory	Riis et al. [67]
2000	Proposes the use of a great variety of Japanese concepts and tools for the statistical control of maintenance processes in a module called “feedback control”	Duffua et al. [20]
2001	Focuses the model to the computer use, expressed in IDEF 0 language (a standard for information exchange)	Hassanain et al. [28]
2002	Glimpses the use of e-maintenance Proposes a guide to analyze the outsourcing convenience as an entry element to the maintenance framework Incorporates both the tacit knowledge and the explicit one and integrates them in a computer database Give special value to the knowledge management	Tsang [79] Waeyenbergh & Pintelon [83]
2006	Suggests the link of tools: quality function deployment (QFD) and TPM into a model	Pramod et al. [64]
2007	Propose a process view in which maintenance contributes to the fulfilment of “external stakeholders” requirements Proposes a model with a methodology of application clearly expressed, oriented to the industrial asset dependability and life cycle cost improvement	Soderholm et al. [73] Crespo [19]

A first group of our check list elements is inspired in ISO standard 9001:2008 [39]. This standard is chosen since it is the international reference for any quality management model, which turns into a generic guide for a process operation in which fulfilment with requirements should be demonstrated, such as in the case of the maintenance function. The elements of this check list are:

- Related to quality management (process approach, sequence and interaction of the processes, description of the elements of each process, generation of documents or records).
- Related to management responsibility (entailment with strategic targets of the organization, objectives definition, senior management commitment, clear definition of responsibilities and authorities, suitable communication).
- Related to resource management (humans beings, materials and infrastructure).
- Related to measurement, analysis and improvement (audits, studies of the internal client satisfaction, information analysis, corrective and preventive actions, continuous improvement approach).

A second group of the check list elements is elaborated considering the “support tools and techniques for maintenance management” mentioned in the selected models. Some of them are:

- Techniques dealing with economic or financial aspects of maintenance;
- CMMS;
- Techniques about human resources management;
- Application of operations research or management sciences;
- Life cycle analysis;
- TPM;
- RCM;
- Simulation, inventories models;
- Reliability theory;
- Expert systems;

Finally, we have explored whether the model is presented with a methodology for its implementation. This is a key question. As we mentioned above, some models limit themselves to enumerating the elements that must conform a maintenance management model, without explaining model dynamics. Nevertheless, an organization that wishes to initiate an implementation of a maintenance management model, may not find enough information concerning the steps to follow.

There are relatively few models defining a clear methodology to be implemented and a way to become operative model. Due to this reason, this criterion becomes a key appreciation in this work.

Briefly discussing the results of the comparative analysis carried out we can say:

- *Concerning the management model.* Declarative models do not regularly have an input–output process approach and do not consider a clear methodology for

its implementation either. In general, these models do not mention in detail the advanced quantitative techniques to be used in maintenance management.

- *Concerning the management responsibility.* All the models define objectives for the maintenance function; however, not all of them link these goals with strategic company targets. In addition, most of the models do not make a clear reference to principles of responsibility, authority and good communication. Maybe this could be because these elements are considered as an initial assumption.
- *Concerning the maintenance support.* Approximately half of the models incorporate the use of support techniques such as operational research techniques or management sciences techniques. TPM and RCM are the most mentioned and they tend to appear together in management models. Also CMMS is mentioned as an indispensable tool in the majority of the models. Recent models include other techniques such as the use of e-maintenance, expert systems, etc.
- *Concerning the management of resources.* The majority of models mention something on the matter, though in several schemes this topic is omitted. An explanation could be that this subject is also considered to be an assumption. For example, almost a third of models do not mention techniques for inventory management and purchase control. Curiously, in earlier models, a major emphasis in aspects related to the human resources management is appreciated.
- *Concerning measurement, analysis and improvement.* All the models consider different phases for maintenance evaluation, analysis and improvement. Although few, more than half of them mention literally the concept “continuous improvement”, this trend has grown especially in the last years.
- *Concerning the methodology and the operation of the model.* A very important attribute of some models is the inclusion of an application/implementation methodology which stimulates continuous improvement. Few clearly incorporate this feature.

After presenting these state of the art maintenance management models, is it possible to identify some key aspects or elements that should be considered when designing and implementing a maintenance management model in a company? We think that this study demonstrates that at least we should consider the following features:

- Input–output process approach.
- Clear implementation methodology.
- Generation of documents and records analysis.
- Clear objectives entailment.
- Incorporation of support technologies (TPM, RCM, etc.).
- Orientation to CMMS.
- Flexibility against rapid structural changes.
- Inclusion of maintenance material management.
- Inclusion of human and information resources management.
- Focus on constant improvement.

- Evaluation and improvement.
- Cyclical operation.

Nevertheless, whatever model an organization adopts, it has to adapt easily to changes in business, communications and industry. A key to achieve this could be the incorporation of modern tools and platforms which are known as “next generation manufacturing practices” (NGMS). This implies the use of e-maintenance as a sub-concept of e-manufacturing and e-business. In this way, e-maintenance would have to be integrated to maintenance management models looking for new ways of working involving collaboration and availability of knowledge and intelligence any time and any place, perhaps changing also the entire business process.

The use of new technologies provides companies new competitive advantages in maintenance management. According to this technology adoption and integration idea, we propose to define three new maintenance management generations or maintenance management maturity levels. Each generation can be in accordance with the CMMI [16] definition for management levels and with the application of the ICTs for maintenance management purposes, as follows:

- I. *Defined maintenance*. The maintenance process is planned and performed in agreement with the company guidelines. The process is reviewed and evaluated to verify whether it fulfils the requirements. Maintenance is set as a key function integrated inside the company philosophy, and the maintenance function uses ICT to automate and manage the dispersed, duplicated and unrelated data.
- II. *Quantitative managed maintenance*. Afterwards and focusing on the technical-operational aspects. The trend was to automate through the use of information systems and the integration of inter-departmental systems. Since then, the amount of information generated thanks to the ICT evolution made easy to apply statistic techniques. Then the maintenance process is controlled and evaluated using quantitative techniques.
- III. *Optimized maintenance*. Maintenance is a process managed with the objective of improving it within a global environment and considering the common causes of variability of the processes. E-maintenance strategies were implemented thanks to the development of Internet and communications technologies. These advances facilitate the globalization of the companies. Systems and communication networks assist the distribution and generation of maintenance knowledge.

Following this definition, Fig. 2.10 plots in the form of a graph, as a practical exercise, reviewed maintenance management models according to their considerations related to the new maintenance management maturity stages or generations. Frontiers among generations are not very clear because some authors have anticipated the use of techniques considered within a higher generation.

Finally, if we now concentrate on utilities networks maintenance management models, we can say that the majority of the papers found in literature only cover the management function for an individual and specific type of network (water, gas, electricity and telecommunications) [1, 7, 31, 42, 57, 69, 84]. Also these

works try to cover specific aspects of network maintenance management (reliability assessment, network monitoring, network risk analysis, etc.) rather than comprehensive network management models [3–5, 8, 9, 13, 22, 25, 27, 30, 32, 45, 48, 49, 51, 52, 71, 74, 78, 81, 82, 87, 88]. A global maintenance framework for these kinds of enterprises has not been found in the literature yet.

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Chapter 3

Measuring the Maintenance Contribution

3.1 Defining the Scope of Maintenance Contribution

3.1.1 Introduction

The assessment of the maintenance impact, in terms of value added to the company, has been a relevant topic within maintenance literature and for many years [11, 51]. Authors agree on the fact that maintenance performance should be measured not only in relation to the degree of fulfillment of certain technical standards [63], but also considering, its relationship and impact in other company departments, as well as its impact in the relationship with customers and on market competitors [66], and even on the environment [46]. Therefore, we must develop an internal but also an external vision of the maintenance management contribution [22] and then we have to be able to measure maintenance management according to that vision.

Different authors have summarized maintenance management contributions using historical data and coherent aggregation of indicators [38, 48, 50]. By doing so, they agree on the fact that companies may also facilitate the management of change, defining cause-effect relationships attached to all managerial action plans. Then, indicators can be utilized as a support for decision making and to guide the development and the improvement of processes.

In this chapter we first present different key performance indicators (KPIs) that can be taken into account to measure the maintenance management contribution in network utilities. Then we pay special attention to the correct consideration of different maintenance costs, for the proper evaluation of maintenance efficiency in the company, but also from the perspective of an asset life cycle costing.

This chapter also serves as an introductory material to the third part of the book where we deal in detail with all these topics. The idea is to offer a framework for proper utilities network maintenance management assessment. That is, a framework for evaluating the effectiveness and efficiency of network maintenance management.

3.1.2 Key Performance Indicators

An indicator could be defined as a measurable quantity at different moments in time, which allows us to compare and evaluate the evolution of a given variable [21, 22, 30]. An indicator should have a procedure attached, showing how it is measured and how it can be used to measure:

- Effectiveness, measuring the degree of activities in different and interesting contucess comparing the results obtained with the ones expected.
- Efficiency, assessing the extent of resources utilization to produce the results obtained.

We can find different classification of indicators according to business sectors and functions. A consistent set of business indicators is presented by Kaplan and Norton [36] by grouping them in four main subsets corresponding to key business aspects, which according to this author are: finance, customer relationships, business processes and innovation and organizational learning and development. They have shown the importance of going beyond traditional financial indicators to track business performance and to set up business strategy.

In the case of maintenance we can find interesting contributions related to indicators. For instance, Wireman [66] presents indicators grouped in several categories: corporate, financial, effectiveness and efficiency, tactical and functional performance indicators. Parida and Chattopadhyay [48] published extensively in the area of maintenance indicators to properly measure maintenance management. He suggests a framework for multi-criteria hierarchical maintenance performance measurement, considering various issues (relating indicators concerning equipment, maintenance task, cost, customer satisfaction, learning and growth, health and security, and employee satisfaction) to measure the contribution of maintenance towards total business goals, and integrating indicators at different business levels, the subsystem/component level, plant level and corporate level are linked with the MPis for the organizational objectives and strategy. Recently, an important move in this area is lead by the European Federation of National Maintenance Societies (EFNMS) together with the Society of Maintenance and Reliability Professionals (SMRP). Both institutions have been collaborating since 2006 with the aim of harmonizing maintenance key performance indicators (KPI). This process takes as a starting point EN 15341:2007 standard besides the well-known SMRP best practices in this field. Three groups of indicators are being considered: economic, technical and organizational. The characteristics of Maintenance KPIs introduced in the EN 15341:2007 standard are the following:

- Simple and synthetic
- Defined in a clear and unambiguous way
- Measurable on homogenous basis
- Linked on one or more factors
- Suitable in management maintenance process.

Fig. 3.1 Groups of maintenance contribution indicators



The EN 15341:2007 standard considers KPIs as tools available to be used in order to:

- Measure the status.
- Compare internal and external benchmarks.
- Verify the technical organizational attitude.
- Diagnose every situation: strength and weakness.
- Define a strategy, objectives, targets and actions.
- Check the results and the progress of a plan.
- Visualize the success and share the achievements.
- Measure needs and changes continuously.

These works try to follow rules derived from best practices to define proper indicators. Some other authors have published about similar rules, for instance Maskell [42] presents seven rules to reach suitable indicators definition:

1. The measurements should be directly related to the company strategy.
2. They must also adopt non-financial measures.
3. The measurements change according to location, not all measurements are valid in all departments and enterprises.
4. Circumstances can change measurements.
5. The measurements should be simple and easy to use.
6. The measurements should provide fast feedback.
7. They have to be designed for continuous improvement rather than only monitoring purposes.

Taking into account previous paragraphs in this work we have selected the following categories of maintenance indicators for network utilities maintenance (Fig. 3.1): management and organization, financial—economics, business and production, quality, security and sustainability, development and improvement.

These indicators also try to offer a characterization of the maintenance and maintenance management contribution to the company. Let us now briefly discuss each one of the areas selected:

- *Management and organization.* In the maintenance context, the management is a compulsory discipline that consists in the aptitude of leading activities and the involved resources (ability, knowledge, skills, responsibilities, commitment, etc.) towards the defined objectives, in order to facilitate their control and proper knowledge management.
- *Finance–Economics.* Maintenance, from a practical point of view of the business, involves techniques to generate significant savings in the companies (optimizing costs), reducing the consequences of the failures and pursuing business effectiveness [46].
- *Business and production.* Maintenance ensures dependability of equipment and therefore production effectiveness or serviceability performance for a given capacity of our production facility or network, respectively.
- *Quality and customer relations.* Maintenance can strongly contribute to customer capture, satisfaction and retention [19]. Especially for services, the customer perspective is actually the best functionality test of the service (product) provided.
- *Security (including safety, confidentiality and integrity) and sustainability (development with a rational use of resources and preserving the environment).* Maintenance actively plays an important role in the protection of internal company resources as well as in the protection of external resources and environment. Many early maintenance contributions highlighted this role of maintenance [26, 64], it is also compulsory to rely on current existing legal norms and regulations regarding these matters.
- *Development and Improvement.* From the maintenance experience, products and services can be perfected, through the continuous improvement of processes and systems, continuously looking for the value generation and the capability to quickly react to changes.

All the aspects referred to above have to be translated into a proper selection of key performance indicators. These KPIs help the decision making processes within maintenance, reducing uncertainty and serving as a qualitative guiding reference. Notice that the more reliable and available data and information about a process that we possess, the greater the potential to reduce our decision making errors within said process.

At the same time, attached to each indicator, we have to provide information helping the managers in the interpretation of the measure we try to capture (the what, how, when, who, etc.). For instance, we must specify the type of measured variable, a metering mode, a period of measurement (metric), a follow-up period (to measure its evolution), someone to be responsible for the indicator measurement and revision, a target value, a unit of measurement and any other factor relevant for the indicator use and interpretation.

Indicator target values have to be discussed within the maintenance management teams. For instance, target values for those activities deemed critical and

non-critical (regular) may differ, according to most important improvements to foster for a given period of time [16].

Cross reference or connections among indicators is another valid practice permitting the vision of cause-effect laws that may apply in a given business scenario.

All the previous aspects will help us later in the design of a suitable control panel, a dashboard, measuring the progress of maintenance activities on a regular basis, at the different business activities levels (strategic, tactic and operational).

3.1.3 Performability Concept and Scope for Maintenance

In the case of network utilities we refer to the provision of user-oriented or system-oriented services through a given distribution network. The “performance” of the network may refer to some aspect of the service provided, for example quality, assuming the systems in place (network) are correct. Performance of the distribution network can be therefore understood [24] as an indicative of “how well the network, assumed to perform correctly, works”.

Some authors may even consider [35] dependability measures as a special class of performance measures. This is because we cannot forget that regardless of network dependability, service performability, in the context of computer and communication system may also depend on other measures such as through-put, response time or resource utilization. Notice that all our measures should be able to track incorrect behaviour of the network due to either design faults (mistakes made by humans or automated tools in the process of specifying, designing, implementing, or modifying a system) or operational faults (physical or human-made faults that subsequently occur during system operation) [39]. Moreover, and especially when service provided performance is degradable, measures of performability are needed to simultaneously address issues of both performance and dependability [44].

In Network Utilities, due to these special characteristics, we may then say that maintenance has not only a network or system dependability scope but a more generic service performability scope. Maintenance really interacts with customers and with a large number of dispersed network elements to manage. The whole maintenance contribution to the performability of the service will be evaluated measuring the performance of maintenance activities plus the analysis of their impact and interaction with equipment, departments, services, customers and ultimately with the society and the environment.

We have, as an exercise, built a figure describing a possible hierarchical vision of the relationship amongst these terms (Fig. 3.2). In the bottom of this figure there are two groups of attributes, one group conditions availability, and other, security. Some authors consider that availability influences security, based on the idea of service continuity. But we could say that dependability performance depends on both groups of features, at the same time, when dependability deals with the consideration of

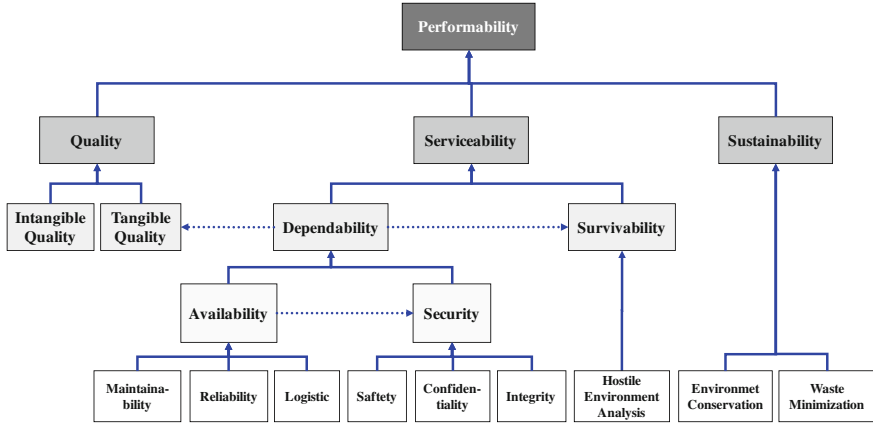


Fig. 3.2 Terms of performance in maintenance

hostile environments the term is scaled to the so-called survivability. Considering both, dependability and survivability, we can reach tangible features of services related to their capacity to be accessible (accessibility) and to be kept in use (retainability), which is equivalent to the serviceability concept of the service. Then, as well as previous factors, performability can be obtained as a combination of the serviceability performance, the service quality performance, and the service sustainability performance.

Finally, notice that all these indicators can be somehow formalized using probabilistic mathematical models [8, 13], of the random variables involved. Therefore we will dedicate time to explore modelling and measuring of performability using typical probabilistic risk analysis tools [11, 46, 47].

3.1.4 Maintenance Can Really Make a Difference in Cost

Besides performability, maintenance contribution should be translated into terms of costs [14]. *The knowledge of quality costs aids managers to justify investments in quality improvement and assists them in monitoring the effectiveness of the effort made* [15]. According to Wilson [65] the cost of the failures could even reach ten times the purchase cost.

We therefore have to properly classify the different cost of equipment maintenance along their lifecycle (Fig. 3.3), where cost is understood as the employed resources, in monetary terms, to achieve a certain objective. Costs can be classified according to different criteria, for instance [6, 12, 33, 40]:

- *Variable and fixed costs.* Depending on whether they are, or are not, dependent on the usage/volume of goods, or services, respectively.

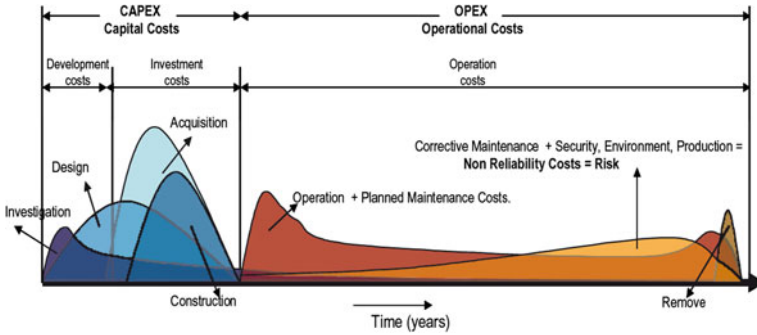


Fig. 3.3 Lifecycle costs of equipment

- *Direct costs and indirect costs.* Depending on whether their relationship with the measured item is, or is not, directly accountable respectively.
- *Tangible costs and intangible costs.* Depending on whether their nature is objective or subjective respectively.
- *Historical and expected cost.* Depending on whether they are objectively based on historical records or previously estimated to achieve the goal.

In network utilities maintenance we have fixed costs such as the rent of facilities or buildings, depreciations, etc. Others costs are variable, changing with the number of customers, for instance the consumption of different utility services or service related labour. Also, there are direct costs such as material costs and there are indirect costs such as cleaning, security or computing services. In addition, maintenance costs such as warranties, indemnities, reparations, penalties, etc., can be classified as tangible or intangible such as loss of image, loss of customers, internal motivation, etc.

Pintelon and Gelders [52] proclaimed the need to assess the benefits and losses caused by the maintenance activities (or the lack of them). That is to say, including not only the tangible costs but also the intangible costs. This is represented by an iceberg in Fig. 3.4.

The application of one or another criterion in order to classify and allocate maintenance costs will depend on the cost accounting system implemented by the company. The cost accounting system will be essential to guide the company towards efficiency improvements.

In the following sections we will review some Cost Accounting Systems (CAS) methods that may facilitate maintenance managers' decision-making processes and that also help measuring the maintenance impact on cost properly. Regulators require that these methods apply [56] valuation principles which are similar to well-known and accepted accounting principles, facilitating audits according to legal requirement. These principles are:

- *Causality.* Employed costs and revenues should be allocated to the activities.
- *Objectivity.* Allocation methodologies should be objective.

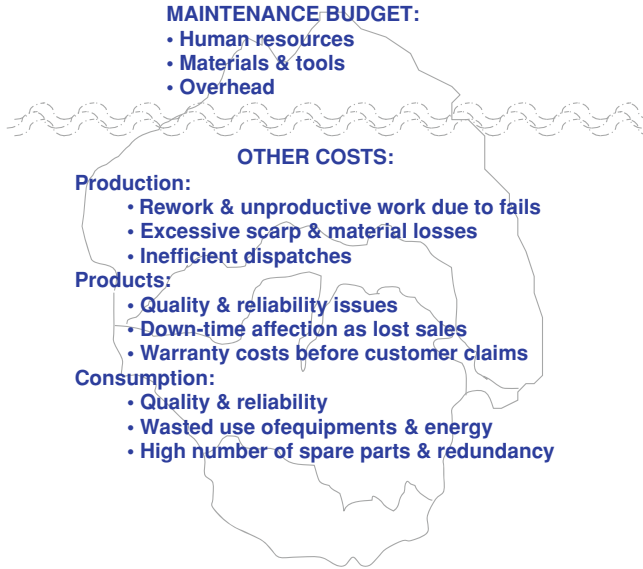


Fig. 3.4 Maintenance costs (modified from [52])

- *Consistency.* Allocation criteria should remain constant over time, to the extent that is feasible and reasonable.
- *Transparency.* Allocation methodologies should be clear and discernible.

3.2 Maintenance Contribution in Terms of Cost

3.2.1 The Nature of the Maintenance Costs

The American Society for Quality Control [7] and the British Standard Institute [9, 10] formally classified costs from a quality perspective, according to the scheme PAF [28, 53], Prevention, Appraisal and Failure. Similarly, in Spain following the Spanish Association of Accounting and Business Administration [4] in the document “Quality costs” and consistent with the Spanish Association for Quality (1991), costs can be summarized into the same three types. Therefore, the maintenance costs could be categorized escalating this up to a performability perspective as:

- Prevention costs are realized costs through remedial activities to achieve the desired level of performability.
- Appraisal costs are applicated costs to supervise and control the obtained level of performability.

- Failure costs are those incurred due to failures and damages that lead to poor performability. These in turn may be due to internal failure or external failure depending on whether the damage is caused inside or outside the organization.

Also, the costs of prevention and evaluation can be labelled performability costs, for example necessary costs to make maintenance activities right corresponding to the planned costs, while the costs of failure can be called non-performability costs or incurred costs because of the improper execution of maintenance activities, corresponding to the unplanned and penalty costs.

Depending on the company and sector [15, 34], there are different classification studies about costs, some of them classify the weight for each type of cost, 65% of the failure costs, 30% of the appraisal costs and 5% of the prevention costs [2, 55]. Thus, according to Crosby [14], in distribution network maintenance we could determine that the level of performability is *the price of non-conformance*, in virtue of the products and services are not perfect.

Then, in terms of performability, maintenance directly contributes to the company through prevention and appraisal costs, for example, to improve network reliability, and indirectly provoking failure costs, for example, absence of a proper maintenance execution. Therefore, in distribution network maintenance, some typical costs following the PAF scheme could be:

- *Prevention costs.* Here we could include costs for quality and maintenance management, material costs in prevention of health and safety, personnel learning and training, security operations, certification and calibration of equipment and tools, preventive maintenance, research and improvements, insurance, etc.
- *Appraisal costs.* Within which we can consider the surveillance and monitoring of services and features, resources and tools used for evaluation, the measurements of customer satisfaction and market perception, and evaluation of the personnel motivation, etc.
- *Failure costs.* Here we could distinguish rework, poor planning, non-productive times due to production failures, time to examine the causes and consequences of failure, loss of work and accidents, obsolete or inadequate use of resources and services, supplier delays, inefficiencies in information systems or communications, contract or billing errors, loss of income or returns, debt, deductions, incurred costs to resolve customer or internal complaints, penalties, loss of market reputation, etc.

The sum of the prevention and appraisal costs is, therefore, inversely proportional to the failure costs, it is assumed that increasing prevention and evaluation, less errors will be generated [32]. Then maintenance has to search a break point or zone where the prevention and appraisal costs are balanced so the failure costs are at an acceptable level, not increasing the total costs (see Fig. 3.5). Thus, maintenance can save budget by making things better, and also be prepared for a better performance even under further irregularities that may occur.

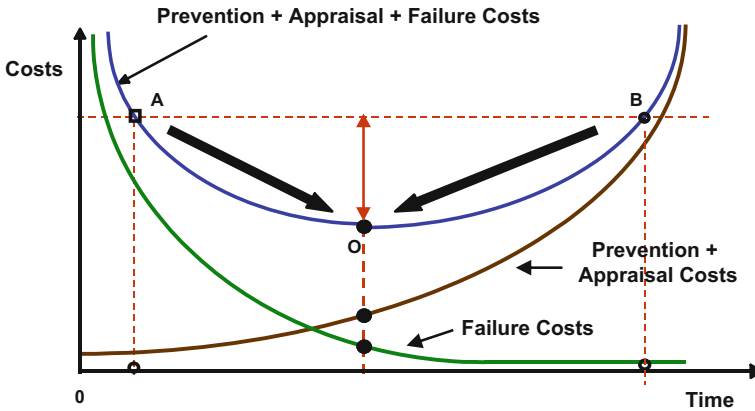


Fig. 3.5 Prevention, appraisal and failure proportionality

The ratio of prevention and appraisal costs against the failure costs has been considered in several studies to predict maintenance budget. In line with this, there is a worldwide relation between prevention and failure activities from 40 to 60% [45]. Although there is always a 5% of unpredictable remedial activities that can not be avoided through preventive programs [45]. These proportions can vary among countries, sectors, type of equipment and management, technology application, etc.

Therefore, the contribution of maintenance evaluation requires the assessment of the planned costs (prevention and appraisal costs) as well as the failure costs (unplanned and penalty costs) including negative and positive consequences of maintenance activities.

At the same time, costs in a service sector company could be classified within four main categories [40], that when adapted to the maintenance costs of network utilities would be as follows:

- *Direct labor costs.* Are personnel costs incurred to provide the services, distinguishing between frontline management and support or secondline management.
- *Direct materials costs.* Are the consumable resources and supplies used in each type of service. In this category we could find raw materials, supplies and minor tools and accessory materials.
- *Technology costs.* Are all the costs relating to the acquisition and use of equipment and information technology, including the implied labor such as assets and measuring equipment and devices (owned or leased), spare parts of assets, commissions, duties, taxes, insurance, transport service, installation, upgrade/overhaul of assets, add-ons to original assets, perfective projects, withdrawal of assets, systems and software licenses, labor costs related to machine setup-validation-testing or training.
- *Overhead costs.* Are the other resources in the organization providing support to field personnel such as administration, occupancy and travel costs, costs related to 'outsiders', such as security, etc.

On the other hand, depending on the type of expenditure, if the invested cost improves or maintains the operating ability of the asset, the items of the maintenance budget are also allocated on capital expenditures (CAPEX) or operational expenditures (OPEX) respectively [41]:

- *CAPEX*. These are the employed expenditures to acquire or upgrade physical assets such as equipment, vehicles, buildings, etc.
- *OPEX*. These are the employed expenditures to operate the assets such as consumable parts, auxiliary utilities, licence fees, rent, etc.

3.2.2 Cost Accounting Principles and Time Values

From an accounting point of view, asset value is its purchase price (opening cost) or its production cost, including costs for the asset to be commissioned, such as transportation, installation, configuration, etc. This asset value has to be updated in each accounting period.

Therefore, achieving asset valuation and representation is not an easy task, because the asset value changes over time due to issues related to technology, its technical performance or its physical condition. So the asset value can suffer from a depreciation or revaluation at the end of a given period:

- *Depreciation*. Decreasing the purchase or production value of the asset according to a reasonable degradation of it and up to a residual value in its withdrawal or replacement.
- *Revaluation*. Increasing the purchase or production value of the asset according to a reasonable potential of it to produce future profits.

These changes of asset value have to be considered both in tangible and intangible assets, to update the future values to present time. Both the depreciation and revaluation can behave with the time or the usage (km, units produced, run time, etc.). In addition, we can find mixed changes with an estimated part according to the time and another part according to the usage.

Consequently, assessing current asset value is a useful tool in managerial decisions. For example, maintenance activities should be evaluated each period in relation to the observed depreciation and/or revaluation of assets. This is because maintenance affects asset conditions and their operational risks. Taking into consideration these facts and the accounting point of view, maintenance contribution could be reflected on the network asset values and the goodwill (intangible asset due to internal organization knowledge) values at the end of each period. But what are the mathematical expressions to calculate the depreciation and revaluation effects in assets? This question is studied in the next section.

3.2.3 Asset Depreciation

Asset obsolescence has to be emended in its present value as a negative correction. This process consists in a deduction of the asset value in each accounting period, discounting the determined depreciation to the opening cost. Cost should be adjusted during the asset life cycle, where the performance is made. Although we have to consider the possible residual cost of the asset and the end of its life, in this procedure, we have to write off the difference between the opening cost and the residual cost. Then, the total accumulated depreciation is equal to the difference between the opening value and the residual value of the asset.

The mathematical equations for the different forms of depreciation in an asset life cycle of n periods (from 1 to n) are different according to the method selected. We now formalize equations for the following depreciation methods: Straight line depreciation, accelerated depreciation, frontloading depreciation and activity depreciation.

- *Straight line depreciation (SLD)*. Uniform or constant depreciation in each accounting periods.

$$\text{SLD per accounting period} = \frac{(\text{opening value} - \text{residual value})}{\text{asset life cycle in periods}}$$

$$\text{Accumulated SLD in each period} = \sum_{i=1}^n \frac{(\text{opening value} - \text{residual value})}{n}$$

- *Accelerated depreciation (ACD)*. Higher depreciation in initial periods than in the last. As an example, we now introduce the digits method:

$$\text{ACD in the accounting period } (i) = \frac{(\text{opening value} - \text{residual value})}{(n^2 + n)/2} \times (n - i + 1)$$

$$\text{Accumulated ACD in each period} = \frac{(\text{opening value} - \text{residual value})}{(n^2 + n)/2} \sum_{i=1}^n (n - i + 1)$$

- *Frontloading depreciation (FLD)*. Higher depreciation in the last than in the first accounting periods. For instance, using the digits method:

$$\text{FLD in the accounting period } (i) = \frac{(\text{opening value} - \text{residual value})}{(n^2 + n)/2} (i)$$

$$\text{Accumulated FLD in each period} = \frac{(\text{opening value} - \text{residual value})}{(n^2 + n)/2} \sum_{i=1}^n (i)$$

- *Activity depreciation (ATD)*. Depreciation per period is calculated according to the level of production or usage that takes place in that period to all the expected production or usage for the asset life cycle.

$$\text{ATD per accounting period} = \frac{(\text{opening value} - \text{residual value})}{\text{expected total asset production} \times \text{actual production}}$$

$$\text{Accumulated ATD in each period} = \frac{(\text{opening value} - \text{residual value})}{\text{expected total asset production}} \sum_{i=1}^n \text{production}^i$$

But let us see a basic example of the four depreciation methods. Imagine a truck with a useful life of 4 years that has 10,000 currency units as its opening value, assume that its residual life is 2,000 currency units, also assume that the expected total production of the truck will be done in 200,000 km and that the kilometres made each year are 70,000, 50,000, 30,000 and 50,000 kms respectively. Then, calculations for the different types of depreciation would be as follows (see Table 3.1).

Figure 3.6 shows the tendency of each type of depreciation in the previous example.

3.2.4 Asset Revaluation

Asset enhancement has to be emended in its present value as a positive correction. This process consists of an addition to the asset value in each accounting period, increasing the determined revaluation to the opening cost. Profits should be adjusted to present time, where the measurement is made. In order to carry out this asset actualization, the use of an interest rate to support the mathematical calculation of present asset value objectively is recommended. In this case, residual values are not considered at the end of the asset life cycle.

Thus, the mathematical equations for the revaluation and actualization to present time are:

- *Present value (PV)*. Current value of a future value (FV) and it is calculated using an interest rate (r) as discounting estimation in all the accounting periods. Where FV_i is the FV in the period i .

$$PV = \frac{FV_i}{(1+r)^i} \Rightarrow FV_i = PV \cdot (1+r)^i$$

- *Accumulated present value*. Current value of a set of future values estimated in a series of accounting periods (from 1 to n).

$$\text{Accumulated PV} = \sum_{i=1}^n \frac{FV_i}{(1+r)^i}$$

Table 3.1 Example of depreciation in a truck

Accounting period	Depreciation	Accumulated depreciation	Accounting value
<i>Straight line depreciation</i>			
1	$(10,000 - 2,000)/4 = 2,000$	2,000	8,000
2	$(10,000 - 2,000)/4 = 2,000$	4,000	6,000
3	$(10,000 - 2,000)/4 = 2,000$	6,000	4,000
4	$(10,000 - 2,000)/4 = 2,000$	8,000	2,000
<i>Accelerated depreciation</i>			
1	$4 * (10,000 - 2,000)/10 = 3,200$	3,200	6,800
2	$3 * (10,000 - 2,000)/10 = 2,400$	5,600	4,400
3	$2 * (10,000 - 2,000)/10 = 1,600$	7,200	2,800
4	$1 * (10,000 - 2,000)/10 = 800$	8,000	2,000
<i>Frontloading depreciation</i>			
1	$1 * (10,000 - 2,000)/10 = 800$	800	9,200
2	$2 * (10,000 - 2,000)/10 = 1,600$	2,400	7,600
3	$3 * (10,000 - 2,000)/10 = 2,400$	4,800	5,200
4	$4 * (10,000 - 2,000)/10 = 3,200$	8,000	2,000
<i>Depreciation according to the production</i>			
1	$(10,000 - 2,000) \times (70,000 / 200,000) = 2,800$	2,800	7,200
2	$(10,000 - 2,000) \times (50,000 / 200,000) = 2,000$	4,800	5,200
3	$(10,000 - 2,000) \times (30,000 / 200,000) = 1,200$	6,000	4,000
4	$(10,000 - 2,000) \times (50,000 / 200,000) = 2,000$	8,000	2,000

$$\begin{aligned}
 \text{If } FV_i = \text{fixed value} &\Rightarrow \text{Accumulated PV} = FV \cdot \left[\frac{1 - \frac{1}{(1+r)^n}}{r} \right] \\
 &= FV \cdot \left[\frac{(1+r)^n - 1}{r \cdot (1+r)^n} \right]
 \end{aligned}$$

If we are interested in knowing the future result of a series of future values that are all the same in each accounting period, the mathematical equation is:

$$\text{If } PV = \text{fixed value} \Rightarrow FV = PV \times \left[\frac{(1+r)^n - 1}{r} \right]$$

However, benefits or future costs should be allocated taking into account the probability of their occurrence. Therefore, expected benefit or cost per accounting

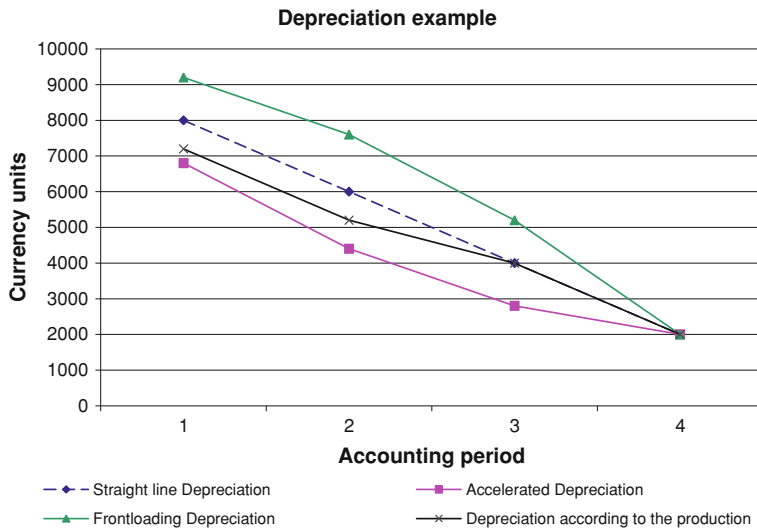


Fig. 3.6 Accounting value due to depreciation in the truck

period must be multiplied by its estimated probability of occurrence. In line with [25] we can present the mathematical equations for the revaluation and present time considering the probability of several choices (m) of each FV_{ij} with a probability P_j of its occurrence, then the PV and accumulated PD would be obtained as:

$$PV = \sum_{j=1}^m \frac{FV_{ij} \times P_j}{(1+r)^i}$$

$$\text{Accumulated PV} = \sum_{i=1}^n \sum_{j=1}^m \frac{FV_{ij} \times P_j}{(1+r)^i}$$

As example of application of these equations, suppose that a company has a building whose value according to the market is 100,000 currency units and that it increases its value at a discount rate of 5% per year. Its future value in four year time will then be:

$$FV_i = PV \times (1+r)^i = 100,000 \times (1+0.05)^4 = 121550.625$$

At the same time, if we have an asset whose operation produces a profit of 1,000 currency units annually during four years, the expected future value of that money at an annual interest rate of 5%, at the end of the four years period will be:

Table 3.2 Example of present value with several possibilities per period

	Accounting period					
	1.00		2.00		3.00	
	Future value	Probability	Future value	Probability	Future value	Probability
Option I	1000.00	0.35	1750.00	0.25	1500.00	0.45
Option II	5000.00	0.15	1500.00	0.30	3500.00	0.20
Option III	500.00	0.50	1250.00	0.45	2500.00	0.35
Sum of future values	1350.00		1450.00		2250.00	
Interest rate	0.05		0.05		0.05	
Present values	1285.71		1315.19		1943.63	
Accumulated present value	4544.54					

$$FV = PV \times \left[\frac{(1 + r)^n - 1}{r} \right] = 1,000 \times \left[\frac{(1 + 0.05)^4 - 1}{0.05} \right] = 4310.125$$

Obviously, in the last case, the accumulated present value of all the future values will be less than the sum of all the future values since they will be devaluated because of the interest rate.

$$\begin{aligned} \text{Accumulated PV} &= FV \times \left[\frac{1 - \frac{1}{(1 + r)^i}}{r} \right] = 1,000 \times \left[\frac{1 - \frac{1}{(1 + 0.05)^4}}{0.05} \right] \\ &= 3545.95 \end{aligned}$$

Let us put forward an example where there are three possible choices for each accounting period future values, with their corresponding probabilities. Table 3.2 presents that scenario.

3.2.5 Understanding Customer Life Cycle Value

In services companies, the most important source of income is the one that comes from the customers’ check books. Although some other possibilities could be available (for instance the regulator could offer grants to deploy certain services), customers are the major justification for the business sustainability.

Understanding customer value, as per the entire period of a commercial relationship with a company, is important and an extremely relevant research topic nowadays.

Customer Present Value (CPV) can be formalized including total income from a customer during a period of time minus all the costs required to serve that customer, such as selling costs, acquisition costs, retention costs and advertising costs [29]. CPV can be used as a reference benchmark, not only to support the generation of

future profits, but also to improve the customer segmentation process. According to Gupta and Lehmann [29] the equation for CPV could be expressed as follows:

$$CPV = \sum_{i=1}^n \left[\frac{(pa_i - c_i) \cdot p_i}{(1 + r)^i} \right] - AC$$

Where,

- $i = 1, \dots, n$ is the accounting periods.
- pa_i = price paid by a customer during the period i .
- c_i = direct cost of servicing the customer during the period i .
- r = discount rate or cost of capital for the company.
- p_i = probability of customer survival in the period i .
- AC = acquisition costs.

Consequently, the CPV can be increased if the servicing costs are decreased, the pricing is increased or the probability of retention is improved. So, thanks to the good disposition and work of the maintenance department, the company income can be increased reducing costs of operation, increasing customer satisfaction demanding new services and reducing the customer abandonment due to bad services. In addition, if the quality of the service is well accepted in the market, then the company reputation will be increased, so the acquisition of new customers will be facilitated.

Based on this, we could evaluate the efficiency of the network investments through the produced CPVs, that is to say, each asset support services for a determined group of customers, so the return on assets (ROA) is the total CPVs in relation to the total value of the operative assets considering the accumulated depreciation (VOA).

$$ROA = \left[\frac{\text{Total CPVs}}{\text{Total VOAs}} \right]$$

3.3 Maintenance Impact on Customer Value

It is a fact, however, that the value of good network maintenance is very rarely calculated from the customer/external point of view. Maintenance contributes to increasing the value provided to the customers and therefore the value that the customers represent to the company from several perspectives [67]:

- Customer relations, increasing service fidelity and reception through good and quick attention [18, 43].
- Price and quality, thanks to better availability and cost reduction, the price of the service decreases and customer satisfaction is improved.
- Internal motivation, image and external reputation [62].

Then, in response to the above, it is crucial to assess the impact of service quality due to maintenance in the customer perception and to estimate customer behaviour in front of a delivered service. It is also important to estimate relative value perception compared to the competition. Perception produces different behaviours, for example, when facing a problem, the customer can wait (asking for a monetary refunding) or start legal action. In both situations, the customer could stop buying our services or could be making negative propaganda and keep the bad experience in mind for the next purchase. Sometimes, however, the customer could remain [37].

A study showing the importance of complaints and the effects of poor quality in the customer relationship was carried out by the Technical Assistance Research Program (TARP), Washington EE.UU [27]. The study is about customer claims in more than 400 large corporations providing services in the United States, Japan and Europe. This TARP study determined that:

- Acquiring a new customer costs five times more than keeping one.
- Increasing customer retention 5% increases the profits between 25–80%.
- A satisfied customer recommends our company to an average of five partners, and an unsatisfied customer transmits the bad experience to an average of between nine to ten partners.
- Not all unsatisfied customers create claims, many drop out without communication, 96% of unsatisfied customers never complain.
- Only 17% of the unsatisfied customers intend to continue with the services.
- 80% of customer satisfaction is due to delivering services correctly, and the remaining 20% is due to resolving claims or problems.

Therefore, maintenance influences customer perception through service quality, increasing the satisfaction with a high quality and quick problem resolution, while customer satisfaction may decrease when maintenance cannot control failures and/or when the resolution of problems is not efficient.

Therefore a crucial issue is linking network dependability to possible levels of customer relationships affection due to violation of existing standard market service level agreements (SLA) with customers. According to Keaveney [37], service failures generate 44% of customer losses and as a consequence important changes in the “customer life-cycle value”. Changes that are due to the loss of customers or simply due to additional extra-expenses required for customer retention.

3.4 Maintenance Contribution in Terms of Intellectual Capital

Besides the impact on current network utility service cost, service performability and customer value, the maintenance organization can contribute to the potential future of the organization by generating knowledge. Knowledge generation will ensure future ability of the organization to perform its functions in a sustainable manner [59].

The knowledge of the maintenance staff is a strategic asset for the company and it can be considered as a competitive advantage within the service sector [20]. This organization knowledge value can be assessed by tangible and intangible aspects in the market. For instance, by the year 1977, the American Accounting Association (AAA) described the utilization of non-financial indicators to facilitate decision-making on the assessment of the human capital.

Many other contributions refer to the assessment of knowledge, illustrating the difficulty of this practice when it is applied to intangible issues which are hard to measure and to reproduce [57, 58].

Intellectual capital has been defined by the Euroforum [23] as *the set of intangible assets of an organization that, in spite of not being taken into account in traditional financial statements, at the present time indeed produce value or have the potential to produce it in the future*. That is to say, the true value of intellectual capital of an asset is not equal to its cost, but it is equal to the involved cost to replace or implement it. A reasonable value of any asset in the market is, therefore, equivalent to the cost of replacement [17].

It can be shown that intellectual capital is present besides the physical assets in three ways [23]:

- Human Capital or due to the employees of the company such as knowledge, experience and creativity.
- Structural Capital which considers the organizational skills to perform its function, such as information sources, technological and systems capital, industrial properties, government franchises, organizational culture, process efficiency, quality of service, innovation capability, etc.
- Relational Capital of the company with agents of its environment (customers, suppliers, public organizations, etc.) as trademark valuation, clients' portfolio, customers' relationships, etc.

According to this classification: How can we assess the contribution of maintenance to this intellectual capital? What is the value of intellectual capital of maintenance?

The International Accounting Standards Committee [30], the International Federation of Accountants (IFAC), and the Asociación Española de Administración y Dirección de Empresas (Spanish Association of Business Management, [5]) have submitted reports on the accounting of intellectual capital where in order to assess intellectual capital there must be a transaction and the value to consider must be the purchase price and the fact that they are going to produce future profits for the company and therefore could be amortised or depreciated. But these reports are not dealing with the accounting of industrial properties, organizational cultures, customer portfolios, etc., when these are generated inside the company.

Most of the consulted methods for the valuation of intellectual capital [20, 23, 54, 57, 58] use as reference equation the sum of different types of capital weighted by some efficiency indexes.

Consistent with this, and aligned with the calculation method developed by Edvinsson and Malone [20], in order to get the valuation of the intellectual capital of an organization, this can be expressed as the sum of the human capital, the

Table 3.3 Economic value of intellectual capital by type

Absolute values	Types of capital
Earnings	Human
Investment in training	Human
Investment in information technology (IT)	Structural
Investment in innovation and development	Structural
Costs in quality and in processes management	Structural
Prevention costs	Structural
Appraisal costs	Structural
Investment in customer relationship	Relational

structural capital and the relational capital multiplied by some efficiency indexes specific for each type of capital.

$$CI = i_H \cdot CH + i_S \cdot CS + i_R \cdot CR$$

where:

- CH, CS and CR stand for the economical value of human capital (CH), the structural capital (CE) and the relational capital (CR).
- i_H , i_E , i_R stand for the efficiency indexes of the intellectual capital, and in the case of more than one index in each type of capital existing, the index to be used in the calculation will be the arithmetic mean or average $[(i_1+i_2+\dots+i_n)/n]$ of all of them.

This equation captures Tobin's structure [60, 61] about intellectual capital, formalized as the product of a coefficient q by the economic value of the capital, trying to show (using q) the replacement value of the asset in the market.

$$q = \frac{\text{Purchase value of the installed Intellectual Capital}}{\text{Replacement value of the Intellectual Capital in the market}}$$

Thus, for values of $q > 1$, the capital value is greater than its material value, so assets fulfilling this are far more profitable than the mean of the market. However, if $q \leq 1$, the value of capital is less than or equal to its material value, so the valuation of assets is below market value.

Due to this comparison, the evaluation of intellectual capital makes sense if we use it in relation to other values of the sector in order to facilitate decision making, because a maintenance department that archives a 99% of availability is not the same as another maintenance department that obtains it exploiting resources and accelerating the assets degradation.

In addition, the different methods for intellectual capital evaluation, previously mentioned, employ some performance indicators as reference indexes such as: personnel motivation, customer satisfaction, personnel performance, rate of training, grade of ICT implementation, un-planned activities, efficiency in the use of resources, etc.

In the following Tables 3.3 and 3.4, we focus on the metrics that can help in the assessment of intellectual capital in maintenance.

Table 3.4 Economic value of intellectual capital by type

Efficiency indicators	Types of capital
Index of staff motivation	Human
Percentage of staff performance according to the individual goals	Human
Training index as percentage of the trained people inside the staff	Human
Percentage of developed skills pursuant to defined competences	Human
Efficiency of staff productive hours against the total hours	Human
Staff efficiency in the execution as percentage of tasks compliance against standards	Human
Percentage of use of IT by staff	Structural
Employment of IT by staff and forecasting techniques in critical assets as percentages against the totals	Structural
Proactivity, proactively versus reactively activation tasks	Structural
Efficiency of productive tasks against the total hours, unproductive tasks are those devoted to troubleshoot errors, rework and redundant tasks	Structural
Efficient use of assets as a percentage of their capacity	Structural
Percentage of scheduled tasks against the total tasks	Structural
Index of R&D efficiency as $1 - (\text{R\&D investments} / \text{expected future income})$	Structural
Market coverage	Relational
Index of customer satisfaction	Relational
Repetition rate of purchases by customers	Relational
Percentage of lost customers	Relational
Probability of occurrence of human accidents	Relational
Probability of occurrence of environmental accidents	Relational
Efficient customer service as a percentage of calls handled	Relational
Efficiency in the resolution of client requests	Relational
Contract and regulation fulfilment as $1 - (\text{costs of warranties and penalties} / \text{total costs})$	Relational

Taking into account these factors, we can use them to gather the right data and information regarding maintenance costs and indicators, helping us to value the intellectual capital of maintenance, providing a comparison with other firms in the sector.

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Chapter 4

Techniques and Tools for Maintenance Management

4.1 Decision-Making in Maintenance

In maintenance management, we must take many times decisions under high levels of uncertainty and that is not normally an easy task [20]. In this section we first introduce some basic and important concepts related to decision-making in management science and then we somehow discuss about the application of those concepts to the management of maintenance.

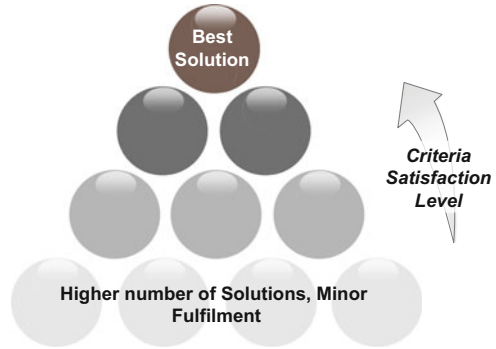
It is a fact that human rationality, when making decisions with limited knowledge and information, to employ emotional criteria to search for satisfactory solutions. This has been shown in different studies on human decisions that we can find in the literature [2, 3, 73, 76, 84]. Many contributions discuss about problems to optimize human decision-making processes when experiencing situations of high uncertainty [31, 63]. Especially for strategic decision-making, a deeper rational analysis in decision-making processes of this nature can be found in the literature [103, 104].

Real decision-making situations are often characterized as follows [51]:

- Seize dominates analysis in complex situations a one-dimensional analysis is not always the best solution.
- The decision process is interactive; it should be upgradeable as improvement continues.
- There are conflicts of value and interest.
- Complex methods of decision-making are difficult to accept.
- Decision methods are not the panacea, but they should be implemented within an adequate information and situation.

In many situations, decision makers may choose directly from the alternatives space but this contributes to the emergence of prejudices. On the other hand, when there are many alternatives, the ability of decision makers is overwhelmed. According to [48], in environments of uncertainty, people take decisions beyond

Fig. 4.1 Multicriteria decision-making



objective criteria, with heuristic shortcuts that can be modeled with patterns. Also, uncertainty in decision-making is present not only to predict something, but also to replicate reality. In this activity it is essential to use effectiveness analysis methods in order to minimize mistakes in estimating the future [5]. It is recommended to use formalized methods to improve decision capacity and consistency:

- Properly reflects value judgements of decision makers.
- Can incorporate consensus among decision makers.
- Are suitable to level of information required and available data.
- Are easy to understand and use.
- Allow a sensitivity method of analysis.

According to this, a decision-making process can be seen as the choice of what is “better” among what is “possible” [106]. The final decision will consist in the selection of one or more alternatives, based on information provided by the decision maker about their preferences on attributes and/or alternatives. Care must be taken not to devote too much importance or commitment to the prediction, but appropriate to the necessary implementation, seeking the Pareto Optimum, the better ranked alternative when it is not possible to improve more goals without harming others and rejecting those that seem “bad” (see Fig. 4.1). Although we will face different decision situations depending on the definition of what is best and what is possible.

In decision-making processes, we count on active people participation; therefore we must also consider motivational factors toward the decision goal. According to Maslow, humans are governed by five motivational factors [64, 65] that we should always take into consideration:

1. Physiological: those that seek primarily to maintain physical equilibrium and survival.
2. Security: they search for protection against danger or threat.
3. Social needs: to be accepted by others to feel part of social groups.
4. Recognition: to be appreciated and valued by others.
5. Need for fulfillment: to push people toward a total development and to use their full potential.

Besides this, notice that although people in general, and managers in particular, rely on information to make better decisions, there are also inputs related to their experience, knowledge and skills that will be of great value to decision-making and therefore cannot be forgotten [81]. This is even more important in problems for which effective resolution or troubleshooting methods are not in place [68].

If we now concentrate on maintenance management decision-making, there are several circumstances that we need to take into consideration:

- There are different organizational levels (strategic, tactical or operational) that should be aligned to the company goals.
- The maintenance department is always balancing [11] the highest quality with the lowest possible cost.
- Decision-making in maintenance is usually characterized by handling conflicts in conditions of stress and uncertainty. However, in maintenance it is unreliable or unjustifiable to use “intuition” or “smell”.
- In practice, the maintenance technicians often feel more confident with their experience, and this would influence their decision.
- The decision will be conservatively based on levels of satisfaction instead of being optimal. Also, according to our point of view, it is necessary to obtain and share knowledge to keep new information updated to maintain networks of service, due to its own characteristic of attention.
- As mentioned above, the maintenance decision-making is a complex problem that must be addressed following a well-structured method that takes into account the different criteria, alternatives and priorities [97].
- Maintenance managers deal with methods that are characterized to search for an easy way of implementation, discrete volume of alternatives and values.
- Modeling techniques should foster anticipation regarding developments of non-controlled variables, based on their historical evolution individually, or on their relationship with other variables. In addition, criteria can be aggregated and totally independent or stochastic.

According to all previously mentioned points and other recommendations found in the literature [10, 79], a method for decision-making in maintenance management should fulfill the following requirements:

- Aligned with company strategy.
- Model future predictions.
- Employ quantitative and qualitative information.
- Elicit and rationalize information from experts.
- Hierarchical structure.
- Easy to understand and to scheduled.
- Consensus among participants.

Consequently, to face environment uncertainty surrounding maintenance, quantitative techniques and tools are very much appreciated to seek for greater precision in the choice of strategies. We will now review the most common tools and techniques that have been used for decision-making in maintenance

management, facilitating the process management and ensuring on-going maintenance improvement. Many maintenance management tools are derived from quality and operations management techniques, while others come directly from the reliability analysis, prediction and improvement fields. In this chapter we will review the techniques following that precise order.

4.2 Management Techniques and Tools in Maintenance

Inside the reviewed international standards, best practices and maintenance management models, we have found the following recommendations about the use of the management tools in maintenance:

- Pintelon with Gelders [97] and Parodi-Herz [89] propose the employment of the subsequent techniques and tools:
 - Total productive maintenance (TPM)
 - Maintenance management information systems (MMIS)
 - Reliability Centered Maintenance (RCM) and reliability theory
 - Failure modelling
 - Statistics process control (SPC) and math programming (MP)
 - Check sheets
 - Decision theory and decision diagrams
 - Multi-criteria decision-making methods (MCDM)
 - Queuing theory
 - Time-motion study (TMS) Universal maintenance standards (UMS) (UMS for job processing times)
 - Project evaluation and review technique (PERT)
 - Work-flows diagrams
 - Project planning and scheduling
 - Manufacturing resource planning (MRP)
 - Optimized production technology (OPT)
 - Justin-time (JIT)
 - Deterministic optimization techniques
 - Markov decision theory and renewal theory
 - Simulation models
 - Inventory models (maintenance, repair and operating (MRO) supplies)
 - Game theory and decision theory.
- Wireman [111, 112] defines the techniques and tools arranged into five level of management such as:
 - Total productive maintenance (TPM)
 - Computerized maintenance management systems (CMMS)
 - Reliability centered maintenance (RCM)
 - Technical and interpersonal training

- Statistical financial optimization
- Work-flow diagrams.
- Garg and Deshmukh [29] highlights the next techniques and tools:
 - Total productive maintenance (TPM)
 - Reliability centered maintenance (RCM)
 - Staffing outsourcing services
 - Condition-based maintenance (CBM)
 - Prediction techniques
 - Balance score card (BSC)
 - Pareto analysis
 - Quality function deployment (QFD).
- Crespo [16] also defines the techniques and tools application according to the correspondent management level:
 - Total productive maintenance (TPM)
 - Expert systems and e-maintenance
 - Techniques of knowledge management (KM)
 - Reliability centred maintenance (RCM)
 - Balance score card (BSC)
 - Criticality analysis (CA) and Priorization matrix
 - Critical path method (CPM)
 - Programme evaluation review technique (PERT)
 - Process capability analysis
 - Queuing theory
 - Materials requirements planning/manufacturing requirements planning (MRP/MRP II) technique
 - Failure root cause analysis (FRCA)
 - Risk-cost optimization (RCO)
 - Life cycle cost analysis (LCCA)
 - Operational reliability analysis (ORA)
 - Replacement models
 - Markovian optimization models
 - Monte Carlo simulation models
 - Statistical inventory models.
- And there are other tools, unmentioned above, imported from quality and operations studies that could be useful to relate information for decision-making [36, 45] (Japan Management [46]) [80, 61]:
 - Arrow diagram (AD) and Activity network diagram (AND)
 - Interrelationship diagram (ID)
 - Strengths, weaknesses, opportunities, threats (SWOT)
 - Human resources management techniques (HRM)
 - Process decision program chart (PDPC)
 - Time series

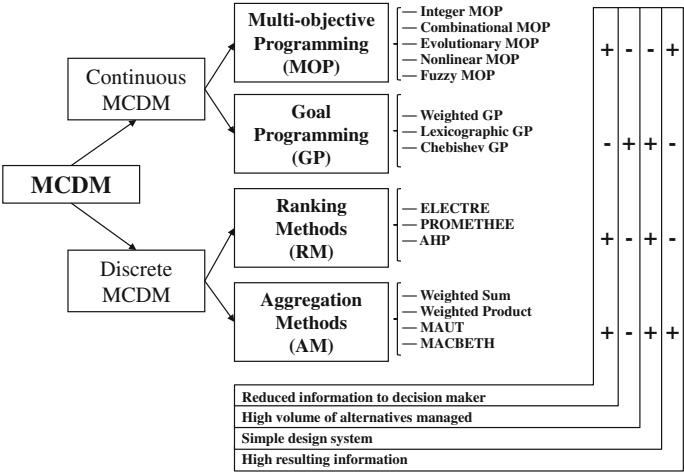


Fig. 4.2 Multi-criteria decision-making methods

- Causal models
- Quality lost function (QLF)
- Brainstorming
- Quality circles (QC)
- Run diagram
- Control charts
- Fishbone
- 5 Whys and 2 hows
- Histogram
- Pareto analysis.

We do not want to finalize this section without including a brief description about the multi-criteria decision-making methods (MCDM), which could be employed in both managerial levels in consort with other techniques and tools and with the ability to relate as much quantitative as qualitative criteria because in maintenance managers have to consider multiple subjects and constraints in a hierarchical form, some strategic, some tactical and some operative, prioritating or ranking the possible alternatives.

To select a proper MCDM, numerous articles compare them from different perspectives [7, 24, 28, 43, 71, 85, 86, 100, 115, 116, 118]. The appropriate method to use depends on the nature and the characteristics of the problem [118]. Based on the mentioned authors, we could classify the MCDM methods depending on the handled information as alternatives (see Fig. 4.2) into the following types and subtypes:

- *Continuous MCDM*. from an infinite set of alternatives, where efficient solutions could be reached by optimization of goals according to determined restrictions,

Multi-objective Programming; or simply satisfactory solutions could be sought by approximations of the set of goals to predefined aspiration thresholds, goal programming. These groups in turn could be divided in categories:

- *Multi-objective programming (MOP)*. Where the main found methods are Integer MOP, Combinational MOP, Evolutionary MOP, Nonlinear MOP and Fuzzy MOP; according to the mechanism to make scalar the problem and derive the effective Pareto Optimal solution.
- *Goal programming (GP)*. Within it the following methods are distinguished, Weighted GP, Lexicographic GP and Chebishev GP; depending on the way to prioritize the various goals approximation by respectively: weights, lexicographic minimization and individual goals minimization (MINIMAX).
- *Discrete MCDM*. from a finite set of alternatives, where reasonable solutions could be Ranked or Aggregated, employing criteria by weights of importance or preferences prioritation.
- *Ranking methods (RM)*. Mostly employed in the revisions are elimination and et choice translating reality (ELECTRE), preference ranking organization method for enrichment and evaluation (PROMETHEE) and analytic hierarchy process (AHP).
- *Aggregation methods (AM)*. Above all, these are the more applied methods: weighted sum, weighted product, multi-attribute utility theory (MAUT), and measuring attractiveness by a categorical-based evaluation technique (MACBETH).

To make the correct decision is not always easy, it must be made by managing problems correctly in all their size, and analyzing benefits or saving that leads to execution [70]. In this sense, the reviewed author's recommendations emphasize each type of MCDM depending on necessary conditions of the problem as volume of information required to decision maker, volume of alternatives managed, facilities for execution on computers and volume of resulting information.

Thus, MOP methods are preferable when there is little information regarding the decision maker and the volume of resulting information is high. GP methods are simplest when the volume of alternatives is high but involves high information of the decision maker. Ranking methods are selected when the required information to the decision maker is minimized and the resulting information could be easily obtained in order by computer, and aggregation methods as before mentioned obtaining ranking or choices as well.

Finally, it is important to remember that for many decision-making activities, the use of a simple reference framework to structure and facilitate decisions can become a key issue. In those scenarios, methods with an easier implementation can be of great application [50].

4.3 Reliability-Based Techniques and Tools in Maintenance

According to International Electrotechnical Commission [44] reliability is “*the ability of an item to perform a required function under given conditions for a given time interval*”, so this ability is expressed in terms of probability to fulfill the required function inside a determined level of performance and outside this level the functioning is considered a failure. As the reliability has to take into account the given operational conditions, this ability in network utilities should be tracked carefully because the operational conditions not only change with the supported capacity but also with the environmental conditions. Attributable to non-optimal conditions of operation, failures often occur in the assets ranging over the years by environmental factors (cleanliness, fastening, temperature, etc.), by operational changes (configurations, preventive maintenance, undue handling, etc.) and by defects in the assets (design imperfection, implementation errors, quality of materials, etc.) [16, 92]. Therefore, the main causes of failures can be classified as follows:

- Physical when the physical cause is the reason why the asset failed, the technical explanation on why things broke or failed.
- Human when they are produced as a result of human errors (omission or commission) resulting in physical roots.
- Latents when they are induced by deficiencies in the management systems that allow the human errors to continue unchecked (flaws in the systems and procedures).

In general, due to the uncertainty of the failure appearance, the study of failures used to be realized by stochastic process versus time over random variables characterized by probability density functions [27, 32] that let us to obtain approaching methods in order to dismiss the error in predictions about their appearance. Therefore, it is crucial to know the behavior of the random variables through historical data of failures, as in the own experience as well as based on accepted compilation databases or books about them such as EuReDatA [24], PERD [90] or OREDA [84].

Reliability centered maintenance (RCM) is the most widespread methodology to study the assets behavior before failures that could affect their main functions in determined conditions of operation, analyzing the causes [72], quantifying the risks [98] and evaluating the remedial measures to detect, avoid or prevent the failures [11].

Before beginning with the reliability-based techniques we will introduce the basic reliability concepts on which they are based.

4.3.1 Introduction to Reliability Functions

Therefore, in a given time interval $(0, t)$, the reliability function $R(t)$ is the opposite of the probability of the occurrence of any failure as we will see in the following basic formulae ($t \geq 0$):

- Probability density function of failures (PDF) ($f(t)$):

$$f(t) = -\frac{dR(t)}{dt} \quad (4.1)$$

- Cumulative distribution function of failures (CDF) ($F(t)$):

$$F(t) = \int_{t_1}^{t_2} f(t) \cdot dt \quad (4.2)$$

- Reliability function ($R(t)$):

$$R(t) = 1 - F(t) \quad (4.3)$$

- Mean time between failures (MTBF)

$$\text{MTBF} = \frac{\text{operation times}}{\text{number of failures}} = \int_{-\infty}^{+\infty} t \cdot f(t) \cdot dt = \int_0^{\infty} R(t) \cdot dt \quad (4.4)$$

- Frequency of failures (δ_f):

$$\delta_f = \frac{1}{\text{MTBF}} \quad (4.5)$$

- Mean time to repair (MTTR):

$$\text{MTTR} = \frac{\text{repair times}}{\text{number of failures}} \quad (4.6)$$

- Failure rate ($\lambda(t)$) or conditional probability of failure:

$$h(t) = -\frac{R'(t)}{R(t)} = \frac{f(t)}{R(t)}; \text{ where solving } R(t) = R(t=0) \cdot e^{-\int_0^t h(t)dt} \quad (4.7)$$

- Cumulative failure rate ($\lambda(t)$):

$$H(t) = \int_0^t h(t) \cdot dt = -\text{Ln} \left(\frac{R(t)}{R(0)} \right) \quad (4.8)$$

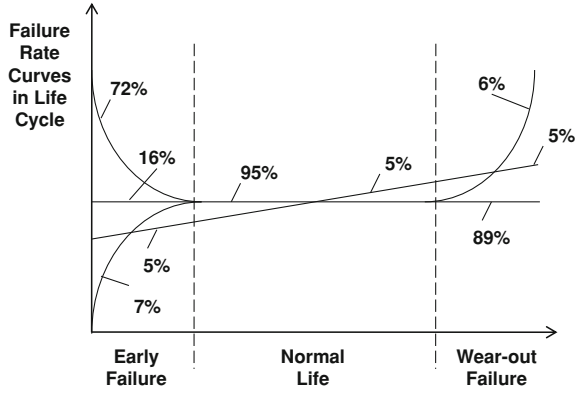
- Mean availability:

$$A = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}} \quad (4.9)$$

- Mean residual life (MRL) is the expected value of the remaining life of surviving after a time t :

$$\text{MRL}(t) = \frac{\int_t^{\infty} R(t) \cdot dt}{R(t)} \quad (4.10)$$

Fig. 4.3 Typical curves of failure rate (modified from [87])



MRL fulfils that [75]:

$$h(t) = \frac{\text{MRL}'(t) + 1}{\text{MRL}(t)}; \quad \text{if } \text{MRL}'(t) \geq -1 \quad (4.11)$$

Some approximations of MRL could be in determined mild conditions [6]:

$$\text{MRL}(t) \approx \frac{1}{h(t)} \quad (4.12)$$

Some of the most broadly used probability density functions in reliability are the exponential and Weibull, being the former for allowing easier calculations while the latter for allowing the representation of other probability functions such as exponential, Rayleigh or normal probability functions as well. In this sense, the most frequent curves for the failure rate according to Nowlan and Heap [82], see Fig. 4.3, could be described either a single one, or by modifications [74, 78], or by a combination of several Weibull curves ones [47, 108] for the three typical phases of asset life cycle: early failure, normal life and wear-out failure.

- Reliability formulae of the exponential probability function:

$$\begin{aligned} f(t) &= \lambda \cdot e^{-\lambda t}; & F(t) &= 1 - e^{-\lambda t}; & R(t) &= e^{-\lambda t}; \\ h(t) &= \frac{f(t)}{R(t)} = \lambda; & H(t) &= \lambda \cdot t; & \text{MTBF} &= \frac{1}{\lambda}; & \text{MRL}(t) &= \frac{1}{\lambda} \\ \text{Var}(t) &= \frac{1}{\lambda^2}; & \text{Median life} &= \frac{1}{\lambda} \cdot \ln(2) = 0.693 \cdot \frac{1}{\lambda} \end{aligned} \quad (4.13)$$

- Reliability formulae of the Weibull probability function (with origin in $\gamma = 0$):

$$\begin{aligned}
 f(t) &= \left(\frac{\beta \cdot t^{\beta-1}}{\alpha^\beta} \right) \cdot e^{-\frac{t^\beta}{\alpha^\beta}}; & F(t) &= 1 - e^{-(t/\alpha)^\beta}; & R(t) &= e^{-(t/\alpha)^\beta}; \\
 h(t) &= \frac{\beta}{\alpha} \cdot \left(\frac{t}{\alpha} \right)^{\beta-1}; & H(t) &= \left(\frac{t}{\alpha} \right)^\beta; & \text{Median Life} &= \alpha \cdot \ln(2)^{1/\beta}; \\
 \text{MTBF} &= \alpha \cdot \int_0^\infty t^{1/\beta} \cdot e^{-t} \cdot dt = \alpha \cdot \Gamma\left(1 + \frac{1}{\beta}\right); \\
 \text{Var}(t) &= \alpha^2 \cdot \left[\Gamma\left(1 + \frac{2}{\beta}\right) - \Gamma\left(1 + \frac{1}{\beta}\right)^2 \right]
 \end{aligned} \tag{4.14}$$

$$\text{MRL}(t) = \frac{\alpha}{\beta} \cdot e^\tau \cdot \Gamma\left(\frac{1}{\beta}\right) \cdot \left[1 - I\left(\frac{1}{\beta}, \tau\right) \right] \tag{4.15}$$

$$\text{Using } \tau = \left(\frac{t}{\alpha}\right)^\beta; \quad \Gamma(z) = \int_0^\infty x^{z-1} \cdot e^{-x} \cdot dx; \text{ and } I(z, \tau) = \Gamma(z)^{-1} \cdot \int_0^\tau x^{z-1} \cdot e^{-x} \cdot dx$$

The definition of the MRL for the Weibull function is extracted from Nassar and Eissa [77],

where the Weibull parameters are the Shape parameter (β) which represents the slope of the line describing the rate change of the rate of occurrence of failures (with normalized values) [25] and the scale parameter (α) or characteristic life, spreading of the distribution over time (with normalized values). The characteristic life parameter represents an age such as the probability of that age being exceeded is 36.79%, or, equivalently that the probability of not exceeding that age is 63.21%.

If $\beta = 1$ the Weibull distribution becomes the exponential distribution, for values between 1.5 and 2.5 indicate stress deterioration, for values greater $\beta = 3.2$ the Weibull curve approaches a normal distribution.

Both formulae, given a number of observations (N) with the data for $[t_1, t_2, \dots, t_N]$, must be solved to find the parameters λ in the case of exponential distributions or α and β in the case of weibull distributions (see representations for both in Fig. 4.4). In order to do so, the computation will be done using an estimation method the MLE, with the function of maximum likelihood that estimates the most likely values of the parameters for a given set of data, being better the convergence of the estimation following the estimation a normal probability distribution when the number of observations N becomes bigger. This function is hard to handle due to be compound by joint density function for all observed data, in order to simplify the computation; it is used to take logarithms on both sides, obtaining the logarithmic maximum likelihood probability function which is easier to deal maximizing logarithmic L through partial derivatives taken with respect to λ or α and β

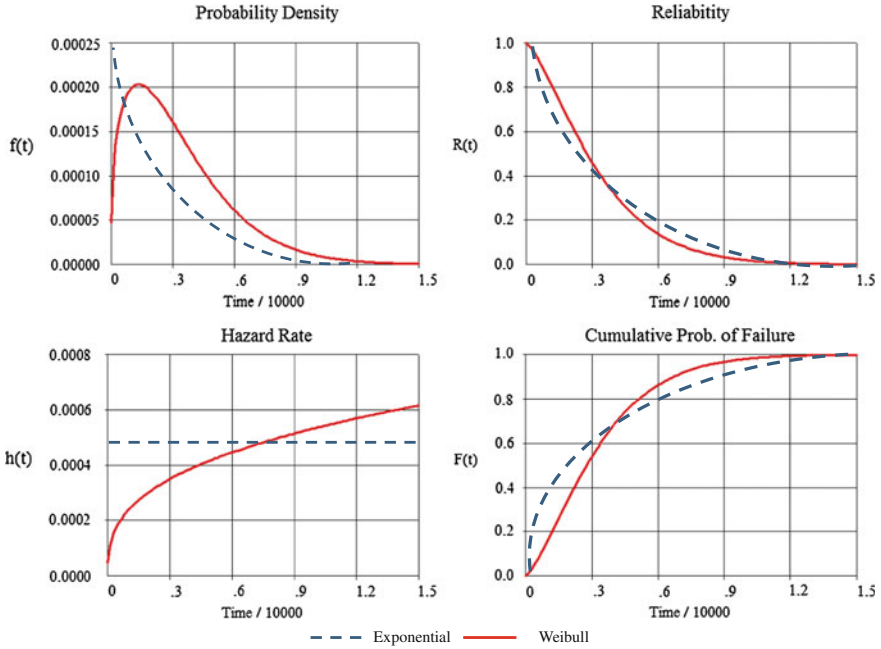
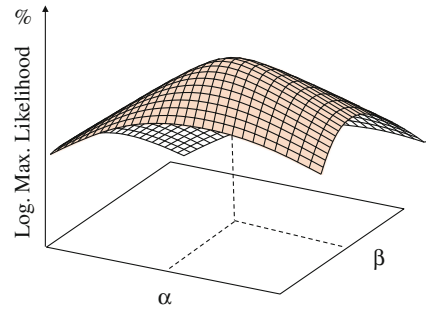


Fig. 4.4 Representations of basic functions for exponential and weibull cases

Fig. 4.5 MLE representations for the Weibull case



equal to 0, respectively (see graphical representation of log-likelihood in the case of weibull in Fig. 4.5).

- MLE estimation of Exponential probability function:

$$\begin{aligned}
 L\{\text{data}|\lambda\} &= L\{t_1, t_2, \dots, t_N|\lambda\} = \prod_{i=1}^N f(t_i|\lambda) \\
 &= L = \lambda^N \cdot \exp\left[-\lambda \cdot \left(\sum_{i=1}^N t_i\right)\right]
 \end{aligned}
 \tag{4.16}$$

Through partial derivatives of logarithms taken with respect to λ equal to 0:

$$\frac{\partial \ln L}{\partial \lambda} = -N \cdot \lambda + \lambda^2 \cdot \sum_{i=1}^N t_i = 0 \Rightarrow \lambda = \frac{N}{\sum_{i=1}^N t_i} \quad (4.17)$$

- MLE estimation of Weibull probability function:

$$\begin{aligned} L\{\text{data}|\alpha, \cdot\beta\} &= L\{t_1, t_2, \dots, t_N|\alpha, \cdot\beta\} = \prod_{i=1}^N f(t_i|\alpha, \cdot\beta) = L \\ L &= \prod_{i=1}^N \alpha^{-1} \cdot \beta \cdot \left(\frac{t_i}{\alpha}\right)^{\beta-1} \cdot \exp\left[-\left(\frac{t_i}{\alpha}\right)^{\beta}\right] \\ &= \alpha^{-N} \cdot \beta^N \cdot \prod_{i=1}^N \left(\frac{t_i}{\alpha}\right)^{\beta-1} \cdot \exp\left[-\left(\frac{t_i}{\alpha}\right)^{\beta}\right] \end{aligned} \quad (4.18)$$

Through partial derivatives of logarithms taken with respect to α and β equal to 0:

$$\begin{aligned} \frac{\partial \ln L}{\partial \alpha} &= \sum_{i=1}^N \left[\left(\frac{1}{\beta} + \ln\left(\frac{t_i}{\alpha}\right) \right) \cdot \left(1 - \left(\frac{t_i}{\alpha}\right)^{\beta} \right) \right] = 0; \\ \frac{\partial \ln L}{\partial \beta} &= -N \cdot \frac{\beta}{\alpha} + \frac{\beta}{\alpha^{\beta+1}} \cdot \sum_{i=1}^N t_i^{\beta} = 0 \end{aligned} \quad (4.19)$$

Then:

$$\begin{aligned} \dot{\alpha} &= \left(\frac{N}{\sum_{i=1}^N t_i^{\beta}} \right)^{\beta}; \\ \dot{\beta} &= \frac{N \cdot \sum_{i=1}^N t_i^{\beta}}{N \cdot \sum_{i=1}^N t_i^{\beta} \cdot \ln(t_i) - \sum_{i=1}^N t_i^{\beta} \cdot \sum_{i=1}^N \ln(t_i)} \end{aligned} \quad (4.20)$$

where $\dot{\alpha}$ and $\dot{\beta}$ are estimations of the parameters α and β with MLE.

In this way, the resulting system of equations has the same number of variables and equations, so the system can be solved, although sometimes there is no easy way to solve it and statistical methods, such as Newton–Raphson method, are used due to the greater rate of convergence. The goodness of the solution can be analyzed according to the Fisher matrix [67].

$$\Sigma = \begin{bmatrix} \text{var}(\alpha) & \text{cov}(\alpha, \beta) \\ \text{cov}(\alpha, \beta) & \text{var}(\beta) \end{bmatrix} = \begin{bmatrix} -\frac{\partial^2 \ln L}{\partial \alpha^2} & -\frac{\partial^2 \ln L}{\partial \alpha \partial \beta} \\ -\frac{\partial^2 \ln L}{\partial \alpha \partial \beta} & -\frac{\partial^2 \ln L}{\partial \beta^2} \end{bmatrix}^{-1} \quad (4.21)$$

Table 4.1 Exponential and Weibull linearization

	Exponential	Weibull
$\dot{F}(t)$	$1 - \exp(-\lambda \cdot t)$	$1 - \exp\left(-\left[\frac{t}{\alpha}\right]^\beta\right)$
$y = a \cdot x + b$	$-\ln(1 - F(t)) = \lambda \cdot t$	$\ln[-\ln(1 - F(t))] = -\beta \ln(\alpha) + \beta \ln(t)$

4.3.2 Fitting Failure Distribution Functions

Focusing on reliability analysis and due to the stochastic behavior of failures, we have to estimate probabilities based on historical data, one obtained from real failures and others experimented stressing the equipments in specific conditions of operation. Consequently, based on historical or tested experience and correlating these with information about operation, condition or the environment; frequencies and tendencies of failures could be deduced pursuing the prediction and optimization about resources and activities in maintenance.

In order to determine the reliability behavior, the historical data could be considered according to the possibility of censored or uncensored observations [23]. In maintenance evaluations sometimes the failure time of some equipments has not yet been produced or simply it is unobserved, then the observation is censored, right censored in the first case because the analysis time is truncated before its occurrence and left censored for the last case when the occurrence of the failure is known but not the precise moment before a determined time. Another type is the interval censored when the occurrence of the failure is known but inside a determined interval of time. Taking into consideration the correct documentation of failures, we will focus on right censored data for our calculations so as to simplify. Thus, for a set of n data orderly in pairs $[(t_1, d_1), (t_2, d_2), \dots, (t_n, d_n)]$ with:

- $d_i = 1$, if t_i is the time that the individual i has survived or has failed.
- $d_i = 0$, if t_i is the time in which the individual has not survived.

Both exponential and weibull distributions could be estimated easily with actuarial and graphical approaches through their linearization in a straight line (see Table 4.1) using different methods to approximate the $F(t)$ (in a life table or in a non linear vertical axis in a probability plot, respectively) according to a geometrical sequence sample (in the horizontal axis) [21, 38].

The estimation of $F(t)$ is shown as $\dot{F}(t)$ in the approaches, in which the parameters a and b are obtained by a linear regression.

- Midpoint plotting position: $\dot{F}_i = (i - 0, 5)/(n + 1)$
- Median plotting position: $\dot{F}_i = (i - 0, 3)/(n + 0, 4)$
- Mean plotting position: $\dot{F}_i = i/(n + 1)$
- Kaplan–Meier algorithm: $\dot{F}_i = \dot{F}_{i-1} \cdot \frac{n-i}{n_i}$; with $\dot{F}_0 = 1$

and n_i is n minus the accumulated failure and censored data at the moment i .

Table 4.2 Weibull estimation of linearized parameters a and b

Sample data			$\hat{F}_i$ Estimation				$\text{Ln}(-\text{Ln}(1 - \hat{F}_i))$			
Time	$\text{Ln}(t_i)$	i	Mean	Mid	Median	K-M	Mean	Mid	Median	K-M
106	4.662	1	0.029	0.015	0.021	0.030	-3.511	-4.182	-3.855	-3.481
411	6.020	2	0.059	0.045	0.051	0.061	-2.803	-3.068	-2.952	-2.772
500	6.215	3	0.088	0.076	0.081	0.091	-2.382	-2.541	-2.473	-2.351
689	6.535	4	0.118	0.106	0.111	0.121	-2.078	-2.188	-2.142	-2.046
773	6.650	5	0.147	0.136	0.141	0.152	-1.838	-1.920	-1.886	-1.806
901	6.803	6	0.176	0.167	0.171	0.182	-1.639	-1.702	-1.676	-1.606
915	6.819	7	0.206	0.197	0.201	0.212	-1.467	-1.517	-1.497	-1.434
1,034	6.942	8	0.235	0.227	0.231	0.242	-1.316	-1.355	-1.339	-1.281
1,123	7.024	9	0.265	0.258	0.260	0.273	-1.179	-1.211	-1.198	-1.144
1,290	7.162	10	0.294	0.288	0.290	0.303	-1.055	-1.080	-1.070	-1.019
1,300	7.170	11	0.324	0.318	0.320	0.333	-0.939	-0.960	-0.951	-0.903
2,227	7.708	12	0.353	0.348	0.350	0.364	-0.832	-0.848	-0.841	-0.794
2,650	7.882	13	0.382	0.379	0.380	0.394	-0.730	-0.742	-0.737	-0.692
2,690	7.897	14	0.412	0.409	0.410	0.424	-0.634	-0.642	-0.639	-0.594
2,733	7.913	15	0.441	0.439	0.440	0.455	-0.541	-0.547	-0.545	-0.501
3,011	8.010	16	0.471	0.470	0.470	0.485	-0.453	-0.455	-0.454	-0.411
3,121	8.046	17	0.500	0.500	0.500	0.515	-0.367	-0.367	-0.367	-0.323
3,524	8.167	18	0.529	0.530	0.530	0.545	-0.283	-0.280	-0.281	-0.238
3,624	8.195	19	0.559	0.561	0.560	0.576	-0.201	-0.196	-0.198	-0.154
3,758	8.232	20	0.588	0.591	0.590	0.606	-0.120	-0.112	-0.115	-0.071
3,929	8.276	21	0.618	0.621	0.620	0.636	-0.039	-0.030	-0.034	0.012
4,328	8.373	22	0.647	0.652	0.650	0.667	0.041	0.053	0.048	0.094
4,704	8.456	23	0.676	0.682	0.680	0.697	0.121	0.136	0.130	0.177
5,000	8.517	24	0.706	0.712	0.710	0.727	0.202	0.219	0.212	0.262
5,053	8.528	25	0.735	0.742	0.740	0.758	0.285	0.305	0.297	0.349
5,473	8.608	26	0.765	0.773	0.769	0.788	0.369	0.393	0.383	0.439
5,569	8.625	27	0.794	0.803	0.799	0.818	0.458	0.485	0.474	0.533
5,886	8.680	28	0.824	0.833	0.829	0.848	0.551	0.583	0.570	0.635
5,946	8.691	29	0.853	0.864	0.859	0.879	0.651	0.689	0.673	0.747
6,018	8.703	30	0.882	0.894	0.889	0.909	0.761	0.808	0.789	0.875
6,200	8.732	31	0.912	0.924	0.919	0.939	0.887	0.948	0.922	1.031
7,203	8.882	32	0.941	0.955	0.949	0.970	1.041	1.129	1.091	1.252
8,760	9.078	33	0.971	0.985	0.979	1.000	1.260	1.433	1.352	0.000

The first two estimation approaches used to be applied in the case of a reduced number of samples, while the last two used to be employed with a large number or with censored data. We will see an example of this fitting the weibull distribution for a set of failures in Table 4.2.

As a result, once the historical data have been ordered chronologically and ranked for the calculation, using the linear regression for the four cases mean, mid, median and K-M, we have obtained the Weibull parameters in Table 4.3. Those values are also compared with the maximum likelihood method estimation (MLE) $\beta = 1.4095$ and $\alpha = 3,656$ and then represented in Fig. 4.6.

Table 4.3 Weibull estimation of parameters α and β

	MeanP		Mid		Median		K–M	
Coeffs	1.111	−9.166	1.228	−10.101	1.174	−9.669	1.080	−8.903
Stderr	0.037	0.291	0.037	0.290	0.037	0.289	0.049	0.384
R ² , Sy	0.967	0.212	0.973	0.211	0.970	0.210	0.940	0.280
F, DofF	895	31	1098	31	1014	31	483	31
RSS, SSE	40.19	1.39	49.07	1.38	44.86	1.37	37.97	2.43
	β	A	B	α	β	α	β	A
	1.111	3822.2	1.228	3739.65	1.174	3775.4	1.08	3815.8

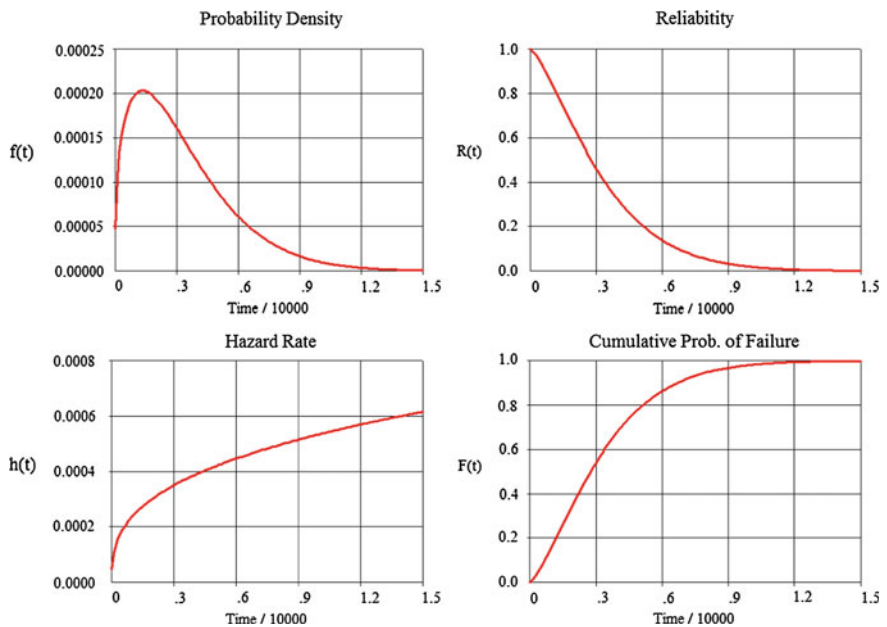


Fig. 4.6 Weibull functions representation for the MLE

The assessment of the goodness of fit is the next step. In order to do so we can follow the Kolmogorov–Smirnov test, consisting of the following steps:

- (a) Select the distribution to assess;
- (b) Determine the significance level of the test (α usually at 1, 5, 10 or 20%), which is the probability of rejecting the hypothesis that the data follows the chosen distribution assuming the hypothesis is true;
- (c) Determine $F(t_i)$ using the parameters assumed in step (a);
- (d) From the failure data compute the $\hat{F}(t_i)$ using the median ranks or the Kaplan–Meier estimate if applicable;
- (e) Determine d , where
$$d = \text{Max}_i \{ |F(t_i) - \hat{F}(t_i)|, |F(t_i) - \hat{F}(t_{i-1})| \}; \text{ P}[d \leq d_\alpha] = 1 - \alpha;$$

Table 4.4 Weibull estimation of parameters α and β

Critical values (Cv)	α			Adjusting value (Av)
	0.1	0.05	0.01	
General	1.224	1.358	1.628	$\sqrt{n} + 0.12 + 0.11/\sqrt{n}$
Exponential	0.990	1.094	1.308	$\sqrt{n} + 0.12 + 0.11/\sqrt{n}$
Normal	0.819	0.895	1.035	$\sqrt{n} - 0.01 + 0.85/\sqrt{n}$
Weibull $n = 10$	0.760	0.819	0.944	$\sqrt{n}$
Weibull $n = 20$	0.779	0.843	0.973	$\sqrt{n}$
Weibull $n = 50$	0.790	0.856	0.988	$\sqrt{n}$
Weibull $n = \infty$	0.803	0.874	1.007	$\sqrt{n}$

(f) If $d > d_\alpha$, where d_α is obtained from the K–S statistic table (see [66]), we reject the hypothesis that data can be adjusted to the distribution selected in step (a).

We can also estimate d_α with a confidence level of $1 - \alpha$ as $d_\alpha = Cv/Av$ following Law and Kelton [58], adjusting for the different probability distribution functions and significance levels (see Table 4.4), where Cv are the critical values and Av are the adjusting values depending both on the distribution hypothesis:

For the previous example, results for this Kolmogorov–Smirnov test are, in the case of the maximum likelihood method estimation (MLE) $\beta = 1.4095$ and $\alpha = 3.656$, $d = 0.119943$ (using median rank and the approximate significance level of 0.01) and $d_\alpha = 0.973/\sqrt{n} = 0.973/\sqrt{33} = 0.16938$, so the hypothesis of weibull distribution for the data is true because $d = 0.119943 < d_\alpha = 0.16938$.

Other useful approximation is the estimation of the mean residual time. This function is represented above in a parametric estimation, being more difficult to calculate for distributions apart from exponential distributions. Therefore, we will present non-parametric estimations of monotonic MRL that provide a good accuracy when the set of data is large. On simple non-parametric approximation of the MRL is the presented by Yang [113, 114] in the case of a right-censored and ordered set of n data [33, 102]:

$$\text{MRL}(t) = \frac{1}{n-i} \cdot \left[\sum_{j=1+i}^n t_j - t \right]; \quad 0 \leq t < t_n \quad \text{and} \quad \text{MRL}(t) = 0 \quad \text{for } t \geq t_n \quad (4.22)$$

Thus, with the data of the last example, we could estimate MRL using the Yang's formulae. The estimated MRL could also be employed to obtain the Weibull parameters α and β by optimization through the solve function of excel, as a nonlinear regression among the Weibull parameters and the estimations of MRL in each observed time (see Table 4.5).

This estimation of Weibull parameters is not as accurate as the estimation based on the cumulative distribution function (CDF) due to the difficulty to approximate to the real MRL function of Weibull, working out the standard error ($\text{SE_of_}\hat{m}(t) = 270.9$) and the correlation index or coefficient of determination ($R^2 = 0.897$).

Table 4.5 MRL estimation through Yang's formulae, optimization and MLE

i	t_i	$\text{Sum}(t_j)$	$\hat{m}(t)$	T	$\text{opt_}\hat{m}(t)$	Upper Lim	Lower Lim	$\text{MLE_}\hat{m}(t)$
0	0	0	3347.00	0.00	3791.03	4344.33	3237.72	3328.67
1	106	106	3342.46	0.00	3688.11	4241.41	3134.80	3308.46
2	411	517	3134.84	0.01	3417.80	3971.10	2864.49	3216.81
3	500	1,017	3147.79	0.02	3345.92	3899.23	2792.62	3187.50
4	689	1,706	3060.92	0.03	3201.21	3754.52	2647.90	3123.94
5	773	2,479	3083.18	0.04	3140.29	3693.60	2586.98	3095.54
6	901	3,380	3064.60	0.05	3051.30	3604.61	2497.99	3052.49
7	915	4,295	3167.81	0.05	3041.76	3595.07	2488.45	3047.76
8	1,034	5,330	3170.39	0.06	2963.11	3516.42	2409.81	3008.07
9	1,123	6,452	3210.53	0.08	2907.20	3460.51	2353.89	2979.05
10	1,290	7,742	3175.64	0.10	2806.21	3359.51	2252.90	2925.01
11	1,300	9,042	3309.48	0.10	2800.33	3353.64	2247.02	2921.81
12	2,227	11,269	2495.85	0.28	2338.45	2891.76	1785.15	2649.17
13	2,650	13,919	2176.58	0.40	2170.42	2723.73	1617.11	2540.10
14	2,690	16,609	2249.16	0.41	2155.69	2708.99	1602.38	2530.29
15	2,733	19,343	2328.35	0.42	2139.88	2693.18	1586.57	2519.70
16	3,011	22,354	2171.08	0.51	2043.28	2596.59	1489.97	2454.03
17	3,121	25,475	2189.53	0.54	2007.08	2560.39	1453.77	2428.95
18	3,524	28,999	1905.90	0.69	1884.28	2437.59	1330.97	2341.94
19	3,624	32,623	1934.95	0.73	1855.90	2409.21	1302.60	2321.40
20	3,758	36,382	1939.33	0.78	1819.02	2372.33	1265.71	2294.46
21	3,929	40,311	1916.21	0.85	1774.03	2327.34	1220.72	2261.20
22	4,328	44,639	1,654.60	1.02	1676.25	2229.56	1122.94	2187.41
23	4,704	49,343	1406.54	1.20	1592.95	2146.25	1039.64	2122.85
24	5,000	54,343	1234.20	1.35	1532.63	2085.94	979.33	2075.09
25	5,053	59,396	1329.32	1.38	1522.35	2075.66	969.05	2066.86
26	5,473	64,869	1038.55	1.61	1444.56	1997.87	891.25	2003.74
27	5,569	70,438	1100.27	1.67	1427.93	1981.24	874.62	1990.04
28	5,886	76,324	939.28	1.86	1375.13	1928.43	821.82	1946.07
29	5,946	82,270	1098.93	1.89	1365.54	1918.84	812.23	1938.01
30	6,018	88,288	1369.36	1.94	1354.24	1907.54	800.93	1928.47
31	6,200	94,488	1781.36	2.05	1326.44	1879.75	773.14	1904.85
32	7,203	101,691	1557.28	2.74	1190.77	1744.08	637.46	1786.32
33	8,760	110,451						

$$\text{MRL}(t) = \frac{\alpha}{\beta} \cdot e^{\tau} \cdot \Gamma\left(\frac{1}{\beta}\right) \cdot \left[1 - I\left(\frac{1}{\beta}, \tau\right)\right] \quad (4.23)$$

In Table 4.5 we use three ways for calculating the MRL: one directly from Yang's formulae ($\hat{m}(t)$); another using optimization through the solve excel function $\text{opt_}\hat{m}(t)$, and finally a last one substituting in the equation of Nassar and Eissa [77] the Weibull parameters obtained by maximum likelihood estimation ($\text{MLE_}\hat{m}(t)$).

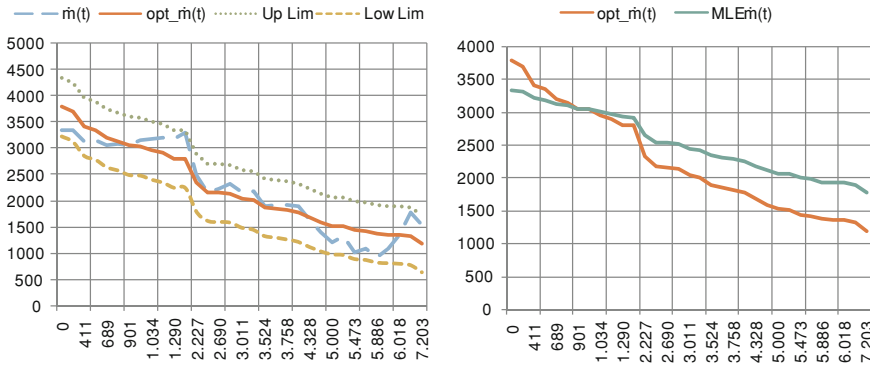


Fig. 4.7 Weibull MRL estimations

In addition, the different estimations of MRL have been presented graphically in the Fig. 4.7, where in the left side there are the Yang’s MRL compared with the optimized MRL and the upper and lower confidence limits (95%), while in the right side the optimized MRL is compared with the derived MRL from the Nassar and Eissa formulae and the MLE Weibull parameters.

4.3.3 Relationship Among the Failure Cause and the Environment

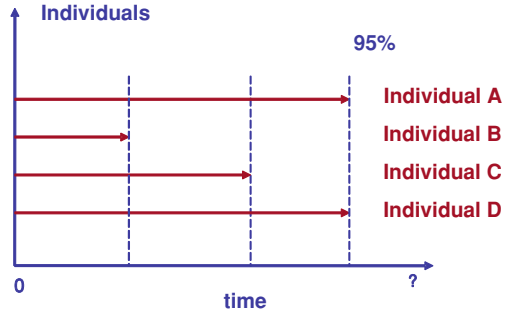
Sometimes, the data sample that we have to derive the distributions could be influenced, besides operating time, by assets operative environment variables. For instance, the reliability of an engine could be influenced by the operating temperature in addition to the operating time. That means that we could expect different reliability, in two engines working at different temperatures, during their life cycle. Therefore, the characterization of these factors contributing to reliability and their identification as prognostic variables could be extremely useful for us.

Data and information about these contributing factors could be decisive to obtain, and even to update over time, reliability estimations. This information could be then used to work out the real risk of an asset.

An example of this is the “Survival Data Analysis”, focused on a group of individuals and how they react to failure after certain length of time [14, 54, 55, 88] (see Fig. 4.8).

There are several techniques to solve survival estimations [49, 60, 105], in which typical failure distribution functions are asymmetrical (to the right). In these cases it is important not only the survival data per individuals, but also about the failed individuals. These techniques could be parametric when the hazard distributions are known or non-parametric when this is not necessary to be specified [1, 9, 13, 35, 39, 40, 59]. Examples are:

Fig. 4.8 Survival representation with censored data



- Non parametric: simpler and less efficient showing relationships among the life cycles and other variables.
 - Kaplan–Meier.
 - Logrank.
 - Cox regression.
- Parametric: more accurate and efficient showing other relationships.
 - Exponential distribution.
 - Weibull distribution.
 - Lognormal distribution.

The influence of these environmental factors may obey different patterns, though in order to simplify we will focus on proportional contributions of the environmental factors to the hazard risk and without time dependency, i.e. the relationship among the hazards of two similar assets with different environmental factors is proportional. This is the case of the Cox's proportional hazard model (PHM, [15], consequently, we would have a time-dependent component of hazard and another component based on a lineal-polynomial expression of the environmental factors (independent among them), where:

- $h(t)$ is the failure rate or conditional probability of failure at time t .
- $H(t)$ is the cumulative failure rate at time t .
- x_i is the environment factors or covariates that contribute to the hazard risk, considering constant to make simpler.
- γ_i is constant coefficients that model the contribution of each z_i .

$$h(t, x) = h_0(t) \cdot e^{\sum_{j=1}^k \gamma_j \cdot x_j} = \gamma_1 \cdot x_1 + \dots + \gamma_k \cdot x_k \quad (4.24)$$

$$H(t, x) = H_0(t, x) \cdot \exp \left[\sum_{j=1}^k \gamma_j \cdot x_j \right] \quad (4.25)$$

$$R(t) = R(t=0) \cdot \exp \left\{ - \int_0^t h(t,x) dt \right\} \quad (4.26)$$

$$\begin{aligned} R(t,x); &= \exp(-H(t,x)) = \exp[-H_o(t,x)]^{\exp \left[\sum_{j=1}^k \gamma_j \cdot x_j \right]} \\ &= R_o(t,x)^{\exp \left[\sum_{j=1}^k \gamma_j \cdot x_j \right]} \end{aligned} \quad (4.27)$$

A partial likelihood function quasi-likelihood model (QLM) based on a conditional probability of failure is recommended by Cox in order to estimate the γ_i coefficients, taking into account only the samples where an occurrence has been produced (m), instead of considering all the samples (N), as in the normal MLE [39].

$$\begin{aligned} L\{\text{data} | \vec{\gamma}\} &= L\{t_1, t_2, \dots, t_m | \vec{\gamma}\} = \prod_{m \in F} P(\text{failure in } t_i | \vec{x}_i) \\ &= \prod_{m \in F} \frac{h(t_i, \vec{x}_i)}{\sum_{l \in R(t_i)} h(t_i, \vec{x}_i)} = \prod_{m \in F} \frac{h_o(t_i) \exp \left\{ \sum_{j=1}^k \gamma_j \cdot x_{ij} \right\}}{\sum_{l \in R(t_i)} h_o(t_i) \exp \left\{ \sum_{j=1}^k \gamma_j \cdot x_{lj} \right\}} \\ &= \prod_{m \in F} \frac{\exp \left\{ \sum_{j=1}^k \gamma_j \cdot x_{ij} \right\}}{\sum_{l \in R(t_i)} \exp \left\{ \sum_{j=1}^k \gamma_j \cdot x_{lj} \right\}} \end{aligned} \quad (4.28)$$

Through partial derivatives of logarithms taken with respect to all the γ_i equal to 0, we get:

$$\begin{aligned} \text{Ln}(L); &= \text{Ln} \left(\prod_{m \in F} \frac{\exp \left\{ \sum_{j=1}^k \gamma_j \cdot x_{ij} \right\}}{\sum_{l \in R(t_i)} \exp \left\{ \sum_{j=1}^k \gamma_j \cdot x_{lj} \right\}} \right) \\ &= \sum_{m \in F} \left[\sum_{j=1}^k \gamma_j \cdot x_{ij} - \text{Ln} \left(\sum_{l \in R(t_i)} \exp \left\{ \sum_{j=1}^k \gamma_j \cdot x_{lj} \right\} \right) \right] \end{aligned} \quad (4.29)$$

Then

$$\frac{\partial \ln L}{\partial \gamma_k} = \frac{\partial}{\partial \gamma_i} \sum_{m \in F} \left[\sum_{j=1}^k \gamma_j \cdot x_{ij} - \text{Ln} \left(\sum_{l \in R(t_i)} \exp \left\{ \sum_{j=1}^k \gamma_j \cdot x_{lj} \right\} \right) \right] = 0 \quad (4.30)$$

$$\frac{\partial \ln L}{\partial \gamma_k} = \sum_{m \in F} \left[x_{ik} - \frac{\sum_{l \in R(t_i)} x_{lk} \cdot \exp \left\{ \sum_{j=1}^k \gamma_j \cdot x_{lj} \right\}}{\sum_{l \in R(t_i)} \exp \left\{ \sum_{j=1}^k \gamma_j \cdot x_{lj} \right\}} \right] = 0 \quad (4.31)$$

where:

- F is the set of the samples where a failure has occurred (m).
- R is the set of samples (l) in risk at time t_i .
- The statistic of QLM is: $-2 \cdot \log(L(\vec{\gamma}))$. The more the statistic of QLM is closer to 0, the more closer the likelihood function will be to 1, and therefore the estimation will be better.

Diverse statistical softwares can implement Cox's regression estimation. Another option is producing a simpler estimation employing optimization through the solve excel function or using a multiple linear regression estimation, based on the log equation of the failure rate, and taking the $h_0(t)$ for the case that all the covariates are zero:

$$\text{Ln}(h(t, x)) = \text{Ln}(h_0(t)) + \sum_{j=1}^k \gamma_j \cdot x_j \Rightarrow \frac{\text{Ln}(h(t, x))}{\text{Ln}(h_0(t))} = \sum_{j=1}^k \gamma_j \cdot x_j \quad (4.32)$$

Instead of employing the Cox's regression estimation, we can utilize parametric accelerated testing methods (for individual factors) such as Arrhenius, Eyring and Inverse Power Rule Models, based on exponential or Weibull distribution functions. In the accelerated life testing (ALT) methods [93], the environmental factors are represented through time acceleration, as follows:

$$t = A_F \cdot t_i \Rightarrow \dot{f}(t) = \frac{1}{A_F} \cdot f\left(\frac{t}{A_F}\right); \dot{F}(t) = F(t/A_F) \quad \text{and} \quad \dot{h}(t) = \frac{1}{A_F} \cdot h\left(\frac{t}{A_F}\right) \quad (4.33)$$

Applying this to the exponential and weibull distributions, we got:

- Parametric exponential ALT estimation:

$$\dot{f}(t) = \lambda \cdot e^{-\left(\frac{\lambda t}{A_F}\right)}; \quad \dot{F}(t) = 1 - e^{-\lambda t/A_F}; \quad \dot{R}(t) = e^{-\lambda t/A_F}; \quad \dot{h}(t) = \frac{\lambda}{A_F} \quad (4.34)$$

- Parametric Weibull ALT estimation:

$$\begin{aligned} \dot{f}(t) &= \left(\frac{\beta \cdot t^{\beta-1}}{(A_F \cdot \alpha)^\beta} \right) \cdot e^{-\left(\frac{t}{A_F \cdot \alpha}\right)^\beta}; \quad \dot{F}(t) = 1 - e^{-\left(\frac{t}{A_F \cdot \alpha}\right)^\beta}; \quad \dot{R}(t) = e^{-\left(\frac{t}{A_F \cdot \alpha}\right)^\beta}; \\ \dot{h}(t) &= \frac{\beta}{A_F \cdot \alpha} \cdot \left(\frac{t}{A_F \cdot \alpha}\right)^{\beta-1} = \left(\frac{h(t)}{A_F^\beta}\right) \end{aligned} \quad (4.35)$$

4.3.4 Reliability Blocks Diagram

Once the basic reliability functions have been introduced for a single element, we can estimate the reliability of groups of independent elements (systems) as association of them in series and in parallel, or a combination of both [41, 94]. In series, the occurrence of just one failure in a single element provokes the failure of the system, while in parallel all the elements have to fail for the entire system failure.

$$R_{\text{series}} = R_a \times R_b \times \dots R_n$$

$$R_{\text{parallel}} = 1 - (1 - R_a) \times (1 - R_b) \times \dots (1 - R_n); \text{ and in the case of 3 elements}$$

$$R_{3\text{parallel}} = R_a + R_b + R_c - R_a R_b - R_a - R_c - R_b R_c + R_a R_b R_c$$

$$R_{m \text{ out of } n} = R_{m/n} = \sum_{i=m}^n \binom{n}{i} \cdot R^i (1 - R)^{n-i}, \text{ in the case of } n \text{ independent elements.}$$

The correspondent formulae for series and parallel reliability blocks, when considering the exponential distribution, are shown in Table 4.6 together with the case of m elements out of a parallel combination of n elements. In this table we use the mean down time (MDT) as the expected time in which the element is down including not only corrective but also preventive downtime.

In case of exponential distributions, other interesting case to study is the parallel standby (ps) system in which the (n) redundant elements are only activated in sequence and when one of them fails, and in order to guarantee operations (we suppose perfect switching activation).

$$R(t)_{\text{ps}} = e^{-\lambda \cdot t} \cdot \left(1 + \lambda t + \frac{\lambda^2 t^2}{2!} + \dots + \frac{\lambda^{n-1} t^{n-1}}{(n-1)!} \right); \quad \text{MTBF}_{\text{ps}} = n/\lambda \quad (4.36)$$

$$R(t)_{\text{ps}} = e^{-\lambda \cdot t} \cdot \left(\sum_{i=0}^{n-1} \frac{\lambda^i t^i}{i!} \right) \quad (4.37)$$

$$R(t)_{m/n \text{ for ps}} = e^{-\lambda \cdot t \cdot m} \cdot \sum_{i=m}^n \frac{(\lambda t)^{i-m}}{(i-m)!} \quad (4.38)$$

$$R_{m \text{ out of } n} = R_{m/n} = 1 - \sum_{i=0}^{m-1} \binom{n}{i} \cdot R^i (1 - R)^{n-i}, \text{ in the case of } n \text{ independent elements.}$$

Besides, in the parallel standby system the switching activation could have a determined reliability R_{sa} , then the influence of this will be for two elements:

$$R(t)_{\text{ps}} = e^{-\lambda \cdot t} \cdot (1 + R_{\text{sa}} \cdot \lambda t) \quad (4.39)$$

Table 4.6 Series, parallel and some out of parallel reliability formulae for exponential distribution

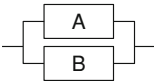
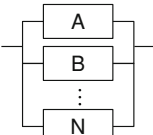
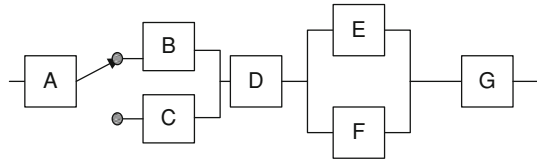
In series of two elements	In parallel of two elements	In parallel m_out_of_n elements
		
$\lambda_s = \lambda_A + \lambda_B$	$\lambda_p = \lambda_A \cdot \lambda_B \cdot (MDT_A + MDT_B)$	$\lambda_{m/n} = \frac{n!}{(n-m)!(m-1)!} \lambda^m \cdot MDT^{m-1}$
$MTBF_s = \frac{MTBF_A \cdot MTBF_B}{MTBF_A + MTBF_B}$	$MTBF_p = \frac{MTBF_A \cdot MTBF_B}{MDT_A + MDT_B}$	$MTBF_{m/n} = \frac{MTBF^m}{\frac{n!}{(n-m)!(m-1)!} \cdot MDT^{m-1}}$
$A_s = A_A \cdot A_B$	$A_p = A_A + A_B - A_A \cdot A_B$	$A_{m/n} = 1 - \frac{n!}{m!(n-m)!} (1-A)^m$
$R_s = e^{-\lambda_A \cdot t} \cdot t e^{-\lambda_B \cdot t}$	$R_p = 1 - (1 - e^{-\lambda_A \cdot t}) \times (1 - e^{-\lambda_B \cdot t})$	$R_{m/n} = \sum_{i=m}^n \binom{n}{i} \cdot (e^{-\lambda \cdot t})^i \cdot (1 - e^{-\lambda \cdot t})^{(n-i)}$

Fig. 4.9 Practical case of a complex system based on exponential distributions

As a practical case, we could obtain the reliability of a system at 1,000 h of operation, which is composed by seven components (see Fig. 4.9), from A to G. We consider the failures rates as exponentially distributed and statistically independent for all components (see Table 4.7), and the switching activation reliability $R_{sa} = 1 - 10^{-5}$ for the connected in parallel standby components (B and C).

We would like to highlight the reliability calculation of the parallel subsystem integrated by E–F, and the parallel standby subsystem compounded by B–C.

$$R_{EF}(t) = 1 - (1 - R_i(t))^2 = 1 - (1 - e^{-2 \cdot 10^{-6} \cdot t})^2 \Rightarrow R_{GH}(1000) = 0.9999$$

$$R_{BC}(t) = e^{-\lambda \cdot t} \cdot (1 + R_{sa} \cdot \lambda t) = e^{-2 \cdot 10^{-4} \cdot 1000} \cdot (1 + (1 - 10^{-5}) \cdot 2 \cdot 10^{-4} \cdot 1000) = 0.9825.$$

Thus, we can now calculate the total reliability of the elements A, B–C, D, E–F and G, in series configuration.

The overall reliability for 1,000 operating hours will be:

$$R(1000) = R_A \cdot R_{BC} \cdot R_D \cdot R_{EF} \cdot R_G = 0.999 \cdot 0.9825 \cdot 0.998 \cdot 0.9999 \cdot 0.970 = 0.95.$$

Table 4.7 Reliability of all components

Component	A	B, C	D	E, F	G
λ	10^{-6}	$2 \cdot 10^{-4}$	$2 \cdot 10^{-6}$	$2 \cdot 10^{-6}$	$3 \cdot 10^{-5}$
$R(t = 1,000)$	0.999	0.9825	0.998	0.9999	0.970

4.3.5 Human Reliability Analysis

As we have mentioned above, human errors contribute importantly to equipment's lack or reliability. Reason [99] defines human error as *"a generic term to encompass all those occasions in which a planned sequence of mental or physical activities fails to achieve its intended outcome, and when these failures cannot be attributed to the intervention of some chance agency"*.

From a performability point of view, circumstances of stress, time constraints, worries, etc., may cause people errors and failures with important consequences [12, 17, 96]. Therefore we must always assume that there is always a non-zero probability of human errors human error probability (HEP) [8, 37].

The relative importance of the human errors has been addressed in the literature. There are studies [17] for different sectors showing that, at least, between 20 and 30% of the total number failures in equipments were produced by human errors. Moreover, according to the Institute of Nuclear Power Operations, human factors could lead up to the 70% of atomic plant failures [110]; and even to a higher percentage, 80–90% [34, 42, 99], of the failures affecting people safety.

The appearance of human errors (lapses, mistakes or violations, [62, 99] depends on diverse performance shaping factors (PSF) influencing an operational context such as task complexity, ergonomics, stress and motivation. In the case of maintenance activities, the main causes of human error used to be [19, 99] deficient maintenance documentation and procedures, inadequate training and experience, poor equipment design, improper work tools, poor work environment and layout, fatigued maintenance personnel, deficient planning or scheduling, miscommunication, deficient supervision and policy problems.

The classification of the factors affecting human errors can be made dividing these factors into the following three groups [57]:

- Personal factors such as the responsibility, skill, motivation, experience, attitude, physical ability, eyesight, self-discipline and training.
- Environmental factors such as temperature, humidity, noise, lighting, vibration, wind and noise.
- Conditional factors that represent the influence of the operating environment such as complex task, inadequate tools or procedures and poor design.

The majority of the reviewed studies [22, 107] about human reliability analysis (HRA), just like for asset reliability analysis, consider the stochastic nature of the human errors. Then the approach to tackle these problems is based on probabilistic risk assessment (PRA) [8], departing from the collection or compilation of human

error databases. At this point, the difficulty is to identify the different sorts of errors, their frequencies, performance shaping factors as well as the valuation (quantitative or qualitative) of their consequences. Some international databases can be a good supporting tool for this purpose [107], such as the Reporting Procedures Manual for the Nuclear Plant Reliability Data System (NPR) [83], the Aviation Safety Reporting Program [4], or the computerised operator reliability and error database CORE-DATA [30, 52] developed by the Birmingham university.

Consequently, from a probabilistic point of view, human error probability (HEP_G) of a generic task and the human reliability ($R_h(t)$) could be described of view as:

$$HEP_G = \frac{\text{number of errors}}{\text{number of tasks carried out}} \quad (4.40)$$

$$R_h(t) = 1 - HEP_G \quad (4.41)$$

Besides this, if we consider the presence of a series of performance shaping factors (PSF) that can increase the HEP, we may model the influence of the operational context, using PSF, as in Eq. 4.42:

$$HEP = HEP_G \cdot \prod_{k=1}^N PSF_k \cdot W_k \quad (4.42)$$

where W_k is the weight of each PSF and HEP_G is the human error probability of a generic task.

In the literature reviewed, we have found diverse techniques to identify errors, frequencies, PSF and the valuation of the consequences [19, 52, 69]:

- Flow charts and check sheets
- Interviews and group of experts
- Simulation such as systems analysis of integrated networks of tasks (Micro-SAINT), human cognitive reliability (HCR) or maintenance personnel performance simulation (MAPPS) Model
- Markovian Models
- Fault tree analysis (FTA) and Probability tree method (PTM)
- Root cause analysis (RCA) and Cause-and-effect diagram (CAED)
- Qualitative risk matrix (QRM)
- Preliminary hazard analysis (PHA)
- Hazard and operability studies (HAZOP)
- Error-cause removal program (ECRP)
- Failure modes-effect analysis (FMEA) and criticality analysis (FMECA)
- Technique for human error rate prediction (THERP)
- Human error assessment and reduction technique (HEART)
- Generic error modeling system (GEMS)
- Systematic human error reduction and prediction approach (SHERPA)

Table 4.8 HEP of generic task (HEP_G) HEART method [110]

Generic task and proposed nominal human unreliability (5–95th percentile bounds)	
(A) Totally unfamiliar, performed at speed with no real idea of the likely consequences	0.55 (0.35–0.97)
(B) Shift or restore system to a new or original state on a single attempt without supervision or procedures	0.26 (0.14–0.42)
(C) Complex task requiring high level of comprehension and skill	0.16 (0.12–0.28)
(D) Fairly simple task performed rapidly or given scant attention	0.09 (0.06–0.13)
(E) Routine, highly practised, rapid task involving relatively low level of skill	0.02 (0.007–0.045)
(F) Restore or shift a system to original or new state following procedures, with some checking	0.003 (8.10^{-4} – 9.10^{-4})
(G) Completely familiar, well designed, highly practised, routine task occurring several times per hour, performed at the highest possible standards by highly motivated, highly trained and experienced person, totally aware of the implications of failure, with time to correct potential error, but without the benefit of significant job aids	0.0004 (8.10^{-5} –0.009)
(H) Respond correctly to system command even when there is an augmented or automated supervisory system providing accurate interpretation of system stage	0.00002 (6.10^{-6} –0.009)
(M) Miscellaneous task for which no description can be found (Nominal 5–95th percentile data spreads were chosen on the basis of experience available suggesting log normally)	0.003 (0.008–0.11)

- Human reliability management system (HRMS)
- Potential human error and cause analysis (PHECA)
- Technique for retrospective analysis of cognitive errors/human error retrospective analysis (TRACer/HERA).

From this list, one of the most employed techniques is the Human Error Assessment and Reduction Technique (HEART) method [110]. This method has been especially used in the nuclear sector, with a precision of 76.7% in its applications [54]. In this method, generic probabilities of human error depend on the activity nature (Table 4.8) and typical PSFs (Table 4.9) are defined. HEART method is easy to apply and understand, although its precision depends on expert's knowledge [101]. Also, this method describes a list of remedial recommendations (see [110]), per each PSF, in order to reduce the probability of human error. Thus:

$$\dot{HEP} = HEP_G \cdot \prod_{k=1}^N ((EPC_k - 1) \cdot APOA_k) + 1 \quad (4.43)$$

where HEP_G is the human error probability of the generic activity, error producing condition (EPC) is the Maximum associated affection to the PSF_k and the assessed proportion of affection (APOA) is the experts estimate weight of the PSF (its value has to be between 0.05 and 1).

Table 4.9 HEP in HEART method [110]

Error producing condition (EPC) and Maximum Predicted Amount by which unreliability might change, going from good conditions to bad

(1) Unfamiliarity with a situation which is potentially important but which only occurs infrequently, or which is novel	X17
(2) A shortage of time available for error detection and correction	X11
(3) A low signal to noise ratio	X10
(4) A means of suppressing or overriding information or features which is too easily accessible	X9
(5) No means of conveying spatial and functional information to operators in a form which they can readily assimilate	X8
(6) A mismatch between an operators model of the world and that imagined by a designer	X8
(7) No obvious means of reversing an unintended action	X8
(8) A channel capacity overload, particularly one caused by simultaneous presentation of non redundant information	X6
(9) A need to unlearn a technique and apply one which requires the application of an opposing philosophy	X6
(10) The need to transfer specific knowledge from task to task without loss	X5.5
(11) Ambiguity in the required performance standards	X5
(12) A mismatch between perceived and real risk	X4
(13) Poor, ambiguous or ill-matched system feedback	X4
(14) No clear, direct and timely confirmation of an intended action from the portion of the system over which control is exerted	X4
(15) Operator inexperience	X3
(16) An impoverished quality of information conveyed procedures and person–person interaction	X3
(17) Little or no independent checking or testing of output	X3
(18) A conflict between immediate and long term objectives	X2.5
(19) No diversity of information input for veracity checks	X2
(20) A mismatch between the educational achievement level of an individual and the requirements of the task	X2

(continued)

Table 4.9 (continued)

Error producing condition (EPC) and Maximum Predicted Amount by which unreliability might change, going from good conditions to bad

(21) An incentive to use other more dangerous procedures	X2
(22) Little opportunity to exercise mind and body outside the immediate confines of the job	X1.8
(23) Unreliable instrumentation	X1.6
(24) A need for absolute judgements which are beyond the capabilities or experience of an operator	X1.6
(25) Unclear allocation of function and responsibility	X1.6
(26) No obvious way to keep track or progress during an activity	X1.4
(27) A danger that finite physical capabilities will be exceeded	X1.4
(28) Little or no intrinsic meaning in a task	X1.5
(29) High-level emotional stress	X1.3
(30) Evidence of ill-health amongst operatives, especially fever	X1.2
(31) Low workforce morale	X1.2
(32) Inconsistency of meaning of displays and procedures	X1.2
(33) A poor or hostile environment (below 75% or health or life-threatening severity)	X1.15
(34) Prolonged inactivity or highly repetitious cycling of low mental workload tasks	X1.05 for each hour, and X1.1 for 1st half-hour
(35) Disruption of normal worksleep cycles	X1.1
(36) Task pacing caused by the intervention of others	X1.06
(37) Additional team members over and above those necessary to perform task normally and satisfactory	X1.03 per additional man
(38) Age of personnel performing perceptual tasks	X1.02

We will now present a simple example to estimate the HEP according to the HEART method.

Let us suppose a generic activity of type (C), complex task requiring high level of comprehension and skills with the proposed nominal human unreliability 0.16, in circumstances of:

- (15) Operator inexperience,
- (23) Unreliable instrumentation and
- (31) Low workforce morale.

For these factors, the expert technicians have decided to weigh each FSP with $\text{APOAA} = 0.2$, $\text{APOAB} = 0.2$, $\text{APOAC} = 0.6$. Then, the estimations $[(\text{EPC} - 1) * \text{APOA}] + 1$, and remedial measures per each factor are:

- (15) Operator inexperience, EPC (X3): $[(3-1)*0.2] + 1 = 1.4$
- (23) Unreliable instrumentation, EPC (X1.6): $[(1.6-1)*0.2] + 1 = 1.12$
- (31) Low workforce morale, EPC (X1.2): $[(1.2-1)*0.6] + 1 = 1.12$

$$\text{HEP} = \text{HEP}_G \cdot \prod_{k=1}^N ((\text{EPC}_k - 1) \cdot \text{APOA}_k) + 1 = 0.26 \cdot 1.4 \cdot 1.2 \cdot 1.2 = 0.28.$$

This is the estimation of human error probability for these three factors, and the remedial measures for each one are (from the list in [110]):

- (15) *Operator inexperience*. Remedial measure: Personnel criteria should contain specified experience parameters relevant to the task.
- (23) *Unreliable instrumentation*. Remedial measure: Regrettably it is a fact that when instrumentation is found to be unreliable operators will cease to trust its indications to the extent of ignoring valid information and preferring to believe their own interpretations, despite overwhelming evidence to the contrary. If instrumentation is thought likely to be unreliable it should be withdrawn from service, and more reliable instrumentation substituted—no doubts should exist about its suitability.
- (31) *Low workforce morale*. Apart from the more obvious ways of attempting to secure high morale by way of financial reward, for example, other methods involving participation, trust and mutual respect, often hold out at least as much promise. Building up morale is a painstaking process, which involves a little luck and great sensitivity. Employees must be given reason to believe in their employer and themselves and this can be accomplished by a battery of activities, such as joint preparation of work plans and objectives, maximal delegation of authority, reward for effort and results, provision of subsidised fringe benefits and firmness of resolve and openness. It is not achieved to any great extent by appeals to workforces to stick by management—the respect necessary to make morale rise is earned not enforced. A sensitive, caring management would be unlikely to encounter such problems.

4.4 Simulation Techniques

4.4.1 Introduction

Simulation: definition is attempting to predict aspects of the dynamic behavior of the system the model represents (see the free on-line dictionary of computing at <http://www.vensim.foldoc.org/>).

Traditionally, the formal modeling of systems to predict their behavior has been via a mathematical model which attempts to find analytical solutions enabling the prediction from a set of parameters and initial conditions. For many systems,

however, simple closed form analytic solutions are not possible. This is the point at which *computer simulation models* come into play. *Computer simulation* is often used as an adjunct to, or substitution for, modeling systems for which these analytic solutions are not possible. It generates a sample of representative scenarios for a model in which a complete enumeration of all possible states would be prohibitive or impossible.

In this book we will see some *computer simulation modeling* which is extremely well suited to study systems that are dynamic and interactive as well as complicated. This technique has been in use in management science since the early 1950s and its methods have gradually evolved alongside general developments in computing science ever since [95].

An important aspect to take into account is that “simulation should imitate the internal processes and not merely the results of the thing being simulated”. That is to say that a simulation model should somehow capture the *structure of a system* in order to predict aspects of its *behavior*, with the purpose of *solving a certain problem*.

Dynamic mathematical models used in computer simulation are typically represented with differential equations (the relationship involving the rates of change of continuously changing quantities modeled by functions) or difference equations (relating a term in a sequence to one or more of its predecessors in the sequence). There is a clear reason for this that is related to the nature of the system being modeled. Some industrial systems or processes, like many process plant processes, occur continuously in time. Others, such as certain manufacturing processes, occur more discretely in time. Even though data collected from continuous processes are by necessity taken at discrete time intervals, model predictions based on these data assume temporal continuity and are commonly written in the form of differential equations. By contrast, discrete-time processes are modeled using difference equations, equations that take into account the discontinuous nature of these processes.

The actual process of computing the model state and producing the state values as the simulation time is advanced in the computer is called model execution [91]. A key design element in model execution is the time advance mechanism [117]. Most common time advance mechanisms are:

- *Time-stepped*. Time is advanced in fixed time increments and the system state is updated (recalculated) at each increment.
- *Discrete-event*. Different part of system state evolve at their own timescales, using the concept of events. Each event signals the specific instant in simulation time at which a particular part of the system is to be updated.
- *Time parallel*. In this case simulation time is partitioned in multiple segments, and each segment is executed independently from each other.

The model execution normally requires the consideration of three different time axes [91]:

- *Physical time.* Time in the physical system that is being modeled. For instance, and assuming units of time in weeks, from week 1 to week 45 of the year 2008.
- *Simulation time.* Representation of the physical time for the purpose of the simulation. Corresponds to the simulated time period of the physical system. For instance, number of weeks since the beginning of the year 2008.
- *Wallclock time.* Ellapsed real time during execution of the simulation, as measured by a hardware clock. For instance, number of miliseconds of computer time during execution.

Many of the models included in this book contain stochastic elements. The first implication of this is the need for a more careful treatment of model results [58]. The result of every model simulation (experiment) has to be considered as samples and these samples depend on the random number of streams used to produce them. Different random numbers will transform into different samples, and simulations will produce different results. In order to reach confidence in these results it is important to produce a certain set of results (samples) and that those results are analyzed using suitable methods. The greater the number of results (runs) the greater the confidence that the results are representative. Another important feature when using stochastic simulation is the fact that when comparing different policy options each option should be compared using the same random numbers. This ensures fair comparison of alternatives. A third important point [95] is that design of experiments is required. Analysis of experiments is a statistical field which may require modeler attention.

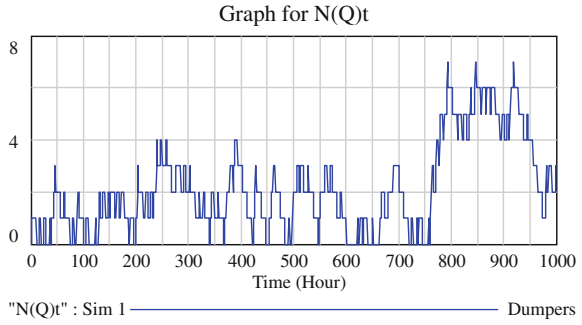
4.4.2 Case Study

Imaging a utility company that has a great number of identical generator sets, each of which uses the same model of diesel engine. There are no companies offering technical services for the mentioned engines near by. The maintenance of the engines has become a problem that the utility company has to face. The company has also decided to carry out a study of the future maintenance needs in order to evaluate the personnel requirement for the accomplishment of this maintenance. The company has decided to hire a pair of technicians initially and would like to know:

1. Their expected utilization of the maintenance technicians;
2. The average number of generators unavailable because they are waiting to be repaired;
3. The average downtime of a generator that waits to be repaired.

Time between failures and time to repair failures was researched and it was found that both times were exponentially distributed. The average time between failures in the fleet was found to be one failure every 10 h of engine operation ($\lambda = 1/10$), and the mean time to repair was 8 h ($\mu = 1/8$). In this case study we will use VENSIM [109], which has special features to facilitate Monte

Fig. 4.10 Sample simulation results for $N(Q)_t$



Carlo type of simulation experiments, and to provide confidence interval estimations. Notation of the model will be as follows:

System status information related variables:

$N(Q)_t$ Gen sets waiting to be repaired at t ;

$N(S)_t$ Gen sets being repaired at t ;

NAR_t Arrival or warning (1 yes, 0 no) for a new gen set to be repaired in period t ;

STR_t A gen set starts to be repaired (1 yes, 0 no) in period t ;

FNR_t A gen set repair is finished (1 yes, 0 no) in period t ;

RN_t Random number, within the interval (0, 1), generated in t ;

EUC_t Total equipment unavailability cost per unit time in t ;

MPC_t Total manpower cost per unit time in t ;

TC_t Total cost per unit time in t ;

Model parameters

λ Failure rate of the entire fleet of gen sets;

μ Service rate of team with a pair of technicians;

M Number of teams in the maintenance crew;

uc Unavailability hourly cost;

mc Manpower hourly cost;

The process first requires modeling the number of gen sets waiting to be repaired $N(Q)_t$:

$$N(Q)_t = N(Q)_{t-1} + NAR_t - STR_t, \text{ with the following initial condition: } \quad (4.44)$$

$$N(Q)_0 = 0 \quad (4.45)$$

Then we model the number of gen sets being repaired $N(S)_t$:

$$N(S)_t = N(S)_{t-1} + STR_t - FNR_t, \text{ with the following initial condition: } \quad (4.46)$$

$$N(S)_0 = 0 \quad (4.47)$$

Fig. 4.11 Convergence of the $N(Q)t$ average value over time

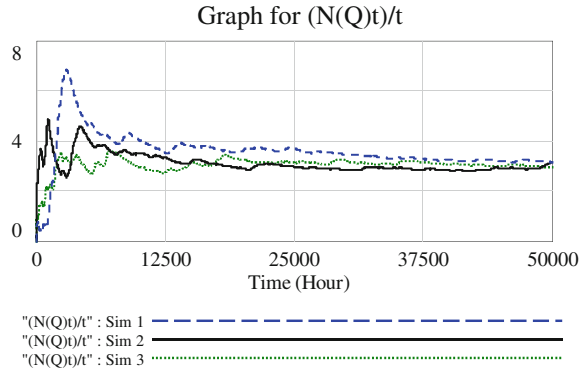


Table 4.10 Sample of statistics provided by the software for the $M = 1$ case

Variable	Min	Max	Mean	Median
$N(Q)t$	0.00	16.00	3.11	2.00
$N(S)t$	0.00	1.00	0.79	1.00
$EUCt$	0.00	640.00	124.21	80.00
$MPCt$	60.00	60.00	60.00	60.00
TCt	60.00	700.00	184.21	140.00

Table 4.11 Mean variable values for different values of M

M	1.00	2.00	3.00	4.00	5.00	6.00
λ	0.10	0.10	0.10	0.10	0.10	0.10
μ'	0.13	0.25	0.38	0.50	0.63	0.75
$N(S)t$	0.80	0.40	0.27	0.20	0.16	0.13
$N(Q)t$	3.20	0.27	0.10	0.05	0.03	0.02
$MPCt$	60.00	120.00	180.00	240.00	300.00	360.00
$EUCt$	124.21	10.67	3.88	2.00	1.22	0.82
TCt	184.21	130.67	183.88	242.00	301.22	360.82

A new gen set will need repair when the following condition is fulfilled:

$$NAR_t = \begin{cases} 1, & \text{if } \lambda \geq RN_t \\ 0, & \text{Otherwise} \end{cases} \quad (4.48)$$

A gen set starts to be repaired when the following condition is fulfilled:

$$STR_t = \begin{cases} 1, & \text{if } N(Q)_t \geq o \text{ and } N(S)_t = 0 \\ 0, & \text{Otherwise} \end{cases} \quad (4.49)$$

A gen set repair will be finished according to the following equation:

$$FNR_t = STR_{t-1/\mu'}, \quad \text{with } \mu' = M \cdot \mu \quad (4.50)$$

Finally cost equations can be defined as follows:

$$\text{EUC}_t = \text{uc} \cdot N(Q)_t \quad (4.51)$$

$$\text{MPC}_t = \text{mc} \cdot 2 \cdot M \quad (4.52)$$

$$\text{TC}_t = \text{EUC}_t + \text{MPC}_t \quad (4.53)$$

Notice that MPC_t is considered constant for each situation in this model.

Figure 4.10 shows the behavior of the queue of gen sets waiting to be repaired over time. Notice that for this particular example of data, the queue never reached a number higher than 8 gen sets to be repaired.

In Fig. 4.11 we can see the convergence of the mean value of that variable over time, for a simulation horizon of 50,000 h and for three different seeds for the random numbers generation.

Numerical values for the variables after 50,000 h, and for the $M = 1$ case, are presented in Table 4.10, where we can see which were the maximum and minimum values for the variables considered during the simulation horizon, as well as their mean and median values during that time.

In Table 4.11, mean values for the main variables considered during the simulation horizon are presented. The best solution in cost is obtained for $M = 2$, and for an expected value of 130.67 monetary units.

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Part II
Processes and Tools to Manage
Maintenance in Network Utilities

Chapter 5

Defining Maintenance Management Framework

5.1 Defining the Maintenance Management Framework

In the previous chapters, we have studied the specific characteristics of maintenance in network utilities, we reviewed existing standards and best current management practices. Of course, to do that we have assumed that these companies have similar objectives and management problems and that these organizations need to conduct certain activities managing similar resources.

The majority of the maintenance models that we have reviewed offer an operational guide to reach certain specific maintenance and organizational goals. Others have the final purpose of developing a computerized system, and some of them simply try to assess the maintenance function. Many of these maintenance models were already applied in a wide variety of industries (see for instance: Pramod [47, 48, 50]) with different types of systems, using different technologies and having different life cycles.

Although all maintenance management models try to continuously improve maintenance performance, maintenance deals with highly diverse problems even in firms within the same productive sector, as a result it is very difficult to use a generally applicable methodology. There is therefore a possibility to delve deep in the search for more specific frameworks or models depending on the maintenance management problem (see Fig. 5.1).

Most of the research contributions that we have found in the literature tend to cover the management function for an individual and specific type of network utility. Moreover, they are centered on one particular maintenance objectives or criteria (such as reliability, monitoring, risk analysis, etc.). We will now try to define a suitable ad hoc framework to serve as a supporting structure and to facilitate maintenance management and its implementation in these types of companies.

From a purely scientific point of view, it is important to notice the difference between the concepts “framework” and “model”. From now on, in this work we will use these two concepts according to the following definitions:

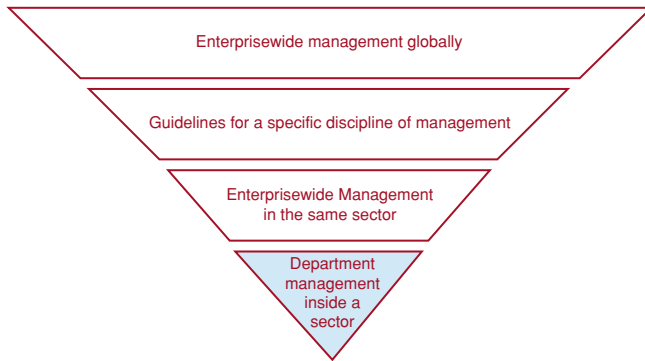


Fig. 5.1 Levels of management standards

The notion “framework”, as in standard conceptual computing models [28], will now be used referring to a conceptual structure used to transmit or address complex issues about some area of knowledge through a generic outline or approach. On the other hand, the term “model” is used when concerning an abstract representation of entities and their relationships involved in the scope of a problem [17] and implemented by a computer language such as UML. From a computational context, the reference models constitute the conceptual basis for the development of specific models and the implementations in a determined scope [25]. In this sense, both models and frameworks serve as templates for the development and comparison of specific models in a given scope.

Therefore, the aim of this book is to propose a framework (instead of a model) consisting of processes, ICT tools, engineering methods and techniques supporting maintenance management continuous improvement toward excellence. This framework definition includes a system structure as a reference basis for future developments of maintenance systems.

Our framework is searching:

- For good balance between internal fulfillment and external requirements; strategy, operational and tactical decisions [32].
- To transform resources and services into customer satisfaction, fulfilling specifications and restrictions on management and cost [41].

Therefore, the framework should not be determined only top-down, from general management to technicians; it has to coordinate human capital with business strategy, it shall generate positive attitudes toward innovation and knowledge sharing [3]. Our intention is that this model will become a baseline leading maintenance management to levels where network utilities risk is minimized (see Fig. 5.2).

In this work we propose an integrated framework, based on process and information systems, with a clear intention to structure better the management in these services companies. Our suggested framework has a fundamental orientation

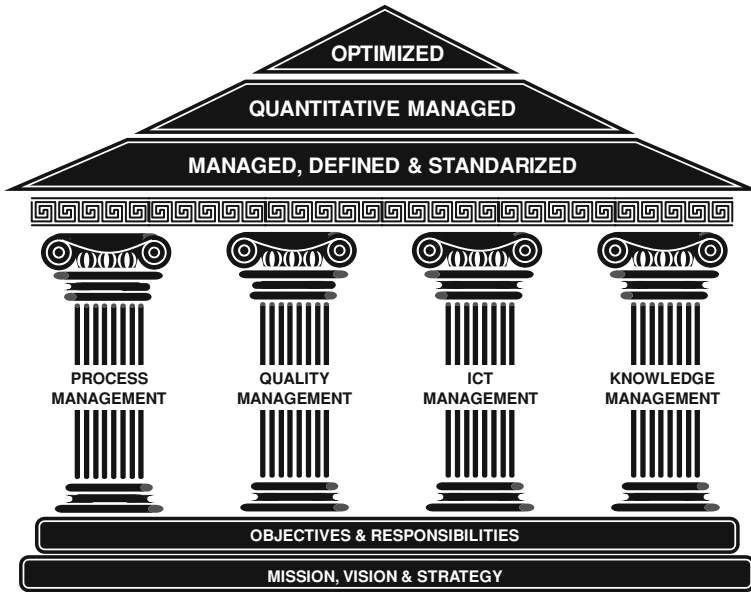


Fig. 5.2 Structure of the maintenance management framework

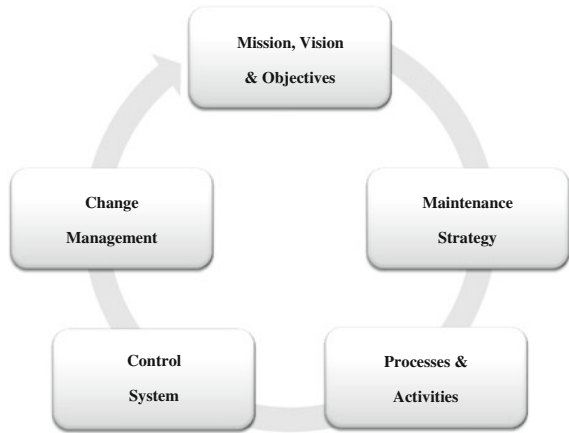
toward ensuring service quality and internal knowledge, thus we try to cover the four following approaches (as suggested by Dandois and Ponte [9]):

- *Strategic*. We try to align business strategy with management guidelines and action plans.
- *Structural*. We redesign processes when required.
- *Technology*. We are taking advantage of ICTs.
- *Human touch*. We foster knowledge management development.

Based on the four pillars in Fig. 5.2, our framework for maintenance management in network utilities can be implemented assuming that an initial maintenance management process redesign will be normally required (according to Davenport [10], Hammer and Champy [21] “Rapid Re”), followed by a continuous improvement of that process. By doing so sustainable standardization and proper knowledge management will lead to superior levels of maintenance organization maturity over time.

If we redesign our maintenance management process, we want this process to be properly standardized and prepared for future improvement. When we redesign we pursue radical business optimization and, in order to do so, we have to concentrate on critical activities adding value to the business. To identify those activities we have to review our current strategy and also take into consideration our customers point of view [37]. At the same time we have to remember the definition of maintenance management [23]: *All the activities of management that*

Fig. 5.3 Maintenance reference framework phases



determine the objectives or priorities of maintenance (defined as objectives assigned and accepted by the management and maintenance department), strategies (defined as a method of management to achieve the goals of maintenance), and responsibilities and implement them into practice by means such as maintenance planning, control and supervision of maintenance, and several methods of improvement including the economic aspects of the organization.

Once redesigned, we want our maintenance management process and framework to be structured as a closed loop type of processes framework (following TQM and EFQM axioms for greater continuous improvement potential).

According to previous considerations we propose a maintenance management framework for network utilities consisting of five phases/processes, as follows (see Fig. 5.3).

- *Phase 1.* Vision, mission, objectives and responsibilities definition.
 - Defining a comprehensive management philosophy throughout the department and at all levels (strategic, tactical and operational).
 - Establishing objectives to guide the management to improve quality and results.
 - Leading the organization to comply with the responsibilities.
- *Phase 2.* Maintenance strategy establishment.
 - Strategy is defined as a set of planned functions to be performed, accomplishing the mission and responsibilities, to reach the desired future objectives.
 - In the strategy implementation, methods, tools and resources are considered in an integral way under restrictions, estimating the time, costs and implications for the service, organization and resources.

- *Phase 3. Process and activities implementation.*
 - Managing resources, processes and their relations toward the improvement of key performances for the strategy.
 - Guiding activities for satisfaction of demanded services.
- *Phase 4. Control and knowledge management system.*
 - Establishing the necessary support on ICT.
 - Evaluation management based on performance indicators and defined practices in each process, verifying the main causes that determine the actual performance of processes.
 - Assessment of knowledge maturity regarding the produced, standardized and exchanged knowledge.
- *Phase 5. Change management.*
 - Guiding the learning and continuous improvement based on results toward efficiency.
 - New management modifications are developed, considering consequences and avoiding or minimizing risks in the organization.

In the following sections we will first describe this framework and then (in subsequent chapters) we will concentrate on its implementation in network utilities, stressing different aspects related to process, quality, knowledge and ICT approaches that we have previously referred to. We are sure that this will help us to reach customer satisfaction by improving our service and resources utilization at the minimum risk.

5.2 Phase I: Mission, Vision, Objectives and Responsibilities

Although the maintenance management definition of the European standard for maintenance terminology [23] is quite comprehensive, we want at this point to develop this concept more extensively and precisely for the case of network utilities companies. Let us then review the definition of the maintenance management vision, mission, objectives and responsibilities that can be established in these types of companies.

The vision of our maintenance department (management) represents a desired, and realistic, maintenance management maturity level ensuring specific maintenance contribution to the company for a given point in time in the future. In network utilities, we have seen how maintenance impacts on important aspects related to company image, reputation, customer expectations, organizational culture and values, etc. We can therefore redefine our vision as “a specific (optimum addressable level for a given period of time) maintenance management maturity level ensuring a required maintenance contribution to the company in terms of network conservation and therefore risk minimization (in customers, society, staff

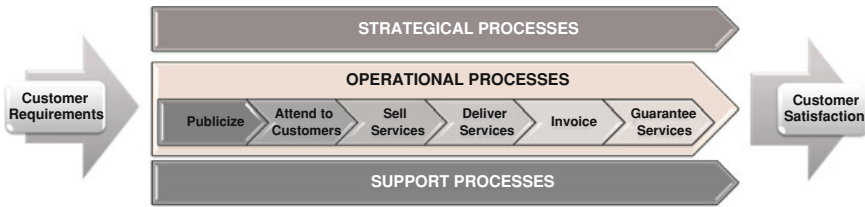


Fig. 5.4 Main operational value chain in network utility

and environment), in line with the organization's culture and values (customer-oriented, multipurpose and flexible teamwork) and reached through technological and organizational (knowledge) continuous improvements".

The path to achieve the vision, in a long-term, is delimited by what we call the mission. In network utilities this maintenance management mission could be defined as: "To *guarantee* the service, i.e. to ensure proper delivery of services provided through the efficient use of the network". Besides this mission statement, we must specify desired objectives to fulfill to reach our vision, and we must assign responsibilities to people involved in the different parts of our mission.

The maintenance department, as any other, must operate according to the business plan, and contribute to the scope of company objectives complying requirements [43, 46]. Maintenance objectives have to be in line with those contained in the company's business plan [44] related to other topics such as production, expenses and investments, quality and security, time of implementation, etc., and they should be SMART [11], i.e. specific, avoiding difficult interpretation, measurable quantitatively, achievable reasonably, realistic according to results and timely. Some of the objectives could be imposed as restrictions determining the internal and external level of performance or satisfaction, delimiting the growth and sustainable development of the organization [41, 52], for example budget constraints or regulations limits. Typical objectives for maintenance management [44] in many organizations have been classified into three groups [18, 19]:

- *Technical objectives.* These depend on the business sector operational imperatives. In general, operational imperatives are linked to a satisfactory level of equipment availability and people safety.
- *Legal objectives/Mandatory regulations.* Usually it is a maintenance objective to fulfill all these existing regulations for electrical devices, protection means, etc.
- *Financial objectives.* To satisfy the technical objective at the minimum cost. From a long-term perspective global equipment life cycle cost should be a suitable measure for this.

In network utilities, which are extremely customer-oriented companies, in line with Earl's Processes Topology [13], maintenance must be considered as a core process, as a primary activity forming part of the main value chain of the company and affecting the central objective of the organization in connection with the

customer satisfaction. Definitely, in these organizations, maintenance cannot only be considered as a “rest-of-departments’s (internal customers) support process”. Notice that we place maintenance as a main operational process [42], with a clear impact on customer satisfaction (see Fig. 5.4, to appreciate where the mission of maintenance fits within the network utility value chain). Thus, we could determine the overall objectives of maintenance in network utilities assigning quantitative measures to the:

- a. Improvement of service quality in response to customer expectations and competition.
- b. Cost efficiency of resources, in quantity over time.
- c. Security in maintenance operations.

That is to say that the maintenance department, aligned with the rest of the organization, must seek to achieve operational objectives of cost and time reduction without compromising the network operation and providing the best possible customer service.

Once the vision, mission and objectives are established, the department responsibilities should be defined to fulfill the mission and to reach these objectives both, internally with other departments and externally with the customers, suppliers and society. We consider that some of these responsibilities are related to, or could be some how classified as:

- Keeping the services and the network in the best possible condition as long as possible, maintaining a solid and operational network that ensures services, according to the requirements set by the organization.
- Managing changes, continuity, availability and security. The maintenance department is responsible for the network conditions/status, and therefore, it has to evaluate the risks and to authorize any action to be carried out in the network.
- Acceptance, upgrading, activation or dismissal of elements. Commissioning each and every new network element prior to regular service [39].
- Managing network configuration properly, to reach the desired capacity.
- Keeping inventory and knowledge updated [24].
- Studies and support in the field, participating in decisions concerning new technologies especially in the preparatory/design phase of the network [44, 49].
- Controlling spare parts, tools and resources.
- Implementing prognosis and analysis, avoiding or predicting problems, minimizing their effects [27].
- Implementation of quality levels of the network and services.
- Managing the relationship with the administration and other organizations, fulfilling the legal regulations about health and safety, environment, etc.

To finalize this section, we can use the IDEF standard to represent the maintenance relationships with the rest of the enterprise entities, we can now represent the maintenance process (in our case maintenance management), as a box (see also [12]) defining a set of inputs/entries (in our case we selected: requested services and escalated incident), outputs (guaranteed services and perfective proposals),

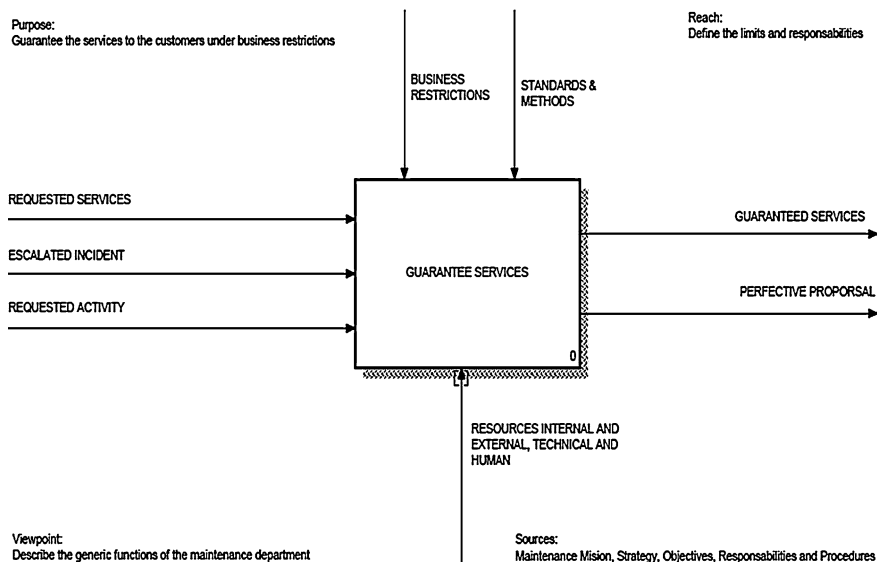


Fig. 5.5 Maintenance management process

fulfilling a set of conditions (business restrictions, standards and methods) and resources. The resources that will be used in the process will be internal and/or external, technical and human. According to this, overall maintenance management process can be represented as in Fig. 5.5.

5.3 Phase II: Maintenance Strategy

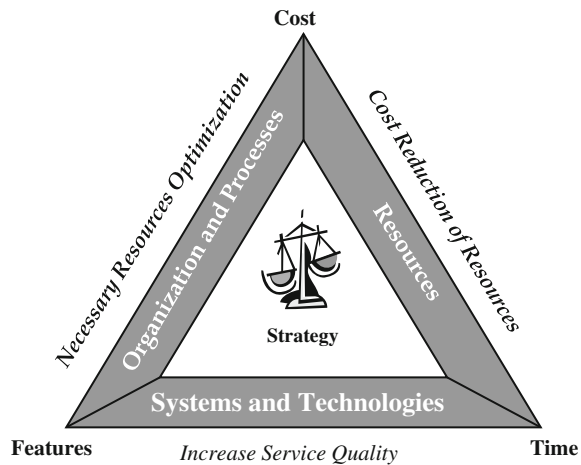
The second phase is to establish a strategy or a plan to ensure this mission and to achieve the vision and objectives of the company.

The term strategy can be defined as a unique and valuable position, involving a different set of activities [45]. Strategy is also a pattern that integrates the goals and policies of an organization [38] to achieve previously defined objectives. A strategy involves developing policies, plans, projects and programs. Later, the implementation of the strategy will imply control, reaction capability and guidelines to measure the department evolution and results.

A maintenance strategy balances three interrelated and in-conflict viewpoints: improvement of services quality, cost reduction and optimization of resources (see Fig. 5.6). We have to:

- *Improve quality.* Investing in more resources and time to make things right.
- *Ensure network operation as long as possible.* Investing in more activities, and therefore, increasing time and costs.

Fig. 5.6 Points of view of maintenance strategy



- *Provide services.* Increasing internal and external customer satisfaction which may sometimes cause the network to work in a high risk status.

As we will see, in order to set out maintenance strategy we will have to analyze potential opportunities, environmental threats, as well as our strengths and weaknesses [2]. Especially for network utilities, because of the high customer interaction and large number of elements to manage, we need a strategy that helps us to lead our maintenance toward a very high level of maturity, as specified in our vision.

According to our framework, maintenance strategy will also be based on the four mentioned supporting pillar (see examples of strategy targets according to these supporting pillars in Fig. 5.7):

- *Quality.* Taking customer satisfaction as a value and trying to meet their needs and expectations in order to retain them and attract new customers from the competitors. At the same time when we increase market share, the staff motivation also increases.
- *Processes.* Evolving through reengineering as continuous improvement and achieving a common department culture measuring process performance toward excellence.
- *Knowledge.* Enhancing the development of human resources (intellectual capital) in a sustainable manner and in line with the evolution of the network and services. The successful development of this pillar contributes to the involvement and competitiveness of the workforce.
- *ICT.* Applying valid technologies for the effectiveness of the three previous pillars. Through proper management of information, decision-making and relationships with suppliers, customers and other departments are facilitated.

Once the maintenance strategy is defined, the next step is to implement it taking into consideration the restrictions of the organization.

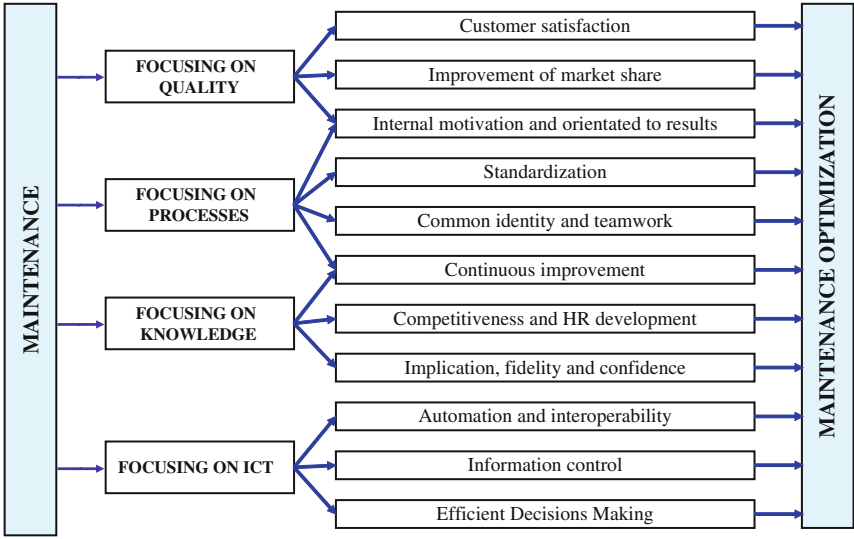


Fig. 5.7 Sample maintenance strategy targets

5.4 Phase III: Processes and Activities Implementation

In this phase we develop all maintenance functions to implement the strategy. We define processes and activities searching maintenance optimization through continuous improvement and customer focus [20]. This is therefore a process-oriented management that will ensure proper decision-making by controlling process performance as a way to pursue effectiveness and efficiency in the fulfillment of the objectives.

The process reengineering should analyze the key functions, relationships and resources, as well as the business constraints. The implementation of the processes is accomplished through a set of activities, producing transformations to achieve a desired result or simply to control the execution of other activities. Processes and activities should be characterized according to their contribution to the strategy, also processes and activities must be optimized in terms of their resources and costs requirements (ABC, “Activity Based Costing”, [30]) (ABM, “Activity Based Management”, [31]). In the following sections we briefly study the main process to consider for the maintenance of network utilities.

5.4.1 Process Map Statement

The overall maintenance management process can be divided into blocks—see for instance divisions made by Crespo [8] or Duffua et al. [12]—according to different

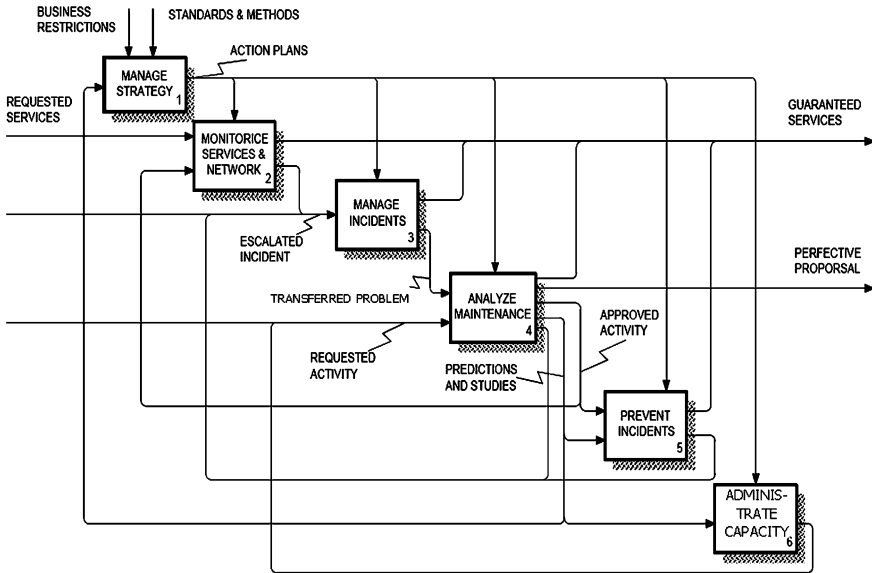


Fig. 5.8 Strategic and operational maintenance processes

levels of business activities: strategic, tactical and operational. Then, the activities have to be categorized by their strategic value, but also by their consumption of resources and/or by their contribution to the quality of the service. Another important consideration here is to understand the type of knowledge required to properly manage the activity as well as their different control possibilities.

As we will see later, many activities will be related to three important terms that we need to define: incident, failure and problem:

- *Incident.* A set of events triggered or linked by the same cause altering the situation where a service was previously [6], that is, the service quality.
- *Failure.* Termination of the ability of an item to perform, under some given conditions and at a given time, the required function [23].
- *Problem.* An unknown underlying cause of one or more incidents [26].

Although superficially they could mean the same, we can specify the difference among them as follows: failure will be used to indicate functional disorders of elements, incident will represent conditions of functional disorders of the whole experience surrounding the service and a problem is considered as failure with unknown solutions.

According to this definition, an incident is also the communication of customer dissatisfaction or disagreement with regards to the provision of a service. Because of this, the maintenance department will have an important role and contribution, as we previously mentioned, to ensure proper customer relationship and satisfaction.

In Fig. 5.8 we summarize standards and best practices that we have reviewed in Part I of this book, and we present a network utilities maintenance management process composed of two building blocks: strategic (and tactical) and operational [8, 12, 15]. More precisely, we consider a strategic process, that we name “to manage strategy” and then five operational processes, named as follows:

- *Monitoring.* Dealing with an alarm and what to do about it, in a proactive sense.
- *To manage incidents.* Facing an incident and how to correct it.
- *To prevent incidents.* Network has to be conserved.
- *To administrate capacity.* Whose responsibility is to occupy resources without saturation.
- *To analyze maintenance.* This reflects continuous improvement using predictive, prognosis and perfective maintenance.

As the reader may surmise, these are different processes representing different value for a particular network utility, they need different resources, and contribute in a different manner to the service quality, according to the type of knowledge they are based on and to their control possibilities.

But how do different blocks in our process map—in Fig. 5.8—interact?

From a high level perspective, the overall maintenance process must guarantee the requested services, dealing properly with incidents, activities to carry out in the network, meeting business restrictions (internal and external environmental restrictions) and following recommendations included in standards and methods. There will also be situations where existing resources will not be sufficient to guarantee the services and some improvement proposals for the network will be required.

On a second level of detail, the process named “guarantee services” is now divided into six sub-processes. The first one (No. 1 in Fig. 5.8) is aimed to coordinate all the maintenance functions interaction. This process, named “manage strategy”, also leads and guides the maintenance management interactions with the rest of the company management functions through action plans, and enforces network documentation and knowledge management. Following this first process we have five operational sub-processes (Nos. 2–6 in Fig. 5.8) which are organized as a value chain providing direct support to the network services in two different ways:

- Proactively (through alarms, analysis, forecasts and activities) preventing incidents, problems and adjusting the network. Services are monitored to detect incidents, some of which could not be detected and so they are inspected in the field or estimated using analysis and predictions.
- Reactively (through analysis and monitoring, implementation and control) protecting the network performance from scaled incidents and managing requested activities (The impact of the said scaled incidents and requested activities on the network service must be studied, disclosed and properly authorized under strict supervision).

Interactions among these processes, from an operational point of view and according to the responsibilities involved, could be as follows:

- A first possible interaction is when a service is demanded by a customer. A maintenance intervention request could be generated to configure the network in a way the service can be provided. Also, the monitoring of that service could be activated to somehow guarantee the service. This type of network configuration actions could also be requested by the “analyze maintenance” or “administrate capacity” processes, for example, when trying to increase or to modify the network capacity to avoid network saturation.
- A second possible interaction is in case of an incident. Maintenance has to solve it to guarantee the network provided services. The incident would be escalated by:
 - The customer.
 - Other department.
 - Inside the maintenance department, detecting it while proactively monitoring the network, or during preventive inspections, expert analysis or predictions.
 - Notice that not all the incidents have a known and fast solution. When the solution is not obvious the problem needs to be analyzed by expert technicians in a collaborative way with the engineering department and with the suppliers involved.
- Maintenance department activities are also requested when there are equipment modifications and certain impact analysis has to be carried out. This is frequent in cases of network capacity addition, etc. Current knowledge and the presence of network maintenance technicians sometimes trigger their interaction for other activities such as, for instance, quality tests or prototypes execution. Many of these types of activities are sometimes approved to be accomplished during preventive maintenance actions.

Each one of these processes will be described in-depth in the following chapters showing practical applications and examples of them.

5.4.2 Process Activity Map

As we have previously mentioned, the main processes and activities are similar in network utilities. Based on the processes and functions described above, we can distinguish different individual activities required to fulfill them. Table 5.1 presents basic and common activities in network utilities. In this Table, the relationship with the process governing the activity is presented, although the technical implementation of each activity could vary in different companies.

Once the activities are defined and limited, their volume, time and resources consumptions should be estimated as well as the necessary budget to carry them out. This can be done based on historical data and forecasting methods, and it is a necessary step prior to scheduling the activities.

The previous table have briefly presented the basic maintenance activities in network utilities, the way in which they are carried out is unique and different for

Table 5.1 Basic maintenance activities in network utilities

Activity	Description	MS	Mtz	MI	AM	PI	MC
Scheduling	Plan of all maintenance activities	✓	✓	✓	✓	✓	✓
Documentation	Elaboration and compilation processes, procedures, reports, etc.	✓	✓	✓	✓	✓	✓
Budgetary control	Management of budget allocated to maintenance	✓	✓	✓	✓	✓	✓
Resources management	Control of resources, tools, staff, vehicles, etc.	✓	✓	✓	✓	✓	✓
Learning	Internal and external learning/training	✓	✓	✓	✓	✓	✓
Logistic	Stores management and spares	✓	✓	✓	✓	✓	✓
Security	Control Security and labor risks with regulations on safety and health	✓	✓	✓	✓	✓	✓
System maintenance	Maintenance of systems	✓	✓	✓	✓	✓	✓
Legal communications	Contacts about maintenance with administration, government or other companies	✓	✓	✓	✓	✓	✓
Monitoring	Monitoring alarms and status of infrastructure and services		✓		✓		✓
Configuration	Setup of any service		✓	✓	✓	✓	✓
Incident management	Management of all kinds of incidents			✓	✓	✓	
Preventive	Preventive maintenance					✓	
Elements modification	Verification of changes in infrastructure		✓			✓	✓
Measures	Evaluation of infrastructure quality, including QoS and SLA				✓	✓	✓
Facilities	Environment conservation, e.g. cleaning, disinfection, humidity, etc.					✓	
Support in field	Support in field on demand to other internal departments					✓	
Predictive	Predictive analysis to avoid or minimize future impacts				✓		✓
Problem management	Problem analysis and repetitive failures or without settlement				✓		
Impact analysis	Study possible impacts and risks on infrastructure	✓	✓	✓	✓	✓	✓
Perfective	Improvement plans or tasks to optimize infrastructure and services				✓		✓
Change management	Coordination activities that could affect services		✓		✓	✓	
Capacity administration	Capacity analysis in infrastructure						✓

Note MS manage strategy, Mtz monitoring, MI manage incidents, AM analyze maintenance, PI preventive incidents, MC manage capacity

each organization and determines the degree of achieved optimization and competitive advantage in the sector. It is important to assess the maintenance management degree of maturity of the organization at a given moment, and to ensure the continuous improvement over time.

5.5 Phase IV: Control and Knowledge Management System

With this phase, our framework tries to fulfill Juran's Quality Trilogy [29]: once the first phases have focused on the processes standardization, the aim is to center on achieving significant improvements, in order to schedule the implementation of these improvements under an ordered control. That is to say: we evaluate first in order to control later and to define the trend of future developments in pursuit of the network maintenance efficiency. With this idea in our mind, we define this phase where a maintenance control system should be established. The maintenance control system will then include "smart decisions and activities undertaken to assess and manage the efficient performance of maintenance to satisfy its objectives over time".

Maintenance decision-making will rely importantly on information and knowledge, both must be reliable and controlled. The more reliable and available information about the network operation and performance, the greater the potential to reduce errors in our decision-making process. At the same time knowledge will be a critical issue in this phase, it has to be managed continuously because the network is constantly evolving and subjected to different operative procedures that can change its performance and which need to be documented to pursue continuous improvement.

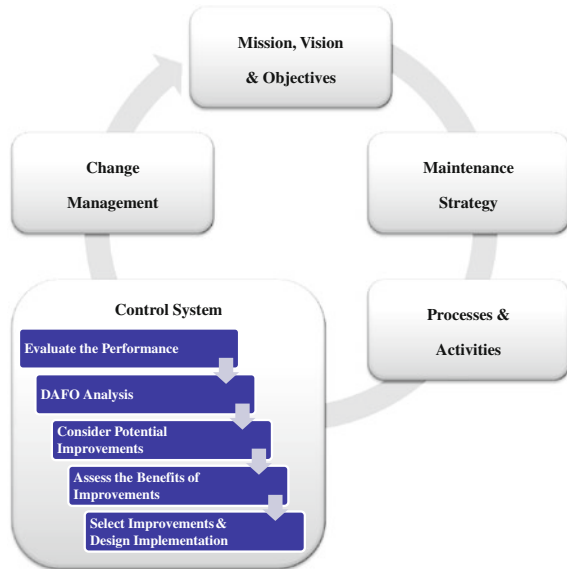
In subsequent chapters of this book we will show how we can increase control and knowledge in maintenance management through the use of:

- Proper performance indicators;
- Suitable modeling techniques (e.g. for network simulation);
- Information systems (including advanced "e-maintenance" systems) and;
- Knowledge maturity assessment, regarding the produced, standardized and exchanged knowledge. For this reason, we will try to define an *evolutive path* for maintenance optimization, describing a few steps or levels of maturity according to the achievement of certain objectives.

Another good practice, as we will see, is that maintenance assessments are structured hierarchically in layers, for each organizational level, and as a single and official data source, internally and externally.

If we follow all these suggestions our maintenance management framework will be implemented to develop plans with clear goals and objectives. In case that any deviation is produced, we will be able to immediately measure it. By doing so, we will facilitate the management of changes and maintain a high precision in our

Fig. 5.9 Continuous improvement evaluation



choice of maintenance strategies. We will easily link effects in the network to their initial root causes. Furthermore, we will also be able to identify possible deadlocks that could stagnate the continuous improvement. This can be done through an analysis considering the internal strengths and weaknesses, and the external opportunities and threats relative to the maintenance framework. We must avoid conformism; searching potential improvements as starting point of a new redesign that could enhance the evolution of results and the maturity of the organization (see Fig. 5.9).

5.6 Phase V: Change Management

The development of a management framework cannot be carried out without peoples' involvement. A new way of doing things involves personnel adaptation to the changed processes. Therefore it is recommended to facilitate the transition, because humans, by their own nature, are resistant to changes.

In a changing environment, success depends on responsiveness [22], it is important to mentally prepare people about the need for change because, as we have seen in Chap. 1, any improvement in the maintenance activities involves major improvements to the organization.

The most used transition models define three involved phases [4, 36]:

- Present or ending phase where the objective is to accept that the old concepts are ending.
- Transition or neutral phase where the changes occur, implying uncomfortable situations for people but empowering creative solutions.

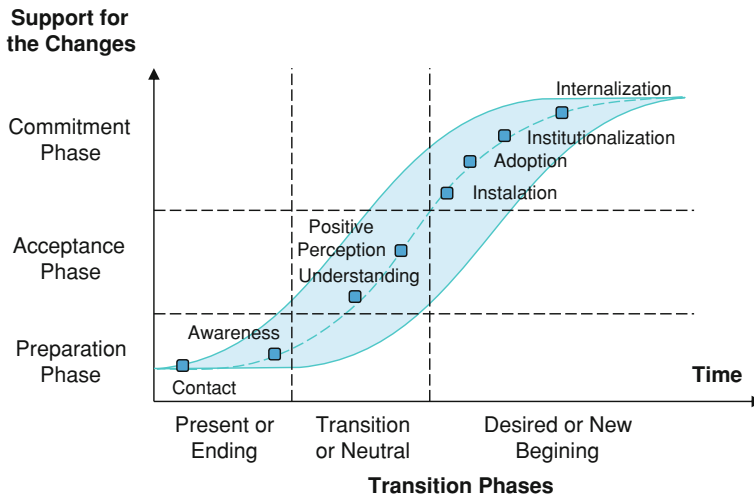


Fig. 5.10 Phases of transition models (based on [4, 7])

- Desired or new beginning phase where people have to commit themselves to the new concepts.

Therefore, when organizations try to plan changes, they will search the internal and external commitment in order to support the implementation of the changes, preparing previously with adequate contact and awareness of change; accepting the changes through understanding and positive perception; and finally committing internal and external agents themselves with the installation, adoption, institutionalization and internalization of the new methods due to changes (see Fig. 5.10).

According to Mumford and Beekma [40] and Bruss and Roos [5], to work with our suggested maintenance management framework implies important changes for everyone over time. Before the changes, the staff does not feel comfortable [7, 22, 33], therefore their communication and implementation must be led by managers and performed with special care, where many cases will require a period of learning/training (transition) to settle and integrate the changes with the minimum possible effect.

There are many physiological theories about the effect of changes on humans, and emotional cycle models are investigated, for example Kübler-Ross [35] defined five emotional stages before negative changes: denial, anger, bargaining (to avoid negative impacts), depression and acceptance (of reality). Other authors have included additional stages to the Kubler-Ross model, such as: Adams et al. [1], who added the following five stages, relief and shock at the beginning, and experimentation, discovery and integration at the ending of the cycle; and Conner [7], who described the cycle as follows; stability (status quo stage), immobilization (before the change), denial, anger, bargaining, depression, testing (new situation), and acceptance. On the contrary, in the case of positive changes, Conner [7]

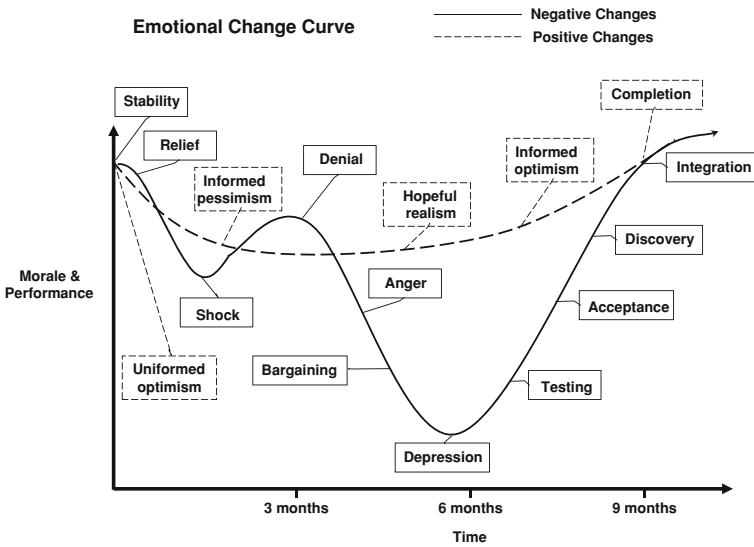


Fig. 5.11 Emotional cycle of changes (based on Kübler-Ross [1, 7, 35, 51])

enumerated five principal stages: uninformed optimism at the beginning, informed pessimism when the goals seem unattainable, hopeful realism when the goals begin to seem realistic, informed optimism based on results and satisfied completion. Even though, some authors have indicated a temporal scale inside the emotional change curve [51], but in spite of defining timescales, human behavior is unpredictable in time, anyone could wander over this cycle overlapping the stages and with different duration or regressing as time goes on.

According to these researchers we could employ the following cycle to describe the emotional impacts of changes, see Fig. 5.11:

So for continuous improvement in maintenance, we need to take into account that planning a correct transition is important, it is a learning process-oriented to fulfill the desired goals. With this purpose, in our framework we could employ the Kotter's eight steps [34] when implementing changes:

- Establish a sense of urgency.
- Create a guiding coalition.
- Develop a vision and a strategy.
- Communicate the change vision.
- Empowering broad-based action.
- Generate short-term wins.
- Consolidate gains.
- Anchor new approaches in the culture.

Besides using this approach, to ensure business continuity [26], the necessary mechanisms to minimize potential risks and consequences due to the changes have

to be established, enhancing skills to reinforce the organization both internally (shareholders and employees) and externally (customers, suppliers, other agencies, etc.).

When producing changes we move to a transitional state that could generate a possible regression if an incorrect decision is executed. Consequently, simulation of changes could help to support the decisions and the adequate implementation of them, empowering the commitment of the involved agents and reducing variations testing in accordance with the plan and requirements. Consequently, tools in this phase could be useful for releasing a new reengineering to tune up the framework implementation, adapting it to new circumstances.

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Chapter 6

Managing Maintenance Strategy

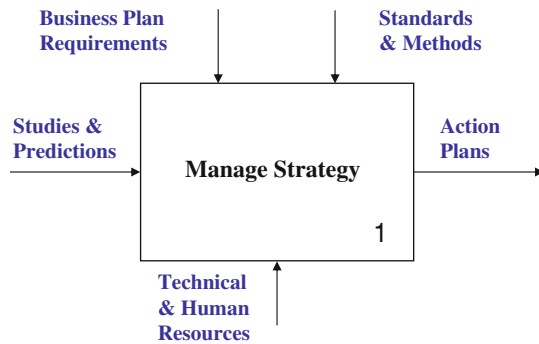
6.1 The “Manage Strategy” Process

The “manage strategy” process coordinates and integrates maintenance management activities in order to achieve the department objectives [20, 32]. This process also analyzes that these objectives are aligned with company’s goals (see Fig. 6.1).

This process is also responsible for leading and controlling continuous improvement and knowledge management for the of the maintenance objectives. By managing strategy we provide consistency to the maintenance actions, ensuring proper department orientation and viability according to predictable and/or unpredictable circumstances that may happen to the network. Therefore, this process ensures that maintenance decisions are unified and ordered to be more proactive and perfective than reactive and conservative. This process produces actions plans [12] to address all operational functions establishing the processes operation patterns [28, 29], the coordinating mechanisms, standardizing the staff skills, the work tasks, the rules and the results. For planning purposes, actions plans should at least contain the following issues in line with Crespo Márquez [8]:

- Involved human and material resources organization, responsibilities and updating knowledge and skills of the staff, accordfulfillment to their intervention in processes and performance standards.
- Description of maintenance activities, at the required detail level, for each specialty.
- Frequency and timing of the activities based on relevant data.
- Action and security procedures oriented toward the satisfaction of the demanded services.
- Necessary tools, equipments and devices to support the proper execution.
- Performance evaluation measurements for the intended purpose.
- Consequences, risks and corrective or remedial measures.
- As “the maintenance documentation is an asset” [34], information management has to be specified (its storage, model, use, origin, source, etc.).

Fig. 6.1 To manage maintenance strategy process



- Flowcharts as guide for decision making ahead of different alternatives.
- Internal and external constraints to fulfill in the activities (costs, legal, etc.).

Inside the actions plans, and in order to ensure proper knowledge management and continuous improvement, all the maintenance functions and relations have to be documented in procedures governed by service level agreements (SLA's) and a control system. Remember that “*standardization and quality are two wheels of the same car*” [19].

The “manage strategy” is a strategic process that must lead and conduct the elaboration of maintenance procedures, standardizing and harmonizing the operations based on best practices, and maintaining all documentation updated, agreed and approved in the so-called *maintenance library* [5]. This library should include: actions plans, procedures, manuals and guides, regulations and legislation, technical instructions, specifications, formats and databases.

In network utilities, the following actions plans can be considered as key plans to keep in the library:

- *Maintenance culture* (vision, mission, objectives, organization and responsibilities). Consisting of documents which guide the organization behavior and performance around the same goals and motivations.
- *Budget management plan*. Composed of documents where the rules for managing all types of costs are established.
- *Quality plan and measurement of the maintenance contribution and its impact*. Documents that implement quality improvement, measuring the performance potential maintenance improvements.
- *Internal and external relations management plan*. Guidelines for the exchange of information, rights and obligations with other internal or external entities.
- *Resources management plan*. Includes documents for the management of maintenance human resources, tools/equipment, devices and the spare parts.
- *Health and safety plan*. For the prevention of occupational hazards and the environment. Documents based on duties, responsibilities and concerns about risks over the staff and the environment are considered.
- *ICT management plan*. It integrates documents focused on the systems management and communications within the maintenance department and about their integration with other external systems.

- *Network and services security plan.* It describes the protection levels established for information, human resources and operating models security. Protection is established against security threats, for example, unauthorized access to sites or usage of services.
- *Service availability and continuity plan.* Adoption of measures to prevent, mitigate or remedy any network risks. This means acting before risks appear and otherwise defining practical guidelines to minimize the impacts when they occur.

Based on the defined objectives and on the action plans, the “manage strategy” process has to establish concrete policies and procedures to focus the efforts on critical issues, determining priority in activities execution and assets conservation. In line with the above-mentioned standards, the following issues have to be taken into account when defining procedures:

- Name and codification;
- Version and changes control;
- Authors and reviewers;
- Approved by, approval date;
- Mission, objectives; scope;
- Overview;
- References and terminology;
- Responsibilities and relations;
- Tools and devices;
- Description;
- Workflows and decision patterns; and
- Indicators and deliverables.

Notice how managing knowledge becomes a critical issue for network utilities maintenance managers. They need to handle knowledge from a wide variety of disciplines: economics, human resource management, technical systems, risk assessment, etc. In the following sections we will review some of these disciplines to which the network utility managers do probably devote more effort: organization and integrated management of human resources, outsourcing, prioritising and budgetary accounting.

6.2 Defining the Maintenance Organization

6.2.1 Key Factors

At the time of establishing the maintenance organization, there are certain key issues to consider when determining the required flexibility and efficiency to achieve the marked goals, such as:

- a. Types of internal and external relationships with suppliers, other departments, agencies, clients, etc.

- b. Hierarchical levels for operation and management activities.
- c. Centralization and decentralization of operations and management.
- d. Use of internal or external resources (subcontracting) for the activities development.
- e. Measurement and development of human skills and specialization.
- f. Degree of support on information and communication technologies.

We will now briefly discuss these topics in the next paragraphs.

In network utilities, the high number of internal relationships would seem to recommend a centralized maintenance management model. However, according to the existing high number of external relations in the field, as well as the dispersed number of assets, a decentralized management model could represent a logical choice from a quality perspective, focusing on customer satisfaction.

Similarly, according to TPM, through hierarchical decentralization the staff participation and involvement in decision making are powered at all levels and it is a key factor in maintenance improvements [35]. The organization design in more or less hierarchical levels influences the processes performance. The organization could be more horizontal when the ability to control processes is higher, reducing intermediate levels between managers and technicians that administrate the information in both ways. In addition, the flexibility to adapt to changes is facilitated when the hierarchical levels of the organization are reduced.

The decision to develop activities with internal or external resources is also a strategic decision [7] and it should be aligned with the business to impact positively on the goals. This is a key choice which is normally based on strategic criteria and not only on technical and/or economic criteria. At the same time, this decision can also change over time or could be scheduled in advance.

The management of human resources skills is a well known topic within strategic planning [4, 20, 25] (Norton and Kaplan [24]). The evolution of organizational capabilities to develop different activities has to be measured both, globally and individually. Competence management dealing with the necessary skills and abilities for the maintenance activities implementation has proven to really be a must in network utilities. The maintenance managers' role is very important at this point. They can actively contribute to leading the improvement of human resources through their responsibility and commitment, defining training plans and considering the relations with the rest of the organization and external parts [35].

In summary, we do consider that knowledge management and competence management are the basis for the definition of the maintenance organization. There is no ideal organization, and according to the above criteria one or another type of organization can be defended depending on the importance of each criterion, although, in the case of network utilities, our experience reflects that (see Fig. 6.2):

- When the number of competences involved is high, then knowledge management intensity is normally high too, and therefore the delegation and decentralization to facilitate new knowledge generation within the organization. As a consequence, the organization becomes more flexible to changes.

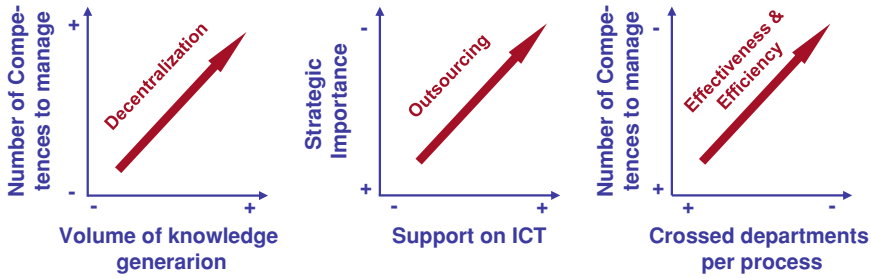


Fig. 6.2 Decentralization, outsourcing, effectiveness and efficiency in the organization

- When the strategic weight of activities in the business is low, and the support on ICT is advanced (in control and monitoring), then the involvement of external resources can be more efficient, especially when selecting more specialized partners.
- When the number of cross organizations operating per process is low and the number of competences is also low, then complications and potential errors in relationships and in information exchange will also be reduced, and therefore the efficiency in process implementation will increase.

6.2.2 Designing the Organization

In order to design maintenance organizations in network utilities we have to consider issues mentioned above as well as other possible legal and geographical requirements to provide the services in the concerned distribution areas. Then, depending on how the decision making authority, responsibility and financial resources are distributed inside the organization and according to classical types of administrative decentralization [38, 46], three types of organizations for network utilities maintenance could be defined, namely: Deconcentration, delegation and devolution organizations (see Fig. 6.3):

- *Deconcentration.* Where the decision making authority in budgetary and technical considerations is strongly centralized to supervise the operation in the distributed areas. It implies a directive style of management with hierarchical levels of procedures, bureaucracy and authorizations.
- *Delegation.* Where the decision making authority in budgetary and technical considerations is delegated but ultimately controlled by the central organization. Being goal-oriented managed and budgetary-technical coordinated, between deconcentration authorizations and local decisions of devolution.
- *Devolution.* Where the decision making authority in budgetary and technical considerations is quasi-transferred to the distribution areas, although collaborations among areas could be established in some issues. The organization is more customer-oriented and reinforces local knowledge and training.

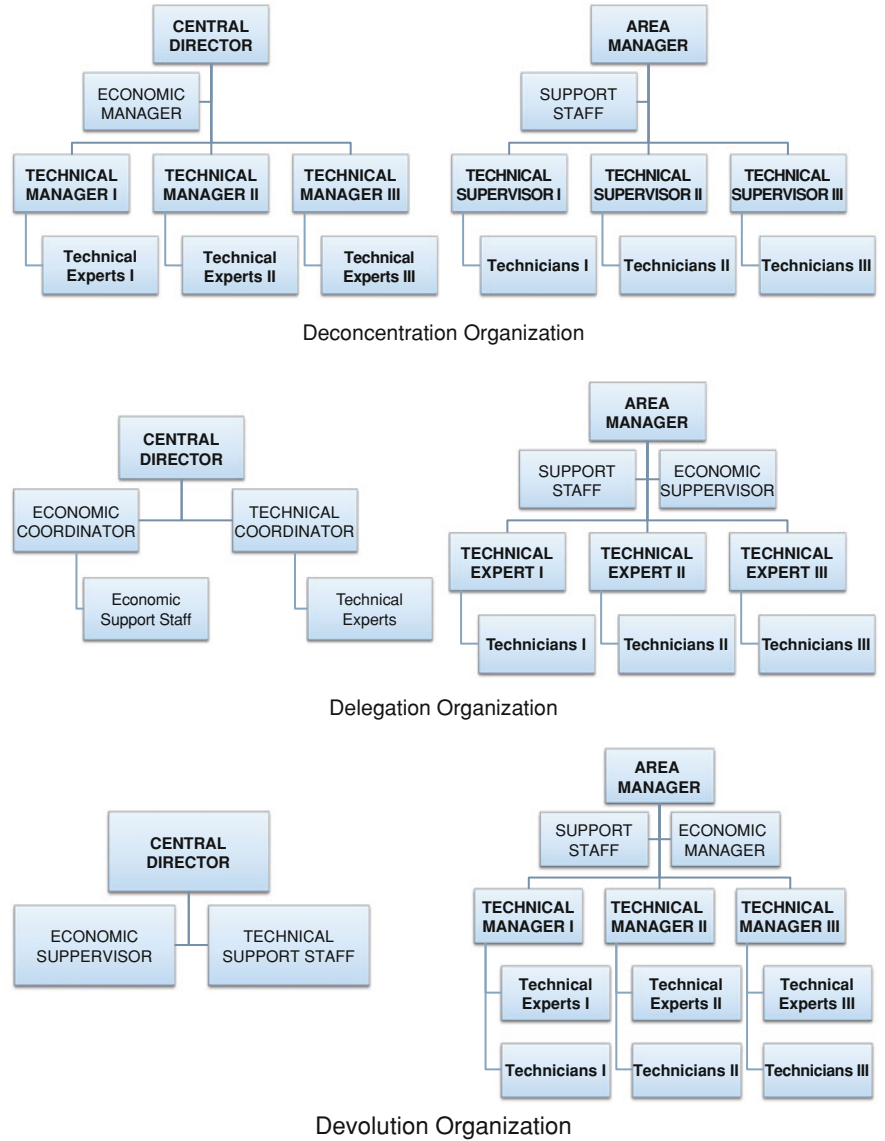


Fig. 6.3 Organizations from a strategic point of view

In network utilities, two operative levels of knowledge are habitually distinguished as much in remote activities as in field activities; frontline or first operative level where technicians are dedicated to perform massively common activities; and backline or second operative level where expert technical knowledge is required to perform particular atypical activities.

In the same proportion as the level of decentralization is increased, the size of the central organization is reduced, collaboratively coordinating or integrating the areas organizations; and the technical expert knowledge is dispersed among areas.

In addition to this, the operational implementation of the maintenance functions could condition the organization in three ways, based on technical operations, based on processes implementation or based on matrix direction [3, 8, 17]. Therefore from an operational point of view we could define three types of organization (see Fig. 6.4):

- *Functional*. When the personnel are organized by technical groups, guided by supervisors and lead by the same manager. Knowledge-effective organization searching specialization.
- *Processes-oriented*. When the personnel are organized by their participation in the processes independently according to their skills and capacities. The supervisors guide their specific execution inside the processes and managers control the results of the processes according to goals and restrictions. Facilitate knowledge exchange, teamwork and control by levels of service.
- *Matrix*. When the personnel are organized with a double dependency, one from functional capacities and other depending on their participation in the processes or projects. Therefore each technician has two managers.

In the extent that the operative organizations move from functional to matrix types, the level of required personnel management and control is greater, causing in some cases conflicts among guidelines, although flexibility and multi-purpose learning is favoured.

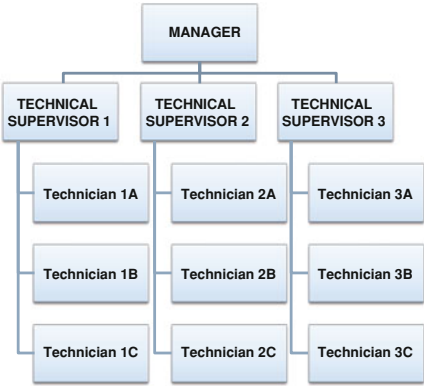
In conclusion, not one type of organization is better than other, that is to say, the appropriate design of an organization depends on all the above-mentioned issues and a type could work better in one company and worse in another.

6.2.3 Sizing Organizations

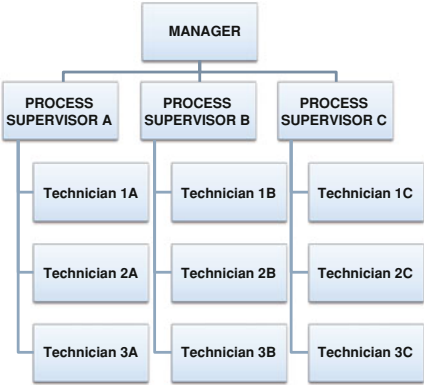
This problem has been addressed by several authors searching for world class maintenance organizations [8, 18, 44]. In the following paragraphs we follow the suggestions of these authors to propose practical guidelines to help in the definition of the size of maintenance organizations in network utilities.

In order to do so, the sizing problem is divided into two different problems, one that focuses on the estimation of the managerial levels, and a second one that concentrates on the design of the operative workforce. Qualitative and quantitative prediction techniques have been applied in both cases:

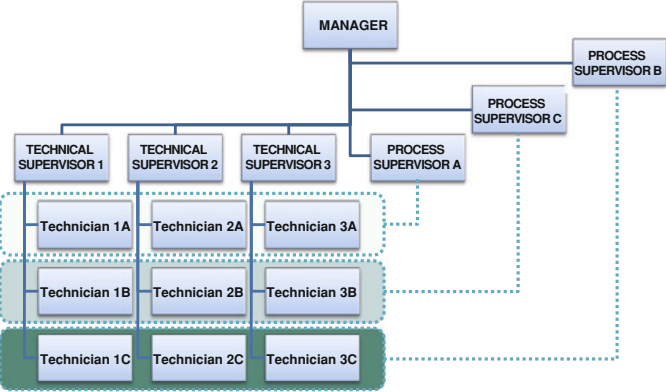
- Qualitative techniques are based on experience or measuring against a benchmark, and they are mainly employed for a rapid implementation.
- Quantitative techniques are based on tendencies (including proportional relations with performance indicators) or statistical modeling, and they are mainly used when there is enough historical data and information.



Functional Organization



Processes-oriented Organization



Matrix Organization

Fig. 6.4 Organizations from operative point of view

Due to the different configurations of maintenance organizations and the diverse mental workload in managers according to the level of management maturity, we will define basic guidelines to apply in all of the managerial cases, employing qualitative estimation. Otherwise, as part of a fulfillment of service quality level, we will describe quantitative estimation for the operative workforce.

Therefore, the proposed steps to size maintenance organizations are the following (see Fig. 6.5):

1. Decide the organization type from an operative point of view.
2. Define the basic operative activities.
3. Analyze the human competences to perform the activities.
4. Define job specification according to competences.
5. Allocate activities to job specifications.
6. Classify activities regarding their type of execution: urgent or scheduled/delayed.
7. Estimate quantitative workload at the first operative level in terms of activities forecasting and their run-time. In the case of urgent activities the estimation is usually carried out using queuing theory, and in the case of scheduled activities using planning techniques and methods such as reliability centred maintenance (RCM), MRP for manufacturing and maintenance resource planning (MRPII), arrow diagram (AD), activity network diagram (AND), PERT or critical path method (CPM).
 - a. Urgent activities estimation using queuing theory where the variability of the generation is recognized and the idea is to deduce the maximum number of activities that could be performed without excessive delay. Poisson forms are the usual distributions of activities arrival and for activity run-time negative exponential distribution are employed. Besides, the queues of maintenance activities use to be characterized as unlimited (with an infinite source) and with attention following the rule first come–first served (FCFS) or first in–first out (FIFO). This is the known queue M/M/C. Then the equations to deduce the number of resources are (see Table 6.1): Thus, the number of resources to perform the activities could be deduced using these formulas, there is also software that facilitates the calculus and tables where knowing the values of λ , μ and L_q the value of c is obtained.
 - b. Scheduled activities and resources estimation is possible, for instance, using planning techniques such as the material requirements planning or manufacturing resources planning (MRP/MRPII) Technique. By using RCM besides MRP/MRPII we can plan and schedule maintenance resources needs of critical items according to their existing capacities. Readers can refer to Shenoy and Bhadury [42] for a detailed method application and also for the presentation of case studies. In this section we will present a vision of the global scheme of the system (see Fig. 6.6) and characterize the changes required for the utilization of MRP for maintenance resources planning purposes.

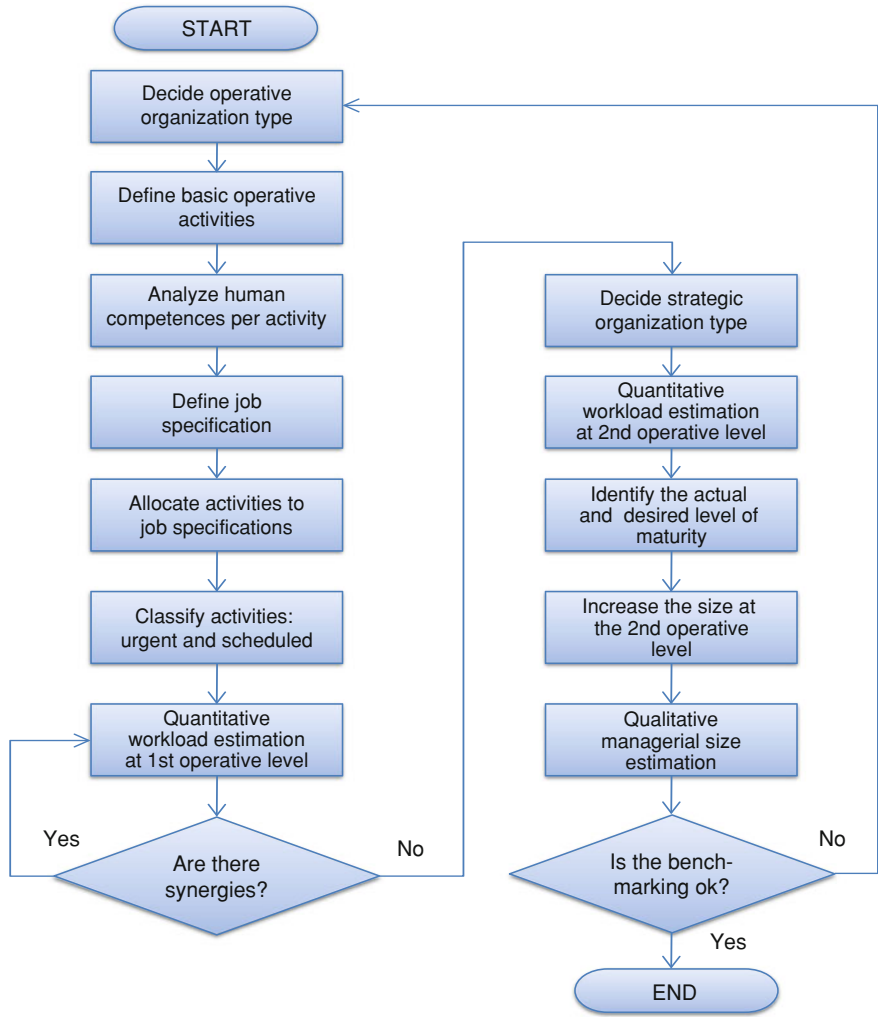


Fig. 6.5 Proposed steps to size maintenance organizations

In this regard, the main modifications to accomplish are as follows (adapted from Shenoy and Bhadury [42]):

- For all preventive maintenance activities to be carried out to end-items—where “end” refers here to the maintenance indenture level—where we will have to detail the resources they need (preventive maintenance bill of materials and work load).
- Preventive maintenance activities will be planned, scheduled and carried out at the scheduled time.
- Inventory records will be modified to allow handling of maintenance manpower resources.

Table 6.1 M/M/C equations

Poisson distribution (x is the generated activities per unit of time)	$P(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$
Negative exponential distribution (x is the activity run-time)	$P(x) = e^{-\mu \cdot x}$
Number of technicians	c
Arrival rate	λ
Service rate	μ
Utilization rate	$\rho = \frac{\lambda}{c \cdot \mu}$
Probability of having zero activities in process (if $c \cdot \mu > \lambda$)	$P(0) = \frac{1}{\left[\sum_{n=0}^{c-1} \frac{1}{n!} \cdot \left(\frac{\lambda}{\mu} \right)^n \right] + \left[\frac{1}{c!} \cdot \left(\frac{\lambda}{\mu} \right)^c \cdot \frac{c \cdot \mu}{(c \cdot \mu - \lambda)} \right]}$
Probability of having n activities in process	$P(n) = \begin{cases} \left(\frac{\lambda/\mu}{n!} \right)^n \cdot P_0 & \text{If } 0 \leq n \leq c \\ \left(\frac{\lambda/\mu}{c! \cdot c^{n-c}} \right)^n \cdot P_0 & \text{If } n \geq c \end{cases}$
Probability of an arriving activity will be queued	$P_w = \frac{1}{c!} \cdot \left(\frac{\lambda}{\mu} \right)^c \cdot \frac{c \cdot \mu}{(c \cdot \mu - \lambda)} \cdot P_0$
Expected average number of activities in process	$L_s = \frac{\lambda \cdot \mu \cdot \left(\frac{\lambda}{\mu} \right)^c \cdot P(0)}{(c-1)! \cdot (c \cdot \mu - \lambda)^2} + \frac{\lambda}{\mu}$
Expected average queue length of activities	$L_q = L_s - \frac{\lambda}{\mu}$
Expected average total time in process	$W_s = \frac{L_s}{\lambda}$
Expected average waiting time in the queue	$W_q = \frac{L_q}{\lambda}$

Notice that this method, with very light modifications, is that which is embedded in most of the CMMS systems dealing with the scheduling of maintenance activities:

8. Review the size of the first operative level searching possible synergies for the execution of both the urgent and scheduled activities by the same resources.
9. Decide the organization type from strategic point of view.
10. Estimate quantitative workload at the second operative level (expert technicians) in terms of activities forecasting and their run-time. In the case of urgent activities the estimation is usually realized using queuing theory, and in the case of scheduled activities using planning techniques and methods. Although Higgins et al. [18] suggest qualitatively one expert technician to support from 40 to 70 technicians, and [44] the ratio paperwork-and-meetings time versus production tasks time could be about 25%.
11. Identify the level of maturity in management and the desired future level.
12. Increase the operative organization size at the second level taking into consideration the additional dedication to standardize and improve the operation.

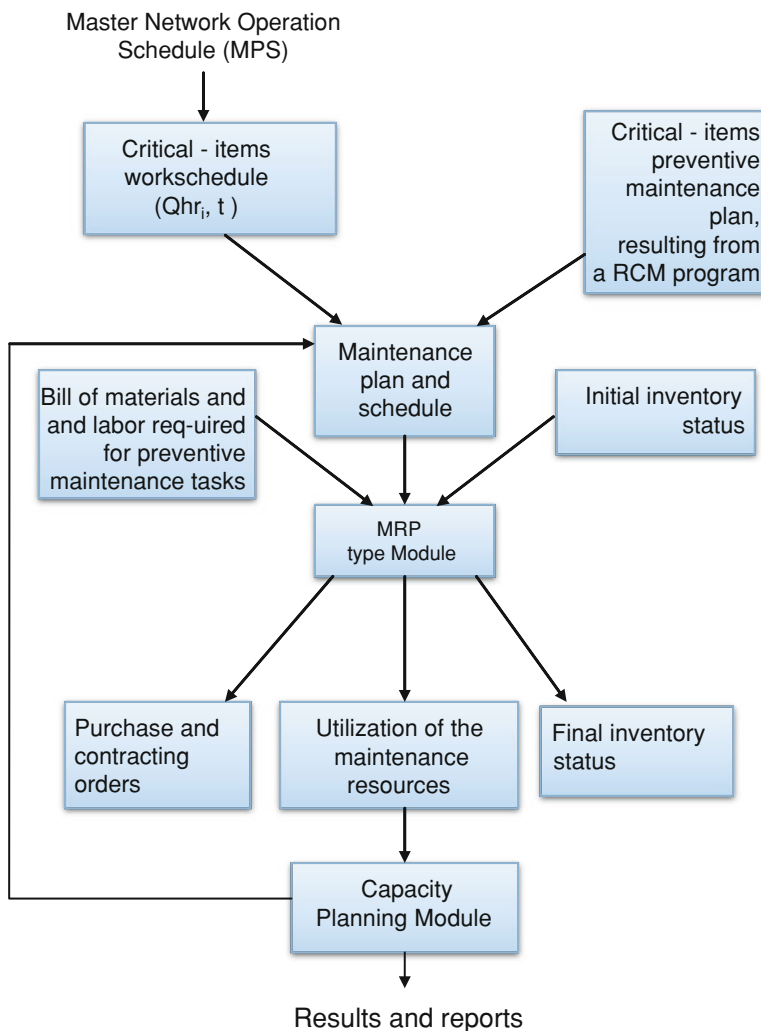


Fig. 6.6 Adapting MRP/MRP II for maintenance resources management in scheduled activities

The ratio paperwork time versus productive works time may be increased in leaps of 10% depending on how ambitious the improvements were. For example, if the annual labor time is 1,760 h, the dedication to improvements will be 170 h (around a month) as a technical course.

13. Qualitatively estimate the managerial organization size depending on operative organization size, the actual maturity level and the necessary dedication to improve it. In line with both the previous authors, there are some known ratios about maintenance workload of managers such as:

- a. One supervisor is necessary to control between 8 and 15 technicians.
 - b. For managers, the ratio of time between their managerial tasks and productive tasks is about 75%.
 - c. If the intention is to increase the maturity level of management, one facilitator of standardization could be added as a training coordinator and two office clerks could be necessary to support the derived administrative tasks per each 80 technicians.
14. The final estimation of the organization size has to be confirmed with the typical human resources budget in the sector.

Now, we will look at a practical case to design the maintenance organization in a network utility based on the proposed steps.

6.2.4 An Organization Design Case Study

Following the proposed steps, a maintenance organization will be designed in a water network utility with geographical presence in areas with different legal restrictions and technical particularities.

1. *Decide the operative organization.* Searching the technical specialization the designated operative organization is functional.
2. *Define the basic operative activities.* For all the areas, we will suppose two basic activities, one urgent focuses on corrective actions, and the other scheduled focuses on preventive actions.
3. *Analyze the human competences to perform the activities.* Trying to simplify, the competences analysis is concentrated on two different technical skills, mechanical and electrical knowledge.
4. *Define job specifications.* Therefore electrical and mechanical jobs are distinguished.
5. *Allocate activities to job specifications.* Both corrective and preventive activities could be produced for mechanical and electrical equipments.
6. *Classify activities on urgent or scheduled/delayed.* Previously described above.
7. Quantitative workload estimation at the first operative level.
 - a. Urgent activities estimation for mechanical and electrical corrections fulfills the queue M/M/C with the unique difference that the number of resources to perform an electrical correction are 1.5 times the resources in a mechanical correction. If the arrival rate is 5 and the service rate is 6, we will analyze the possible number of technicians to provide a service level of around 90% in the probability of not waiting in the queue (see Table 6.2). Accordingly, in this table, derived from the previous equations, the number of necessary technicians to provide probability of waiting less than 10% is 3 or more. Consequently, in this organization

Table 6.2 Analysis of M/M/C for $\lambda = 5$ and $\mu = 6$

<i>c</i>	1	2	3	4	5
P_0	16.67%	41.18%	43.21%	43.43%	43.46%
P_w	83.33%	24.51%	5.77%	1.10%	0.17%
L_q	4.1667	0.1751	0.0222	0.0029	0.0003
L_s	5.0000	1.0084	0.8555	0.8362	0.8337
W_q	0.8333	0.0350	0.0044	0.0006	0.0001
W_s	1.0000	0.2017	0.1711	0.1672	0.1667

Fig. 6.7 Bill of materials and labor (MRP's BOM) requirements for preventive maintenance

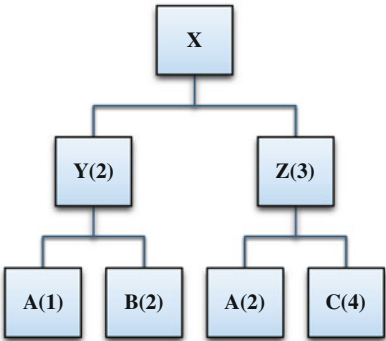


Table 6.3 Level, lead time and lot size of different preventive maintenance resources

Item	Level	Lead time (LT)	Lot size (LS)
X: Network revision x	0	1	1
Y: Replacement of branch	1	2	1
Z: Revision of connexion	1	3	1
A: Spare part	2	2	7
B: Technician type I	2	0*	1
C: Technician type II	2	1*	1

- 3 or more technicians are necessary to provide 90% of service level for corrective activities in mechanical equipment and $3 \times 1.5 = 4.5$ technicians in electrical equipments (we would suggest 5).
- b. Scheduled activities estimation using planning techniques. In this case study, and as an example, we develop the planning required for the accomplishment of a scheduled revision (item X) in the network. This work needs to be finished in ten weeks ahead of the current date. The work consists of five maintenance tasks: two replacements of network branches (item Y) and three connexions revisions (item Z). All referred task have a lead time (LT), and require resources to be carried out. To simplify the case study we assume that items Y and Z require resources according to Fig. 6.7, at the same time, lead time and resources are presented in Table 6.3. Notice that some of the labor resources (Item C) require time (lead time *) to arrive at the maintenance work location and therefore they have to be contacted in

Table 6.4 Explosion of resources requirements

<i>Weeks</i>	1	2	3	4	5	6	7	8	9	10	Level
Gross requirements										1	0
Scheduled receipts											Item
On hand											X
Net requirements										1	LT
Planned-order receipts										1	1
Planned-order releases									1		
<i>Weeks</i>											Level
Gross requirements										2	1
Scheduled receipts											Item
On hand											Y
Net requirements										2	LT
Planned-order receipts										2	2
Planned-order releases							2				
<i>Weeks</i>											Level
Gross requirements										3	1
Scheduled receipts											Item
On hand											Z
Net requirements										3	LT
Planned-order receipts										3	3
Planned-order releases							3				
<i>Weeks</i>											Level
Gross requirements						6	2				2
Scheduled receipts											Item
On hand	3	3	3	3	3	3	4	2	2	2	A
Net requirements						3					LT
Planned-order receipts						7					2
Planned-order releases					7						LS=7
<i>Weeks</i>											Level
Gross requirements							4				2
Scheduled receipts											Item
On hand											B
Net requirements							4				LT
Planned-order receipts							4				0
Planned-order releases							4				
<i>Weeks</i>											Level
Gross requirements						12					2
Scheduled receipts											Item
On hand											C
Net requirements						12					LT
Planned-order receipts						12					1
Planned-order releases					12						

advance. Also notice that some of the material resources required (Item A—spare parts) are as stock already in the warehouse, and also have a lot size in

Table 6.5 MRP output for the maintenance planning

Amount of work () and purchase []	1	2	3	4	5	6	7	8	9	10	Final inventory week 10
X: Network revision x									(1)		
Y: Replacement of item							(2)				
Z: Revision of item						(3)					
A: Spare part					[7]					6	
B: Technician type I							[4]				
C: Technician type II						[12]					

case of an order placement. In Table 6.4 we present the so-called requirements explosion which includes time off-set for activities according to existing lead times. Results of Table 6.4 are summarized in Table 6.5 to reflect outputs of the MRP system according to Fig. 6.6. resources would be 4 and 12 of type I and II technicians respectively and for the time activities Y and Z are being carried out. Also notice that we assume that no capacity constraint is reached for these resources, or that they can be outsourced appropriately when needed.

8. It is decided that no possible synergies exists.
9. Decide the strategic organization type. Due to the different geographic legal restrictions, a devolution organization is decided.
10. Quantitative workload estimation at the second operative level (expert technicians). In this level, to shorten, the level of expert technicians is fixed at 25% of the first operative level, taking in consideration that the future dedication to increase the maturity level of management. Let us consider that for the previous examples in points 7a and 7b, technician type I and II are the mechanic and electrician respectively, and that requirements of those technicians for planned maintenance would be constant over time, then the suggested number of expert technicians per speciality would be $[(3 + 4) \times 0.25] = 1.75$ (we would suggest 2) and $[(5 + 12) \times 0.25] = 4.25$ (we would suggest 5).
11. Identify the level of maturity in management and the desired future level. If the real maturity level expressed in CMMI terms is about 2 and the desired is the 3 level, then we will reserve additional time to develop the improvements as we have included in the 25% ratio of expert technicians per speciality.
12. Increase the operative organization size at the second level. This influence has been considered in step 10 in our suggestion.
13. Quantitatively workload estimation in the managerial levels may be defined as follows:
 - a. In central organization there are:
 - One central director.
 - One economic supervisor with an administrative clerk.
 - Two technical coordinators as technical support staff (one per technology).

Devolution Organization

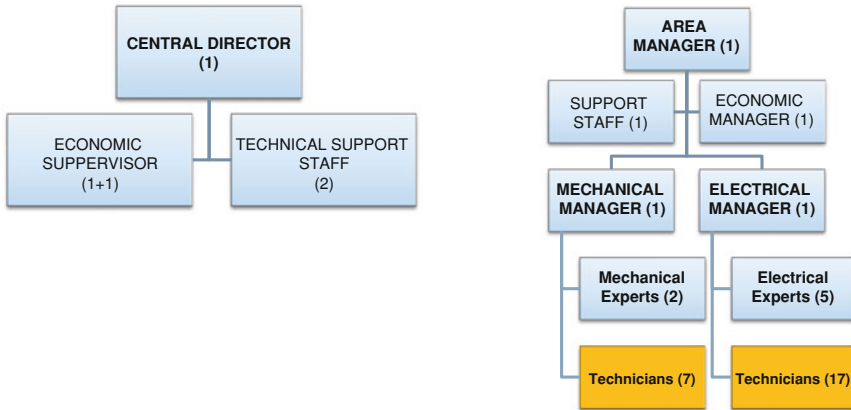


Fig. 6.8 Proposed steps to size maintenance organizations

b. In area organization results:

- One area manager.
- One economic manager.
- One administrative clerk as support staff.
- One mechanical manager with 7 technicians and 2 mechanical experts.
- One electrical manager with 17 technicians and 5 electrical experts.

14. The final estimation of the organization size has to be confirmed with the typical human resources budget in the sector. Avoiding an analysis of the final organization in term of costs, this could be represented as in the Fig. 6.8.

This sequence of steps has the intention of being a basic guideline for the size of a maintenance organization in a network utility, but other combinations of (the strategic and operative) organizations or methods of sizing could be utilized. Here we have tried to describe the most representative in this type of companies.

6.3 Outsourcing Services in Maintenance

Outsourcing is defined as the delegation of business functions totally or partially to another company along with part of the administrative and operational control. Therefore, a contractual relationship governed by service agreements is established between two companies, a supplier and a customer.

Outsourcing in maintenance is a practice that is being increasingly used (Elfing and Baven [11]) (with a tendency of about 37% in Spain AEM [1]), and especially in utilities companies. The decision to outsource is not a simple choice, it impacts the entire company and therefore it becomes a strategic issue

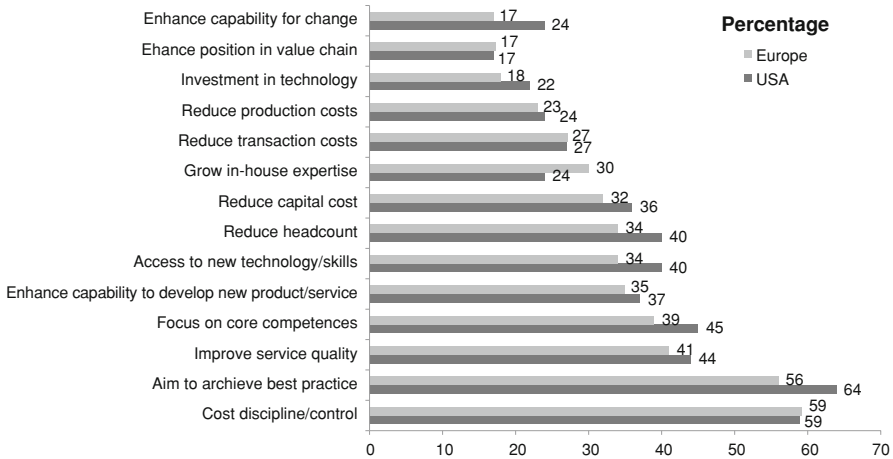


Fig. 6.9 Outsourcing reasons [22]

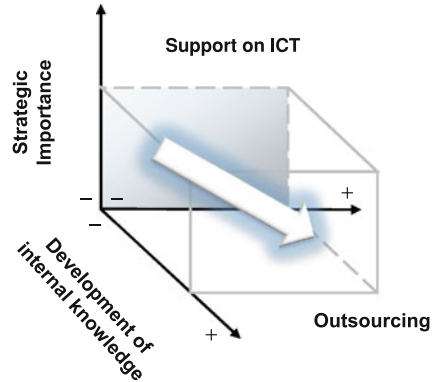
inside the company [7, 36]. Proper alignment with the business strategy is required to impact positively on the achievement of the organization's objectives. There are different strategic reasons why people decide to execute processes of outsourcing. Kakabadse and Kakabadse [22] present a survey of European and US based organizations in which 747 responses to questionnaires showing the main reasons about companies outsourcing (see Fig. 6.9) are presented.

According to this survey many of the main factors to trigger an outsourcing decision are in conflict, for instance searching for the best quality, lowest cost, highest capacity available service. An outsourcing process looks for the specialization in activities which are not, normally, core activities of the organization [11]. Example of these activities are those related to systems, accounting, buildings, human resources, call centers, engineering, logistics, etc. At the same time, organizations often outsource those activities having fluctuating work patterns, in their burden and performance. Maintenance often fits within this type of activities profile.

Outsourcing normally becomes a strategic decision and it must contribute to reach the objectives of the organization by improving quality and security, reducing cost and optimizing resources [10]. Outsourcing allows the organization to focus its efforts on improving those functions that are source of competitive advantages and more profitable to the core business.

Outsourcings have several advantages and disadvantages. Within the advantages we can list, for instance [2, 15, 21, 43]:

Fig. 6.10 Outsourcing criteria



- Cost reduction, at the same quality to employ a more specialized supplier.
- Costs restructuring, changing fixed costs by variable costs in terms of provided services.
- Stimulates the local employment through contracts with local firms.
- Obtaining rapid budget by selling assets.
- Quality improvement for higher specialization.
- Access to outside expert knowledge.
- Standardization and access to scale economies.
- Flushing resources for other purposes.
- Improving company focus.
- Improving management of difficult functions to handle.
- Optimizing routine tasks.
- Share the risk by flexibility of demand with the supplier company.
- Providing legal guarantee in services.
- Developing relationships between financial aspects and levels of service.
- Starting point for changes in the organization.
- Speed through reengineering.

We also have to consider potential risks and disadvantages which affect any outsourcing plan:

- Unfulfilled or questionable expectations, in the developed scenario to generate the process of outsourcing.
- Changes in the quality due to breach of the service agreements, either by the knowledge or capabilities of the supplier company, or by errors in the definition of the same company.
- Loss of knowledge or skills through transfer to the supplier, where it is more difficult to retain and improve; this happens frequently.
- Loss of control over the externalized functions, source of learning for the internal staff.
- Dependence by the supplier could cause adverse consequences for the client (extraordinary investments).

- Loss of security by transferred staff to the supplier, by hoax and illegal transmission of knowledge and information to the competence.
- Public and internal opinion due to outsource jobs to another company.
- Loss of staff motivation, because it can create a feeling of alienation within the client company that their jobs are valueless.

In network utilities, maintenance plays an important role within the value chain, as we have been able to appreciate in previous chapters; therefore maintenance outsourcing will always be a strategic issue. In this type of companies outsourcing requires a careful evaluation in order to keep, and properly manage, knowledge and professional experience in networks maintenance that the company owns. To that end, outsourcing decisions should take into account maintenance managers and engineers' opinions and considerations.

As a starting point of these outsourcing processes, it is recommended to define the mechanisms to manage the change. To plan a correct transition until the supplier can fulfill agreed service levels. Also, to ensure business continuity in outsourcing and a possible process reversion in the transitional phase, at any time, or at the end of contract. Mechanisms to manage change are basically related to knowledge management and to company's information systems, as key supporting tools for an effective outsourcing process management [15] (see Fig. 6.10).

The decision about which activities have to be outsourced, the scope of the outsourcing process, will come next. Managers have to identify the key knowledge and skills to empower department internally, and outsource those controlled activities which are redundant or with less strategic importance [13, 14]. To do this, activities have to be categorized by their:

- Strategic value;
- Cost and resources consumption;
- Contribution to quality;
- Involved knowledge; and
- Control possibilities.

It is also important not to progress beyond the adequate level of outsourcing, because there is a point of "irreversibility" where it would be impossible to react and with inadmissible actions in resources and time to correct the consequences, once they have occurred. A good practice is to outsource progressively in levels, increasing external services once the management and control of such activities is achieved.

Once the scope of the outsourcing is clearly defined, the following step is the supplier selection and the planning of outsourcing implementation. There are many considerations to take into account during this phase and negotiation to avoid the risks listed above. Probably the main point is to have the ability to launch a proper collaborative win-to-win process with the supplier, where they are considered as a key strategic partner. It is advisable to guide suppliers to offer services based on their different knowledge levels, and thus avoid concentrating only on cost reduction oriented approaches. Aspects most favoured to select a supplier could be:

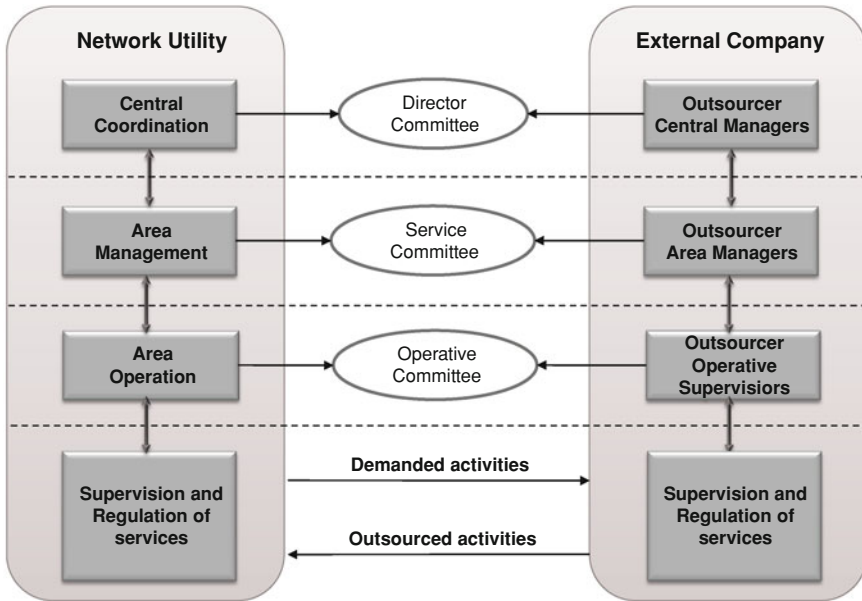


Fig. 6.11 Relationship model committees

- Experience in the sector.
- Flexibility on demand for services.
- Confidence.
- Technical and economic solvency.
- Will of collaboration oriented to services as strategic support.
- Transparency of suitable pricing.

For a proper control of outsourced maintenance resources there is not a unique type of maintenance contract, this will depend on the functions to be developed by the supplier and on the form in which they will be rewarded. For instance, measuring each specific work can be based on a catalog of prices or on a fixed established price per service that is carried out during a predetermined period of time.

With the idea of serving as a guideline on preparation of maintenance contracts, the European Standard EN 13269 [12] describes the basic sections that a maintenance contract should include: heading; contract objective; useful definitions; scope of the works; technical, commercial, organizational and legal considerations.

Another good practice, besides a detailed maintenance service contract definition, it is to define a two-way relationship model to track performance of the contracted services. By this we mean a simple model where tracking rights and liabilities of different players from both organizations (in-house and contracted staffs) such as:

- Organization chart including roles and responsibilities at each company level.
- Procedures, restrictions, recommendations and standards to apply as reference.
- The communications methods and normal periodicity.
- The meetings rules and normal periodicity.
- Urgent communications and meetings.
- Reports and KPI to exchange and escalate (internally or between companies).

Network utilities organizations are normally composed by a central structure plus several area structures. This fact could complicate the relationship with the suppliers' staff and as a good practice companies create a structure of committees to evaluate the contract performance from a collaborative point of view, not only at operative levels but also at managerial levels. Consequently, a proposed structure will be divided in levels (see Fig. 6.11), as follows:

- I. Two area committees oriented to levels of the offered services in activities performance and technical issues.
 - The Operative Committee and
 - The Service Committee.
- II. One central committee oriented to global objectives and from contract perspective.
 - The Director Committee.

The Operative Committee should be concentrated on the service level in each type of outsourced activities, mainly from a technical and specialized perspective and with a meeting periodicity at less twice a month. Then it has to be attended by the operative levels of both companies. In this committee particularities in the area activities and KPIs should be analyzed, trying to discover cause of deviations, weak points and risks, perfective actions, repetitive or non-productive activities, and possible process improvements. In the case of great deviations in service level or the necessity of large investments to improve the results and the conservation of the network, these have to be escalated to the upper committee in order to be approved.

The Service Committee has a main liability of analyzing the received service in an aggregated way over all the types of outsourced activities at less once a month. Therefore the area managerial levels should attend to this committee reviewing the results of the operative committee, the global service levels and the expenses inside the area, filtering the deviations, improvements and investments issues toward the uppers committees. In addition, qualitative evaluation of the external company must be realized over the shown behavior or manners, such as implication and representative image of the network utility.

The Director Committee will be composed of the authorized representatives at large of the central organization if the contract is global for many areas or of the area organization if the contract is local inside a single area with a decentralized legal authorization. The meeting periodicity has to be at less once a quarter. Inside their functions as cascade of the previous committees it reviews the escalated

issues from them and analyzes the global service level identifying the legal severities and exceptional circumstances according to the contract. Besides continuity, extension, renewal or closure of the contract is under its responsibility. Budgetary fulfillment is the key in this evaluation as basis of previsions and tendencies. It has to clarify any disagreements or misinterpretations of the contract and to apply the penalties and bonuses.

6.4 Maintenance Budget

The budget of a department is the main constraint for the organization. As we have seen, the economic criteria is totally in conflict and inversely proportional to the level of quality and reliability. Therefore, maintenance budget has to be tuned following the standards of quality and reliability, and optimized continuously according to the circumstances.

In our maintenance management framework, the accounting system becomes a critical issue. We foster to base this system on activities parameterization according to the “activity based costing” (ABC) method. If we do so, we can properly optimize resources and costs in different activities following an “activity based management” (ABM) system. This is known as implementing an activity-based accounting system [26]:

- a. Selecting the cost basis,
- b. Tracing resources to activities,
- c. Determining the activity measure,
- d. Assigning secondary activities,
- e. Calculating the activity cost, and
- f. Determining the activity performance measurement.

Besides this, costs and activities have to be allocated to the different services based on causality relationships this will determine the tariff regulation [37]. This allows and facilitates the financial audits from the regulator concerning all financial statements (balance sheets, profit and loss accounts, cash flow statements, and so on) as well as non-financial information (economic, environmental, users’ consumption, technical indicators related to efficiency and quality, and so on), which means:

- A directly traceable cause-and-effect relationship with provision of the service;
- A verifiable relationship between the equipment and the output of the service; and
- A direct causal relationship with a pool of common costs or revenues on the basis of a relevant, reliable, and verifiable factor such as relative use.

In this section we will proceed to describe the typical sections or parts within a maintenance budget in a network utility company (see Table 6.6). The reader will realize that we use the PAF scheme (Prevention, Appraisal and Failure costs)

Table 6.6 Basic maintenance budget in network utilities

Maintenance costs in network utilities	P	A	F	C	O
Labor costs					
Staff					
Administration	X	X	X	X	X
Operation	X	X	X	X	X
Surveillance	X	X	X	X	X
Specialized support	X	X	X	X	X
Management	X	X	X	X	X
Learning, training and documentation	X			X	X
External resources					
Outsourcing (external services)	X	X	X	X	X
External personnel	X	X	X	X	X
Specialized technical assistance	X	X	X	X	X
Facilities	X		X		X
Material costs					
Raw materials	X		X		X
Supplies and minor working tools	X		X		X
Accessory materials	X		X		X
Technology costs					
Hardware					
Equipments	X		X	X	
Instruments and devices to measure		X		X	
ICT equipments and devices	X	X	X	X	X
Spare parts of equipments	X		X		X
Transport service	X	X	X	X	X
Installation	X			X	
Upgrade/overhaul of asset	X			X	
Add-ons to original asset	X			X	
Asset rent	X	X	X		X
Software and licences	X	X	X	X	X
Withdrawal of assets	X		X		X
Perfective projects	X			X	
Internal impacts					
Idle capacity losses			X		X
Opportunity losses			X		X
Business interruptions			X		X
Quality affections			X		X
Overhead costs					
Occupancy and travel costs					
Soils and spaces for rent	X				X
Not our own rent of infrastructure	X		X		X
Consume of non ICT services	X		X		X
Consume of ICT services	X		X		X
Furniture	X				X
Office supplies	X				X
Travel and vehicle expenses	X	X	X	X	X

(continued)

Table 6.6 (continued)

Relations to outsiders					
Commissions, duties, licences, taxes, insurance and fees	X	X	X	X	X
Guarantees	X		X		X
Penalties			X		X
Damages due to outsiders			X		X
Environment affection			X		X
Damages provoked to outsiders			X		X
Security					
Health and safety	X	X	X		X
Access control	X	X			X
Security affection and stealing			X		X

P prevention cost, *A* appraisal cost, *F* failure cost, *C* CAPEX, *O* OPEX

presented in Sect. 3.2 to represent maintenance costs and the different divisions for maintenance budget in service companies. This PAF scheme division of the budget depends on the nature of the costs: direct labor costs, direct materials costs, technology costs, and overhead costs. Also every type of cost inside each division can also be classified in capital or operational expenditures when the realized cost improves the value of the asset or simply it is necessary to operate it, respectively.

The main divisions of the budget in Table 6.6 are:

- Labour costs are composed by:
 - Staff, which includes all those necessary costs evolved with internal human resources distinguishing among different levels of expertise and improvement (i.e. learning and training).
 - External resources, containing human resources costs outside the organization.
- Materials costs, where raw materials, supplies, minor tools and accessory materials are allocated.
- Technology costs, taking in consideration industrial, information and communication technologies:
 - Hardware which includes all those necessary costs evolved with acquisition and use of the equipments and measuring devices.
 - Software cost, including purchase, modifications and maintenance.
 - Withdrawal of assets that involve adjustments in the life cycle and their amortization, distinguishing the cause of their retirement.
 - Perfective projects generating improvements in the assets to mitigate errors or risks.
 - Internal impacts as representation of the consequences due to a bad operation of the department in the use of the capacity, the lost opportunities, and losses provoked by business interruptions and quality affections. The idea is to facilitate the decisions about investments in preventive or corrective activities according to the impacts.

- Overhead costs providing maintenance support, composing of:
 - Facilities, furniture, and services, constituting the operational environment and the transportation ways inside the company.
 - Relations with outsiders due to legal restrictions, contracts or commitments, not only through business relationships, but also legal, social, customer and environmental relationships. This group includes mandatory costs such as licensing, environmental protection, costs of guarantees and penalties, etc.
 - Security costs invested to protect the operation and the information, integrating the protection of human resources, the environment and the network.

All these figures will later be measured and controlled, according to the occurrence of each type of cost involved in each process or activity [9, 23], and those costs will be linked to the network and customers, as a way to improve decision making, that can now take into account activities value or impact on maintenance effectiveness and efficiency. Thus, the activities can also be classified as preventive, appraisal or failure and translated to the budgeted costs according to their nature described above.

For example, in Table 6.6 maintenance labor costs can be classified as preventive (P), assessment (A), or failure/corrective cost (C) while other costs can only be of a unique kind. This is the case of training costs which are always a prevention (P) cost because they allow enhancing technical knowledge to solve or prevent failures. In addition, failure/corrective costs can also be classified regarding to the consequences of maintenance errors [45]: environmental impact, penalties, loss of opportunities, etc.

Table 6.6 also allows us to classify costs as Capex (C—when the costs are invested for initial operation) or Opex (O—when the costs are produced during the normal operation). Notice how some of them are always Capex (C) or Opex (O), but others can be Capex or Opex depending on the occasion.

Using the Table 6.6, it is easy to set the distribution of each type of costs according to industry best practices. For instance Moubray [31] considers a reasonable costs distribution balance with prevention costs around 25%, the appraisal costs near 35% and the failure costs close to 40% (this does not considers intangible costs due to possible risks). In addition, according to the Spanish Maintenance Association [1] a regular proportion between staff costs and external resources in these type of companies could be 60 and 40%, respectively. Other authors, for instance Bertolini [6], discuss about the proportion of each principal divisions of the maintenance budget for a general company as a 44.5% of labor costs, 31% of material costs, 16.5% technology costs and 8% of overhead costs. Of course these proportions can only be used as guidelines, check points, because they could change among companies of different sectors or even within the same sector. It is clear to us that the maintenance budget proportions are a peculiar characteristic in each company, according to the existing internal management structure.

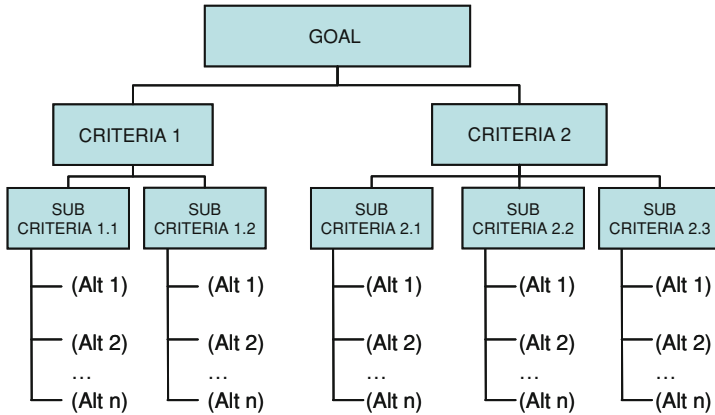


Fig. 6.12 Example of decision hierarchy

6.5 Scheduling Maintenance Work

In this section we present several aspects which are important when scheduling maintenance work in a network. We first deal with the problem of determining asset's criticality within the network and then we discuss a method to establish task priority considering other factors like the need urgency and the customer rank.

6.5.1 Asset Criticality

A method that may be used to prioritize assets according to their criticality is the analytical hierarchy process (AHP). This approach was developed at the Wharton School of Business by Thomas Saaty [40] and allows the decision makers to model a problem in a hierarchical structure showing the relationship of the goal, objectives (criteria), sub-objectives and alternatives. Alternatives that are deficient with respect to one or more objectives can compensate by their performance with respect to other objectives. AHP was composed departing from several previously existing but unassociated techniques and concepts such as hierarchical structuring of complexity, pairwise comparisons, redundant judgments and the eigenvector method for deriving weights and consistency considerations. The resulting method may crucially help when dealing with complex technological, economic and sociopolitical problems.

AHP is built on a solid yet simple theoretical foundation based on three basic principles:

- Decomposition;
- Comparative judgments; and
- Hierarchy composition or synthesis of priorities.

Table 6.7 Judgments ratio scale [39]

Judgments	Score
Equal	1
	2
Weak	3
	4
Strong	5
	6
Very strong	7
	8
Absolute	9

The decomposition principle is applied to structure a complex problem into a hierarchy of clusters, sub-clusters and so on (Fig. 6.12).

The pairwise comparisons are used to derive local priorities of the elements in a cluster with respect to their parents. AHP uses a set of one-to-one comparisons to evaluate alternatives under each criterion. These pair-wise comparisons are the smallest in decisions. Alternative comparisons and criteria weighting is done in separate steps. Criteria weights combine both objective measures and subjective preferences. AHP aims at quantifying relative priorities for a given set of alternatives on a ratio scale [39] (see Table 6.7).

This scale, from one to nine, is used to give the relative preference between two alternatives, is able to capture a great deal of information, and has proven to be extremely useful due to the fact that the AHP is somewhat scale independent [16]. Reasons defending this scale are the following [40]:

- The human ability to make qualitative distinctions is well represented by five attributes: equal, weak, strong very strong, and absolute. Compromises between adjacent attributes can be made when greater precision is needed. The totality requires nine values and they may be consecutive;
- The human brain has a psychological limit for simultaneous comparisons of 7 ± 2 items. This mental capacity has something to do with the number of fingers. Therefore, a nine-point scale will be sufficient to carry out the comparisons between items.

The principle of hierarchic composition of synthesis is applied to multiply the local priorities of elements in a cluster by the global priority of the parent element, producing global priorities for the lowest level elements (the alternatives) [40].

AHP has many advantages over conventional scoring methods such as an increase in accuracy and consistency, and that the subjective consideration is quantified in a structured framework. However, the major drawback in the use of AHP is the effort required to make all pair-wise comparisons [27]. As the size of the hierarchy increases, the number of required pair-wise comparisons increases exponentially. Also the AHP is complex in terms of higher levels of detail required by the evaluators when asked for their preferences.

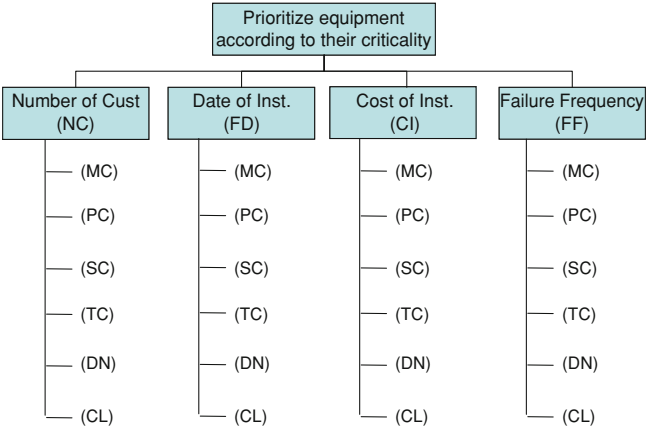


Fig. 6.13 Criteria to set up hierarchy according to AHP

Let us see, in the next paragraphs, details of the procedure to follow in order to carry out an assets criticality analysis following AHP, and for a telecom network maintenance.

1. *State the goal.*

Prioritize the equipment of a network according to their criticality, and in order to ensure proper maintenance.

2. *Define the criteria.*

- Installation date (ID),
- Installation cost (IC),
- Number of clients (NC),
- Failure frequency (FF).

3. *Identify the alternatives.*

- Main centre (MC),
- Primary connection (PC),
- Secondary connection (SC),
- Tertiary connection (TC),
- Distribution network (DN),
- Customer link (CL).

The decision hierarchy structure to set up preventive maintenance in the telecom network can be appreciated in Fig. 6.13:

4. *Define the scale for each criteria.* Notice that the defined scale for each criteria may require a search for certain equipment historical data. For instance, in order to score each equipment regarding FF criteria (see Table 6.8), a review of the equipment failure records will be required.

Table 6.8 Judgments ratio scale

L	NC (#)	DI (years)	CI (€)	FF
9	≥ 220.000	>10 years	>120.000	1 Fail per day
8	[100.000, 220.000]	[10, 8]	[72.000, 120.000]	1 Fail per week
7	[72.000, 100.000]	[8, 7]	[48.000, 72.000]	1 Fail per month
6	[48.000, 72.000]	[7, 6]	[24.000, 48.000]	1 Fail in 3 months
5	[12.000, 48.000]	[6, 5]	[12.000, 24.000]	1 Fail in 6 months
4	[2.000, 12.000]	[5, 4]	[6.000, 12.000]	1 Fail in 9 months
3	[500, 2.000]	[4, 2]	[3.000, 6.000]	1 Fail per year
2	[25, 500]	[2, 1]	[500, 3.000]	1 Fail in 2 years
1	≤ 25	<1 year	≤ 500	1 Fail in >3 years

5. *Alternatives evaluation for each of the selected criteria.* Table 6.9 presents this evaluation.
6. *Quantify judgments on pair alternative criteria and determine the criteria weighting and its consistency.*

The quantified judgments on pairs of criteria alternatives, C_i and C_j , (criteria in our case study are FF, FD, FS and FC) are represented by an n -by- n matrix, known as a judgments matrix. This judgments matrix can be represented by

$$A = (a_{ij}) \quad (6.1)$$

The entries a_{ij} are defined by the following entry rules:

Rule 1. If $a_{ij} = \alpha$, then $a_{ji} = 1/\alpha$, $\alpha \neq 0$;

Rule 2. If the criteria C_i is judged to be of equal relative importance as criteria C_j , then $a_{ij} = a_{ji} = 1$. Obviously $a_{ii} = 1$ for all i .

Thus the matrix A has the form

$$A = \begin{bmatrix} 1 & a_{12} & \dots & a_{1n} \\ 1/a_{12} & 1 & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 1/a_{1n} & 1/a_{2n} & \dots & 1 \end{bmatrix} \quad (6.2)$$

where a_{ij} represents how many times the criteria C_i is more important for equipment criticality than criteria C_j (see Table 6.10 for our case study).

Having recorded the quantified judgments of comparisons on pairs (C_i, C_j) as numerical entries a_{ij} in the matrix A , what is left is to assign to the n contingencies $C_1, C_2, C_3, \dots, C_n$ a set of numerical weights $w_1, w_2, w_3, \dots, w_n$ that should reflect the recorded judgments.

Saaty proved that the eigenvector of the comparison matrix provides the best approximation to the priority ordering (weight) of the different criteria, and the eigenvalue is a measure of consistency (to find the priority vector or the weight of

Table 6.9 Alternatives (assets) evaluation for each criteria

Asset	Criteria							
	NC	$\frac{NC_i}{\sum_i NC_i}$	DI	$\frac{DI_i}{\sum_i DI_i}$	CI	$\frac{CI_i}{\sum_i CI_i}$	FF	$\frac{FF_i}{\sum_i FF_i}$
(MC)	9	0.33	2	0.09	10	0.29	3	0.12
(PC)	7	0.26	6	0.27	6	0.18	2	0.08
(SC)	5	0.19	2	0.09	2	0.06	4	0.15
(TC)	3	0.11	2	0.09	9	0.26	8	0.31
(DN)	2	0.07	2	0.09	5	0.15	6	0.23
(CL)	1	0.04	8	0.36	2	0.06	3	0.12
Total	27		22		34		26	

Table 6.10 The judgments matrix (A) for the criticality analysis

a_{ij}	NC	DI	CI	FF
NC	–	3	1	1
DI	1/3	–	1/2	1/3
CI	1	2	–	1/2
FF	1	3	2	–

each factor included in the priority ranking analysis, the eigenvector corresponding to the maximum eigenvalue is to be determined from matrix analysis).

Consistency in the pair-wise comparison matrix means that when basic data is available, all other data can be logically deduced from them. For instance, if factor A_1 is three times more important than factor A_2 and factor A_1 is six times more important than factor A_3 , then $A_1 = 3A_2$, and $A_1 = 6A_3$. It should follow that $3A_2 = 6A_3$ or $A_2 = 2A_3$ and $A_3 = 1/2A_2$. If the numerical value of the judgment (comparison) in the (2, 3) position is different from 2, then the matrix would be inconsistent.

It is very difficult to identify “ $n - 1$ ” comparisons which relate all factors or activities and of which one is absolutely certain. It turns out that the consistency of a positive reciprocal matrix is equivalent to the requirement that its maximum eigenvalue $\lambda_{\max}$ should be equal to the number of factors “ n ”. Then it is possible to estimate the inconsistency—consistency index (CI)—as follows:

$$CI = \frac{\lambda_{\max} - n}{n - 1} \tag{6.3}$$

This index is further used to calculate the consistency rating (I_R) as detailed in Saaty [41]:

$$I_R = \frac{CI}{RI} \tag{6.4}$$

where RI is the random average value of CI for a n -by- n matrix. Values of RI are shown in Table 6.11 [40].

Table 6.11 RI values for matrices of different order (N)

N	1	2	3	4	5	6	7
RI	0	0	0.52	0.89	1.11	1.25	1.35

Table 6.12 Criteria weights and I_R provided by Expert Choice® (2011)

Criteria	Weight (wi)
NC	0.302
DI	0.110
CI	0.230
FF	0.358
IR	0.02

A consistency rating (I_R) of 0.10 or less is considered acceptable. In the case of inconsistency the process of evaluation of the judgments matrix should be repeated. We have used the software Expert Choice ® (from Expert Choice, Inc. Arlington(VA)-based leading provider of enterprise portfolio analysis software and services) to produce the results for weights and the consistency rating shown in Table 6.12.

7. Determine the final equipment criticality hierarchy.

See Tables 6.13 and 6.14.

6.5.2 Activity Priority

Considering that the number of activities could exceed the size of resources or the time for proper resolution in network utilities, before the confluence of many activities, a dynamic mechanism has to be established to arrange them according to their importance and maximizing the benefits that produce the implicated costs to the maintenance. A mostly accepted method to prioritize maintenance activities is the one known as NUCREC [18] where priority is calculated multiplying three crucial factors with values from 1 (highest importance) to 4 (lowest importance):

- The level of Need Urgency,
- The level of Customer Rank and,
- The level of Equipment Criticality.

The level of Need Urgency is usually classified according to the failure impact on the services [30]:

- Class I, or essential, when the occurrence of any failure will result in a catastrophic or total loss of the service.
- Class II, or critical, when the occurrence of any failure would severely degrade the service in 30% or more.

Table 6.13 Final hierarchy provided by Expert Choice

Asset relative value $\times$ criteria weight					
Asset	$\frac{FF_i}{\sum_i FF_i} \times w_{FF}$	$\frac{FD_i}{\sum_i FD_i} \times w_{FD}$	$\frac{FS_i}{\sum_i FS_i} \times w_{FS}$	$\frac{FC_i}{\sum_i FC_i} \times w_{FC}(4)$	Final asset hierarchy = (1) + (2) + (3) + (4)
	(1)	(2)	(3)		
(MC)	0.01	0.01	0.07	0.04	0.22
(PC)	0.08	0.03	0.04	0.03	0.18
(SC)	0.06	0.01	0.01	0.06	0.13
(TC)	0.03	0.01	0.06	0.11	0.21
(DN)	0.02	0.01	0.03	0.08	0.15
(CL)	0.01	0.04	0.01	0.04	0.11

Table 6.14 Final assets criticality ranking

Assets	Final hierarchy	Ranking
(MC)	0.22	1
(TC)	0.21	2
(PC)	0.18	3
(DN)	0.15	4
(SC)	0.13	5
(CL)	0.11	6

- Class III, or serious, when the occurrence of any failure does not have a dramatic impact on the service.
- Class IV, or scheduled, when the occurrence of any failure has not an immediate impact on the service.

In addition, with reference to the level of Customer Rank, seeking the customer orientation, customer segmentation has to be taken into account in order to prioritize activities, distinguishing among the various types of customers based on their demanded volume of consumption or their contracted service level agreements. So it is important to discriminate the activities over equipment that provides services to rigorous customers than other massive customers. For this classification of customers could use the “Rating ABC or Pareto Analysis or Rule 20–80” with 4 care levels:

- Very Important People (VIP). Primary care level, for vital customers such as hospitals.
- A. Secondary care level, for a few customers who produce high and constant income, requiring special attention such as large companies.
- B. Tertiary care level, for customers that require attention above the standard, such as SMEs.
- C. Quaternary care level, for massive market customers, the majority, for which it is offered the standard care.

Lastly, the Equipment Criticality has to be incorporated as we have explained in the previous section.

Table 6.15 Discrete matrix prioritation example

Prioritisation scheme		Customer rank				Service L. agreement		Customer rank			
Need urgency	Equipment criticality	VIP	A	B	C	Need urgency	Equipment criticality	VIP	A	B	C
1	1	1	1	1	1	1	1	8	8	8	8
	2	1	1	2	2	2	2	8	8	12	12
	3	1	2	3	3	3	3	8	12	24	24
	4	1	3	3	3	4	4	8	24	24	24
2	1	2	2	2	2	1	1	12	12	12	12
	2	2	3	3	3	2	2	12	24	24	24
	3	2	3	3	3	3	3	12	24	24	24
	4	2	3	3	3	4	4	12	24	24	24
3	1	3	3	3	3	1	1	24	24	24	24
	2	3	3	3	3	2	2	24	24	24	24
	3	3	3	3	4	3	3	24	24	24	24
	4	3	4	4	4	4	4	24	72	72	72
4	1	4	4	4	4	1	1	72	72	72	72
	2	4	4	4	4	2	2	72	72	72	72
	3	4	4	4	4	3	3	72	72	72	72
	4	4	4	4	4	4	4	72	72	72	72

As a result, we should have three values from 1 to 4 that could be multiplied to obtain the NUCREC activity priority with values from 1 to 64. Although to simplify the utilization and the customization of the performance, it is usual to restrict the number of possible values (64) integrating multiples values in a common factor such as four to discrete the magnitudes. One example could be, the assignation of the values from 1 to 16 at I priority, from 17 to 32 at II priority, from 33 to 48 at III priority, and from 49 to 64 at the IV priority. Another possibility is to discretise the 64 values using a matrix, where we could deduce the correspondent priority (see Table 6.15).

Thus, we could prioritize all the maintenance activities automatically in relative values of 1–4, with one being the highest priority value, and corresponding to different service levels for resolution of the activity.

Consequently, this prioritization matrix allows, the unification of the control and the internal notification and escalation about their performance at the same time establishing certain conditions to inform the organization and to demand additional support, avoiding too much stress on the organization. For example, you may use the above-mentioned prioritization to report to the managerial levels the occurrence of all those with prioritization one, and to escalate to the expert technicians those activities whose prioritization from 1 to 2 and their service level agreement exceeded. Accordingly, the internal rules to attend the activities in an ordered way distinguishing the operative and managerial levels in the organization would be simple and automate the dispatching without difficulty.

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Chapter 7

Monitoring Services

7.1 Introduction

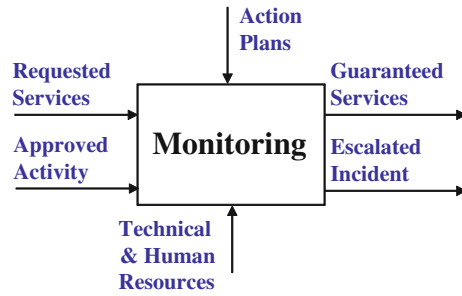
In network utilities, maintenance-related processes and activities require high levels of coordination and integration. In our framework we will design activities and define a standardized process government to be in the best possible position to fulfil agreed internal and external service levels. This requires that, inside the operative processes definition, the main activities will be described with the aim of facilitating their computational implementation.

A first level of performance can be achieved through processes monitoring, incident management, incident prevention and proper capacity administration. An advanced second level of performance can reinforce the first one by analyzing maintenance, searching for the optimization through sustainable knowledge management.

Within the first level of performance processes, in this chapter we concentrate on the process dedicated to monitoring services and the network. As will be discussed later, this process can be considered the core of the proactive network maintenance, and therefore we have to pay special attention to it.

7.2 Monitoring Services and the Network

This process is about dealing with services and network conditions, managing any alarm that may appear, determining what to do about it, always in a proactive sense. Organizations increasingly rely on monitoring systems to control remote equipment and services offered to customers [7]. Depending on the quality requirements, the networks need a service view with a unified management, which must be provided by an integrated management system. This process should ensure the provision of services through the real time network monitoring and

Fig. 7.1 Monitoring process

surveillance, according to internal and external service quality parameters (with different institutions, with the customers, etc.). In short, it is important to link services status with real time information concerning the condition of the equipment which are supporting those services (see Fig. 7.1).

Consequently, this process can be considered the core of the proactive maintenance, employing online sensing, diagnosis and prediction before any customer complains about the services. Proactive maintenance [2, 6], increases the knowledge about network conservation and utilization, recognizing any change in physical and service conditions even from remote sites. In these environments, we name proactive activities as those preventive activities generated as a consequence of equipment monitoring or inspections and previously to any other external driver [4].

Depending on the type of network to monitor, the coverage of this process will be more or less complex and so will the devices required to accomplish this process. We can monitor the entire network, critical equipment or only their critical functions. The real coverage will depend on the cost benefit analysis of the network monitoring program.

In addition, the implementation of network monitoring contributes to human error reduction (see Chap. 1) thanks to the facilitation of Performance Shaping Factors (PSF) such as improving the root-cause detection in a short time, providing means for conveying spatial and functional information to operators (and in a form which they can assimilate), decreasing mismatches between perceived and real risk, and supervising the reliability of the instrumentation and responsibilities allocation.

Two types of monitoring processes can be defined, automatic and under inspection:

- Automatic monitoring processes are based on ICT facilities, and they provide online information automatically and periodically from the network. As a general rule, a centralized supervision of these processes and information is more cost-effective in order to exploit synergies in a Network Operation Center (NOC) (see Fig. 7.2). In these cases a client-server structure to represent the broadcasting of the information is used, and normally this structure offers two possibilities [7, 4]:

Fig. 7.2 Network Operation Center (NOC)



- Active monitoring when the equipment is interrogated to obtain the information.
- Passive monitoring when the equipment emit the information before any defined event.
- Under inspection monitoring processes are selected when the ICT facilities are not possible or they do not prove to be cost-effective. In these cases periodic or continuous inspections have to be developed [2].

Within the monitoring processes, three principal functions will now be described: the monitoring of incidents, the equipment and services configuration and finally implementing service pricing. Some examples will also be presented to illustrate interesting issues concerning these processes.

7.2.1 Monitoring Incidents

This process controls the service and network conditions in real time. It monitors the normal state of the network, but also the approved actions that could modify that normal state. In this way, susceptible events that could generate incidents can be managed (see Fig. 7.3), using proper filtering procedures. The priority of the incident is very importance for this process, so it can be dispatched within the organization conveniently.

The normal process flow begins with the generation of an alarm or a warning in the network, which (after being captured) is filtered and processed according to technical and qualitative criteria. To avoid the flooding of similar incidents, correlation among actual incidents has to be developed which helps to make possible the incident creation or the assignation to one existing incident, in any case the parameterization of the incident is finally carried out to facilitate its management.

7.2.2 Monitoring Network and Services Configuration

Complying with the [3], configuration management must take responsibility for maintaining monitored elements configuration properly and, if possible, to configure them remotely to ensure proper levels of supervision. Thanks to this, the

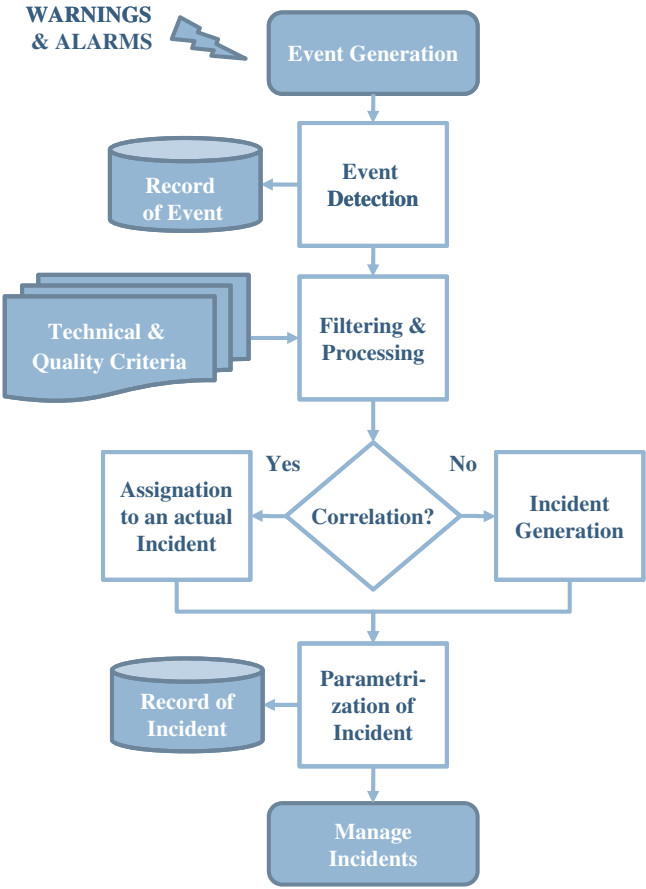


Fig. 7.3 Monitoring of incidents flow

degeneration of the network configuration and performance is prevented, avoiding the existence of false alarms confusing and burdening the organization. The historical network element versions and configurations, along the elements life cycle, have to be recorded, and security measures have to be implemented to limit illicit access.

In this case, the process flow (see Fig. 7.4) begins with a configuration change detection by periodic or under demand inspections. This configuration change detection could be generated by the process *to analyze maintenance* or by the *monitoring* process. In the first case, the configuration change is escalated as an approved activity, so the implementation is carried out and checked. In the second case, the configuration change is handled deriving into a configuration incident and this will need to be analyzed and executed carefully. After the implementation of both cases, the results have to be reviewed, executing a reverse plan when these are not satisfactory, or activating a quarantine period to be sure about the

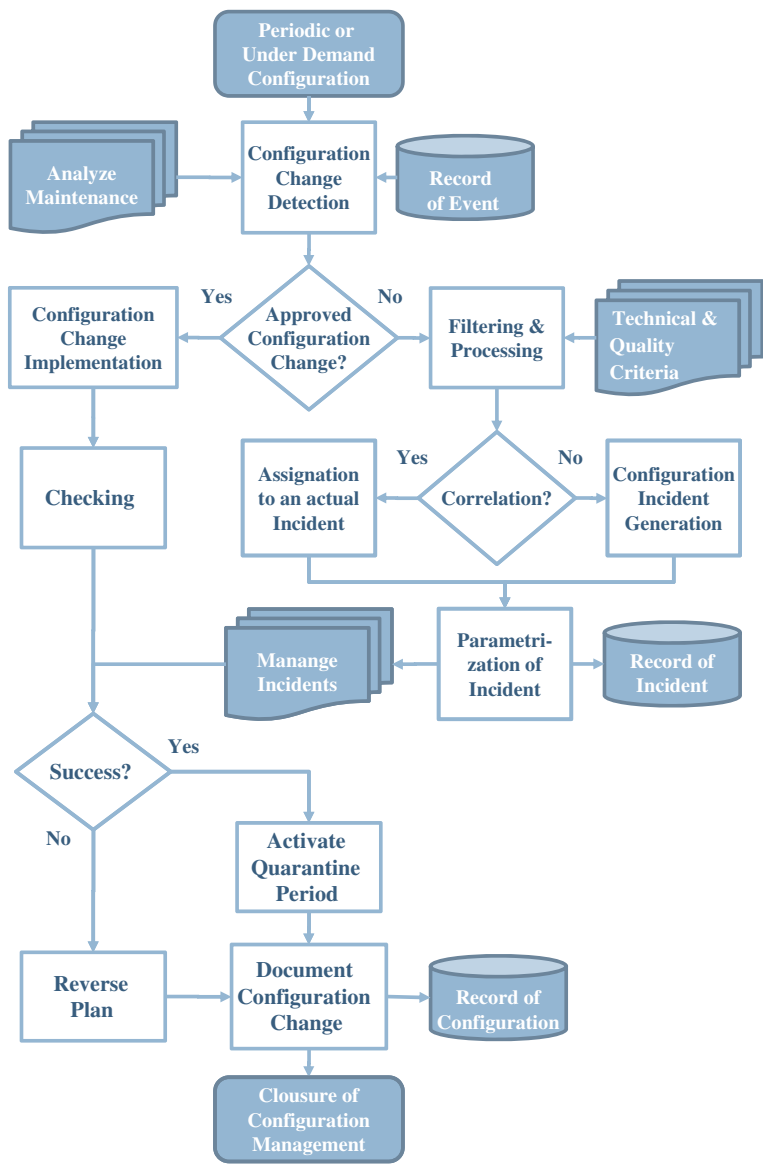


Fig. 7.4 Configuration management flow

configuration stability. Finally, the configuration activity is documented to close the flow.

The ability to properly represent the hierarchical relationship among the different elements, even in geographical maps, is included inside this function.

7.2.3 Services Pricing

One of the most frequently developed functions inside the monitoring process in network utilities is the pricing (implementation of tariffs) of services. Taking part of the customer relationship, the services pricing determines the revenue for each customer in accordance with the service consumption [8]. In some countries the price is controlled by the regulator and sometimes it is imposed or grant-aided. In any case, inappropriate pricing is one of the most frequent sources of customer complaint, and therefore in network utilities it has to be in correspondence with the contracted service level agreements, thereof maintenance again has a great influence in customer perception about the services. Obviously, the adequate consumption of services is supervised in this process alerting of service abuses, waste and security threats.

Modern systems of pricing search a bidirectional relationship to increase the customer satisfaction [1] based on practical tools for remote metering, sophisticated pricing and direct load control, as well as possible new futuristic convenience services, for both the consumer and utility. In this sense, flexible tariff implementation (tariffication) is possible according to the network usage or considering discounts due to network failures (see Fig. 7.5).

Figure 7.5 shows the monitoring contribution to pricing using the ICT advantages for remote metering, periodically and/or under demand. Then in order to realize a flexible tariff pricing, the measures have to be filtered and processed according to the contracted service level agreements. Later, the measures have to be validated depending on the service conditions, recording them with the relationship of any possible service affection.

7.3 Sampling Frequency and Behavior Patterns

A monitoring process is based on identifying physical changes on equipment conditions, their operation and operation environment. A crucial aspect of this process is to identify equipment patterns triggering warning or alarm messages. The objective is to detect equipment degradation from normal conditions; consequently there are two main issues to be identified:

1. Inspection or sampling frequency: This needs to be deduced according to the P-F interval or Failure Detection Threshold (FDT). This interval (see Fig. 7.6) starts when the physical change can be captured (P, Potential Failure) and finishes when the critical event or failure (loss of function) takes place (F, Functional Failure). Inspection frequency has to be a fraction of the PF interval, in such a way that the event or failure could be detected with enough time to be corrected or to be able to reduce its consequences.
2. Behavior pattern: The pattern of degradation depends on the nature of the physical variable and there are diverse international recommendations for each

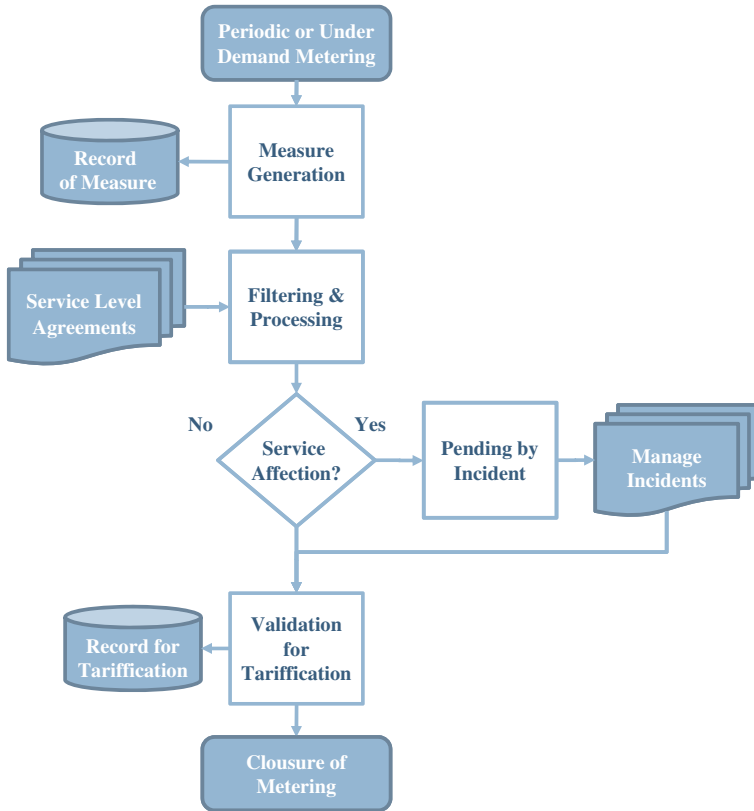


Fig. 7.5 Metering flow for tariffication

type of variable used within the monitoring process: temperatures, pressures, vibrations, amperage, voltage, displacements, humidity, amplitudes, thickness, cracks, presence of chemicals or particles, etc. [5] classifies the monitoring techniques depending symptoms that are produced by a potential failure:

- Dynamic effects, concerning rotating or in motion equipment, in time or frequency domain, such as vibrations or amplitude changes.
- Particle effects, relating to their presence in the operation environment and detected by filtering, sedimenting, light-emitting or magnetic techniques.
- Chemical effects, with reference to the occurrence of the inappropriate quantities of chemicals in the operation environment, which can be perceived by microscopy, spectography or chromatography.
- Physical effects in the appearance and structure of the equipment, distinguishing by ultrasonic, radiology, light-emitting, electrical or fluid techniques.
- Temperature effects mainly detected using thermography, when the potential failure provokes temperature changes in the normal operation.

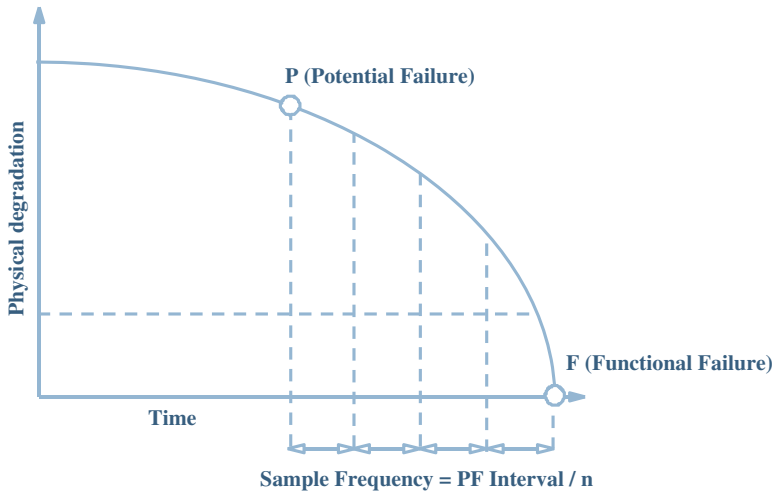


Fig. 7.6 PF Interval and Sample Frequency

- Electrical effects, such as resistance or conductivity variations from the normal operation.

In the following sections we deal with different aspects that must be taken into consideration when estimating inspection frequencies and/or changes in the behavior patterns.

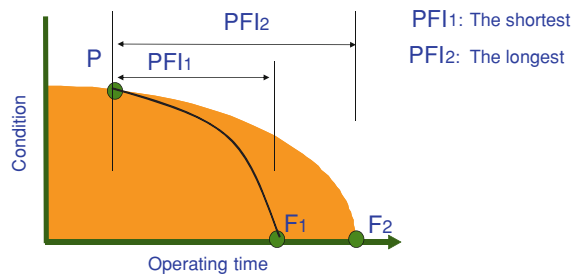
7.3.1 Consistency of the PF Interval

PF interval consistency is an important factor when establishing monitoring frequencies and/or when setting warnings and alarms thresholds within a predictive maintenance program. A PF interval is consistent when its length is repeated over time for the same failure mode, i.e. the degradation of the item for a given failure mode always follows the same pattern and the PF interval is always similar (low dispersion). Many network items may show extremely high consistency in PF intervals for many failure modes, this circumstance is well known and it is used to set up properly inspection intervals in their predictive programs.

In the event of a non consistent PF interval for a failure mode, the analyst must try to get extra data, PF interval samples, and fix the inspection interval considering the shortest observed PF interval (for instance, in Fig. 7.7 the inspection interval could be fixed as a fraction of the shortest PF interval PFI_2 , $II = PFI_2/n$, with n an integer >1).

In some cases, manufacturer pre-selected warnings and alarms thresholds may not match with current item operating conditions and the analyst must reselect these limits according to observed equipment behavior for specific failure modes.

Fig. 7.7 PF Interval dispersion when low consistency



See for instance Fig. 7.8, this figure contains the different vibration measures (in mm/sec) and over several months (notice that the inspection time is set up to a week approximately) of an important water network pump. In this particular case, the original warning threshold is reached when the equipment degradation is in an advanced situation; the slope of the curve linking the different vibration measures is already very high. The maintenance manager decides then to accelerate the sampling process and observes almost a vertical increase in the vibration pattern. Once the next sample is processed and the trend is confirmed, the maintenance department forces of course the equipment stoppage before the alarm threshold is reached.

The inspection of the item offers a terrible condition of some of the elements of the pump (bearing), as shown in Fig. 7.9. This condition could have resulted in a catastrophic failure according to the maintenance team.

For this particular case study, the bearing replacement was done and the new suitable operating conditions were confirmed (See Fig. 7.10), but the most important thing is that the maintenance analyst modified the thresholds for warnings and alarms of this failure mode in the pump as in Fig. 7.11.

Maintenance analyst's intention was both, avoiding a similar level of damage, and enlarging the PF interval, that now includes periods of time with linear, non exponential, deterioration of the item for this failure mode. In this case, the inspection interval is maintained but the sense of a better control of the failure mode is achieved after the incident inspection.

7.3.2 Monitoring of Dynamic Effects in Their Frequency Domain

Observing many dynamics effect within their frequency domain can be a must for many critical network items (especially for rotating equipment). Frequency domain analysis offers extra information and a different perspective of the problems helping in the identification and prognosis of many failure causes that may produce similar item failure modes.

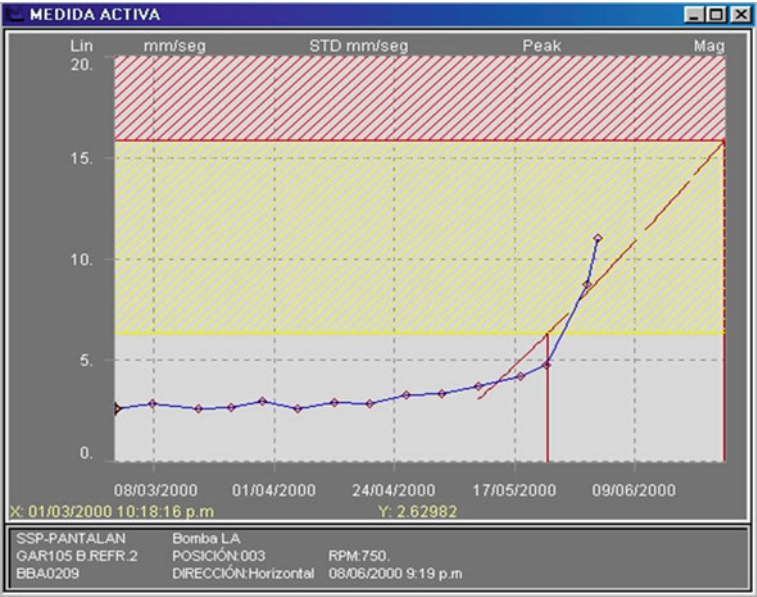


Fig. 7.8 Case study initial warning and alarm thresholds

Fig. 7.9 Case study item conditions observed during inspection



Frequency domain analysis capabilities are present in many monitoring devices using Fourier analysis. Fourier analysis helps in the process of fast identification of the different elements of vibration. Amplitude, speed and/or acceleration of these elements can be isolated and therefore modifications in normal behavior patterns identified (Fig. 7.12). In signal processing, the Fourier transform often takes a time series or a function of continuous time (for instance, overall vibration signal $f(t)$ in Eq. 7.1), and maps it into a frequency spectrum (Fig. 7.12).

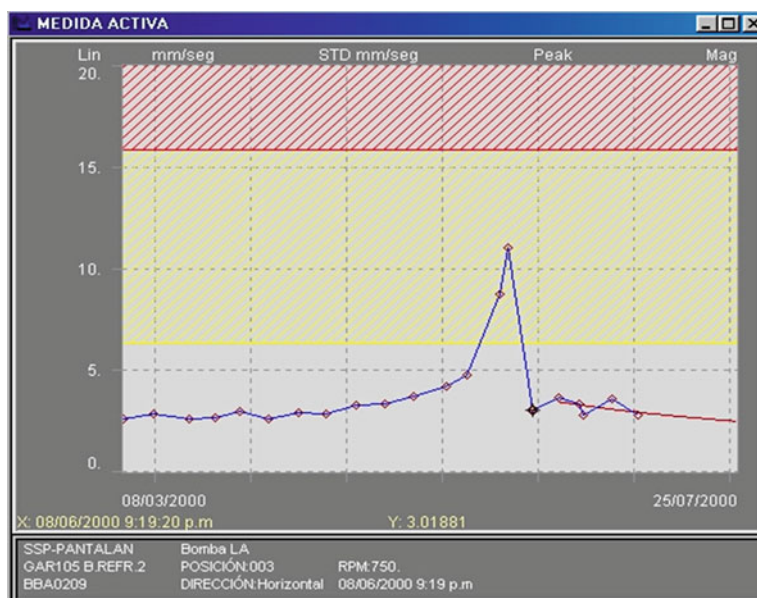


Fig. 7.10 Vibration samples analysis after the bearing replacement

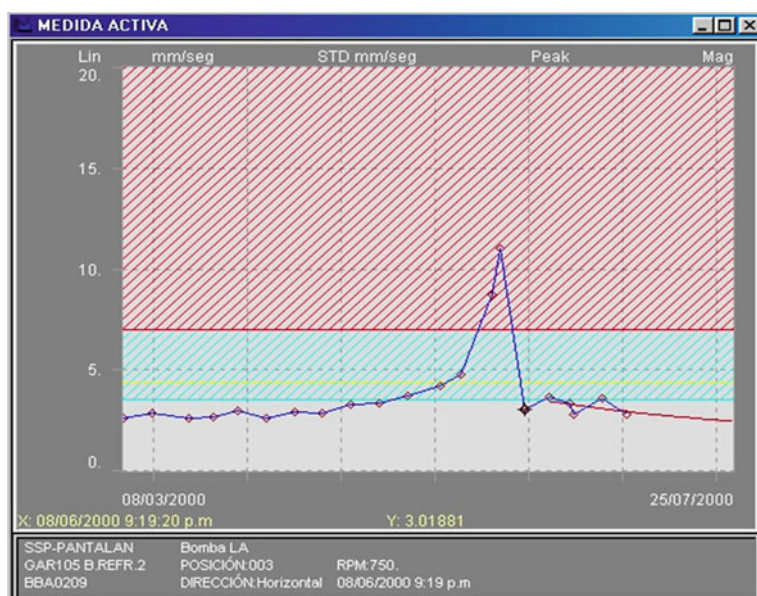


Fig. 7.11 New warning and alarm thresholds

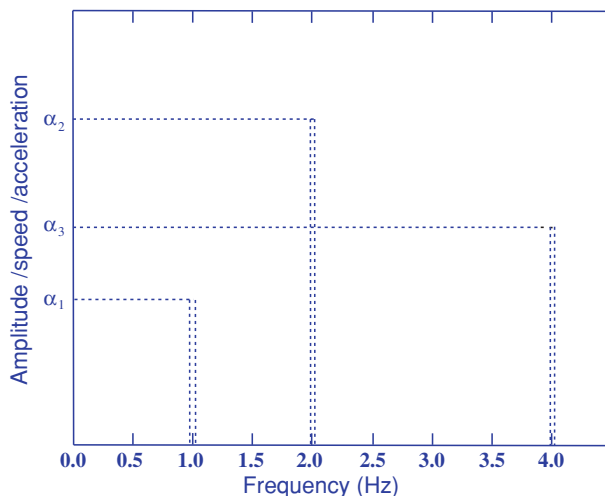


Fig. 7.12 Spectral vision of vibration

$$f(t) = A_0 + A_1 \sin(\omega t + \varphi_1) + A_2 \sin(2\omega t + \varphi_2) + A_3 \sin(3\omega t + \varphi_3) + \dots \quad (7.1)$$

That is to say, it takes a function from the time domain into the frequency domain; it is a decomposition of a function into sinusoids of different frequencies; in the case of a Fourier series or discrete Fourier transform, the sinusoids are harmonics of the fundamental frequency of the function being analyzed. When the function f is a function of time and represents a physical signal, the transform has a standard interpretation as the frequency spectrum of the signal. At this point, for instance the magnitude of the resulting complex-valued function ($A_1 \sin(\omega t + \varphi_1)$) at frequency ω represents the amplitude of a frequency component whose initial phase is given by the phase of this function (φ_1).

Fourier transforms are not limited to functions of time, and temporal frequencies. They can equally be applied to analyze spatial frequencies, and indeed for nearly any function domain. This justifies also their use in branches such diverse as image processing, heat conduction and automatic control.

For the vibration analysis considered in the case study presented in [Sect. 7.3.1](#), global vibration records warning triggered a more specific frequency domain analysis as the one presented in [Fig. 7.13](#). Changes in the normal peak values for specific frequencies, and subsequent investigation of the item generating pulses to that frequency could determine the failure cause as appreciated in [Fig. 7.14](#).

Frequency domain analysis requires constant angular speed of the item and constant load, otherwise frequency values and vibration monitoring values (amplitude, speed or acceleration), respectively, will change.

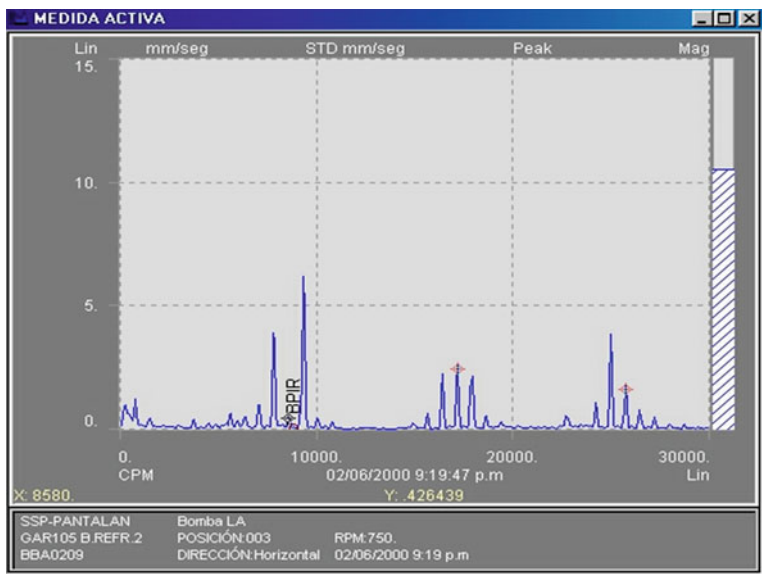


Fig. 7.13 Spectral vision of vibration for the case study in Sect. 7.3.1

Fig. 7.14 Failure cause confirmation for the case study in Sect. 7.3.1



7.3.3 PF Interval and MTTF

A clear differentiation between the PF interval and the mean time to fail (MTTF), for a given failure mode of an item, is also a very important issue. In many maintenance departments this confusion exists, leading to an important number of errors and non-cost-effective decisions.

Consider the example in Fig. 7.15, this figure represents three failures of a given item during certain operating time. Each of these failures has different PF intervals, in this case two of them (Failure 2 and 3) had a PF interval of 3 months

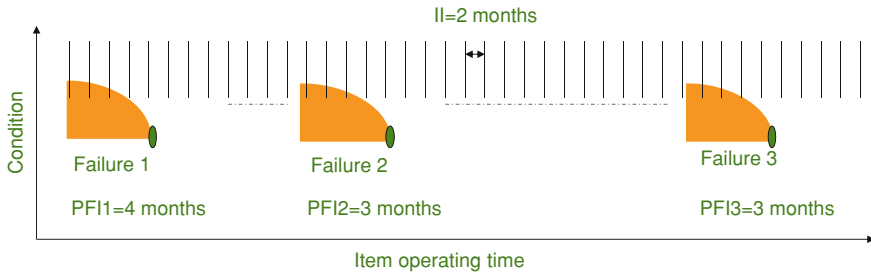


Fig. 7.15 Inspection interval, PF interval and MTTF for a failure mode

while Failure 1 had a PF interval observed of 4 months. According to this data, maintenance managers fixed the inspection interval (II) to 2 months (always less than the minimum PF interval). At the same time the mean time to fail (MTTF) of the item for this failure mode is around 20 months for this example.

Notice that, for management purposes, PF intervals (and not MTTF) are needed to set up inspection intervals. A common mistake is fixing inspection intervals based on MTTF, causing therefore failure events to appear. In the example in Fig. 7.15, fixing $II = 10$ months (since MTTF is 20 months) would never help the maintenance department to ensure reliability of the equipment for the failure mode, and failures would be identified in inspections only by “chance or good luck”.

7.3.4 PF Interval and Technology

Monitoring technology conditions the PF interval of the failure mode, this is due to the fact that different technology offers different accuracy for the measurement of a parameter, and therefore the P point of the PF interval can be identified with different anticipation. As a general rule, more precise equipment will be more expensive but will offer a longer PF interval and therefore inspection intervals can also be longer.

This consideration is important regardless the fact that many technologies are, or can be, complementary for many reasons in many predictive programs. In Fig. 7.16, a graphical representation of different PF intervals is shown for a failure mode monitoring case study.

7.3.5 Linking Monitoring Parameters to Failure Modes

When monitoring services and networks it is important to determine the items and ultimately the parameters which need to be monitored. This depends on the importance of the item function (item criticality) to the service provided through the network, and for a critical item, on the criticality of its failure modes.

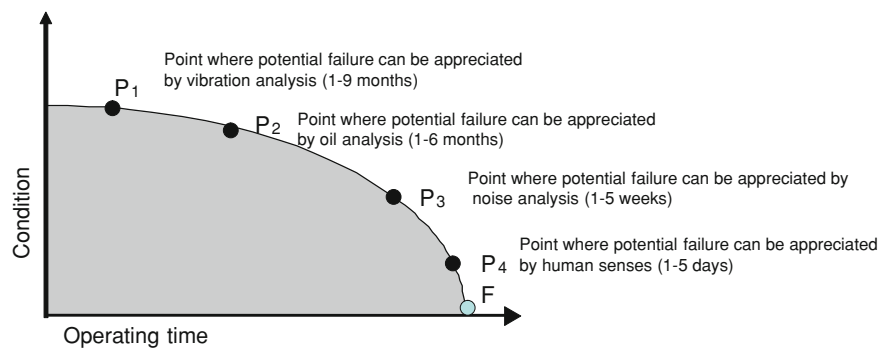


Fig. 7.16 PF interval versus monitoring technology in a case study

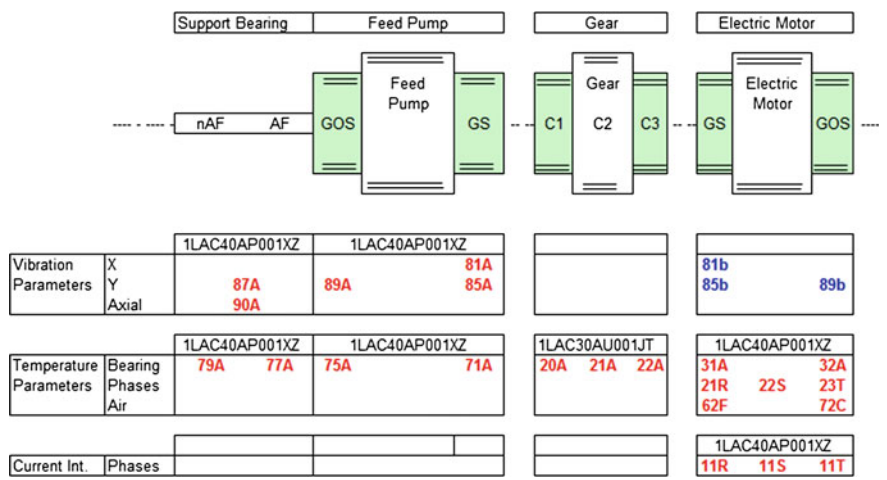


Fig. 7.17 PF interval versus monitoring technology in a case study

Item functional analysis and failure mode analysis (Failure Mode Effect and Criticality Analysis—FMECA) is therefore very convenient in advance to any item monitoring process or even after that process. Identifying parameters required to predict a failure mode is an inductive process that should start with the most critical failure modes of the critical item, FMECA will identify and classify failure modes by their criticality.

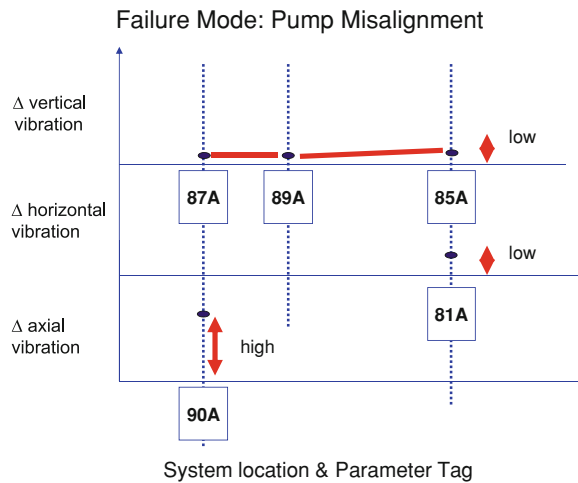
We now introduce a case study with Fig. 7.17, where a map identifying parameters being monitored in a water feed pump is presented. That figure contain the different vibration, temperature and electrical current signal that are currently monitored in this feed pump.

The pumping station control system has certain thresholds pre-established for the different variables according to Fig. 7.18, but no information exist connecting monitored parameters to failure modes. Linking parameters to failure modes

		Parameters Thresholds			
		NORMAL	Warning	Alarm	Failure
Vibration Parameters (mm/s)	X	0 - 1,99	2 - 3,51	3,5 - 4,48	> 4,49
	Y	0 - 1,99	2 - 3,51	3,5 - 4,48	> 4,49
	Axial	0 - 4	4 - 6	6 - 8	>8
Temperature Parameters (Δ T, °C)	Main bearing	0 - 40	41 - 45	46 - 50	>50
	Support bearing	0 - 60	61 - 65	65 -70	>70
	Motor bearing	0 - 30	31 - 35	36 - 40	>40
Current Int.(Amp)	Phases	0 - 30	31 - 50	51 - 70	>71

Fig. 7.18 PF interval versus monitoring technology in a case study

Fig. 7.19 PF interval versus monitoring technology in a case study

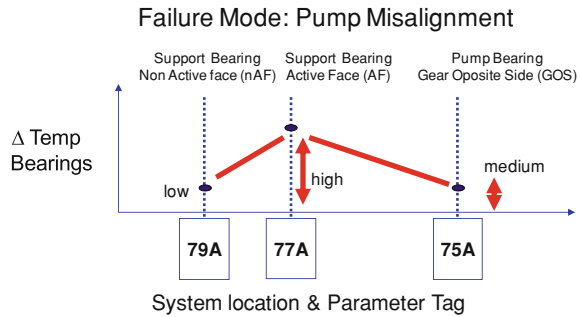


requires of system to capture this information, and to make it operational, as we will discuss later in the book. Of course, this information can later add enormous value to network maintenance planning and control.

For this case study the maintenance department identified, after a FMECA analysis, that misalignment of the pump resulted to be a critical failure mode for this particular scenario. Among all the signals being monitored, the maintenance department has to obtain those really relevant for this failure mode advance identification and control. In Figs. 7.19 and 7.20 expected variations of the monitored parameters for vibration and temperature, when misalignment is produced, are presented.

This information was obtained by eliciting knowledge from the maintenance staff, same can be done for other signals (electrical current). Of course, this is not a trivial process. The information is often locked away in the heads of domain experts and many times the experts themselves may not be aware of the implicit

Fig. 7.20 PF interval versus monitoring technology in a case study



conceptual models that they use. Eliciting knowledge consist in drawing out and making explicit all the known knowns, unknown knowns, etc.

Figures 7.19 and 7.20 show how among all vibration and temperature parameters, only tags 90A (Axial vibration in the support bearing) and 77A (support bearing active face temperature) will have relevant increase when misalignment is produced in the considered pump). Increments in other parameter values will be irrelevant and/or much lower in value. This facts, together with a higher consumption of energy (increments in electrical current), may lead to the consideration of this failure mode to be at risk.

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Chapter 8

Managing Incidents

8.1 Introduction

ITIL terminology defines an incident as any event which is not part of the standard operation of a service and which causes, or may cause, an interruption to or a reduction in the quality of that service. The stated ITIL objective is to restore normal operations as quickly as possible with the least possible impact on either the business or the user, at a cost-effective price. Also ISO 20000 [2] defines an incident (part 1, 2.7) as any event which is not part of the standard operation of a service and which causes or may cause an interruption to, or a reduction in, the quality of that service.

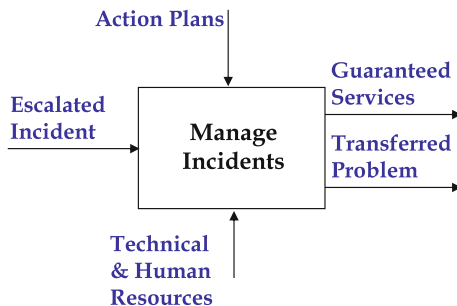
Another opportunity to improve a first level of network maintenance management performance is through the incidents management process. Managing incidents is the process dedicated to ensure the effective resolution of incidents in order to guarantee the level of the involved services (Fig. 8.1).

The resolution of an incident could be executed remotely or in the field. In any case, this process will control, close and document the incident.

8.2 Managing Incidents Process Work Flow

In case of an incident, the first thing to do is to question whether a known solution exists, otherwise it is escalated as a problem (here “problem” is an incident whose behavior or solution is unknown and there is no procedure to deal with it, [3]) to be analyzed in the search of one possible solution (see Fig. 8.2). In order to be more efficient in the implementation of incidents resolution, this process normally implements a corrective plan, where all the remedial actions to restore an optimal service quality in the minimum possible time are discussed. This plan is dynamic and will be the guideline when incidents occur. It will be updated continuously,

Fig. 8.1 The process “To Manage Incidents”



improved and developed. This process normally uses different techniques to analyze incidents according to their root cause, technology, priority, situation in the network, frequency and degree of service affection.

The incidents could be generated either proactively—from the monitoring process—or reactively from internal departments or external customers and other third parties. In order to facilitate diverse communication and ways to receive reactive information about possible incidents, centralized call centers can be used for the treatment of this information, analyzing the proper derivation of a call into an incident. This process is also responsible for the management of third party damages, for example, broken infrastructure, fire, flood, etc.

Following the reception of the escalated incident, this is examined in-depth reviewing its correct generation, scheduling, its treatment according to its category and priority and whether any correct known solution is possible. If there is a correct known solution, then it is implemented according to defined standards or procedures, otherwise the incident is escalated as a problem (see Fig. 8.2) and it is analyzed to find out a suitable solution and the procedure for its proper implementation. To implement the solution, remotely or on the field, the necessary resources have to be identified and collected previously. When the adopted measures to solve the incident are not successful, the incident has to be reviewed again, defining new solutions or escalating it as a problem. Finally the incident is documented and recorded.

In a certain sense, this process provides the first level of corrective action and it involves:

- Reception and analysis of incidents.
- Usually provides 24×7 mode monitoring.
- Testing and diagnosis, searching for the cause of the incident.
- Prioritizing and scheduling of the incidents.
- Assignment and dispatching of incidents.
- Recording and continuously updating the incident degree of progress and expected resolution within the management systems.
- Providing status up to date information and the root cause of the incident.
- According to the difficulty, escalation to specialists.
- Resolution of incidents, ensuring the time implicated in terms of quality.

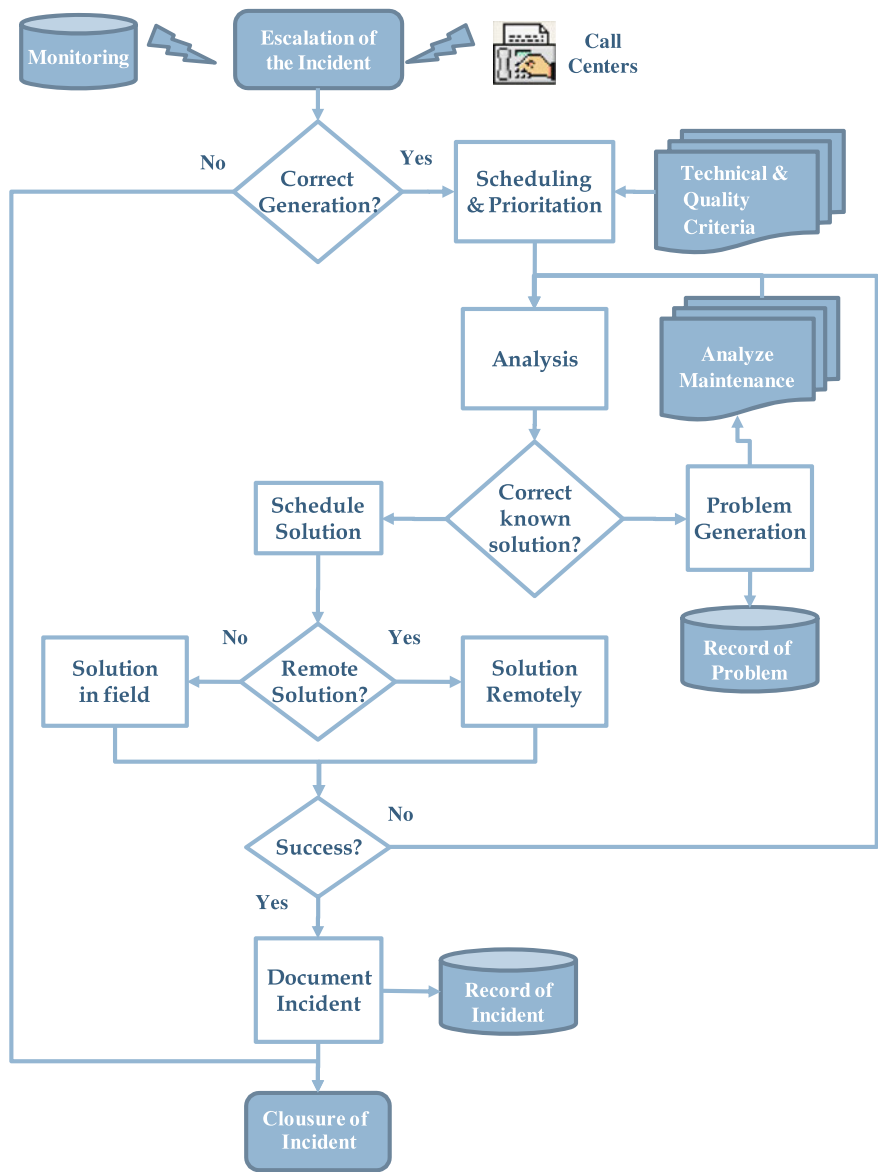


Fig. 8.2 Management of incidents flow

- Verification of the implemented solutions.
- Notification of the incidents and their closure.
- Further analysis and documentation.

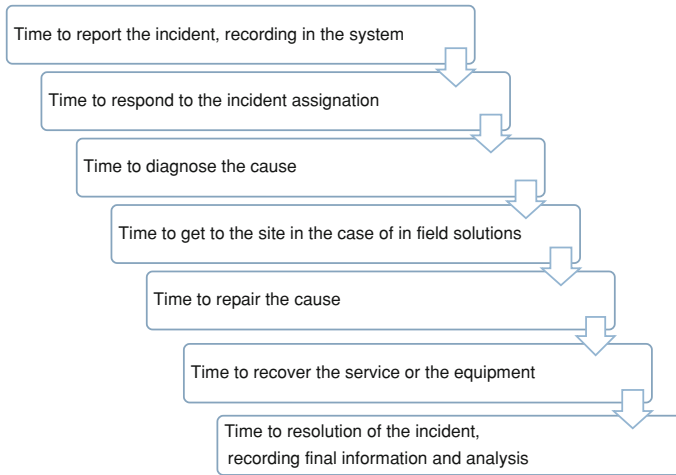


Fig. 8.3 Timescale for the management of incidents

With the intention to become more efficient in incident troubleshooting, all performed corrective actions have to be analyzed in terms of quality and service level agreements, developing and continually improving a corrective plan as a learning guide. An effective corrective plan must include, at least:

- Symptom-cause-solution relationships about incidents.
- Frequency of incident occurrence.
- Incident consequences.
- Method of resolution, indicating the resources required and the average time employed.
- Priority levels and category.

In this process, due to the urgent execution of the corrective activities, the notification and escalation have an important weight because depending on how difficult is to obtain the rule “symptom-cause-solution” and/or to provide the appropriate resources, the incident normally has to be escalated to fulfill the contracted service level agreements, as we have seen in the previous chapter. In addition, incidents may occur unexpectedly, so that their number could require a size in resources, and time for proper resolution, exceeding expectations. Therefore the incidents have to be ordered for their execution in proportion to their level of priority, which must drive the effort priority [5]. The time frame for incident prioritizing, escalation and notification inside the workflow is usually divided into seven different time periods (ITSMF 2010), which somehow show inefficiencies appreciated for the incident resolution (see Fig. 8.3).

In the next sections, two practical examples related to incident management will be presented. The first example attempts to estimate the incident’s time of resolution setting up a priority rule when considering an environment of Poisson distributed inter-arrival intervals (irregular intervals of incident arrivals according

Table 8.1 M/M/C equations with priority

Poisson distribution (x is the generated incidents per unit of time)	$P(x) = \frac{e^{-\lambda} \lambda^x}{x!}$
Negative Exponential distribution (x is the incident run-time)	$P(x) = e^{-\mu x}$
Number of technicians	C
Arrival rate	$\lambda = \sum_{i=1}^N \lambda_i$
Service rate	μ
Utilization rate	$\rho = \frac{\lambda}{C \cdot \mu}$
Probability of having zero activities in process (if $C \cdot \mu > \lambda$)	$P(0) = \frac{1}{\left[\sum_{n=0}^{C-1} \frac{1}{n!} \cdot \left(\frac{\lambda}{\mu} \right)^n \right] + \left[\frac{1}{C!} \cdot \left(\frac{\lambda}{\mu} \right)^C \cdot \frac{C \cdot \mu}{(C \cdot \mu - \lambda)} \right]}$
Expected average number of incidents in process	$L_s = \frac{\lambda \cdot \mu \cdot \left(\frac{\lambda}{\mu} \right)^C \cdot P(0)}{(C-1)! \cdot (C \cdot \mu - \lambda)^2} + \frac{\lambda}{\mu}$
Expected average queue length of incidents	$L_q = L_s - \frac{\lambda}{\mu}$
Expected average total time in process	$W_s = \frac{L_s}{\lambda}$
Expected average waiting time in the queue	$W_q = \frac{L_q}{\lambda}$
Expected average waiting time in the queue per priority	$W_{kq} = \frac{1}{A \cdot B_{k-1} \cdot B_k}$
Expected average total time in process per priority	$W_{ks} = W_{kq} + \frac{1}{\mu}$
Expected average queue length of incidents	$L_k = \lambda_k \cdot W_{kq}$
Where A is	$A = \frac{\lambda}{(1-\rho) \cdot L_q}$
B_k is	$B_k = 1 - \sum_{i=1}^k \frac{(\lambda_i)}{(C \cdot \mu)}$ for $k = 1, \dots, N$ and with $B_0 = 1$

to the Poisson distribution). The second example shows a maintainability estimation considering a negative exponential distribution and the impact of technicians tasks learning.

8.3 Estimating Incidents Resolution Time Assigning Priority

Priory assignment to incidents allow them to be orderly managed according to that value (assume 1 is the highest priority and 4 the lowest priority). Let us assume that the rule is to assign a free technician to the highest priority incident, and that inside the same priority, the rule of the queue will be FIFO.

Let us consider again Poisson distribution in arrival times of incidents and negative exponential distribution for the service time that is accomplished without interruption. No incident will be rejected when a new incident with major priority arrives to the queue. With these assumptions, equations presented in Table 8.1

need to be completed with the following in order to distinguish the expected averaged time per each priority $i = 1, \dots, N$ (see Table 8.1).

Assume an electrical network utility, where the incidents arrive in a M/M/c queue and they are classified in four levels of priority (according to the NUCREC). Assume that there are five resources to solve the incidents, the service rate is 1.5 incidents per hour and the incidents arrival rates per priority are:

- $\lambda_1 = 1$ per h.
- $\lambda_2 = 2$ per h.
- $\lambda_3 = 2$ per h.
- $\lambda_4 = 1$ per h.

We will now calculate the average time of each incident in the queue according to its priority, following equations in Table 8.1.

- The global arrival rate is $\lambda = 1 + 2 + 2 + 1 = 6$ per h.
- The service rate is $\mu = 1, 5$.
- The utilization rate is $\rho = 6/5 \cdot 1, 5 = 0, 8$.
- The expected average time of each priority in the queue and in progress are with $L_q = 2,2165$:

$$A = \frac{\lambda}{(1 - \rho) \cdot L_q} = \frac{6}{(1 - 0,8) \cdot 2,2165} = 13,53$$

$$B_0 = 1$$

$$B_1 = 1 - \sum_{i=1}^1 \frac{(\lambda_i)}{(5 \cdot 1,5)} = 0,866$$

$$B_2 = 1 - \frac{(\lambda_1 + \lambda_2)}{(5 \cdot 1,5)} = 0,6$$

$$B_3 = 1 - \frac{(\lambda_1 + \lambda_2 + \lambda_3)}{(5 \cdot 1,5)} = 0,333$$

$$B_4 = 1 - \frac{(\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4)}{(5 \cdot 1,5)} = 0,2$$

$$W_{1q} = \frac{1}{A \cdot B_0 \cdot B_1} = 0,085 \text{ h, then } W_{1s} = 0,751 \text{ h}$$

$$W_{2q} = \frac{1}{A \cdot B_1 \cdot B_2} = 0,142 \text{ h, then } W_{2s} = 0,808 \text{ h}$$

$$W_{3q} = \frac{1}{A \cdot B_2 \cdot B_3} = 0,369 \text{ h, then } W_{3s} = 1,035 \text{ h}$$

$$W_{4q} = \frac{1}{A \cdot B_3 \cdot B_4} = 1,108 \text{ h, then } W_{4s} = 1,774 \text{ h}$$

The expected average length of the queue of incidents, waiting to be solved and per priority is:

$$L_1 = \lambda_1 \cdot W_{1q} = 1 \cdot 0,085 = 0,085 \text{ incidents}$$

$$L_2 = \lambda_2 \cdot W_{2q} = 2 \cdot 0,142 = 0,284 \text{ incidents}$$

$$L_3 = \lambda_3 \cdot W_{3q} = 2 \cdot 0,369 = 0,738 \text{ incidents}$$

$$L_4 = \lambda_4 \cdot W_{4q} = 1 \cdot 1,108 = 1,108 \text{ incidents}$$

With this information we are able to design resources according to our service level per priority.

8.4 Maintainability Estimation

We have mentioned above, in [Sect. 1.4.2](#), how maintainability is the ability of an item to be retained in, or restored to a state in which it can perform a required function, when maintenance is performed according to given conditions and with prescribed procedures and resources. This includes not only the manner in which a maintenance task is to be performed but also the availability of maintenance resources (people, spare parts, tools, and manuals), the preventive maintenance program, skill levels of personnel and the number of people assigned to the maintenance crew.

Maintainability, normally denoted as $M(t)$, can also be considered as a mathematical function which represents the probability that a maintenance task will be successfully completed before a given elapsed time t :

$$M(t) = P(\text{DMT} \leq t) = \int_0^t m(\tau) d\tau \quad (8.1)$$

where DMT represents the duration of the maintenance task and $m(t)$ is the maintainability density function of DMT. Several distribution functions can be used to model maintainability of different engineering systems and scenarios. In the case that we use the exponential distribution function, the mathematical expression for the maintainability function is [\[4\]](#):

$$M(t) = 1 - e^{-\lambda t} \quad (8.2)$$

where λ is named maintenance task frequency, which is the inverse of the mean duration of the maintenance task $\text{MDMT} = E(\text{DMT})$, with

$$E(DMT) = MDMT = \int_0^{\infty} t x m(t) dt = 1/\lambda \quad (8.3)$$

Equation 8.3 can also be expressed as follows:

$$E(DMT) = MDMT = \int_0^{\infty} t x m(t) dt = 1/\lambda \quad (8.4)$$

This represents the area below the function which is complementary to the maintainability function. MDTM in case of a repair task is named MTTR and λ is named repair frequency.

Consider a case study when a maintenance team has been carrying out different repairs on a given network item and the times (in h) required where:

$DMT_i = TTR_i = 16, 16, 14, 14, 15, 15, 17, 18, 17, 19$.

Assume that we use the exponential distribution function for the mathematical expression for the maintainability function, as in Eq. 8.2. Consider that we want to calculate:

- (1) The probability to repair the item before a time of $t = 4$ h;
- (2) The mean time to repair (MTTR), expectation of the random variable;
- (3) The time to repair (TTR) the item with a probability to complete the task of 20 and 95% ($M(t) = 20\%$, $M(t) = 95\%$).

To solve these questions we first need to calculate the failure frequency λ

$$MTTR = \frac{\sum_{i=1}^{i=n} TTR_i}{n} = 16.10 \text{ h}$$

Thus, $\lambda = 1/MTTR = 1/16.10 \text{ h} = 0.062112 \text{ repairs/h}$.

Solving the first question is calculating $M(t = 4)$

$$M(t = 4) = 1 - e^{-0.062112 \times 4} = 0.22$$

This means that we have a 22% probability for the item to be repaired within 4 h time.

The second question can be solved considering that because of the exponential distribution function of the maintainability, the TTR can be estimated as follows:

$$TTR[M(t)] = \frac{1}{\lambda} [-Ln(1 - M(t))] \quad (8.5)$$

Substituting $M(t)$ values in Eq. 8.5 we have that:

- Case 1: $M(t) = 20\%$ and $TTR(20\%) = 3.6 \text{ h}$, which is the time to repair for a 20% probability of maintenance task completion.
- Case 2: $M(t) = 95\%$ and $TTR(95\%) = 48.2 \text{ h}$, which is the time to repair for a 95% probability of maintenance task completion.

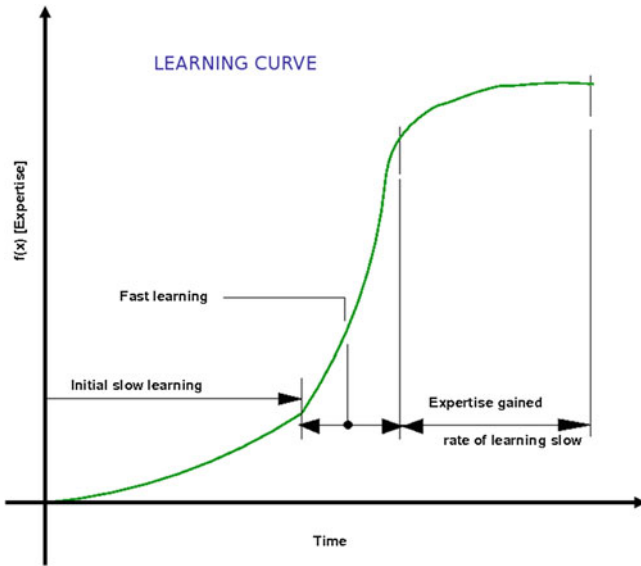


Fig. 8.4 The learning curve

8.4.1 Modeling Task Learning Impact on Maintainability

In any maintenance operational task, various methods, techniques and technologies are applied to solve the problem. In general, the maintenance staff will have varying levels of expertise in these techniques. Moreover, some of the technicians could not know some of the technologies to be used. The learning curve, which gives one's expertise as one learns a subject with time, tells us that initially the rate of learning is slow, then there is very fast learning and finally as expertise has been gained, the rate of (new) learning drops (Fig. 8.4).

Nowadays, in almost all maintenance tasks, some new technology is to be used and some new concepts need to be mastered. As is seen in the learning curve, the initial learning part is difficult. A tight deadline for the task completion makes it worse. The initial learning part becomes longer. In some cases, learning is put aside and all efforts are put toward getting some visible output. What can happen is that the technicians never get past the initial part of the learning curve. This may result in lack of creation of fundamental concepts and a poor quality in new and basic advanced maintenance tasks. This is like building without adequate foundation. The result may vary from a fatal maintenance failure to an overall lack of quality and reliability of the item maintained.

The primary reason for why experience and learning curve effects apply, of course, is the complex processes of learning involved. As discussed in [6], learning generally begins with making successively larger finds and then successively smaller ones. The equations for these effects come from the usefulness of

mathematical models for somewhat certain predictable aspects of those generally non-deterministic processes. They include:

- *Labor efficiency.* Workers become mentally more confident and spend less time hesitating, learning, experimenting or making mistakes. Over time they learn shortcuts and improvements. This applies to all employees and managers, and not just those directly involved in maintenance.
- *Standardization, specialization and methods improvements.* As processes, parts and products become more standardized, efficiency tends to increase. When employees specialize in a limited set of tasks, they gain more experience with these tasks and operate at a faster rate.
- *Technology-driven Learning.* Automated production technology and information technology can introduce efficiencies as they are implemented and people learn how to use them efficiently and effectively.
- *Better use of equipment.* As total production has increased, manufacturing equipment will have been more fully exploited, lowering fully accounted unit costs. In addition, purchase of more productive equipment can be justifiable.
- *Changes in the resource mix.* As a company acquires experience, it can alter its mix of inputs and thereby become more efficient.
- *Product redesign.* As the manufacturers/service providers and consumers/users have more experience with the product/service, they can usually find improvements. As manufacturers/service providers produce/provide more products/services, they learned how to best produce/provide items working for the least money.
- *Network-building and use-cost reductions.* As a product/service enters more widespread use, the consumer uses it more efficiently because they are familiar with it. One example is email accounts; the more there are, the more efficient the network is, the lower everyone's cost per utility of using it.
- *Shared experience effects.* Experience curve effects are reinforced when two or more products share a common activity or resource. Any efficiency learned from one product/service can be applied to the other products/services.

How can we model the task learning effect and its impact on maintainability? In the following paragraph we answer this question by using a formulation that is similar to the Cox proportional hazards model [1] but now applied to maintainability.

As we mentioned before, in the case that we use the exponential distribution function, the mathematical expression for the maintainability function is presented in Eq. 8.2. This now will be named baseline maintainability function $Mo(t)$, with:

$$Mo(t) = 1 - e^{-\lambda t} \quad (8.6)$$

Consider now the function $z(u)$ to be the covariate function that can be expressed as follows

$$z(u) = \gamma z_o(u) \quad (8.7)$$

where γ is a coefficient that we use to adjust the effect of learning and $z_o(t)$ is the learning curve for a specific task that is a representation of the percent of

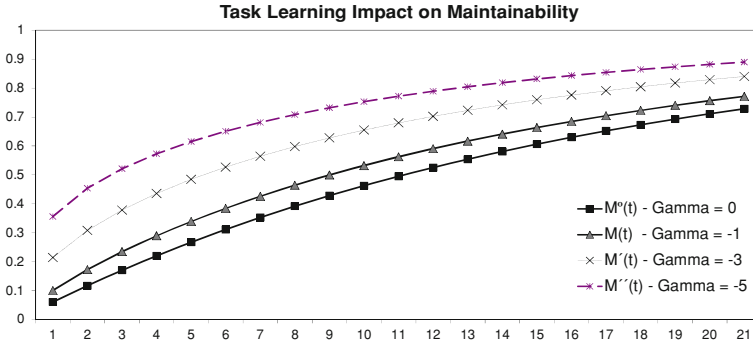


Fig. 8.5 Different graphs for maintainability considering $z_0(u) = 0,2$

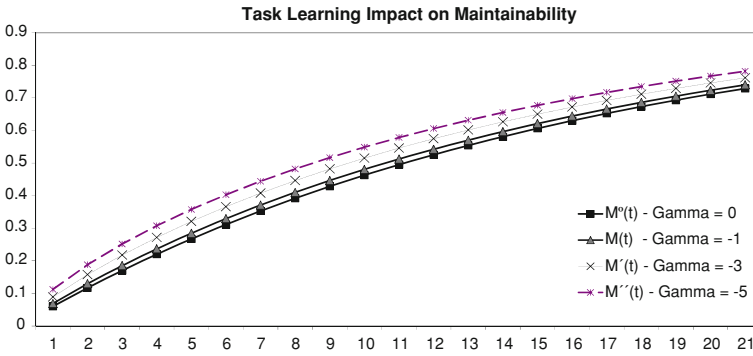


Fig. 8.6 Different graphs for maintainability considering $z_0(u) = 0,02$

improvement registered over the initial duration of the maintenance task (DMT) and over time (or over number of task carried out). In order to produce this effect on maintainability the γ coefficient value must be negative.

Hence, the maintainability function can now be estimated as

$$M(t) = 1 - \left(e^{-\lambda t}\right)^{e^{z(u)}} \quad (8.8)$$

Notice that $Mo(t)$ is the maintainability function (in Eq. 8.8) under $\gamma = 0$. Values for maintainability predictions over time (or number of tasks accomplished) are presented in Figs. 8.5 and 8.6 when we have two different values of the learning curve ($z_0(u) = 0.2$ in Fig. 8.5 and $z_0(u) = 0.02$ in Fig. 8.6). Gamma (γ) coefficient variations are presented within a range $[0, -5]$ to appreciate its impact on maintainability values and to easy model calibration.

Managers can now have a better understanding of time consumption for the maintenance task accomplishment over time.

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Chapter 9

Incidents Prevention

9.1 Introduction

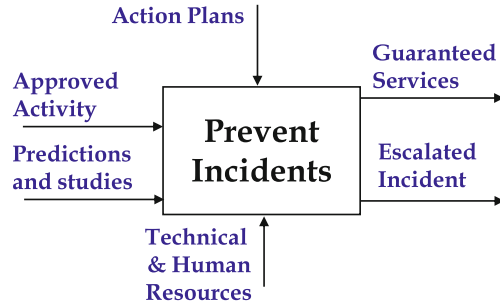
In this process we deal with necessary actions to preserve the network operating conditions. A network is like an operating machine, which inherently degrades progressively (due to the passage of time, to manipulations, to different external actors, etc.) and that requires certain efforts to avoid such degradation.

Preventing incidents is a process that can be defined as the set of activities on elements, necessary to maintain the network in the best possible level of performance to reduce or minimise future incidents (see Fig. 9.1). Our aim is to ensure the conservation of the network in the medium/long term, correcting deterioration and possible lack of service quality just in time. This needs to be done ensuring proper quality and cost trade-off (both in resources and in time).

The key of this process is the implementation of preventive programmed routines to reinforce equipment reliability, and to detect and correct small problems before they affect the service. According to Wireman [36] and Higgins [19], the maintenance tasks planned on a weekly basis should be within 60–80% of the total preventive task. Through this periodic maintenance, we can also distribute the costs of the network degradation over time [13]. Preserving the network also implies carrying out the following activities:

- Keeping the network updated;
- Accepting, upgrading and/or activating elements;
- Continuous training for technicians;
- Detection of damages and frauds produced by outside agents;
- Detection of anomalies, malfunctioning;
- Setting elements control and quality testing;
- Programmed substitution of elements reaching the end of their lifecycle;
- Logistics and control of spare parts, tools and resources;
- Relationship with administrations and other organizations;

Fig. 9.1 To prevent incidents process



- Knowing the network deeper to initiate enhancing actions, feedback reports to the engineering on the reliability and maintainability of the network.

As it was mentioned before, preventive systematic routines are not the only one type of activity considered within this process; we also include other on-demand activities, normally implemented in the field, to encourage network conservation.

On-demand activities can be either maintenance activities triggered by the network monitoring process, i.e. condition-based maintenance activities; or other activities not directly considered as maintenance, but with important maintenance implications in order to avoid future problems (for example replacing or commissioning equipment, equipment modifications, etc.).

Another important part of the incidents prevention process is the management of all the resources to carry out the referred activities. The process *to prevent incidents* is a process responsible for logistics and focusing on providing the right resources at the right time. All the necessary personnel, parts and tools for these activities implementation needs to be properly scheduled.

This chapter is organized as follows: in the following two (Sects. 9.2 and 9.3) we provide information about the workflow to follow when considering network preventive routines and on-demand activities, respectively. Then, preventive maintenance optimisation model are reviewed in Sect. 9.4. Within that section we have selected one of the more complex models (partial preventive replacements with minimal repair) to exemplify the computation process and to obtain the optimum solution for a specific problem. After that, in Sect. 9.5, we discuss about priority setting when dealing with preventive maintenance in networks. In many occasions, optimal preventive maintenance routines cannot be implemented for any single asset of our networks. Due to different possible constraints, and in order to ensure maintenance effectiveness, we have to focus our attention and budget to specific network items. In Sect. 9.5 we present two options to select those items. Finally Sect. 9.6 deals with another important aspect of network preventive maintenance which is the geographical positioning of the different network items. This circumstance may alter, for instance, the preventive maintenance scheduling process. In order to understand implications of assets location in maintenance scheduling we present a method to with tackle the preventive maintenance routing, and therefore scheduling, problem in networks.

9.2 Preventive Systematic Routines

Preventive routines should be done according to a certain criteria and units of measure, so as to review some basic points of the network in order to extend its lifetime and in line with the equipment life cycle costing [26]. As we have seen, these criteria could be based on condition or status, time or frequency, use or number of network clients.

Previous history and present or future diagnostics of the network could be used to determine the required network systematic preventive maintenance.

As we may guess, this management function is not easy, and it should be checked and redefined permanently over time; this is a clear evolutionary process, depending on time and knowledge gained from the network (see Fig. 9.2).

The preventive routines work flow begins with the definition of the criteria used to determine the criticality of the equipment (defined strategically) and the priority of the task. Later, routines are planned according to their priority, the equipment operating time and the required routine frequency. This planning could, however, result neither effective nor efficient sometimes, and therefore it is convenient that the plan is properly evaluated and simulated before its definitive scheduling (to reduce risk and meet the required terms and conditions of maintenance time and cost). After this evaluation, a maintenance plan is used as a guide for the maintenance task scheduling, which is a process that will take into account some others factors like the staff technical knowledge, existing and/or punctually arranged resources, the geographical characteristics of each area, etc. If everything goes well, the preventive routines will be executed, producing information updates, modifying the equipment history, suggesting changes in the existing criteria for the routine planning, etc.

During the execution of systematic routines, other preventive actions could be suggested and/or corrective actions could be required and escalated as incidents to the process *to manage incidents* when they cannot be corrected straight forward.

9.3 Other On-Demand Activities

Sometimes, preventive tasks cannot wait for scheduled times or simply there are other approved activities to be executed or supported by maintenance (such as advanced equipment calibration, supervision in field before any modification of the network, or a checking before any level of risk or unauthorized activity). Preventive maintenance staff can be demanded to inspect or to execute any task that could affect the network or services. These tasks are mainly focused on work homogenisation over the network in order to ease its maintenance, maintainability and required service levels. Other samples of on-demand activities can be:

- Elements activation or commissioning.
- Restoring configuration after withdrawal.
- Controlling access to technical sites of the network.

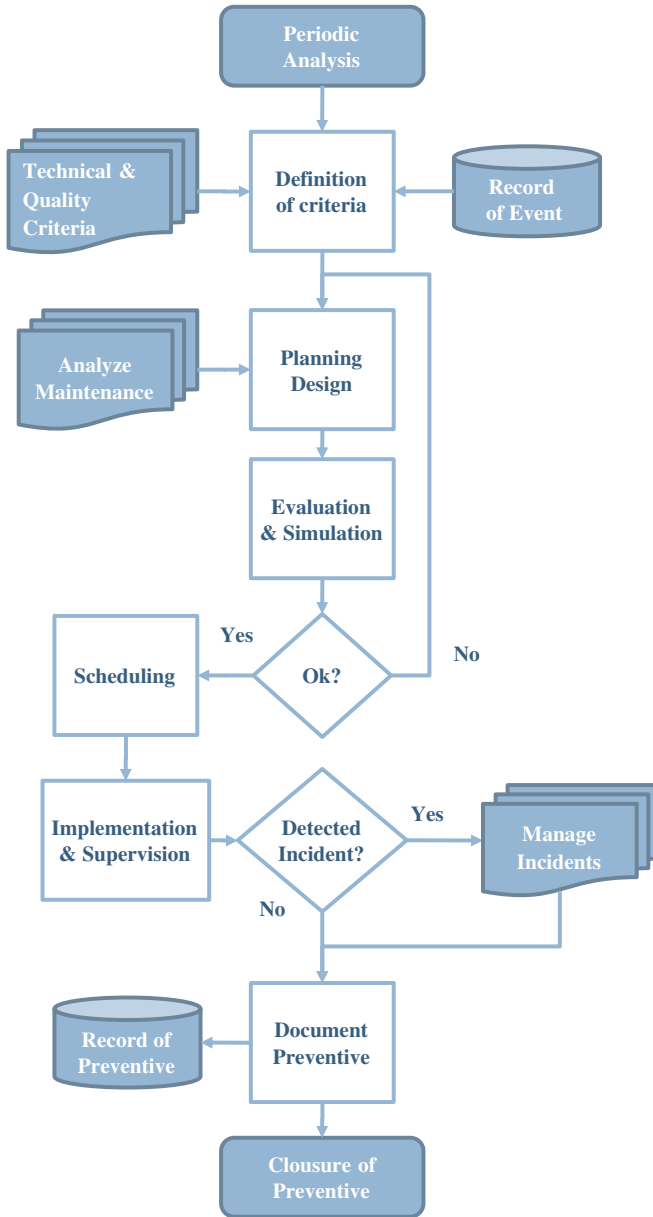


Fig. 9.2 Preventive routines flow

- Inspection and field support.
- Control of changes.
- Operational control of stores, resources and health and safety.
- Relationship with administrations and other institutions.

It is recommendable to control that the cost of unplanned maintenance task and other on-demand activities is under control (good practices suggest this cost to be around 1.5 times the cost of planned maintenance tasks, see [36]). Therefore, the execution of these on-demand tasks should be scheduled in time to avoid excessive replication, reactivity and overtime.

The normal work flow of this process (see Fig. 9.3) starts by assessing the correct generation of the approved on-demand activities and scheduling the performance of these activities considering their priority and according to predefined criteria for existing and necessary resources. In this scheduling process, activities demanding resources just for simple inspections are distinguished from those ones requiring our own specialised personnel to be accomplished. In any case, both are checked to ensure the expected results.

As a precautionary measure in case of a bad performance of an on-demand preventive task, the existence of a “reverse plan” is a must. This plan will be executed to restore the state of the network and services, will generate an incident, and will make sure that those differences found with respect to expected network performance are analysed. Hereafter, like in the case of positive performance, the resulting state is quarantined for a determined period to be sure about the network stability. Finally, the activity is documented to close the flow.

9.4 Preventive Maintenance Optimisation Models

9.4.1 *Introduction to Maintenance Optimisation Modelling*

In the following sections we review different approaches to maintenance optimisation modelling. It is not the intention to present a complete review of the modelling possibilities found in the literature [12, 14, 17, 25, 32, 33, 35] but to offer the reader a variety of examples that can be used to solve the problem. We call reader's attention to the fact that complex problems may require the use of diverse and complex optimisation techniques that need further study. However we will solve some of the more complex examples for the reader to appreciate the computational requirements of the suggested models (see Sect. 9.4.6). The order of the following sections is taken from previous reviews and maintenance model classifications found in the literature.

9.4.2 *Total Replacement Models*

In classic total replacement models [3] we assume that the equipment is always replaced completely, the replacement is done instantaneously, i.e. consumes no time, and the equipment failure is detected as soon as the failure takes place. Normally there are two types of replacement options:

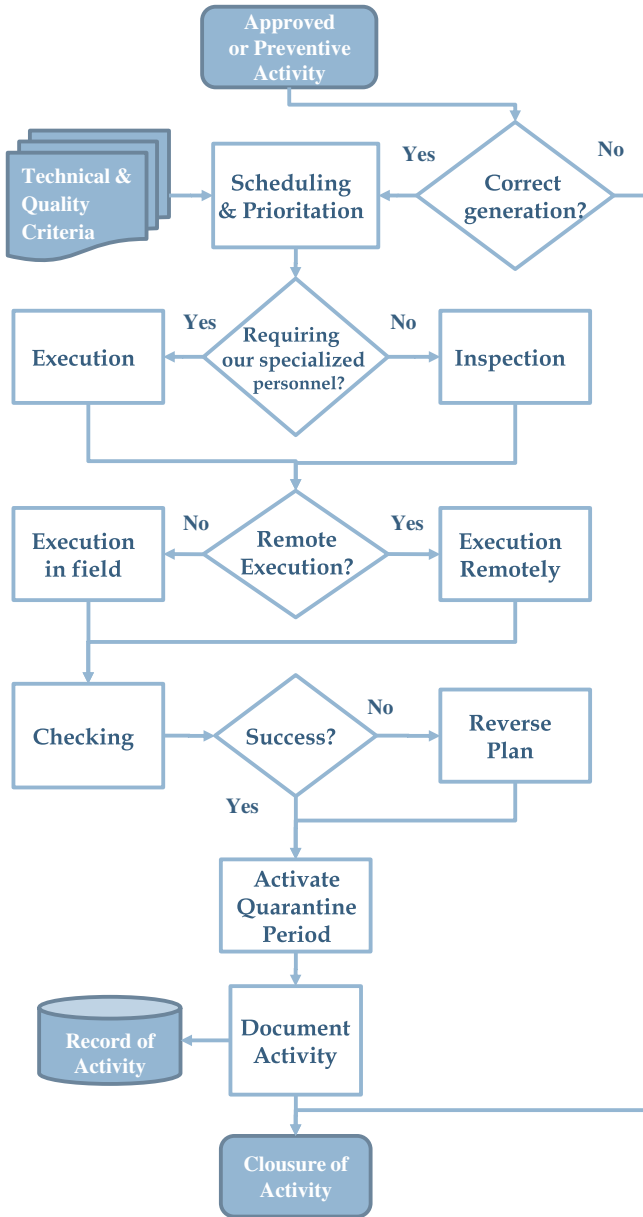


Fig. 9.3 Preventive on-demand activities flow

- Preventive replacement (PR). Following a predetermined preventive maintenance strategy;
- Corrective replacement (CR). Following the equipment failure.

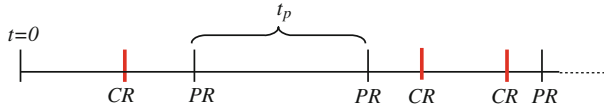


Fig. 9.4 Equipment maintenance over time

Basic total replacement models normally consider the following PR policies:

- Constant interval replacement (CIR). Replacement is done after a certain constant time interval;
- Age-based replacement (ABR). Replacement is done when the equipment reach a certain operating time—age.

Let us now formalise these options in the following sections.

9.4.2.1 Constant Interval Replacement

Replacement is done after the failure (CR) or after a certain constant time interval t_p (PR) (see Fig. 9.4). The model is built to determine the optimal time interval between two preventive replacements. The optimisation criteria is to minimize the total expected cost per unit time.

We will use the following notation:

C_p	PR unit cost;
C_c	CR unit cost;
t_p	PR time;
$F(t)$	Time to failure probability distribution function
$f(t)$	Time to failure probability density function
$N(t_p)$	Expected number of failures ¹ within the time interval $(0, t_p)$;
$TEC(t_p)$	Total expected cost per unit time.

If a failure is produced, it will take place within the time interval $(0, t_p)$, and the total expected cost per unit time $TEC(t_p)$, for the interval t_p , will be as follows:

$$TEC(t_p) = \frac{TEC(0, t_p)}{\text{Length}(0, t_p)} = \frac{C_p + C_c N(t_p)}{t_p} \quad (9.1)$$

If we follow this policy, and the number of failures is appreciable, notice that many preventive replacements could be done when the operating time of the equipment is below t_p , which of course could make this policy less efficient.

¹ Barlow and Hunter (1990) showed that $N(t_p) = \int_0^{t_p} \lambda(t) dt$, where $\lambda(t)$ is the failure rate.

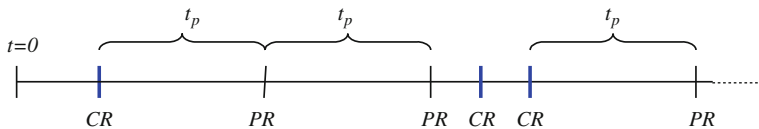


Fig. 9.5 Age based policy and equipment maintenance over time

9.4.2.2 Age Based Replacement

In this case (Fig. 9.5), the PR is done after the equipment reaches a certain operating time—age, t_p . In case of equipment failure a CR is done and the next PR is scheduled after t_p units of time. We again want to calculate the best t_p which minimises $TEC(t_p)$.

This time the equipment may reach the PR time t_p ; this will happen with a probability equal to $R(t_p)$, or fail before that time, with a probability equal to $F(t_p)$. The expected cost for the interval $(0, t_p)$ is now equal to $C_p R(t_p) + C_c F(t_p)$, and the expected length of the cycle is equal to t_p times the probability of the preventive cycle $R(t_p)$, plus the expected length of the failure cycle times the probability of the failure $F(t_p)$.

The length of the failure cycle can be estimated calculating the expected value of the failure distribution now truncated in t_p as follows:

$$M(t_p) = \int_{-\infty}^{t_p} \frac{tf(t)dt}{F(t_p)} \quad (9.2)$$

$$TEC(t_p) = \frac{C_p R(t_p) + C_c F(t_p)}{t_p R(t_p) + M(t_p) F(t_p)} \quad (9.3)$$

Then the optimum of $TEC(t_p)$ will be obtained by minimising Eq. 9.3 with respect to t_p .

9.4.3 Partial Replacement Models

These models are extensions of those presented above in Sect. 9.4.2. Their formulation [16, 23] is motivated by the idea of possible partial replacement of equipment consisting of lower indenture levels.

In these models we will assume that, in our equipment, partial preventive replacements (PPR) of lower indenture level components can be done, at certain operating times T_i , restoring the entire equipment to its initial failure rate. However, it is also common that after a certain number of PPRs, these will be more expensive than a complete preventive equipment replacement (PR).

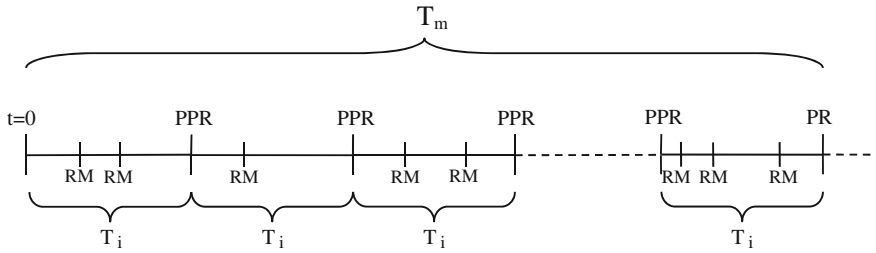


Fig. 9.6 Equipment maintenance with PPR and minimal repairs option

Basic PPR models normally consider two options concerning corrective maintenance:

- Minimal repairs, inexpensive but without failure rate restoration capabilities;
- Normal repairs, more expensive but with failure rate restoration capabilities.

Let us now formalise each of these options in the following sections.

9.4.3.1 PPR with Minimal Repairs

In this policy we assume that the total preventive replacement of the equipment (PR) is done after $(m - 1)$ partial preventive replacements PPR. For equipment subject to $(i - 1)$ PPRs with $(i < m)$, the next PPR will be done after T_i units of operating time since the last PPR (or PR in case $i = 1$). In case of failure, a minimal repair will be carried out, which will take the equipment back to operation but without restoring its failure rate (Fig. 9.6). PPR and PR, unlike minimal repairs, will restore the equipment failure rate to initial conditions.

We will now use the following additional notation:

C_{pp}	PPR unit cost;
C_p	PR unit cost;
C_{mr}	Minimal repair unit cost;
T_i	Time to carry out the PPR;
$\lambda_i(t)$	Failure rate at t for equipment with $(i - 1)$ PPRs;
TEC (m, T_i)	Total expected cost per unit time;
T_m	Total preventive replacement cycle (with $T_m = m * T_i$);
$N_{c,j}(T_i)$	Expected number of failures in j th T_i interval;
$N_c(T_m)$	Expected number of failures along the total preventive replacement cycle;
N_{pp}	Expected number of PPRs in the total preventive replacement cycle;
N_{pt}	Expected number of PRs in the total preventive replacement cycle;
N_{mr}	Expected number of MRs in the total preventive replacement cycle;

The total expected cost per total maintenance replacement cycle will be:

$$C_{pp} * N_{pp} + C_p * N_{pt} + C_{mr} * N_{mr} \quad (9.4)$$

For a total maintenance replacement cycle will have:

$$N_{pp} = m - 1;$$

$$N_{pt} = 1;$$

$$N_{rm} = N_c(T_m),$$

Then Eq. 9.18 can be written as follows

$$C_{pp} * (k - 1) + C_p + C_{mr} * N_c(T_k) \quad (9.5)$$

Where $N_c(T_k)$ can be calculated as follows

$$N_c(T_m) = \sum_{j=1}^m N_{c,j}(T_i) \quad (9.6)$$

with:

$$N_{c,j}(T_i) = \int_0^{T_i} \lambda_j(t) dt \quad (9.7)$$

Here we assume that we can have more than one failure in each T_i interval, but failures are not affecting the failure rate during that interval. Therefore

$$\begin{aligned} N_c(T_m) &= \sum_{j=1}^m N_{c,j}(T_i) = \sum_{j=1}^m \int_0^{T_i} \lambda_j(t) dt \\ &= \int_0^{T_i} \lambda_1(t) dt + \int_0^{T_i} \lambda_2(t) dt + \cdots + \int_0^{T_i} \lambda_m(t) dt \end{aligned} \quad (9.8)$$

Once $T_m = m * T_i$ (with m the number of preventive interventions per cycle), we can obtain the total expected cost per cycle as follows:

$$\begin{aligned} CTE(T_m) &= CTE(m, T_i) = \frac{(m - 1) * C_{pp} + C_{pt} + C_{rm} * N_c(T_m)}{T_m} \\ &= \frac{(m - 1) * C_{pp} + C_{pt} + C_{rm} * \sum_{j=1}^m \int_0^{T_i} \lambda_j(t) dt}{m * T_i} \end{aligned} \quad (9.9)$$

The problem to solve is to find the optimal number T_m of PPRs minimising the total expected cost per unit time $CTE(m, T_i)$ with. $m \in [1, K]$ and $T_i \in [0, T_{\max}]$. Where $T_{\max}$ is the maximum possible duration of the T_i interval and k is the number of functions $\lambda_m(t)$ which are available in our reliability data base.

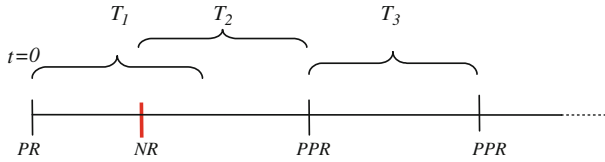


Fig. 9.7 Maintenance activities over time with PPRs and NRs

9.4.3.2 PPR with Normal Repairs

In this policy we again assume that the total preventive replacement of the equipment (PR) is done after $(k - 1)$ partial preventive replacements—PPR. For equipment that went through $(i - 1)$ PPRs, with $i < k$, a normal repair—NR—will be carried out in case of a failure, or another PPR will be done after T_i units of time since the last maintenance (notice that now with that maintenance we restored the failure rate of the equipment), whichever comes first (Fig. 9.7).

We will use the following additional notation:

- C_c NR unit cost;
- C_{ec} NR extra unit cost (exceeding PPR unit cost);
- $Mi(T_i)$ Mean of the truncated distribution in T_i , equipment with $(i - 1)$ PPR's.

If we suppose that

$$C_c = C_{pp} + C_{ec}$$

then the expected unit cost for a PPR for equipment with $(i - 1)$ PPRs since last PR:

$$\begin{aligned} CE(T_i) &= C_c F_i(T_i) + C_{pp} R_i(T_i) = (C_{pp} + C_{ec}) F_i(T_i) + C_{pp} R_i(T_i) \\ &= C_{pp} + C_{ec} F_i(T_i) \end{aligned} \quad (9.10)$$

and the total expected cost for a PR cycle of the equipment will be

$$TEC(k, T_1, \dots, T_k) = (k - 1)C_{pp} + C_p + C_{ec} \sum_{i=1}^k F_i(T_i) \quad (9.11)$$

The length of the PR cycle is

$$L(k, T_1, \dots, T_k) = \sum_{i=1}^k \{T_i R_i(T_i) + M_i(T_i) F_i(T_i)\} \quad (9.12)$$

with

$$M_i(T_i) = \int_{-\infty}^{T_i} \frac{t f_i(t) dt}{F_i(T_i)} \quad (9.13)$$

and the total expected cost per time unit is given by:

$$\text{CTE}(k, T_1, \dots, T_k) = \frac{(k-1)C_{pp} + C_s + C_{eic} \sum_{i=1}^k F_i(T_i)}{\sum_{i=1}^k \{T_i R_i(T_i) + M_i(T_i) F_i(T_i)\}} \quad (9.14)$$

Our problem will be to determine the optimal number of PPRs, and the moment when they should be carried out, in order to minimize the total expected cost in Eq. 9.14.

Notice that models in the last two sections are extensions of the models presented in the previous sections. Also notice that those models are the result of having $k = 1$ in our last formulation.

9.4.4 Replacement Models with Imperfect Maintenance

In the replacement models that we presented in previous sections we have assumed that equipment is “as good as new” after preventive replacements PR or PPR. In many other cases, the failure pattern of the equipment may change after the preventive maintenance activities. Modelling these cases will require that, after the preventive maintenance, we set the failure rate of the equipment to a point between “as good as new” and “as bad as old”. This concept is known as imperfect preventive maintenance (IPM) and the corresponding model is then known as the IPM model.

In the basic IPM model [5], the IPMs are carried out at fixed time h_k ($k = 1, 2, \dots, N-1$) and the equipment is replaced (PR) after $N-1$ IPMs. In case of equipment failure between IPMs, a minimal repair is carried out. We will assume that, if we do the k th IPM at t operating time of the equipment, after that k th IPM the equipment age will be set to b_{kt} , (see Fig. 9.8).

In the IPM model we make the following assumptions:

1. IPMs are carried out at times $h_1, h_1 + h_2, \dots$, where h_i is the length of the i th interval ($i = 1, 2, \dots, N-1$) and the PR of the equipment is carried out after the interval N ;
2. Failures between two IPMs are solved with minimal repairs. After k IPMs equipment life is reduced b_{kt} , with $0 = b_0 < b_1 < b_2 < \dots < b_{N-1} < 1$;
3. After PR the equipment is as good as new and its failure rate is totally restored;
4. $\lambda(t)$ is continuous and strictly increasing with time;
5. Times to carry out IPM, minimal repair and PR are ignored;
6. After N time intervals the cycle ends with a PR.

We have to find the size of the intervals (h_k), and the number of IPMs ($N-1$) before PR to minimise the total expected cost per time unit. We will use the following notation:

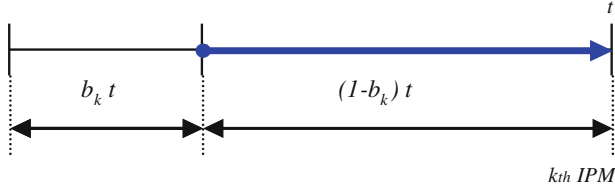


Fig. 9.8 IPM impact on equipment age

y_i equipment age when the i th IPM is carried out;
 C_{ipm} IPM unit cost;
 C_p PR unit cost;
 C_{rm} Minimal repair unit cost

The equipment age when we do the k th IPM is obtained as

$$y_k = h_k + b_{k-1}y_{k-1} \quad (9.15)$$

Therefore, during the k th time interval, the equipment age is within $[b_{k-1}y_{k-1}, y_k]$. The total expected cost per cycle is

$$TC(y_1, y_2, \dots, y_N) = C_{rm} \sum_{k=1}^N \int_{b_{k-1}y_{k-1}}^{y_k} \lambda(t) dt + (N-1)C_{ipm} + C_p \quad (9.16)$$

The cycle expected length will be

$$LE(T) = \sum_{k=1}^{N-1} h_k \quad (9.17)$$

and the total expected cost per time unit is

$$TEC(y_1, \dots, y_N) = \frac{TC(y_1, \dots, y_N)}{LE(T)} = \frac{C_{rm} \sum_{k=1}^N \int_{b_{k-1}y_{k-1}}^{y_k} \lambda(t) dt + (N-1)C_{ipm} + C_p}{\sum_{k=1}^{N-1} h_k} \quad (9.18)$$

9.4.5 Inspection Models

The purpose of these models is to define the optimal inspection schedule, or to find the best moments in time to check the equipment condition. Inspection models will normally consider other possible preventive activities and corrective maintenance. In inspection models, and according to a certain mathematical formulation,

inspections, PR and CR will be combined with the idea of reaching the minimum total expected cost per unit time.

Inspection models were introduced by Barlow et al. [4]. Initial formulation of the problem was an age-based inspection model, where there are no preventive maintenance replacements and the equipment is replaced only after a failure. This model considers the following assumptions:

- Equipment failure is known only after an inspection;
- Inspections do not deteriorate the equipment;
- Equipment cannot fail during the inspection;
- Each inspection has a unit cost C_i ;
- The cost, per time unit, associated to a non-detected failure is C_f ;
- The corrective replacement cost is C_s .

The inspection policy in this model considers that an inspection is done at times $x_1, x_2, x_3, \dots, x_n$, until a failure is found, then the equipment is replaced, and another cycle starts (Fig. 9.9).

When there is a failure between x_{n-1} and x_n , at time t_i , the cost per inspection cycle is

$$C(t_i, x_n) = nC_i + (x_n - t_i)C_f + C_s \quad (9.19)$$

Note that Eq. 9.19 supposes that equipment fails at t_i and then suffers a performance loss, associated with the failure, until it reaches x_n . At x_n the inspection takes place, then the failure will be found and the replacement done. We carry out therefore n inspections and a corrective (total) replacement per cycle.

The expected cost of this policy is expressed in Eq. 9.20.

$$CE(t, x) = \int_{x_{n-1}}^{x_n} [nC_i + (x_n - t_i)C_f + C_s]f(t)dt \quad (9.20)$$

If we now add for all n values, we can obtain the total expected cost as follows:

$$\sum_{n=1}^{\infty} \int_{x_{n-1}}^{x_n} [nC_i + (x_n - t_i)C_f + C_s]f(t)dt \quad (9.21)$$

To achieve this cost per time unit we have to calculate the cycle expected length:

$$LE(x_1, x_2, \dots, x_n) = \int_0^{\infty} tf(t)dt + \sum_{n=1}^{\infty} \int_{x_{n-1}}^{x_n} (x_n - t)f(t)dt + T_s \quad (9.22)$$

where T_s is the replacement time. Then, the total expected cost per time unit is

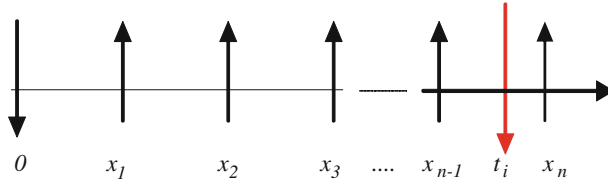


Fig. 9.9 Equipment inspections and failure

$$\text{CTE}(x_1, x_2, \dots, x_n) = \frac{\sum_{n=1}^{\infty} \int_{x_{n-1}}^{x_n} [nC_i + (x_n - t)C_f + C_s]f(t)dt}{\int_0^{\infty} tf(t)dt + \sum_{n=1}^{\infty} \int_{x_{n-1}}^{x_n} (x_n - t)f(t)dt + T_s} \quad (9.23)$$

9.4.6 Practical Models Implementation and Case Study

We will now implement, as an example, the model presented in [Sect. 9.4.3.1](#) for the optimisation of the number of overhauls (PPR) in a pumping system unit, within a water network utility, before its final preventive replacement (PR). The purpose of this section is offering the reader a concise idea of the numerical calculations required to implement such a model for a real life situation. We now try to move from the mathematical formulation presented in the referred section, to a numerical algorithm that we can implement in our computer.

The first think we will be to discretize the time, according to the following steps:

- We chose a time step T ;
- We divide the time to overhaul, selected for our analysis ($T_{\max}$), into N T time units intervals;
- We express T_i as $T_i = T * i$, with $i \in [1, N]$.

The most complicated part in the model implementation is the definition of the expected number of failures in the j th T_i interval

$$(N_{c,j}(T_i)) :$$

$$N_{c,j}(T_i) = \int_0^{T_i} \lambda_j(t)dt, \text{ with } T_i \in (0, T_{\max}], j \in [1, m] \quad (9.24)$$

If we use Newton-Cotes formulation then $N_{c,j}(T_i)$ can be expressed as follows:

$$N_{c,j}(T_i) = \int_0^{T_i} \lambda_j(t) dt = \sum_{k=1}^n \int_{I_k} \lambda_j(t) dt \cong \sum_{k=1}^n \left(\sum_{i=0}^m \lambda_i \alpha_i \right) \quad (9.25)$$

Where we have m different failure rates, $\lambda_j(t)$, $j = 1, 2, \dots, m$. In this case study we do consider a tri-parametric Weibull $(\beta_j, \eta_j, \gamma_j)$ type of failure rate, so:

$$\lambda_j(t) = \frac{\beta_j}{\eta_j} * \left(\frac{t}{\eta_j} \right)^{\beta_j-1}$$

Thus, we have m failure rate functions like:

$$\begin{aligned} \lambda_1(t) &= \frac{\beta_1}{\eta_1} * \left(\frac{t}{\eta_1} \right)^{\beta_1-1} \\ &\vdots \\ \lambda_m(t) &= \frac{\beta_m}{\eta_m} * \left(\frac{t}{\eta_m} \right)^{\beta_m-1} \end{aligned}$$

Let us consider the function: Integration $(0, t_p, n, \lambda(t), \beta_j, \eta_j, \gamma_j)$; as a function implementing the Newton-Cotes formulation in Eq. 9.25, then we can write:

$$N_{c,j}(T_i) = \text{Integration}(0, t_p, n, \lambda(t), \beta_j, \eta_j, \gamma_j)$$

If we now consider the vectors:

$$\bar{\beta} = [\beta_1, \beta_2, \dots, \beta_k], \bar{\eta} = [\eta_1, \eta_2, \dots, \eta_k], \bar{\gamma} = [\gamma_1, \gamma_2, \dots, \gamma_k]$$

Then we can write the following algorithm:

Given $T_{\max}, T, n, C_{pt}, C_{pp}, C_{ic}, \lambda(t), \bar{\beta}, \bar{\eta}, \bar{\gamma}$:

$$N = T_{\max} / T$$

For $m = 1, 2, \dots, K$ calculate

For $i = 1, 2, \dots, N$ calculate

$$T_i = T * i$$

For $j = 1, 2, \dots, m$ calculate

$$N_{c,j}(T_j) = \text{Intergration}(0, t_p, n, \lambda(t), \beta_j, \eta_j, \gamma_j)$$

$$N_c(T_m) = \sum_{j=1}^m N_{c,j}(T_j)$$

$$\text{CTE}(m, T_i) = \frac{(m-1) * C_{pp} + C_p + C_{rm} * N_c(T_m)}{m * T_i}$$

$$\min \text{CTE}(m, T_i) \rightarrow \begin{cases} T_i, \text{opt} = i_{\text{opt}} * T \\ m_{\text{opt}} \end{cases}$$

Table 9.1 Failure rate, per interval, of the pumping units

Time interval	$\lambda_m(t)$
O - 1st PPR	$\lambda_1(t) = Weibull(\beta_1 = 1.6, \eta_1 = 2.4, \gamma_1 = 0)$
1st PPR - 2° PPR	$\lambda_2(t) = Weibull(\beta_2 = 1.9, \eta_2 = 2.55, \gamma_2 = 0)$
2° PPR - PR	$\lambda_3(t) = Weibull(\beta_3 = 2.5, \eta_3 = 2.7, \gamma_3 = 0)$

This is an iterative algorithm, given a certain number of preventive interventions (m), the algorithm estimates CTE(m, T_i) the different possible values of T_i ($T_i = T * i$, with $i \in [1, N]$) obtaining a set of N values CTE($m, 1T$), CTE($m, 2T$), ..., CTE(m, NT). In this model, m is not a constant but an optimization variable, and therefore we must explore other values of m , i.e. $m = 1, 2, \dots, K$. To finally obtain a total of $K * N$ values of CTE(m, T_i). The optimization process finishes when we find out the minimum value of CTE(m, T_i), given the corresponding m & T_i values.

In our case, study the pumps can be overhauled (PPR) certain number of times before the final replacement (PR). During the time between overhauls pumps may suffer failures and we can accomplish minimal repairs for fast equipment fix, although not improving equipment reliability for the failure mode (only when we overhaul or we replace the pump the failure mode failure frequency is reset).

The maintenance management team has decided, according to present equipment knowledge, to accomplish pump overhauls every 6 months and to do a maximum of 2 overhauls before the pump replacement (2 PPR and 1 PR). After the first total preventive replacement cycle (18 months) the maintenance cost results to be high (62230 Euro/month). The maintenance manager decides to explore other strategies. He wants to know the optimal time between overhauls and the associated maintenance cost to that strategy.

Relevant data obtained for the failure mode under analysis, after 18 months of several pump units operation are:

1. Failure rates are distributed as presented in Table 9.1:
2. $C_{pp} = 100,000$ Euro = 100 K Euro
3. $C_p = 250,000$ Euro = 250 K Euro
4. $C_{rm} = 40,000$ Euro = 40 K Euro

Comparing present strategy with other potential strategies means the following calculations:

Current preventive maintenance strategy: (Fig. 9.10)

In this case

$$N_{c,j}(T_i) = \int_0^{T_i} \lambda_j(t) dt = \int_0^{T_i} \frac{\beta_j}{\eta_j} * \left(\frac{t}{\eta_j} \right)^{\beta_j-1} dt = \frac{T_i}{\eta_j} * \left(\frac{T_i}{\eta_j} \right)^{\beta_j-1}$$

Then, if we have select $T = 3$ months and a maximum of $m = 3$ preventive interventions we have the following possibilities for preventive strategies:

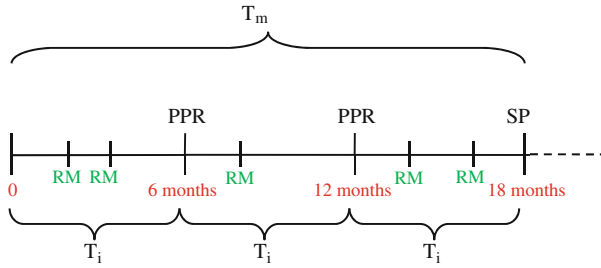
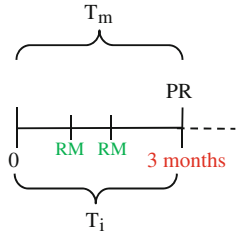


Fig. 9.10 System evolution for current maintenance strategy

Fig. 9.11 System evolution
for $m = 1, T_i = 3$ months



1) $m = 1$

a) $T_i = 3$ months (Fig. 9.11)

$$N_{c,1}(T_i) = \int_0^{T_i} \lambda_1(t) dt = \frac{T_i}{\eta_1} * \left(\frac{T_i}{\eta_1} \right)^{\beta_1 - 1} = 1.4291$$

$$N_c(T_m) = \sum_{j=1}^1 N_{c,j}(T_j) = 1.4291$$

$$\begin{aligned} \text{CTE}(1, 3 \text{ months}) &= \frac{(m-1) * C_{pp} + C_p + C_{rm} * N_c(T_m)}{T_m} \\ &= \frac{250 \text{ K€} + 40 \text{ K€} * 1.4291}{3 \text{ months}} = 102.4 \text{ K€}/\text{month} \end{aligned}$$

b) $T_i = 6$ months (Fig. 9.12)

$$N_{c,1}(T_i) = \int_0^{T_i} \lambda_1(t) dt = \frac{T_i}{\eta_1} * \left(\frac{T_i}{\eta_1} \right)^{\beta_1 - 1} = 4.3321$$

$$N_c(T_m) = \sum_{j=1}^1 N_{c,j}(T_j) = 4.3321$$

Fig. 9.12 System evolution
for $m = 1$, $T_i = 6$ months

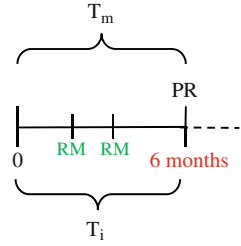
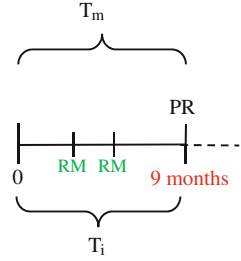


Fig. 9.13 System evolution
for $m = 1$, $T_i = 9$ months



$$\begin{aligned} \text{CTE}(1, 6 \text{ months}) &= \frac{(m-1) * C_{pp} + C_p + C_{rm} * N_c(T_m)}{T_m} \\ &= \frac{250 \text{ K€} + 40 \text{ K€} * 4.3321}{6 \text{ months}} = 70.5 \text{ K€}/\text{mess} \end{aligned}$$

c) $T_i = 9$ months (Fig. 9.13)

$$N_{c,1}(T_i) = \int_0^{T_i} \lambda_1(t) dt = \frac{T_i}{\eta_1} * \left(\frac{T_i}{\eta_1} \right)^{\beta_1-1} = 8.2878$$

$$N_c(T_m) = \sum_{j=1}^1 N_{c,j}(T_i) = 8.2878$$

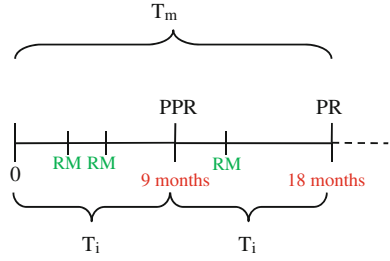
$$\begin{aligned} \text{CTE}(1, 9 \text{ months}) &= \frac{(m-1) * C_{pp} + C_p + C_{rm} * N_c(T_m)}{T_m} \\ &= \frac{250 \text{ K€} + 40 \text{ K€} * 8.2878}{9 \text{ months}} = 64.6 \text{ K€}/\text{month} \end{aligned}$$

2) $m = 2$

a) $T_i = 3$ months (Fig. 9.14)

$$N_{c,1}(T_i) = \int_0^{T_i} \lambda_1(t) dt = \frac{T_i}{\eta_1} * \left(\frac{T_i}{\eta_1} \right)^{\beta_1-1} = 1.4291,$$

Fig. 9.16 System evolution
for $m = 2$, $T_i = 9$ months



$$\begin{aligned} \text{CTE}(2, 6 \text{ months}) &= \frac{(m-1) * C_{pp} + C_{pt} + C_{rm} * N_c(T_m)}{T_m} \\ &= \frac{100 \text{ K€} + 250 \text{ K€} + 40 \text{ K€} * 9.4144}{12 \text{ months}} = 60.54 \text{ K€/month} \end{aligned}$$

c) $T_i = 9$ months (Fig. 9.16)

$$N_{c,1}(T_i) = \int_0^{T_i} \lambda_1(t) dt = \frac{T_i}{\eta_1} * \left(\frac{T_i}{\eta_1} \right)^{\beta_1-1} = 8.2878,$$

$$N_{c,2}(T_i) = \int_0^{T_i} \lambda_2(t) dt = \frac{T_i}{\eta_2} * \left(\frac{T_i}{\eta_2} \right)^{\beta_2-1} = 10.9808,$$

$$N_c(T_m) = \sum_{j=1}^2 N_{c,j}(T_i) = 19.2687$$

$$\begin{aligned} \text{CTE}(2, 9 \text{ months}) &= \frac{(m-1) * C_{pp} + C_{pt} + C_{rm} * N_c(T_m)}{T_m} \\ &= \frac{100 \text{ K€} + 250 \text{ K€} + 40 \text{ K€} * 19.2687}{18 \text{ months}} = 62.26 \text{ K€/month} \end{aligned}$$

3) $m = 3$

a) $T_i = 3$ months (Fig. 9.17)

$$N_{c,1}(T_i) = \int_0^{T_i} \lambda_1(t) dt = \frac{T_i}{\eta_1} * \left(\frac{T_i}{\eta_1} \right)^{\beta_1-1} = 1.4291,$$

$$N_{c,2}(T_i) = \int_0^{T_i} \lambda_2(t) dt = \frac{T_i}{\eta_2} * \left(\frac{T_i}{\eta_2} \right)^{\beta_2-1} = 1.3618,$$

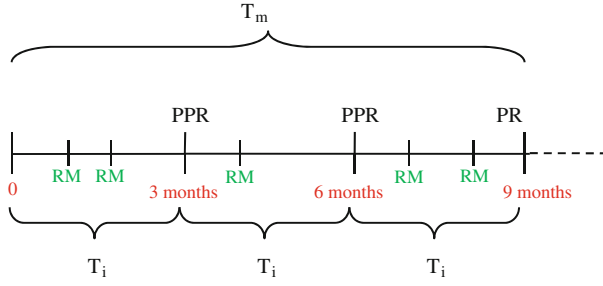


Fig. 9.17 System evolution for $m = 3$, $T_i = 3$ months

$$N_{c,3}(T_i) = \int_0^{T_i} \lambda_3(t) dt = \frac{T_i}{\eta_3} * \left(\frac{T_i}{\eta_3} \right)^{\beta_3-1} = 1.3013$$

$$N_c(T_m) = \sum_{j=1}^3 N_{c,j}(T_i) = 4.0922$$

$$\begin{aligned} \text{CTE}(3, 3 \text{ months}) &= \frac{(m-1) * C_{pp} + C_{pt} + C_{rm} * N_c(T_m)}{\frac{T_m}{9 \text{ months}}} \\ &= \frac{2 * 100 \text{ €} + 250 \text{ €} + 40 \text{ €} * 4.0922}{9 \text{ months}} = 68.19 \text{ K€}/\text{month} \end{aligned}$$

b) $T_i = 6$ months (Fig. 9.18)

$$N_{c,1}(T_i) = \int_0^{T_i} \lambda_1(t) dt = \frac{T_i}{\eta_1} * \left(\frac{T_i}{\eta_1} \right)^{\beta_1-1} = 4.3321,$$

$$N_{c,2}(T_i) = \int_0^{T_i} \lambda_2(t) dt = \frac{T_i}{\eta_2} * \left(\frac{T_i}{\eta_2} \right)^{\beta_2-1} = 5.0823,$$

$$N_{c,3}(T_i) = \int_0^{T_i} \lambda_3(t) dt = \frac{T_i}{\eta_3} * \left(\frac{T_i}{\eta_3} \right)^{\beta_3-1} = 7.3615$$

$$N_c(T_m) = \sum_{j=1}^3 N_{c,j}(T_i) = 16.7759$$

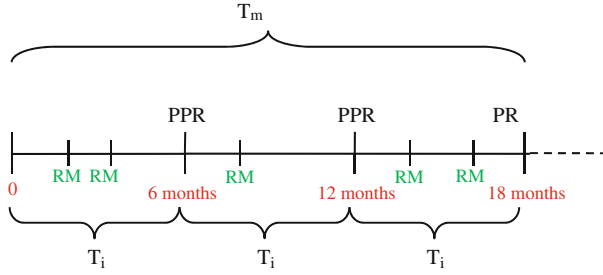


Fig. 9.18 System evolution for $m = 3$, $T_i = 6$ months

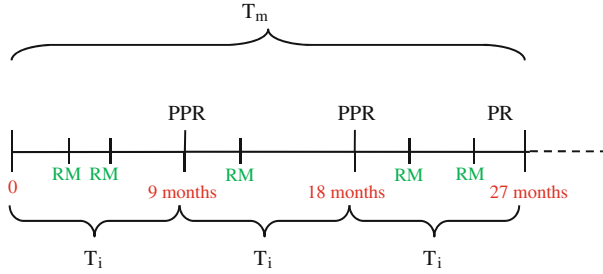


Fig. 9.19 System evolution for $m = 3$, $T_i = 9$ months

$$\begin{aligned}
 \text{CTE}(3, 6 \text{ months}) &= \frac{(m-1) * C_{pp} + C_{pt} + C_{rm} * N_c(T_m)}{T_m} \\
 &= \frac{2 * 100 \text{ K€} + 250 \text{ K€} + 40 \text{ K€} * 16.7759}{18 \text{ months}} \\
 &= 62.23 \text{ K€}/\text{month}
 \end{aligned}$$

c) $T_i = 9$ months (Fig. 9.19)

$$N_{c,1}(T_i) = \int_0^{T_i} \lambda_1(t) dt = \frac{T_i}{\eta_1} * \left(\frac{T_i}{\eta_1} \right)^{\beta_1-1} = 8.2878,$$

$$N_{c,2}(T_i) = \int_0^{T_i} \lambda_2(t) dt = \frac{T_i}{\eta_2} * \left(\frac{T_i}{\eta_2} \right)^{\beta_2-1} = 10.9808,$$

$$N_{c,3}(T_i) = \int_0^{T_i} \lambda_3(t) dt = \frac{T_i}{\eta_3} * \left(\frac{T_i}{\eta_3} \right)^{\beta_3-1} = 20.2860$$

$$\begin{aligned}
 N_c(T_m) &= \sum_{j=1}^3 N_{c,j}(T_i) = 39.5547 \\
 \text{CTE}(3, 9 \text{ months}) &= \frac{(m-1) * C_{pp} + C_{pt} + C_{rm} * N_c(T_m)}{T_m} \\
 &= \frac{2 * 100\text{K€} + 250\text{K€} + 40\text{K€} * 39.5547}{27 \text{ months}} \\
 &= 75.27 \text{ K€}/\text{month}
 \end{aligned}$$

Results show that the most convenient maintenance strategy is to execute PPRs every six months, and for a total of two maintenance preventive interventions (one PPR and one PR). The cost associated to the suggested strategy is 60.54 K€ /month (minimum obtained value), versus the 62.23 K€/month of the current strategy which means savings around 2 K€ /month, for this specific case.

9.5 Setting Preventive Routines Priority

We will present different options for preventive routines priority setting. We will use the presented case study of a telecommunications network utility, in [Sect. 6.6.1](#), to exemplify these options. In this network the preventive maintenance is conducted according to a certain criteria, trying to reach a suitable trade-off or balance between quality and maintenance cost (both in resources and time).

The preventive planning execution is based solely on quantitative criteria, as indicated by suppliers of existing systems, and implies an economic cost that due to the large number of elements is rarely completed.

In this case we are also considering a multi-territory company, and therefore many organisational aspects are difficult to manage and a consensus in budget is very difficult to be agreed. At this point the company is searching for a methodology to: *“Identify equipment needing more preventive maintenance, considering the entire company scope, establishing a priority order”*.

The goal of the company is to concentrate on critical equipment preventive maintenance implementation, assuming just a few elements to be critical.

For easy and understandable decision-making, we will rely on the properties of two methods: the AHP method [29–31] and the linear regression analysis. We will recommend the employment of AHP when there is not enough quantitative information and linear regression otherwise.

In the application of the AHP method, in [Sect. 6.6.1](#), a group of maintenance experts from several hierarchical levels, managers (to assess the strategy) and technicians (to assess the activities), decided that in addition to failure frequency (FF), the following criteria will be considered in order to prioritize equipment deserving preventive maintenance activities:

Table 9.2 Matrices following the AHP method application

Criteria	Weight (wi)
FF	0.358
NC	0.302
CI	0.230
DI	0.110
IR	0.02

- Date of Installation (DI).
- Cost of Installation (CI).
- Number of Clients (NC).

If we first consider the existence of sparse historical data, we can try to apply the AHP method, just like we did in [Chap. 6](#), taking advantages of the criteria pairwise comparisons. In [Table 6.8](#), the scale for each criterion is presented. Finally the result in ranking of importance, following the method explained in [Chap. 6](#), is presented in [Table 9.2](#).

We can see that in this case the criterion with the highest weight is failure frequency with a 35.8%, followed by number of clients with a 30.2% and cost of installation with a 23%. The criterion with less weight is date of installation with a 11%. The decision making quality of AHP application is acceptable if it is less than 0.1 or 10%. The consistency rating was 0.02, considered acceptable.

Thus, as a first option, we can classify the network equipment according to the referred criteria, calculating their priority number as follows:

$$\text{Priority} = 0.11 \cdot \text{DI} + 0.23 \cdot \text{CI} + 0.302 \cdot \text{NC} + 0.358 \cdot \text{FF}$$

In the event that there is enough historical data (in our case up to 01/01/2006), we can correlate the number of produced incidents (Y) with the score of the equipment for each one of the above expressed criteria. In order to do so we can use a linear regression with least squares fitting.

To make it simpler and practical, values (for the four criteria in [Table 6.8](#)) per distribution zone and for a representative universe of 150 equipments are presented in [Table 9.3](#), once discretized (D_FI, D_CI, D_NC, D_FF). In [Table 9.3](#) only a part of the list is presented. In addition, the variable Y is also discretized (D_Y) dividing the produced incidents by 300 and then using the same scale as per DI years.

Now we can use linear regression to express the expected number of incidents as a linear function of the four previous criteria and by doing so obtaining an equipment priority, as follows:

$$\text{Priority} = -0.43 \cdot \text{DI} + 0.604 \cdot \text{CI} + 0.586 \cdot \text{NC} - 0.765 \cdot \text{FF}$$

With a 88.77% of adequate equation consistency, $R^2 = 0.8877$ (correlation coefficient squared) and $|t| > t$ of the Students distribution for 95% confidence interval of probability.

Table 9.3 Partial list of 150 evaluated equipment with linear regression

Zone	FI	CI	NC	FF	Y	D_FI	D_CI	D_NC	D_FF	D_Y
ALA25	25-02-99	29863	6253	1.45	570	6	6	4	2	2
COA0	06-07-99	99964	39275	1.55	1471	6	8	5	2	4
GRA0	30-07-99	107930	43164	3.08	1432	6	8	5	1	4
JAA23	15-08-00	20257	1648	4.23	256	5	5	2	1	1
GRA42	23-08-00	14726	3253	1.43	275	5	5	4	2	1
AGA05	14-04-00	19785	4967	1.83	198	5	5	4	2	1
SEB26	30-09-99	10812	2976	1.70	130	6	4	4	2	1
COB01	20-12-99	43137	12143	2.22	521	6	6	5	2	2
SED04	09-04-99	17032	3853	1.56	324	6	5	4	2	2
COA2	06-03-00	149046	48218	3.33	2606	5	9	6	1	8

After this case study we can conclude that both methods are some of the very diverse possibilities that we have to prioritize maintenance, searching consensus and practical implementation. AHP methodology suggests a quantitative process but allows the consideration of qualitative aspects, like subjective pairwise comparisons or ad hoc criteria scalability. The method has the advantage of fairly accurate assessment of certain people preferences (tangible and intangible) in a consistent manner. On the other hand, when the number of considered alternatives increases, a clear disadvantage is the total number of needed pairwise comparisons, which makes the process extremely tedious. Besides this, the inclusion of any irrelevant criteria may affect the impact of existing relevant criteria.

The linear regression is more practical when there are many decision criteria, although some of them could be correlated and this fact could distort the results. For that reason, sensitive analysis is always essential to confirm the developed models.

Consequently, we suggest that qualitative methods can be employed searching for the consensus and in order to assign criticality among the different hierarchical levels of the network equipment. Quantitative methods could then be used to determine priority among the same hierarchical level equipment, where their characteristics are similar and the historical data base would be more convenient. To conclude, this model can also be used to prioritize the preventive maintenance and to distribute the budget for preventive maintenance among territories with agreed criteria.

Furthermore, the priority for preventive maintenance intervention could be based on geographical aspects related to logistics, as we will see in the next section.

9.6 Preventive Actions and Geographical Displacements

Network preventive activities can also be prioritized and many times scheduled or released using the geographical displacements cost as a fundamental criteria. In order to do so, the geographical positioning of the network elements needs to be considered.

What are the network areas that should be under the responsibility of a given maintenance centre? How many centres should we have? What is the displacement cost involved for a given maintenance strategy? These are the most common problems to solve and not only for preventive maintenance. We have the same problems when we manage any activity requiring displacements in the field and deploying spatial sites in the network to serve as origin of displacement centres (departure-arrival points).

When dealing with this problem in networks, Geographical Information Systems (GIS) are very useful tools to improve decision making [9, 10]. This is due to their capability [2, 20] to integrate different types of information layers (services, cartography, network architecture, etc.) allowing an easy mapping of locations and displacements [8, 18] and the subsequent optimization of travel costs and time response [22].

Using GIS systems we can represent the network topologically-connected as a combination of nodes and arcs with certain attributes [1], then we can try to find the shortest path between two nodes, etc. [6, 7, 37]. To solve this problem, one of the most employed algorithms is the Dijkstra's Algorithm [11, 15, 21, 24, 28] that now can be used to determine the shortest path in a maintenance displacement. Thus, given a network with topologically-connected nodes by arcs, the Dijkstra's Algorithm is developed through a sequence of recursive steps to review all the possible paths discovering the better shortest paths from an initial node [27, 34]. In order to control the recursive steps, four labels are assigned to each node:

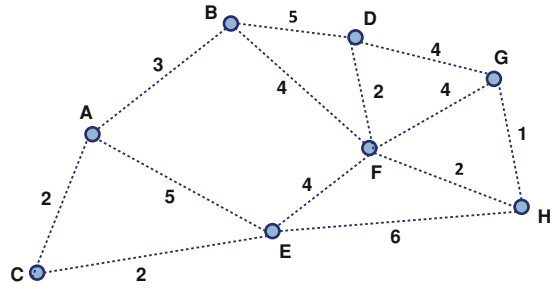
- Distance label, to store the distance value from the initial node.
- Mark label, to show if the distance value is “tentative” or “permanent” as a shortest path.
- Connection label, to mark the connection of the node with the previous one inside the shortest path, that is, the parent in the network.
- State label, to indicate whether the node has been evaluated, or not, using the algorithm (“operative” or “inoperative” otherwise).

Initially, the distance labels are set to infinity in all nodes except in the initial node where the value is set to 0, the mark labels are set to “tentative” except in the initial which is set to “permanent”, the state label is set to “operative” for the initial node and “inoperative” for the rest, and the connection label is set with the initial node for itself and is empty for the rest of the nodes.

Consequently, the sequence is realized recurrently over the operative node until all the mark labels as permanent, as follows:

1. Select all the adjacent and “tentative” nodes to the “operative” node and calculate the tentative distance from the initial node. The adjacent node with the mark label as “permanent” does not have to be evaluated in the sequence.
2. If the calculated tentative distances are lower than existing ones, they will be recorded.

Fig. 9.20 Shortest path example inside a distribution area



3. Evaluate all the tentative distances to find out the smallest and record it as “permanent” and with the name of the operative node as parent in the connection label.
4. Set the “operative” node as “inoperative” and the found node with smallest path as “operative” for the next iteration.
5. Stop when all nodes are “permanent”, otherwise go to the first step.

For example, imagine a network utility within an area of eight cities where one of them is the central location of the maintenance organization, departure point for all maintenance displacements. In order to optimize the preventive routines the maintenance displacement with the shortest paths needs to be calculated using Dijkstra’s Algorithm. In Fig. 9.20 the geographical disposition of the nodes and potential displacements are presented showing the arcs distances. In Table 9.4 the sequence of steps is presented:

We will describe the process for each operative node, beginning with the node A as initial:

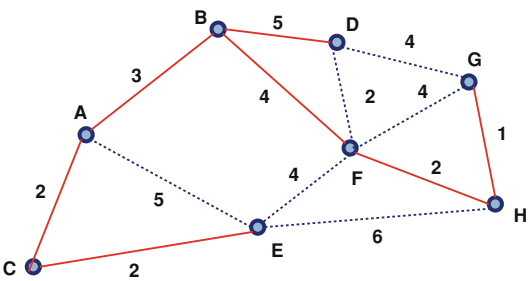
1. The tentative distances from A to the adjacent B, C and E are 2, 3, and 5 respectively; so the smallest path is AC and the next operative node is C.
2. The tentative distances from A to de C’s adjacent node E is 4; so the smallest path is AB (3) < AC (4) and the next operative node is B.
3. The tentative distances from A to the B’s adjacent nodes, D and F are 8 and 7 respectively; so the smallest path is ACE (4) < AF (7) and the next operative node is E.
4. The tentative distances from A to the E’s adjacent nodes, F and H are 8 and 10 respectively; so the smallest path is ABF (7) < ACEF (8) and the next operative node is F.
5. The tentative distances from A to the F’s adjacent node D, G and H are 9, 11 and 9 respectively; so the smallest path is ABD (8) < ABFD (9) or ABFH (9) and the next operative node is D.
6. The tentative distances from A to the D’s adjacent nodes, G is 12; so the smallest path is ABFH (9) < ABFG (11) and the next operative node is H.
7. The tentative distances from A to the H’s adjacent nodes, G is 10; so the smallest path is ABFHG (10) < ABFG (11) and the next operative node is H, but due to all the nodes are permanent, the algorithm ends.

Table 9.4 Iterations of Dijkstra’s Algorithm in the example

S	Op. Node	Path	B		C		D		E		F		G		H	
			T	P	T	P	T	P	T	P	T	P	T	P	T	P
1	A	A	3		2	2	∞		5		∞		∞		∞	
2	C	AC	3	3	–	2	∞		4		∞		∞		∞	
3	B	AB	–	3	–	2	8		4	4	7		∞		∞	
4	E	ACE	–	3	–	2	8		–	4	7	7	∞		10	
5	F	ABF	–	3	–	2	8	8	–	4	–	7	11		9	
6	D	ABD	–	3	–	2	–	8	–	4	–	7	11		9	9
7	H	ABFH	–	3	–	2	–	8	–	4	–	7	10	10	–	9
8	G	ABFHG	–	3	–	2	–	8	–	4	–	7	–	10	–	9

Note S = Sequence, Op.Node = Operative Node, T = Tentative and P = Permanent

Fig. 9.21 Found shortest paths of the example inside a distribution area



In short, the shortest paths for each node from the initial A are represented in the Fig. 9.21.

Consequently, the geographical distribution of elements could be considered as a rule to plan the preventive maintenance routines by optimizing network displacements. The calculation of the shortest paths for a graph of N nodes implies $O(N^2)$ operations, nevertheless this problem will simply be solved once, or in case of a substantial network modification.

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Chapter 10

Network Capacity Analysis and Administration

10.1 Introduction

Network occupation normally triggers the process responsible for the management and administration of network capacity, with the purpose of improving resources utilisation [4]. This process defines thresholds to generate saturation or congestion alarms automatically. In addition, it also develops remote reconfiguration of elements in order to use or to occupy non saturated resources (see Fig. 10.1).

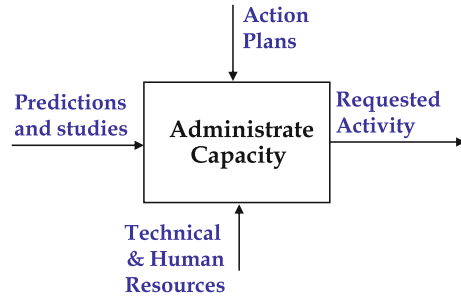
Spare capacity is viewed as “a productive space that needs to be exploited” [3], or as a “factor of efficiency” [10]. However and obviously, sufficient and additional capacity has to be ensured before abrupt increments of demands over an existing network. Therefore, and as a control measure, regulators carry out inspections to evaluate adequate capacity implementation in accordance with actual contractual or legal requirements [7].

In network utilities it is common to use a process named “to manage capacity” which has assigned the responsibility for long-term capacity planning. However, the real time surveillance of capacity levels is normally accomplished through an operative process which administrates short-term network capacity. Consequently, it is convenient to distinguish between two capacity related processes in networks: to manage long-term capacity (that is usually a responsibility of the engineering department), and to administrate short-term capacity (usually, a responsibility of the maintenance department).

Therefore, maintenance has to provide and to supervise proper measures of capacity occupation, managing routing alternatives when facing any potential incident and in order to guarantee the agreed customer service levels. Also, and through the dependability of the process “to analyse maintenance”, maintenance shall evaluate possible alternatives to route the whole, or a portion, of the capacity in critical paths of the network periodically.

In accordance to previous paragraphs, two types of capacity related thresholds are normally defined:

Fig. 10.1 Capacity Administration process



- A prevision of network saturation, triggering a notification to the engineering department concerning the need to modify the network, enlarging available capacity;
- A call for an immediate short-term reliever action.

The process “to analyse maintenance” will determine these thresholds based on capacity predictions and studies, then the “to administrate capacity” has to manage services demand seasonally, controlling possible variation of equipment performance ensuring minimum potential impact on service levels.

When analysing network capacity, engineers must also consider a proper balance of service routing as a method for equipment’s reliability, for instance, it is a common practice to balance the flows among redundant paths or equipment.

Network capacity analysis is therefore a complex process that must be supported by advanced modelling techniques, enabling the accurate calculation of thresholds and the consideration of multicriteria decision-making (customer-oriented, equipment reliability, costs efficiency, etc.). Simulation techniques can play here a very important role, generating new knowledge for network capacity management judging different issues and circumstances. For further reading about this topic, the reader is referred to different contribution addressing these issues, linking capacity with operations and business growth (see for instance [6, 5] or [1]).

10.2 The Capacity Administration Process Workflow

The work flow of the capacity administration process begins with the detection of events activated by the warnings and alarms already configured in the network (see Fig. 10.2).

The event is filtered and processed according to existing technical and quality criteria, to evaluate the correct event generation. Besides this, the event has to be discarded when affected by the presence of any possible incident. Otherwise, the flow continues calling the process “to analyse maintenance” for studies and predictions. In order to avoid repetitive presence of similar events, possible correlation with current capacity changes activities has to be identified. If this is not

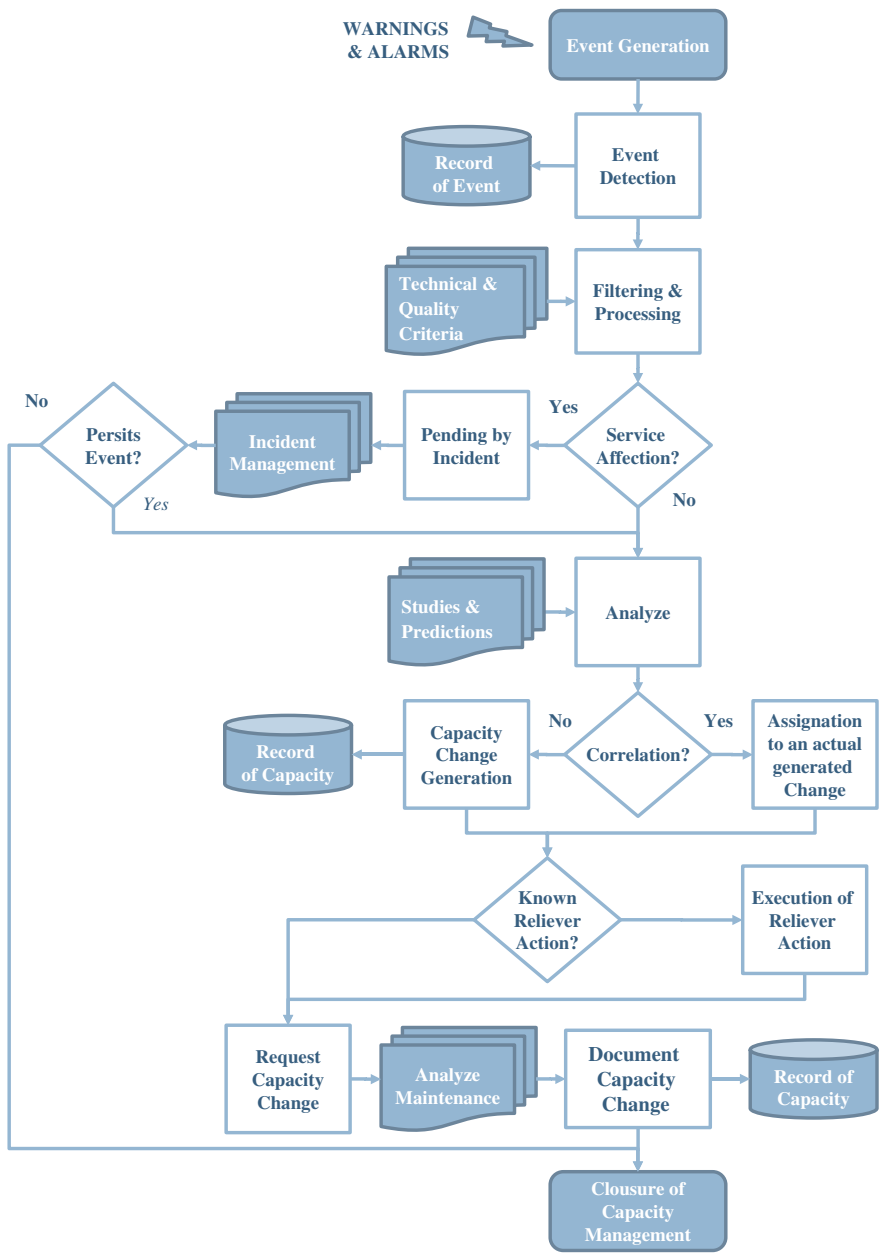


Fig. 10.2 Capacity Administration flow

the case a new capacity change activity is generated. When an immediate reliever action exists (tested according to past experience) it will be executed, otherwise the capacity change activity is escalated as a requested activity to the process “to

analyse maintenance” and the process will wait for the implementation of the capacity change. When a capacity change activity is escalated to the process “to analyse maintenance” and a network modification could be approved, for instance, involving the maintenance department. But another possibility is that a perfective network modification proposal could be considered and evaluated. Finally, once the capacity change is carried out, the workflow documentation is recorded and the process ends.

In the subsequent Sections we use a high level dynamic simulation model to explore the capacity management process and to understand the implications of capacity for business growth. Maintenance efforts are devoted to keep capacity available at the minimum possible cost, therefore the following model also illustrates the importance of maintenance activities and their contribution to business growth.

10.3 Simulating Business Growth and Network Capacity

10.3.1 Introduction to System Dynamics Simulation Technique

In this section we will use System Dynamics methodology. System Dynamics is [9] a methodology for studying and managing complex feedback systems, such as one finds in business and other social systems. In fact it has been used to address practically every sort of feedback system. While the word system has been applied to all sorts of situations, feedback is the differentiating descriptor here. Feedback refers to the situation of X affecting Y and Y in turn affecting X perhaps through a chain of causes and effects. One cannot study the link between X and Y and, independently, the link between Y and X and predict how the system will behave. Only the study of the whole system as a feedback system will lead to correct results.

The basis of the method is the recognition that the structure of any system—the many circular, interlocking, sometimes time-delayed relationships among its components—is often just as important in determining its behaviour as the individual components themselves. There are often properties-of-the-whole which cannot be found among the properties-of-the-elements; in some cases the behaviour of the whole cannot be explained in terms of the behaviour of the parts. The methodology:

1. identifies a problem;
2. develops a dynamic hypothesis explaining the cause of the problem;
3. builds a computer simulation model of the system at the root of the problem;
4. tests the model to be certain that it reproduces the behaviour seen in the real world;
5. devises and tests in the model alternative policies that alleviate the problem;
6. implements this solution.

System Dynamics uses specific tools to build diagrams of our models. In this exercise we use the stock and flow diagrams (SFD)—or level and rate diagrams (LRD)—which are ways of representing the structure of a system with detailed

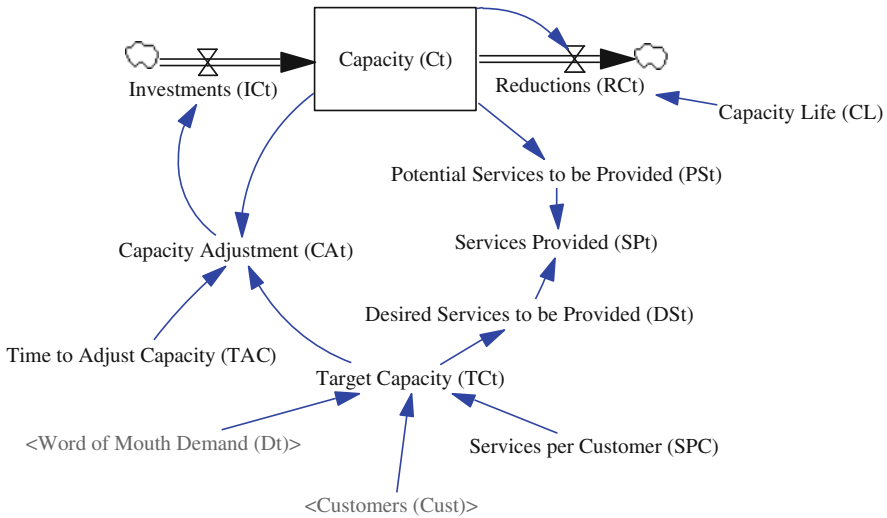


Fig. 10.3 Sample Stock and Flow Diagram (SFD)

information about the variables typology. Stocks or state variables (levels) are fundamental to generating behaviour in a system; flows or transitions variables (rates) cause stocks to change. Stock and flow diagrams contain specific symbols and components representing the structure of a system. Stocks are things that can accumulate—such as inventory, money or capacity—and are represented with boxes. Flows represent rates of change and they are expressed by decision functions—such as reductions in inventory through sales or increases in capacity through investments—and they are represented or drawn as valves. These diagrams also contain “clouds”, which represent the boundaries of the problem or system in question; auxiliary variables, etc. Systems are composed of interconnected networks of stocks and flows, including many information channels, which connect the levels to the decision functions. Modellers must be able to represent the stock and flow networks of people, material, goods, money, energy, etc. from which systems are built. Stock and flow diagrams are the most common first step in writing the executable code of a simulation System Dynamics model because they help to define types of the variables that are important in causing behaviour. Therefore we can say that stock and flow diagrams provide a bridge from conceptual modelling to assigning equations to the relationships between variables.

- Figure 10.3 depicts a very simple structure of a reservoir or level, with an inflow and an outflow. To specify the dynamic behaviour, a system of equations is defined. It consists of two types of equations, which correspond to levels and decision functions (rates). Equations control the changing interactions of a set of variables, as time advances. The continuous advance of time is broken into small intervals of equal length dt . For example the equations describing the state of the levels in Fig. 10.3 is

$$C(t) = C(t - dt) + dt \cdot [IC(t) - RC(t)] \quad (10.1)$$

$$C(t_0) = C_0 \quad (10.2)$$

- Levels in Fig. 10.3 at times t depends on its value at times $t-dt$ and the value going in from $IC(t)$ minus the value going out to $RC(t)$. Notice that it is necessary to give the initial value of it to solve this equation.
- In our next model formulation we will simplify the notation of Eqs. 10.1 and 10.2 as follows:

$$C_t = C_{t-1} + ICA_t - RCA_t \quad (10.3)$$

$$C_0 = TC \quad (10.4)$$

- There will be as many equations as variables. To determine the variables' behaviour, the differential equations system is integrated. This can be done with software that supports this and which uses different numerical integration methods.
- Sometimes, however, it has been found that SFD is a very specific tool “only for analyst” and/or “model builders”. It may create confusion when used as a general purpose tool for model building with business teams, etc. There exists some empirical research [8] showing that even highly educated people may have difficulties in discerning between stocks and flows.

An enormous value of modern system dynamic modelling tools is that they facilitate the process of capturing models of the underlying behaviour structure of organisational systems. The modelling software available on the market today greatly contributes towards achieving that objective by allowing model builders to concentrate on conceptualising the system rather than on the technicalities of model building [2].

For our case study we use one of the most popular commercial software named [11] *Vensim*[®] (Vensim is a registered trademark of Ventana Systems Inc.) which provides high rigour for writing model equations. It adds features for tracing feedback loops. In addition “Causes Tree” and “Uses Tree” features help in debugging the model. Vensim also provides very powerful tools for multiparametric simulation results optimisation which allows the analyst to validate results and model structure as well as to determine most convenient policy options by parametrising these policies.

10.3.2 Model Introduction

The dynamic simulation model that is now introduced links business growth with capacity management dynamics. We use a classical viral approach to represent a service (as a product) introduction into a new market. For the new service (product) introduction, tackling the capacity management problem plays a crucial

role. Proper capacity management allows the company to reach the customers earlier than competitors and therefore ensures a faster growth and early business development. However capacity deteriorates and new efforts (investments) are required to ensure that desired capacity is in place when needed. The speed of capacity deterioration can be reduced using proper maintenance management techniques and therefore impact on these techniques can be measured in terms of business growth.

10.3.3 Model Variables Notation and Equations

In order to characterise the relationship between market growth and capacity management we first present the notation and definition for the main variables and parameters of the model, using traditional Systems Dynamics variables typology:

- *State variables*

CSR_t	Cumulative sales revenue in time t ,
C_t	Capacity in time t ,
Cus_t	Customers in time t ,
$PCus_t$	Potential customer in t ,

- *Transition variables*

SR_t	Sales revenue in time interval t ,
CA_t	Capacity adjustment in time interval t ,
$NCus_t$	New customers in interval t ,
IC_t	Investments in new capacity in time interval t ,
RC_t	Reductions in capacity in time interval t ,

- *Auxiliary variables*

SP_t	Services provided in time t ,
DS_t	Desired services to be provided in time t ,
PS_t	Potential services to be provided in time t ,
CA_t	Capacity adjustment in time interval t ,
IS_t	Initial service sales in time interval t ,
RS_t	Regular service sales in time interval t ,
D_t	Word of mouth demand in time interval t ,
TC_t	Target capacity in time t ,
$CNcC_t$	Contacts of noncustomers with customers in interval t ,
CC_t	Contacts with customers in interval t ,
PCC_t	Potential customer concentration in t ,
$PCus_t$	Potential customer in t ,
Cus_t	Customers in time t ,
TM_t	Total market (persons) in time t ,

- *Parameters*

SPC	Average services provided per customer (number of services),
IP	Average initial price of a service (monetary units),
RP	Average regular price of a service (monetary units),
TAC	Time to adjust capacity (months),
S	Sociability (contact/person/Month),
F	Fruitfulness (person/contact),
CL	Capacity Life (Months),

- *Model state variables equations and initial values are as follows:*

We have a model that builds capacity based on investment with a construction delay, and that depreciates capacity only after the capacity life is used up. This implies that we reduce capacity when it depreciates (see Eqs. 10.3 and 10.4). We now use a very simple viral diffusion model that is now utilised to describe the introduction of a new service in the market (see Eqs. 10.5–10.8, besides Eqs. 10.11 and 10.14).

$$\text{Cus}_t = \text{Cus}_{t-1} + \text{NCus}_t \quad (10.5)$$

$$\text{Cust}_0 = 1000 \quad (10.6)$$

$$\text{PCus}_t = \text{PCus}_{t-1} - \text{NCus}_t \quad (10.7)$$

$$\text{PCust}_0 = 1e + 006 \quad (10.8)$$

In Fig. 10.4 we represent market variations as changes in two level variables (potential customers and customers) through a rate variable named new customers (NCus).

In this model we add service generation dynamics to relate the market to the firm. This will allow us to investigate the use of different strategies to respond to market demand. In Eqs. 10.9 and 10.10 we calculate cumulative sales revenue (CSR), which is also a level variable in our model. Fig. 10.5 represents the business revenue model.

$$\text{CSR}_t = \text{CSR}_{t-1} + \text{SR}_t \quad (10.9)$$

$$\text{CSR}_0 = 0 \quad (10.10)$$

- *Transition and auxiliary variables equations:*

Investments and reductions in capacity are calculated as in Eq. 10.5

$$\text{IC}_t = \text{MAX}(\text{CA}_t, 0) \quad (10.11)$$

With a capacity adjustment (CA_t) that is calculated as a function of the target capacity, current capacity and time to adjust these values. In supplying a new market we stand to loose market share (new customers and customers) because we are supplying the service later than some competitor. What can we do to supply the service (and hence bring on new customers) sooner? We can try to build Capacity

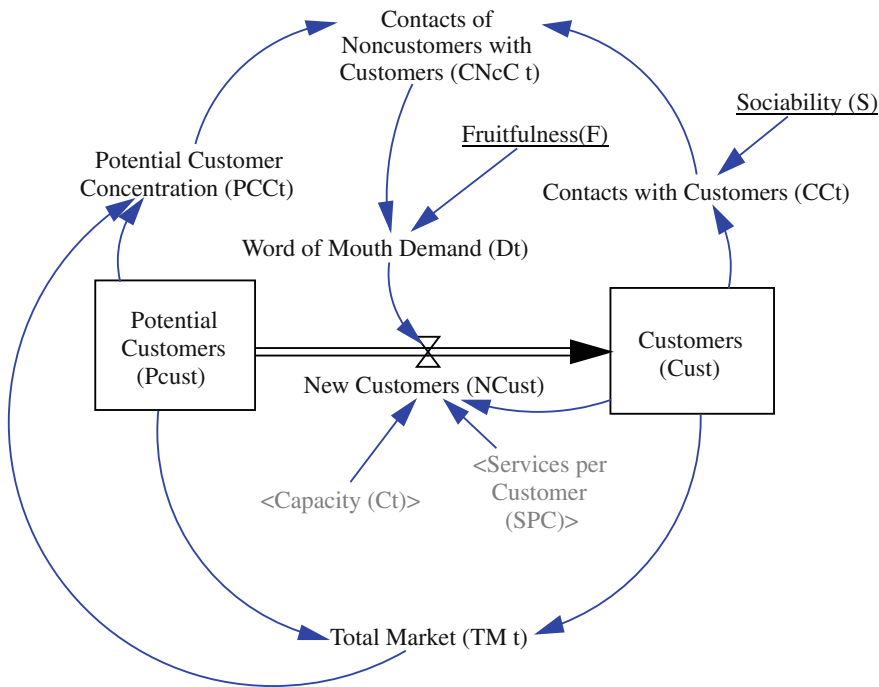


Fig. 10.4 Market Dynamics Stock and Flow Diagram (SFD)

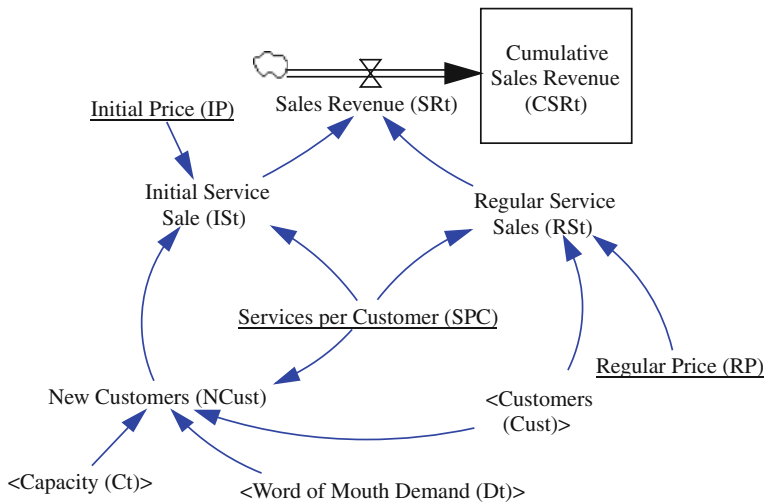


Fig. 10.5 Business revenue model (SFD)

faster by lowering the time to adjust capacity (TAC).

$$CA_t = (TC_t - C_t)/TAC \quad (10.12)$$

And a target capacity obtained as a function of the current existing demand and number of services provided by customer.

$$TC_t = (D_t + Cus_t) * SPC \quad (10.13)$$

Reductions in capacity depend on existing capacity and capacity life.

$$RC_t = CA_t/CL \quad (10.14)$$

Sales revenue for a given time period is calculated as in Eq. 10.15 as a result of new customer's sales plus existing customers sales.

$$SR_t = IS_t * IP + RS_t * RP \quad (10.15)$$

With

$$IS_t = NCus_t * ISP_t * SPC \quad (10.16)$$

And

$$RS_t = Cus_t * ISP_t * SPC \quad (10.17)$$

New customers are a result of demand generation but also of existing capacity and services provided per customer.

$$NCus_t = \text{MIN}(D_t, (C_t/SPC) - Cus_t) \quad (10.18)$$

Desired services to be provided depend on target capacity.

$$DS_t = TC_t \quad (10.19)$$

Services provided depend on demanded serviced and potential services.

$$SP_t = \text{MIN}(DS_t, PS_t) \quad (10.20)$$

With potential services to be provided equivalent to existing capacity.

$$PS_t = C_t \quad (10.21)$$

Contact of noncustomers with customers can be modelled as

$$\text{CNcC}_t = \text{CC}_t * \text{PCC}_t \quad (10.22)$$

With customer contacts depending on their socialising activities.

$$\text{CC}_t = \text{Cus}_t * S \quad (10.23)$$

And a potential customer concentration (PCC_t) depending on existing customers and total market.

$$\text{PCC}_t = \text{PCus}_t / \text{TM}_t \quad (10.24)$$

With

$$\text{TM}_t = \text{Cus}_t + \text{PCus}_t \quad (10.25)$$

Word of mouth demand depends on the fruitfulness of the noncustomers with customers contacts

$$D_t = \text{CNcC}_t * F \quad (10.26)$$

• *Model parameters initial values are:*

$$\text{IP} = 50 \quad (10.27)$$

$$\text{RP} = 10 \quad (10.28)$$

$$F = 0.01 \quad (10.29)$$

$$S = 60 \quad (10.30)$$

$$\text{SP} = 1 \quad (10.31)$$

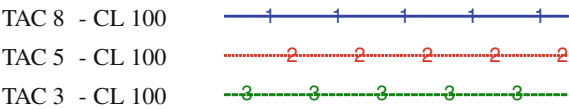
$$\text{CL} = 100 \quad (10.32)$$

$$\text{TAC} = 8 \quad (10.33)$$

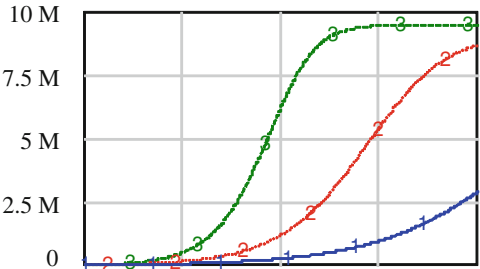
10.3.4 Model Simulation Results

In Figs. 10.6, 10.7, 10.8 and 10.9 we present simulation results for the different model state variables and most important transition variables when the Time to Adjust Capacity changes from 3 to 8 months. Results are in consonance with our

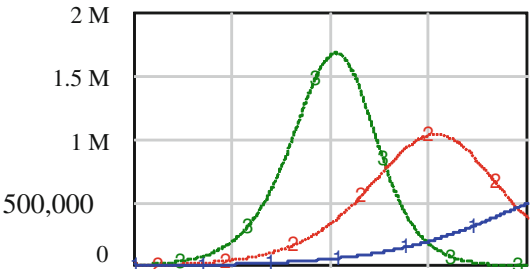
Fig. 10.6 Revenue variables for different Time to Adjust Capacity



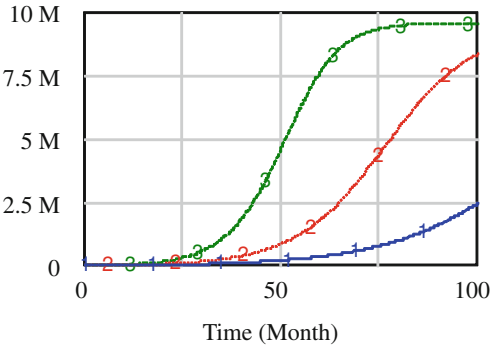
"Sales Revenue (SR t)"



"Initial Service Sale (IS t)"



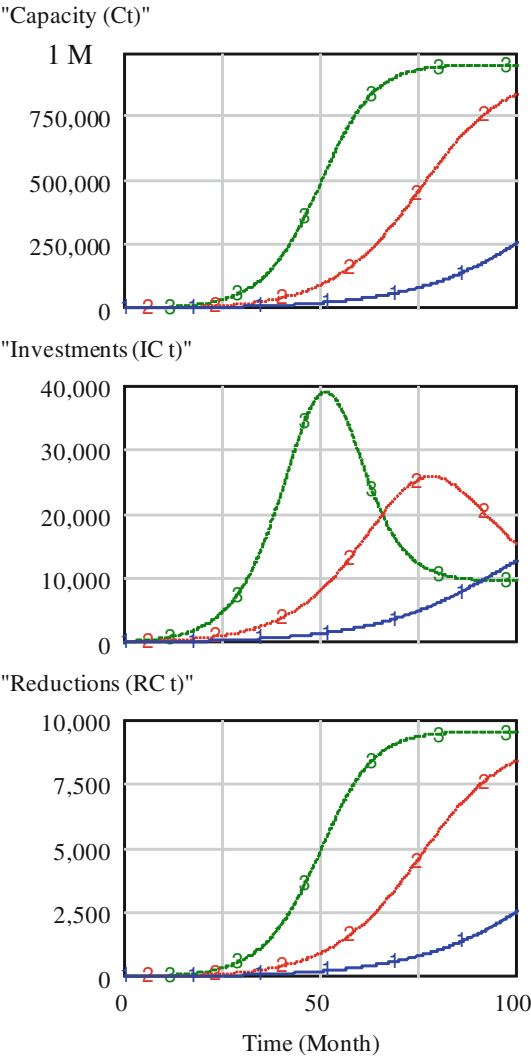
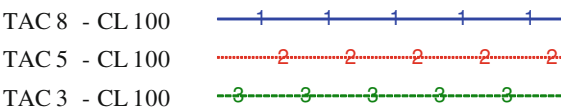
"Regular Service Sales (RS t)"



comments prior to Eq. 10.2 and show the importance of building our capacity fast, especially when approaching new markets and we have tough competition.

Sales revenue is composed by two revenue streams: Initial Service Sales and Regular Service Sales. The difference in Sales revenue after 75 months is impressive for the different time to adjust capacity (TAC), the relationship can be in a ratio 9–1 when TAC ratio goes 3–8.

Fig. 10.7 Capacity variables for different Time to Adjust Capacity



The rate of capacity investments increases with the number of new customers, and the faster we cope with that change the more new customers we can reach (Fig. 10.8), and therefore the earlier we deliver more regular service over time (Fig. 10.9).

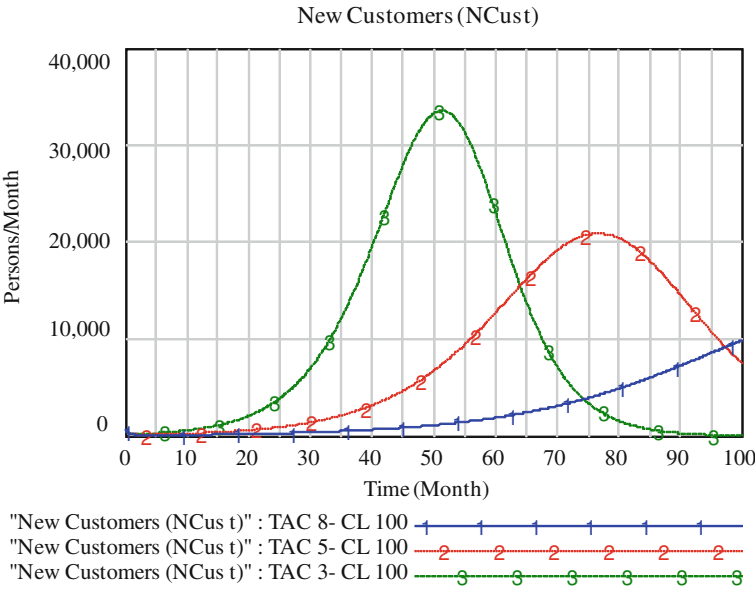


Fig. 10.8 Time to Adjust Capacity impact on New Customers

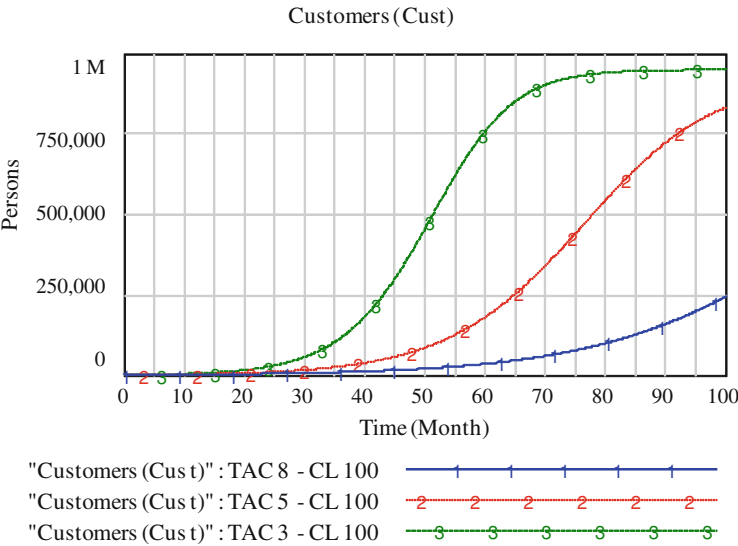
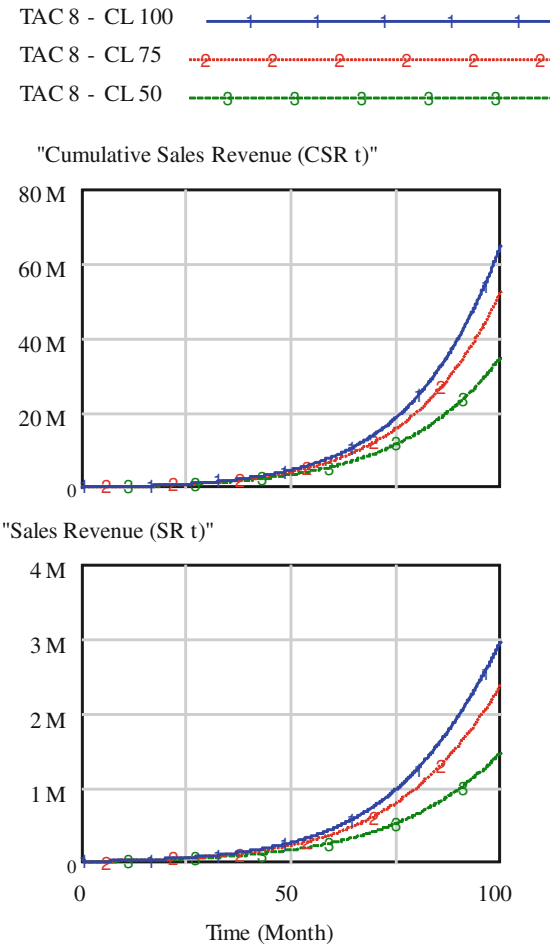


Fig. 10.9 Time to Adjust Capacity (TAC) impact on Ccustomers

Fig. 10.10 Impact of Capacity Life on model revenue variables



These graphs are all obtained assuming a constant capacity life of 100 months but that could not be the case. Capacity could be early reduced by different causes like a bad use, bad operation, of that capacity, or certainly by a bad maintenance of the network. We have observed extremely high reduction of capacity life as a consequence of improper maintenance. Network deterioration can be then very much accelerated after early stages of equipment life cycle causing enormous impact on customer acquisition, or in a general case, serious reduction in service level agreements (SLAs) and customer retention.

The question then is: to what extent may a reduction in capacity life impact business growth? What are the managerial implications of this simulation exercise? We review these points in the following Section.

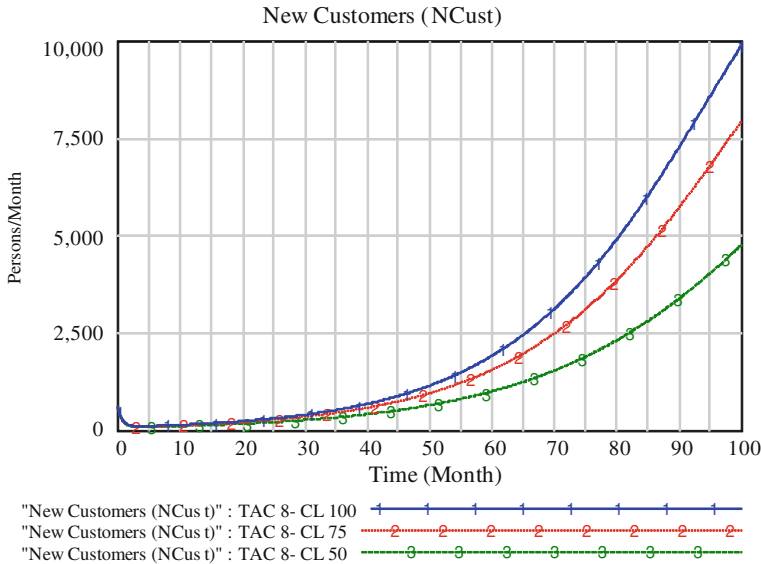


Fig. 10.11 Impact of Capacity Life on customer acquisition

10.3.5 Managerial Implications

To show this point we take the worst case of previous examples (TAC 8 months), assuming that in the other cases the impact of reductions in capacity life will even be more significant.

In our model, graphs show a nonlinear relationship between capacity life and new customers over time. A given percentage reduction in capacity life represents a greater percentage reduction in new customers acquisition (Fig. 10.10) and definitively in sales and cumulative sales revenue (Fig. 10.11).

These results also show that in order to maintain the same level of customer retention we would need a more than proportional investment in capacity over time. Maintenance is therefore critical for the business growth.

As a summary, we demonstrated that the process of adjusting capacity, due to new business needs or due to punctual short-term network saturation problems will always be extremely important for the business. Due to the importance of maintenance contribution to these processes, maintenance pays a crucial role in business growth and in customer retention.

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Chapter 11

Analyzing Maintenance

11.1 Introduction

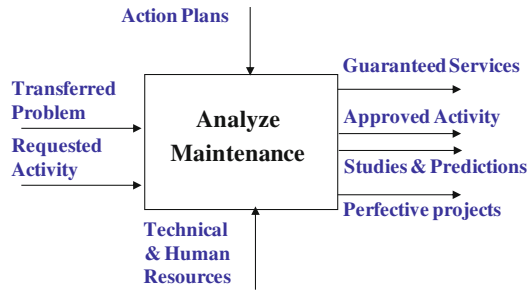
The process “To Analyse Maintenance” provides technical support to maintenance and requires the necessary expertise for the resolution of network and services problems [22]. The process allows maintenance management continuous improvement process [4, 26, 34, 52] by using prognosis, predictive and perfective maintenance (see Fig. 11.1).

The process “to analyze maintenance” includes the accomplishment of two kinds of functions:

- Support functions (for the rest of the operative processes):
 - *Managing problems.* Analysis and resolution of problems to minimize their effects based mainly on root cause analysis. Prognosis and predictive maintenance [26]. To avoid or predict potential problems and incidents.
 - *Network modifications and security administration.* The maintenance department is normally responsible for the network conditions and therefore evaluates and authorizes any action in the network.
- Optimization functions (for network quality and cost efficiency over the life cycle):
 - *Management of network and service quality and reliability levels.* This includes services dependability management through maintenance effectiveness.
 - *Management of assets life cycle cost, including logistic and obsolescence studies, searching efficiency in maintenance performance.*

It is common to find that the amount of work in this process may rise to an important number of labor hours when having ambitious scopes. Our recommendation is to begin focusing on the most critical network equipment,

Fig. 11.1 The process “To Analyse Maintenance”



representing high economic impact and frequency of failures. Once these assets are under control, the process can be extended to other less critical network items.

11.2 Problem Management

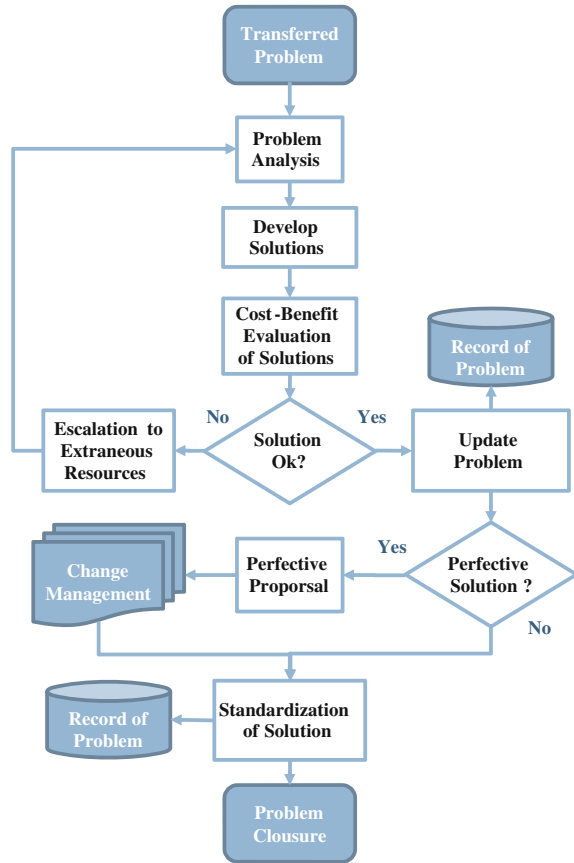
The process “to analyse maintenance” is responsible for problems management over the network life cycle, and relay on techniques such as the ones that we have reviewed in Chap. 4. Of course different incidents require different treatment, which is usually classified into three specialized levels:

- First level of care, *primary or basic*. Where we will try to solve the incident according to a corrective action plan and up to a certain basic level of knowledge or to a given target time.
- Second level of care, *secondary or specialized*. With a higher difficulty than the first level, these incidents are escalated and transferred to specialists, depending on the technology and necessary technical knowledge.
- Third level of care, *tertiary or extraneous resources*. If none of the above levels have been successful in the resolution of the incident, exceeding target time-scales, the incident will be escalated to external suppliers or internal engineering colleagues, in order to, collaboratively, study a possible solution.

The problem management process [22] (see Fig. 11.2) analyses the problems to minimize their effects based on the rule symptom-root cause-solution analysis [40], employing several techniques such as the Ishikawa diagram, 5 Whys and 2 Hows, brainstorming, etc.

The problem management work flow in Fig. 11.2 begins with the analysis of the transferred problem, developing solutions using a given methodology and with internal or external resources. Potential solutions are evaluated using cost-benefit analysis. Once the solution is approved, the problem record is updated. If the solution is perfective, requiring a considerable network modification, it has to be authorized, so it is guided to the *change management* process, outside maintenance department and from where an approved perfective activity will be generated. This activity will follow the standardized process within the organization: will be

Fig. 11.2 Problem management process work flow



scheduled, implemented and its implementation reviewed according to the standards. These perfective proposals can also be called network improvements, updates and/or modernization activities (Monchy [42]).

Inside the first step of the problem management flow, problem analysis, we may generally include the following phases to establish the basis to develop possible solutions [33]:

1. List the symptoms of the problem.
2. Analyze symptoms.
3. Identify possible causes.
4. Formulate theories.
5. Test theories.
6. Identify root causes (physical, human and latent).
7. Evaluate alternatives.
8. Design controls.
9. Implement.
10. Verify performance.

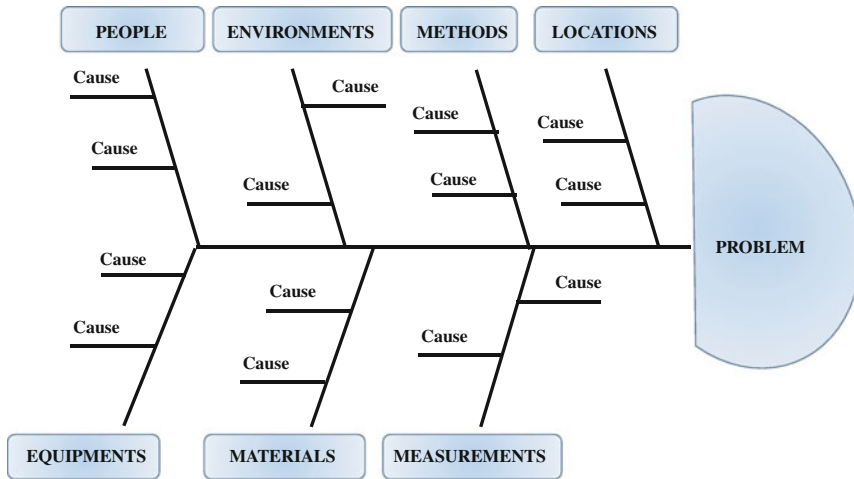


Fig. 11.3 Fishbone diagram

We can find different approaches to obtain the root cause of a problem, another good practice can be the employment of a descriptive technique as 5W2H [27] together with a graphical technique such as Ishikawa diagram [20]. With seven questions we may find out the possible factors leading to the observed problem effects. Causes related to these factors can then be identified and presented in a graph (see Fig. 11.3). These seven questions are:

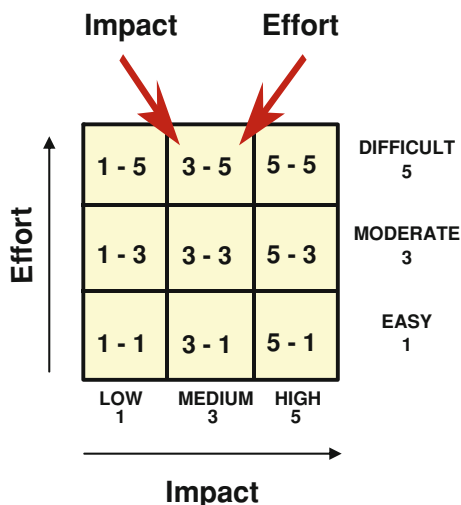
- What? What happens...? About the problem.
- Who? Who is involved...? About people.
- When? When does it happen...? About timing.
- Where? Where is...? About location and equipments.
- Why? Why do...? About the causes.
- How? How is it...? About the methods, measurements, procedures and environment.
- How much? How much does it cost? About measurements, quantities of supplies and the cost of the effects.

It is a good practice to review periodically at least those incidents whose frequencies and consequences are critical (in terms of service impact or in terms of cost). Notice that, although this seems to be a really straight forward procedure, it has been found that only one-third of companies perform some type of root cause failure analysis on equipment problems [55].

The root cause failure analysis method that we propose here should include the following basic steps [11]:

1. Select the team responsible of the failure analysis;
2. Identify the problems;

Fig. 11.4 The problem priority matrix. Adapted from Crespo [11]



3. Determine the significance (impact) of the problems and estimate the expected effort of subsequent steps of your analysis for each problem. Plot results in the priority matrix, defined as in Fig. 11.4. Start the analysis for those problems with highest significance and less expected effort to be solved;
4. Identify the physical cause of the problem. Formulate hypotheses, i.e. the technical explanations on why things failed according to the physical evidence that was found;
5. Determine, for each one of the validated hypotheses, the physical, human and latent root causes of the problem. Use or follow a root cause failure analysis logic tree as in Fig. 11.5;
6. Propose recommendations, corrective actions, that can minimize, eliminate or mitigate the consequences of the failure events. Solutions should be sustained with any cost-benefit analysis;
7. Determine whether corrective actions have been effective in resolving problems. Corrective actions should be tracked to ensure that they have been properly implemented and are functioning as intended. The recurrence of the same or similar events must be identified and analyzed. If an occurrence reoccurs, the original occurrence should be reevaluated to determine why corrective actions were not effective.

Causes of failure can be classified as physical, human or latent [33], see their definitions in Sect. 4.3.

Good practices when doing this type of analysis are the followings:

- Try to capture physical causes of failures as *failure modes* in your CMMS. By doing so an important part of the analysis is accomplished based on failure evidences recorded by the maintenance technicians. The physical cause has to

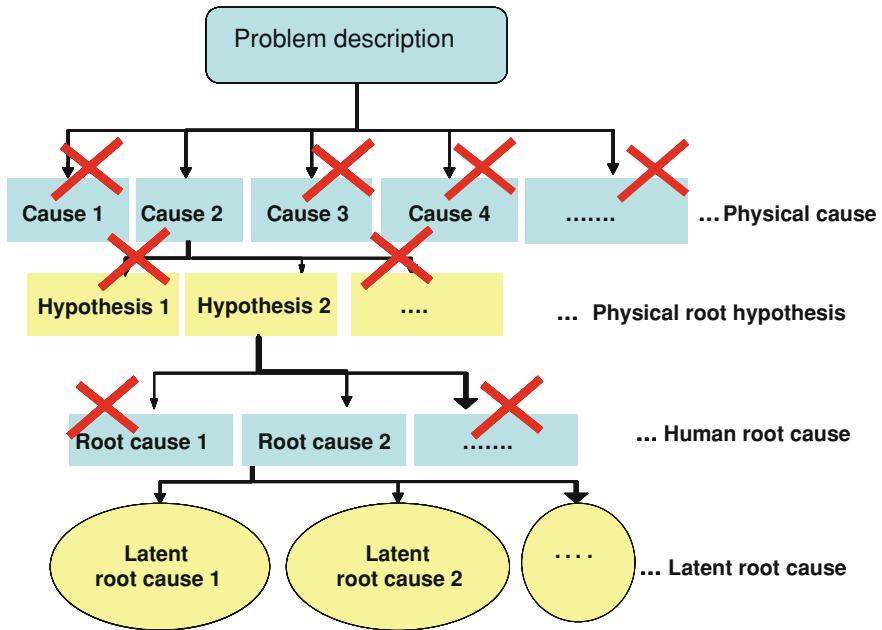


Fig. 11.5 The root cause failure analysis logic tree. Adapted from Crespo [11]

be therefore recorded by the technicians in their day-to-day work at the time of corrective work-order compilation.

- Discuss possible physical root cause of the failure with your team, select most likely hypothesis based on team members experience and test/validate these hypotheses within your operational environment.
- When validating hypothesis try:
 - not to be conditioned by repetitive comments or behaviors (for instance, “...Everybody says security is priority # 1, but we all know cost is priority # 1”);
 - to avoid observations including people beliefs (for example, “...a bypass of ‘upsetting’ alarms had to be done...”);
 - to avoid observations regarding people possibilities at work (for instance, “...when we practice we simply standardize...but we cannot improve the task”);
 - to avoid a bad working environment (for example, excessive leaks, lack of cleaning, excess of oxidation, etc.)
- When validating hypothesis, the following technical information should be collected:
 - Operational variables values (from the Distributed Control System—DCS)
 - Maintenance history records

- Shift logging data (events per shift)
 - Inspection results (visual, NDT/NDA, etc.)
 - Lab results (chemical, metallurgical)
 - Vibration data and specs
 - Purchasing information
 - Maintenance procedures
 - Operational procedures
 - Design data and modifications
 - Personnel training.
- When validating hypothesis, the following people should/could be interviewed:
 - Observers
 - Qualified maintenance technicians
 - Operators
 - Electricity and instrumentation technicians
 - Engineers
 - Suppliers and vendors/buyers
 - Manufacturers (of parts and OEMs)
 - Other similar process departments
 - Deposit and reception personnel
 - Purchasing agents
 - Security personnel
 - Quality personnel
 - External experts.
 - Necessary recommendations to solve the physical, human and latent root causes, fully resolving the contribution of “that branch” to the occurrence of the failure. This kind of recommendations should be the most common one. Examples:
 - Insufficient torque; solution: applying proper torque
 - Non-compliance procedure, lack of training, lack of diffusion; solution: train person, disseminate procedure, talks.
 - Use simple techniques to measure risk and to do risk-cost-benefit analysis (See Fig. 11.6 and subsequent analysis of two possible solutions in Tables 11.1, 11.2, 11.3, 11.4, 11.5).

Table 11.1 describes current situation of the seal problem in Fig. 11.6 in terms of costs and failure frequency. Later, Tables 11.2 and 11.3 present two proposed options considering different seals from two suppliers and with different characteristics. Total annualize risks are calculated for both options and a final risk-cost-benefit analysis is presented in Table 11.4. Finally, the selected option is compared to current situation.

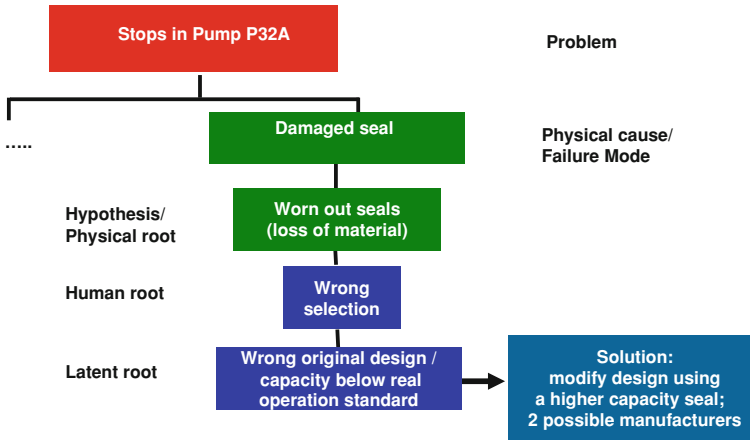


Fig. 11.6 Sample solution identification process

Table 11.1 Current situation risk analysis for problem in Fig. 11.6

Stops in pump P32 A			
Event: mechanical seals failure			
	Min	Max	
1. Failure frequency	9	13	F/y
2. Labor cost	1	2	M\$
3. Materials cost	10	15	M\$
4. Annual repair cost $(2 + 3) \times (1)$	99	221	M\$/y
5. Repair time	8	16	h
6. Production impact	1	1.5	M\$/h
7. Event penalization (5×6)	8	24	M\$
8. Annual penalization $\times$ failure (7×1)	72	312	M\$/y
Total annualized risk: $(4) + (8)$	171	533	M\$/y

11.3 Predictive Maintenance for Proactive Action

Predictive maintenance has to be focused on critical failure modes and those variables/parameters that can offer information about their potential presence. These variables will help us to control instability as much in services and resources as in management issues as the budget [4, 26, 34, 52]. The aim of predictive maintenance is to produce patterns of behavior for all the processes allowing us to act in anticipation facilitating the decision making. Some patterns could be developed to be employed in real time and others under demand or with a determined periodicity depending on the cost effectiveness analysis. In any case, the patterns have to be reviewed periodically to be properly tuned using the same

Table 11.2 Option 1 risk analysis for problem in Fig. 11.6

Stops in pump P32 A			
Proposed solution: replace current seals			
Option 1: Seal A			
	Min	Max	
1. Failure frequency	1	2	F/y
2. Labor cost	1	2	M\$
3. Materials cost	15	22	M\$
4. Annual repair cost $(2 + 3) \times (1)$	16	44	M\$/y
5. Repair time	8	16	h
6. Production impact	1	1.5	M\$/h
7. Event penalization (5×6)	8	24	M\$
8. Annual penalization $\times$ failure (7×1)	8	48	M\$/y
Total annualized risk: $(4) + (8)$	24	92	M\$/y

Table 11.3 Option 2 risk analysis for problem in Fig. 11.6

Stops in pump P32 A			
Proposed solution: replace current seals			
Option 2: Seal B			
	Min	Max	
1. Failure frequency	0.5	1	F/y
2. Labor cost	1	2	M\$
3. Materials cost	25	30	M\$
4. Annual repair cost $(2 + 3) \times (1)$	13	32	M\$/y
5. Repair time	8	6	h
6. Production impact	1	1.5	M\$/h
7. Event penalization (5×6)	8	24	M\$
8. Annual penalization $\times$ failure (7×1)	4	24	M\$/y
Total annualized risk: $(4) + (8)$	17	56	M\$/y

or different statistical methods and including additional information or knowledge from recent experiences. These predictions and studies reinforce the decision making directly in the processes “to manage strategy”, “monitoring”, “to prevent incidents” and “to manage capacity”. Predictive maintenance also can be used together with statistical analysis in some cases [16]:

- Problems with large amounts of information, such as equipment reliability studies.
- Problems requiring rapid decision on critical incidents.
- Problems requiring human reasoning simulation.

Among the main advantages of employing a predictive maintenance program, Mobley [39] describes the following:

Table 11.4 Risk-cost-benefit analysis of both options

Stops in pump P32 A			
Suggested solutions: replace actual seal			
Options 1 and 2:			
	Min	Max	
Scenario after the replacement:			
	Seal A:		
Repair annual cost	16	44	M\$/y
Annual penalization $\times$ failure	8	48	M\$/y
Total expected risk	24	92	M\$/y
	Seal B:		
Repair annual cost	13	32	M\$/y
Annual penalization $\times$ failure	4	24	M\$/y
Total expected risk	17	56	M\$/y

Table 11.5 Risk-cost-benefit analysis current versus selected option

Stops in pump P32 A			
Proposed solution: Replace current seal with Seal B			
	Min	Max	
Current situation:			
Annual repair cost	99	221	M\$/y
Annual penalization $\times$ failures	72	312	M\$/y
Total risk:	171	533	M\$/y
Future situation:			
	Seal B:		
Annual repair cost	13	32	M\$/y
Annual penalization $\times$ failures	4	24	M\$/y
Total expected risk:	17	56	M\$/y
Benefits of the change:			
Repair cost	86	189	M\$/y
Penalization due to failures	68	288	M\$/y
Total benefits:	154	477	M\$/y

- It increases overall profits, process efficiency, product quality, production capacity and the operating asset life cycle. It should generate a return on investment that could be between the range 10:1 and 12:1.
- Elimination of unnecessary maintenance. Based on his survey, the actual costs normally associated with the maintenance operation are reduced by more than 50%.
- Reduction of costs caused by failures (lost production, repair parts inventory, etc.). It could reduce the number of catastrophic, unexpected machine failures by an average of 55%.
- Reduction of actual time required to repair or rebuild plant equipment. The average improvement in mean-time-to-repair (MTTR) was a reduction of 60%.

However, predictive maintenance approaches have received several criticisms due to their complexity when we are in the following scenarios [32, 45]:

- The system contains a large number of sub-systems and components. This case generates a wide variety of maintenance situations that can be handled using different models and methods. Most of the analysis are conducted at a single equipment level, and no infrastructure exists for employing a real-time remote machinery diagnosis and prognosis system for maintenance.
- There is no expertise regarding modeling and statistical techniques. A generic and scalable prognostic methodology or toolbox does not exist. Currently, methods are focused on solving the failure prediction problem. The need for tools for system performance assessment and degradation prediction has not been well addressed.
- The changeable nature of large technical systems will present constant challenges and many developed prediction algorithms have been demonstrated in a laboratory environment, but are still without industry validation. All these difficulties highlight the need to develop special computerized systems that can cope with the management of complex engineering systems.

Once the employment of predictive maintenance techniques is decided, the development and optimization of the behavior patterns will be done at three different levels (see Fig. 11.7).

1. In the first level, the objectives and scope of the analysis are defined, delimiting the inspections and data collection of information as basis of the next level.
2. The second level incorporates the expert knowledge including appropriate criteria and recognizing possible patterns. If there is a possible pattern, it will be modeled taking into account whatever useful correlation. Later, the pattern is documented and represented with the intention to make the knowledge explicit and to facilitate its simulation and verification according to the previous objectives and the historical usage and experience. In case of a negative verification the flow is guided toward an adjustment program where the causes of this negative result are analyzed and documented in order to be considered in a new first level as refinement. When there is not any possible pattern the flow finishes.
3. The third level chases the generation of new knowledge in the organization based on the verified and explicit pattern, which will be standardized and studied. In the case of potential serious repercussions during the implementation, the flow is redirected toward the first level allowing a modification; otherwise the implementation is carried out searching for the process automation. Automation is developed in proportion to the maximum level of intelligence, as a support system or expert system. Finally, in addition to the produced prediction and studies, implementation activities could be generated internally or externally as a demanded perfective proposal and depending on the scope of the changes. After the implementation this process has to be reviewed again over time updating information for a sustainable future.

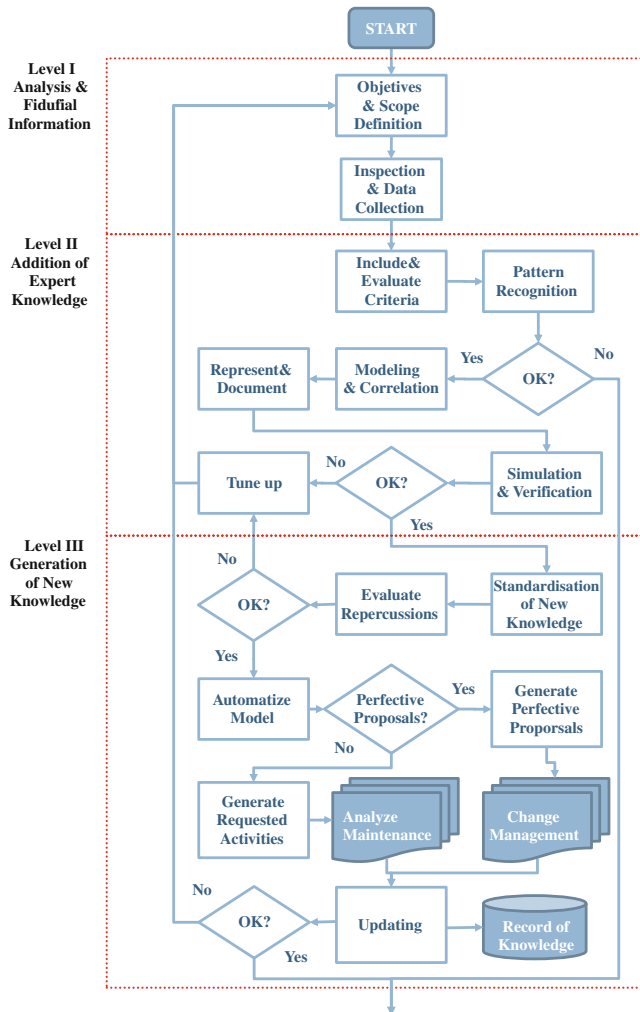


Fig. 11.7 Pattern development and optimization flow

This function will increase reliability with experience, and will evaluate reliability with evidence. Therefore, this workflow fulfills the levels indicated by Russell and Norvig [48] for decision making seeking to minimize the error with the maximum possible rationality, that is to say, to establish the rules for decision making based on expert knowledge to choose the best decision according to the specific circumstances in a given environment. As a result of suitable implementation of predictive maintenance programs we will:

- Have a better problem analysis and identification, providing documented specifications and useful information.
- Collect and incorporate expert knowledge (tacit and explicit), similar to human reasoning, incorporated after consensus which increases satisfaction and motivation.
- Have better analysis consistency with expert knowledge, improving quality and applicability of decisions, in a way that risks decrease.
- Generate new knowledge and refinement of the rules. Redesign the models to increase the value of decisions as continuous improvement.
- Automate the decision making. By automation we improve the responsiveness, we reduce complexity, costs and errors in the processes and also the information is continuously updated, thereby the quality of decision making increases. But we must be careful with our software. Remember that *instead of worrying about whether a particular machine can be intelligent, it is far more important to make a piece of software that is intelligent* [49].

In today's global business environment, due to the developed level of integrated network systems and the synchronization of information and decisions, it is critical the need of a real-time remote machinery prognostics and health management (R2M-PHM) system, as Pintelon and Parodi-Herz [45] indicate distinguishing the following categories:

- *Machine intelligence*. Intelligent monitoring, predict and prevent, and compensation, reconfiguration for sustainability (self-maintenance).
- *Operations intelligence*. Prioritize, optimize and responsive maintenance scheduling for reconfiguration needs.
- *Synchronization intelligence*. Autonomous information flow from market demand to factory asset utilization.

11.3.1 Walking the Path to Proactive Risk Assessment: A Case Study

This case study is concerned with predictive maintenance techniques applied to network repairable assets, helping to model the online reliability and risk of these assets over time.

We focus on physical causes of failures as a consequence of: normal wear during their operating time; resulting states after maintenance intervention could modify—we are concerned about reductions—the useful life of the assets; and the operating environment, since environment and geographical characteristics of the territory could have a great influence on the equipment deterioration.

The purpose of the study that we present here was to deduce a practical method, for computational purposes, offering real-time reliability estimations according to changes in environmental factors, and in cumulative failures/repairs impact on the

asset, over time. Also we wanted to offer a practical vision of how this can be linked to risk when approaching the maintenance scheduling decision-making process. To illustrate this, in the sequel we first define a reliability function building process, and then the methodology to apply that function to estimate the online risk of a failure mode and to schedule specific PM activities later. Finally some conclusions will be presented.

Different techniques could be utilized to obtain a failure pdf and the corresponding impact of maintenance effectiveness on that pdf, but in order to simplify, in this case study we will consider exponential distributions for failures (this is suitable according to the type of electrical equipment considered, and the data provided by the network utility, that basically included number of failures, for a given failure mode and for a given timeframe, without capturing operating times to failure of the asset), and we will measure the maintenance interventions impact considering a conservative perspective of a situation between “as good as new” and “as bad as before” [28] considering that there are possible effects of maintenance that will never be overcome after activities on the equipment.

$$R(t) = e^{-\lambda t}; \quad h(t) = \lambda$$

Therefore we search for a relationship representing the reduction of reliability per accumulated maintenance activity on the equipment. We assume in this work that the decrease in reliability is higher with the maintenance event incidents and that this effect becomes relatively less important when the number of maintenance tasks increases (this, of course, can be very different for each failure mode that is modeled). In the case study we assume that reliability is reduced in a fix percentage (for this we use the parameter p) each time the number of PMs is duplicated:

$$T_n = t \times p^{\ln(n+1)/\ln 2} \quad (11.1)$$

With T_n : virtual operating time after n failures and for a given failure mode; t : real equipment operating time for the failure mode and n : cumulative number of failures per failure mode. With this we assume that reliability reductions follow an exponential behavior. Values for p can vary from 1 to 0.5, assuming that $p = 1$ represents no reliability reduction (see Fig. 11.8). The estimated reliability for considering the two factors mentioned above could be expressed as follows:

$$R(T_n) = e^{-\lambda T_n} = e^{-\lambda t \cdot p^{\ln(n+1)/\ln 2}} \quad (11.2)$$

Conditions of different equipment parameters could accelerate asset degradation, for example, temperature or humidity. This is important for network utilities where calculating the reliability over a specific number of equipment and for a specified amount of time, should consider different environmental conditions such as dust, humidity or temperature. Assuming that maintenance procedures are applied similarly in all of geographic areas, we could use the concept of “location or environment related obsolescence” to represent the specific negative effect of these factors (see [37], applying also this definition to buildings and infrastructure assets).

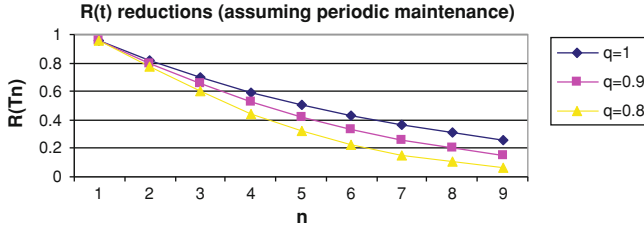


Fig. 11.8 Example of reliability reductions for periodic maintenance interventions

The influence of these environmental factors may obey different patterns. We will consider that the unique effect of a unit increase in an environmental factor (covariate) is multiplicative with respect to the failure rate. To simplify, we will also consider that the effect of an abnormal situation in a given equipment condition (as a consequence of the asset deterioration) will also produce a similar multiplicative effect on reliability (the reader is referred to [54], for the discussion about a more precise considerations of environmental monitored parameters vs. failure consequence monitored parameters). This is the case of Cox's proportional hazard model (PHM, published in 1972), consequently we can express the hazard rate as follows:

$$h(t, x) = h_0(t) \cdot e^{\sum_{j=1}^k \gamma_j \cdot x_j} = \gamma_1 \cdot x_1 + \dots + \gamma_k \cdot x_k \quad (11.3)$$

where: $h(t)$ is the failure rate or conditional probability of failure at time t ; x_i are the environmental factors (with $i = 1 \dots k$) or covariates contributing to $h(t)$; and γ_i are constant coefficients (with $i = 1 \dots k$) representing each x_i contribution. Then:

$$R(t, x) = R_0(t) \exp \left[\sum_{j=1}^k \gamma_j \cdot x_j \right] \quad (11.4)$$

If we now consider:

$$R_0(T_n) = e^{-\lambda T_n} = e^{-\lambda t \cdot p^{\ln(n+1)/\ln 2}} \quad (11.5)$$

Then we obtain:

$$R(T_n, x) = \left[e^{-\lambda t \cdot p^{\ln(n+1)/\ln 2}} \right] \exp \left[\sum_{j=1}^k \gamma_j \cdot x_j \right] \quad (11.6)$$

A partial likelihood function (Quasi-Likelihood Model, QLM) can be used, as recommended by Cox in order to estimate the γ_i coefficients, taking into account only the samples where a failure event occurred. However, for the initial implementation of this case study, we adopted a more empirical and practical approach to speed up the coefficient determination process. A maintenance supervisor's knowledge elicitation process was accomplished following a conservative approach (See the tool used in Fig. 11.9 which was developed using Excelsius, ® SAP Business Object).

This conservative approach means that function loss is considered to be produced [$h(t, x) = 1$] when monitoring parameters (or a combination of them) were reaching well known limit/alarm thresholds values, as agreed with the methodology review team. The monitored parameter values will change and the covariates will also change. We pay special attention to the covariates value, instead of their value trend; we do not use historical data and therefore the computational process of reliability estimation is safe and easy. Therefore, in our case study the risk values are recalculated over time on a permanent basis. The more the covariates (functions of the parameter values) separate from their initial or ideal conditions, the higher the risk changes recalculated for that failure mode.

In order to reduce the computing needs for the network (for the different failure modes of the different equipment of a large network) the following methodology was implemented:

- a. Estimation of the failure rate (λ) for a failure mode. This was done using a database where failures (for each failure mode) are registered over a 4 years timeframe and for similar equipment “classes” (for instance 2500 similar transformers and remember that Weibull with $\beta = 1$ is equal to exponential distribution), without the consideration of environmental factors influence.
- b. Determination of the failure mode criticality (priority) assessing consequences of the failure mode for the network (interruption time and number of customers affected).

Consider the following notation:

- C: Failure mode criticality
 Nfe: Number of failure events of the failure mode
 Ta: Time of affection of the failure mode to a customer
 Nnc: Average number of normal customers affected by the failure mode
 Nsc: Average number of singular customers affected by the failure mode
 Mnc: Maximum numbers of normal customers affected by a failure mode
 Msc: Maximum numbers of singular customers affected by a failure mode
 wnc: Weight of the normal customer
 wsc: Weight of the singular customer.

Then for this particular case study, the criticality of the failure mode is obtained according to the following algorithm:

$$C = Nfe \times Ta \times \left[wnc \left(\frac{Nnc}{Mnc} \right) + wsc \left(\frac{Nsc}{Msc} \right) \right]$$

- c. Determination of the maintenance impact as a percentage (p) and for the failure mode considered, agreed with the review team ($p = 0.9$ in this case).
- d. Selection of parameters, among those monitored, offering proper information regarding the presence of a potential failure (for the failure mode under analysis). The different variables/parameters required for the diagnosis of the failure mode are included, although sometimes a clear early detection of the potential failure is

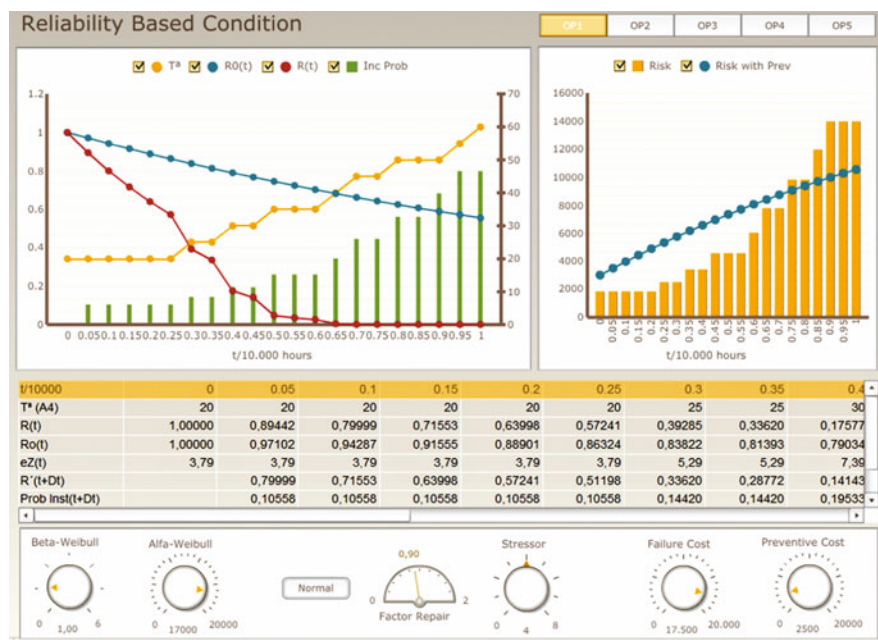


Fig. 11.9 Illustration of the tool used to estimate coefficients and to visualize risk-based maintenance strategies

not possible, or maybe, when possible, this detection cannot be unique for this failure mode. When doing the analysis, the team discovered that some of the parameters were not physically available at that time and therefore they could not be introduced in the expression for the reliability that would be used. Unless otherwise specified, we initially assumed that all the covariates are independent. The influential variables for this failure mode (lack of outflow) are:
Available variables taken into consideration and used in the algorithm:

- A1 (Upper oil layer temperature)
 - A2 (Lower oil layer temperature)
 - A6 (Load current intensity)
- Parameters not available (N/A) at the time of the analysis:
- A3 (Air temperature)
 - A9 (Oil level)
 - A10 (lack of fans' feed)
- e. Elicitation of the parameters coefficients with the review team (see tool in Fig. 11.9).
For each one of the covariates the team had to find out the corresponding coefficients.
- f. Determination of $R(t)$ as in Eq. 11.6.

g. Determination of the online failure mode risk associated to the asset, as follows:

$$\text{Risk}(t, t + \Delta t) = \frac{R(t) - R(t + \Delta t)}{R(t)} C \quad (11.7)$$

where $R(t + \Delta t)$ is the estimated reliability in time $(t + \Delta t)$ assuming no variation in the value of the monitored parameters for the time interval $(t, t + \Delta t)$, and C are the consequences of the failure (direct and indirect consequences of the failure, including customer affection valuation) expressed in economic terms.

h. Determination of the criteria governing the PM scheduling according to risk. In this case study, the team decided that the value obtained for the online risk would be compared with the cost of an alternative strategy of preventively maintaining the equipment for that mode of failure. In order to estimate the cost of this strategy, equipment PM would represent a certain cost, then the equipment would be restored to a condition (that in this case study was the one resulting when updating only covariates values in the reliability function), and later that risk can be again calculated for the new reliability function. When the online cost of the risk of doing nothing exceeds the alternative PM strategy to a certain extent, then PM maintenance is scheduled and accomplished. Decisions are therefore taken based on risk and online.

This methodology was implemented in two important systems of the network, generating online values for risk of every failure mode of these systems. The maintenance department could afterwards schedule preventive maintenance activities on these systems when it was proven to be cost effective (risk effective) compared to “doing nothing” during a certain period.

11.4 Network's Changes Administration and Security

“Changes administration” refers to controlling the implementation of changes in the network, while “changes management” is normally a more general process outside the maintenance boundary and scope, involving many network utility departments, with the intention of controlling and authorizing changes in the network processes and systems. Therefore, when we refer to *network's change management* or to *the management of network's change*, we refer to substantial and perfective changes requiring the analysis, agreement and authorization from other departments, such as new technologies, new systems, etc.

In network utilities, due to the large amount of dispersed elements, information and sites, there has to be a unique organization responsible for the network security. This organization will give the access to the network operative information and sites. This will be a must to avoid any possible interference in services and/or in maintenance activities. A good practice is to place this responsibility in maintenance departments, because the requests for action should be dealt with

efficiency and consistency according to their priority, evaluating and communicating their total impact to all interested parties before implementation. Then the scope of this function encompasses as much the security of services as the operation information security.

If we now look at the work flow for this process, this starts when a requested activity (see Fig. 11.10) is checked previously to its scheduling and prioritising. The need for a more extensive authorization, launching the process “to Change Management” is also checked. The requesting agent of the activity can be inside or outside the maintenance organization and authorization procedures may be different. Once change management authorization and procedures are solved or are needless, the following step is to analyze the activity impact in the operation and also to check the existence of the compulsory reverse plan. Once the activity is approved, the coordination of its execution is developed by the process “to analyze maintenance”. This process will ensure that the documentation produced is recorded, regardless of the internal or external execution, and therefore that the organization learns from the operation impact analysis and from the evaluation of the coordination needs. The proper implementation of an approved activity could be carried out by the maintenance organization in field and/or remotely or by third parties (in this case the supervision of the implementation, in field or remotely, will be required).

Examples of activities launching this process are: requested changes of configuration in monitoring processes (maintenance department), requested changes in the network (because of the Continuity of Operations Plan), requested inspections in the field (third party), changes in the network to solve a problem of capacity saturation (operations department).

11.5 Management of Quality and Reliability Assurance

Inside the process *to analyze maintenance*, the management of quality and reliability levels of the network and service is accomplished. This is done in accordance with the established strategic criteria. This function extends, across all maintenance operations, the quest of effectiveness in performance in order to obtain the right network conservation and services quality, in the right time.

As a result, reliability centred maintenance (RCM) analysis is crucial for this function. The methodology is now very much oriented to capture customers' affection as a consequence of failures. In this sense, reliability and quality are strongly linked in network utilities, and this relationship may derive in recommendations for internal failures prevention, but also (as we can see when we develop a Continuity of Operations Plan) in guidelines showing how to react in case of the occurrence of critical emergencies, internal or external attacks or natural disasters.

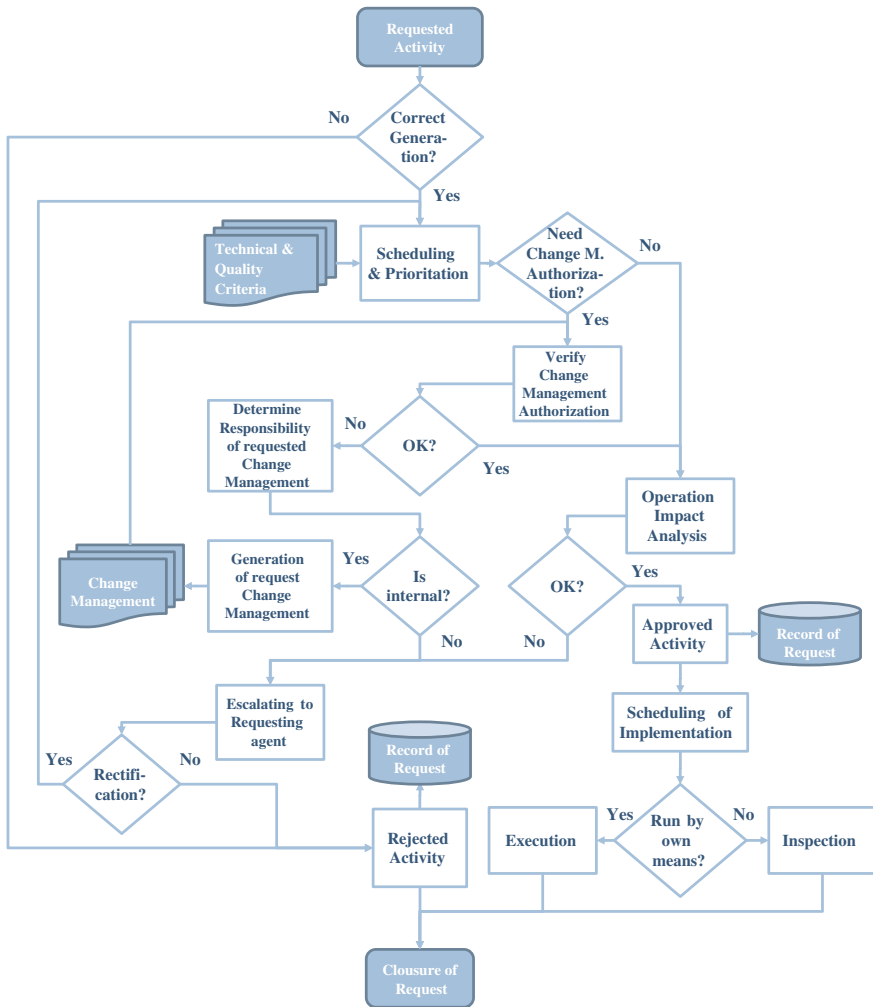


Fig. 11.10 Management of changes and operation security flow

A basic work flow of the RCM analysis is presented in Fig. 11.11 [3, 9, 46], and includes the following steps:

1. System selection and boundary definition determining the rules and constraints that underlie the analysis.
2. Inspection and collection of information.
3. System description and functional or geographical block diagram into sub-systems and items, identifying operational and environmental conditions which may affect the system, subsystems and individual items.
4. System functions and functional failure analysis (FFA).

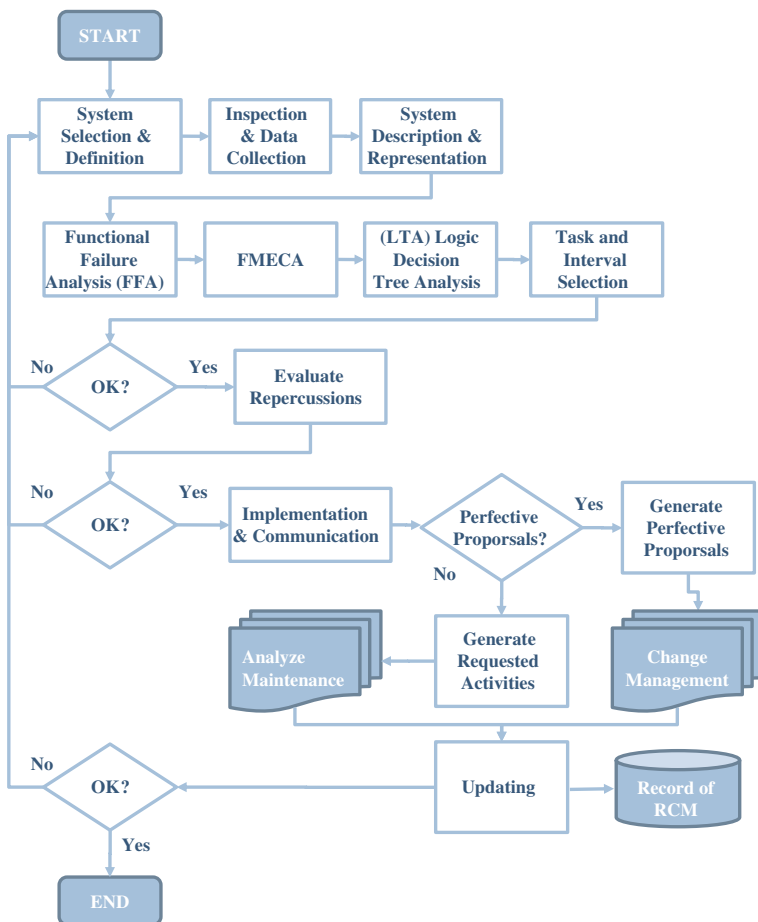


Fig. 11.11 Reliability analysis flow

5. Determination of the failure modes, their effects and estimation of their probability (frequency) of occurrence and criticality analysis according to their consequences and frequency (FMECA).
6. Logic decision tree analysis (LTA) about selection of maintenance actions.
7. Task selection and determination of maintenance intervals.
8. Evaluation of repercussions depending on the profitability of the selected task.
9. Implementation of the maintenance actions.
10. Updating with in-service data collection.

A generic process for the logic selection of the maintenance actions to correct or prevent the occurrence of failure modes could be represented as in the Fig. 11.12. For other more specific maintenance actions addressing hidden failures modes the reader is referred to Moubray [43].

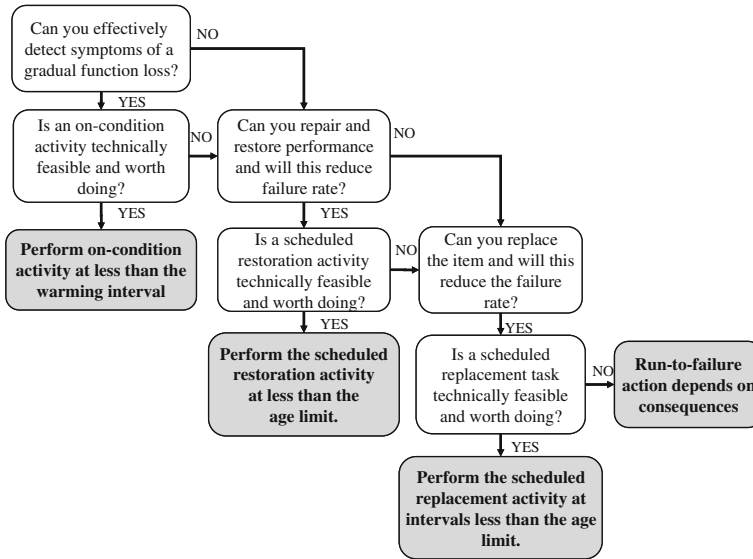


Fig. 11.12 Maintenance activity selection through a logic decision tree

11.6 Managing Continuity and Availability of Services

Although in some companies this responsibility may fall on other departments, implementing the continuity of operation plan (COOP) becomes frequently part of the maintenance department's mission in network utilities. Regarded as an internal procedure, the COOP describes how the rest of the processes should act in case of critical emergencies. Notice that network utilities are normally considered as critical infrastructures [6] due to the high impact of their service disruptions on society. Let us then dedicate some time to study this procedure, look at its main features and review required maintenance department's involvement to implement it.

The COOP has to be integrated inside the business continuity plan (BCP), which develops actions, communications and procedures to maintain the business functions before adverse situations or disasters such as critical disruptions or diminishments by major failures, natural disaster or attacks, including description of contingencies and, disaster recovery or health and safety measures. These terms are frequently used in ICT sector, although their importance is crucial for all types of infrastructures or resources in order to provide an adequate level of availability of them and their supported services.

According to Conrad et al. [8] and Snedaker [51], the classification of adverse situations or disasters can be made according to their causes, and could be adapted considering also human causes, as follows:

- Natural disasters, when they occur naturally in a localized area, such as:
 - *Cold weather-related hazards*. Avalanche, severe snow, ice storm, hail storm, severe or prolonged wind.
 - *Warm weather-related hazards*. Severe or prolonged rain, heavy rain and/or flooding, floods, drought (can impact urban, rural and agricultural areas), fire, tropical storms, hurricanes, cyclones, typhoons, tornado or wind storm.
 - *Geological hazards*. Earthquake, tsunami, volcanic eruption, landslide and land shifting.
- Human disasters, when the cause is provoked by humans intentional or unintentionally such as:
 - *Terrorism*. Bombs, armed attacks, hazardous material release (biohazard, radioactive), biological attack (air, water, food).
 - *Fire and explosion*. Explosive device found, arson or accidental.
 - *Meddling*. Threat or boasting, minor intrusion, major intrusion, total outage, broader network infrastructure impaired, malware or rogue insider, fraud, embezzlement, larceny, theft or extortion.
 - *Civil threats*. Kidnapping, civil disorder, rioting, unrest, protests, broad political or targeted protests, product tampering, workplace violence, radioactive or biological contamination.
 - Human accidents and hazards by inadvertent errors, omissions, lack of knowledge, laziness or carelessness served as a source of disruption: transportation or infrastructure accidents and failures.
- Technological disasters, when the causes are generated from the operation environment such as:
 - *Electricity issues*. Generation failure, power outage, brown-outs, rolling outages, failure of infrastructure.
 - *Gas or oil issues*. Generation failure, outage, explosion, evacuation, collapse of system.
 - *Water or sewer issues*. Generation failure, outage, contamination, shortage, collapse of system.
 - *Communications issues*. Generation failure, outage, collapse, rolling outages, failure of infrastructure.
 - Application or software flaws.

Different organizations and associations are developing standards, frameworks or guidelines about BCP, such as ASIS International, BSI Group, DRII Disaster Recovery Institute International or the International Standards Organization. The most recent proposed standard is ANSI/ASIS/BSI BCM.1 [1] for Business Continuity Management Systems based on the ANSI/ASIS Organizational Resilience standard (ASIS SPC.1 [2]) and the BSI's Business Continuity Management (BS 25999 [5]). Other used standards as basis in more specific areas are:

- ANSI/NFPA 1600 [1] Standard on Disaster/Emergency Management and Business Continuity Programs by National Fire Protection Association and ANSI.
- DRJ [13] Generally Accepted Principles by the DRII.
- ISO/PAS 22399 [25] Societal security—Guideline for incident preparedness and operational continuity management by ISO.
- ISO/IEC 27001 [24] Information technology—Security techniques—Information security management systems—Requirements by ISO.
- ISO/IEC 20000 [23] Information technology—Service management by ISO or the complementary ITIL v3 (2010) Information Technology Infrastructure Library by IT Service Management Forum (ITSMF).

The development of a COOP, for a maintenance department, could be considered as complementary function to those of quality and reliability assurance. In the risk analysis, non-normal disruptions must be taken into account, comparing the consequences of their occurrence with the costs, and benefits, of the remedial or recovery measures. The aim is to restore the services up to an acceptable level (Recovery Point Objective—RPO) in an acceptable time (Recovery Time Objective—RTO) limiting the disruption impact on services.

The COOP focuses on recovery essential operational capabilities at an alternate site and performing those capabilities for an extended period of time before returning to normal operations [8]. Consequently, COOP has to be implemented defining clearly, for these critical cases, the responsibilities and decision authorities, laws and regulations, communications and coordination, tests, resilience strategies, effective responses, alternative measures, additional resources, etc. And in network utilities due to their geographical coverage, a vulnerability analysis is necessary considering the geographical characteristics, network topology, tolerance operation levels, locations of resources and suppliers, etc. Besides previous points, the COOP has to be periodically reviewed, maintained and updated in case of organizational changes, modifications in resources or variations on strategies.

There are a number of reasons why business firms may not have continuity of operations plans [14]: temptation to deny the possibility of low-probability events; up-front costs of planning; the possibility of little or no payback; and the pressing nature of current problems. In relation to the economic impact of the disasters to the operation, the COOP focuses on continuity of the possible resources that could be affected by them [15]:

- Infrastructure and critical operative processes.
- People and the environment, in the event of a crisis securing and protecting employees, visitors, the environment, customers and suppliers.
- Social reputation, including the information systems that could affect supplier connections and customer relationships.

As a result, based on previously mentioned contributions and standards, we could propose a workflow to develop a COOP (see Fig. 11.13).

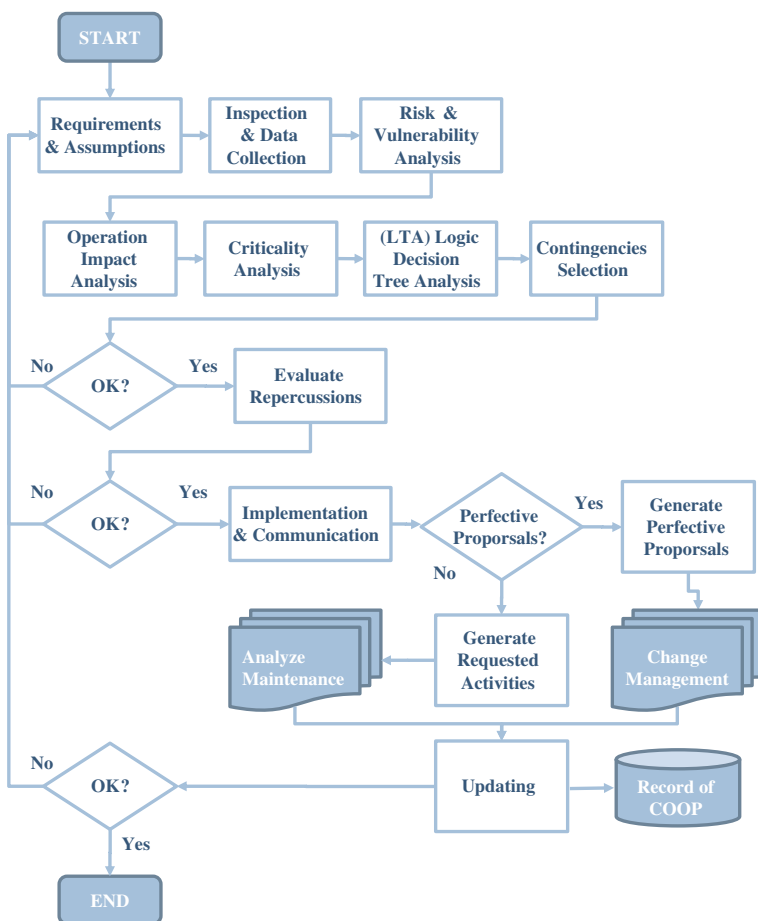


Fig. 11.13 Continuity of operation plan work flow

The work flow starts with the scope and responsibility definition, in accordance with the requirements (RPO, RTO, etc.) and assumptions about possible crisis or disasters. Then it continues with the inspection and data collection of external and internal relationships and resources at risks. Later the risk and vulnerability analysis is accomplished considering the geographical differences in the dispersed network of this type of utilities. This is done in order to determine the business impact analysis together with a criticality analysis according to the consequences and frequency. Hereafter, based on the resulting rank of criticality for each case, a logic decision tree analysis (LTA) is employed to select among the different contingencies. Next, the repercussions of the chosen contingencies have to be evaluated with a cost-benefit analysis, previously to their implementation and communication to the entire organization.

In addition to the produced COOP, some perfective activities proposals could be generated, internally or externally, depending on the scope of the changes. Finally, the COOP has to be updated, as we have mentioned, in cases of organizational changes, modifications in resources or variations on strategies and also after the occurrence of any crisis or disaster to assess the plans' suitability.

As example of this, suppose an earthquake disaster occurs inside a geographical area occasioning a power outage. Let us assume that this does not affect peoples' health, and let us only consider in this example the impact on infrastructure. Concerning the impact on people, it is important that the establishment of alternative worksites and communications, and additional workforce have to be activated to operate in the crisis. In order to reduce the impact on reputation, truth notifications to customers and society and coordination with authorities or third parties are assumed always to be generated in the same way.

Thus, after the activation of a crisis committee, the network contingencies described in the COOP are evaluated to decide the appropriate recovery strategy in the specific case that we are in. If there are different types of effects on the network, the adequate contingencies will be different too in each part of the network:

- In some part the activation of redundancy strategies could be possible;
- In other parts alternative routing may be accessible;
- Or in the worst case scenario, the installation of power generators could be effective, with an acceptable degradation of the service.

The developed COOP has to serve as an easy-to-understand guide with a clear responsibility allocation and support for decision making, decreasing the tension generated by stress in critical situations, highlighting the weaknesses of the network operation and defining the necessary contingency measures based on best practices in order to ensure the availability and continuity of services to the minimum impact. Notice that, as a precautionary measure, it is convenient that the organization is informed and trained in these procedures periodically.

11.7 Evaluation of Asset Life Cycle

In this section we deal with asset deterioration and its implications in terms of life cycle costing. We try to detect the acceleration of this process, to improve our estimations and to take into account its financial implications. At the same time, a benchmarking with similar companies can be important to maintain a competitive position in this area.

Through life cycle costing (or Life Cycle Cost Analysis—LCCA) we may set our maintenance performance to obtain the right cost, at the required reliability level. This estimation of performance, should also consider the assets' obsolescence analysis [21, 35], dealing with four main obsolescence causes:

- Functional factors related to changes in the use of the assets.
- Economic factors referring to the cost of using the assets versus replacing them by alternative ones.
- Technological factors related to the efficiency of the actual technology versus new alternatives.
- Social, legal, political or cultural factors referring to recommendations or obligations concerning assets' modifications, such as changes on health and safety laws or on social ecological trends.

Consequently, the above criteria together with equipment's criticality could affect the decision to repair or replace an asset, even generating warnings when the total costs during the asset life cycle is exceeded or deviated from our best prevision. Then, this analysis determines the real network worth as well as the maintenance activities contribution by reducing deterioration; it may also help to understand the effects of a maintenance contract on the network. In addition, the logistic policy has to take in consideration these criteria.

For those readers interested in early developments of LCCA methodology, some antecedents are the following [31]:

- 1933. The first reference to a "Life Cycle Analysis" appeared in a report by the *General Accounting Office (GAO)* of the Government of the United States. The analysis was related to the purchase of machinery;
- 1950. Lawrence D. Miles defines the concept of Value Engineering at *General Electric*, incorporating aspects related to LCCA techniques;
- 1960. The *Logistics Management Institute* of the United States of America developed research on Obsolescence Engineering for the US Ministry of Defence. As a final result of this research the first Life cycle Cost Manual is published in 1970;
- 1975. The Federal Department of Supplies and Services of the United States of America developed different logistics and acquisition techniques based on the LCCA;
- 1979. The US Department of Energy introduced a proposal (44 FR 25366, April 30 1979) requiring LCCA evaluations to be included in all new constructions and mayor modifications in US government facilities;
- 1980. The American Society for Testing and Materials (ASTM) developed a series of standards and database oriented to ease the search of necessary information for the application of the LCCA;
- 1992. In the University of Virginia, Wolter Fabrycky and B.S. Blanchard developed a LCCA model including a structured process to calculate the costs of failures and equipment degradation (assuming constant rate of failures).

Since 1992 many LCCA models have been published, including different mathematical formats and specific considerations concerning assets degradation, chronological value of money, etc. [10]. In this Section, we will consider one of the most recent and interesting models to understand assets' degradation, to

evaluate the failures cost over the equipment life cycle and finally to estimate equipment virtual valuation over time, under certain conditions of an existing network. We are referring to the generalized renewal process (GRP) model.

11.7.1 Generalized Renewal Process

This methodology allow us to analyze historical reliability and repair data and thanks to this option we can understand and improve the performance of teams, prioritize and make decisions focused on reliability, etc. We will be able, for instance, to compare different maintenance:

- Geographic areas
- Groups of technicians
- Procedures
- Changes in operational or organization
- Teams from different manufacturers
- Environmental conditions and operation, etc.

This model considers the repair time to be negligible so that the processes can be viewed as point processes. This model provides a way to describe the rate of occurrence of events over time such as in the case of data obtained from a repairable system. This model is particularly useful in modeling the failure behavior of a specific system and understanding the effects of the repairs on the age of that system. For example, consider a system that is repaired after a failure, where the repair does not bring the system to an as-good-as-new or an as-bad-as-old condition. In other words, the system is partially rejuvenated after the repair.

There are different formalizations of the GRP process to model equipment evolution over time. In this case, we will try to simplify the implementation of the GRP method based on the study by Mette and Wenbiao [38]. In this study they use the maximum likelihood approach (Maximum likelihood Estimation, MLE) [7, 17], applied to the Weibull distribution function, to reduce historical data points and processing power required for the analysis.

The approach is built on the GRP II by Kijima and Sumita [30], to cover complex systems with multiple equipment and repairs. The resulting method GRP II has improved the cumulative intensity function (CIF) results obtained with GRP I [12] because:

- It is less sensitive to variations in β .
- For elevated number of repairs, shows linear behavior, meaning that the effects of recent (last) repairs are less important for the cumulative intensity function.
- At high failure rates (when repairs grow to infinity), the results are more accurate.

- More accuracy for complex systems, or with data from multiple devices of the same type, is more accurate than the GRP I.
- For use of this method the following hypothesis are considered:
 1. The failure distribution function is a Weibull.
 2. The preventive and corrective times are negligible compared to the times between failures.
 3. All faults are detected.
 4. Equipment may have different initial failure rate and number of errors.
 5. The repair outweighs damage during the time between repairs i and $i - 1$, but also previous damage could be compensated.
 6. The starting age of any equipment is 0 and $t_0 = 0$.

The mathematical format of the model requires the definition of the following variables and parameters:

- The concept and variable of virtual age (V_i) represent the calculated age of the system immediately after the i th repair occurs (with $V_0 = 0$ for $t_0 = 0$).

$$V_{i-1} = y. \quad (11.8)$$

- The times between successive failures (between i and $i - 1$) are denominated t_i con $t_0 = 0$. Then $x_1, x_2, \dots, x_n$ are variables representing the intervals between successive failures.

$$t_i = \sum_{j=1}^i x_j \quad (11.9)$$

- The real age is:

$$S_i = \sum_{k=1}^i x_k; \quad \text{con } k = 1, 2, \dots \quad (11.10)$$

- Repair should compensate the damage suffered between maintenance (between failures i and failure $i - 1$). To show the quality of this repair we introduce the parameter q , measuring the effectiveness of that maintenance activity:

$$\begin{aligned} V_i &= q \cdot (q^{i-1} \times x_1 + q^{i-2} \times x_2 + \dots + x_i) = \sum_{j=1}^i q^{i-j+1} \times x_j \\ &= q \times S_i = q \times \sum_{k=1}^i q^{i-k} x_k \end{aligned} \quad (11.11)$$

- The failure distribution function can be expressed now as follows:

$$F(t|V_{i-1} = y) = \frac{F(t+y) - F(y)}{1 - F(y)} \quad (11.12)$$

And using the Weibull function we obtain:

$$\begin{aligned} F(t_i|V_{i-1} = v_{i-1}) &= \frac{F(x_i + v_{i-1}) - F(v_{i-1})}{1 - F(v_{i-1})} \\ &= 1 - e^{-\left[\frac{(x_i + v_{i-1})^\beta - v_{i-1}^\beta}{\alpha^\beta}\right]} \end{aligned} \quad (11.13)$$

- The failure density function will be:

$$\begin{aligned} f(t_i|t_{i-1}, t_i, \dots, t_1 = f(t_i|V_{i-1})) \\ = \frac{\beta}{\alpha} \left(\frac{x_i + v_{i-1}}{\alpha}\right)^{\beta-1} \cdot \exp\left[-\left[\frac{(x_i + v_{i-1})^\beta - v_{i-1}^\beta}{\alpha^\beta}\right]\right] \end{aligned} \quad (11.14)$$

- The failure rate (ROCOF):

$$\lambda(t_i|V_{i-1}) = \frac{\beta}{\alpha} \times \left(\frac{x_i + v_{i-1}}{\alpha}\right)^{\beta-1} \quad (11.15)$$

- Once we get the failure data (N observations $t_1, t_2, \dots, t_n$) we now try to fit parameters of our distribution function (now α , β and q) using the maximum likelihood estimation (MLE) method [18]. The maximum likelihood estimation function is as follows:

$$\begin{aligned} L\{\text{data}|\alpha, \cdot\beta, q\} &= L\{t_1, t_2, \dots, t_n|\alpha, \cdot\beta, q\} \\ &= \prod_{i=1}^n f(t_i|\alpha, \cdot\beta, q) = L \end{aligned} \quad (11.16)$$

$$\begin{aligned} L\{\text{data}|\alpha, \cdot\beta, \cdot q\} &= f(t_1) \times f(t_2|t_1) \times f(t_n|t_{n-1}) R(T|t_n) \\ &= \alpha^{-n} \times \beta^n \times \left[e^{-\left[\frac{(T-t_n+v_n)^\beta - v_n^\beta}{\alpha^\beta}\right]} \right] \\ &\quad \times \prod_{i=1}^n \left(\frac{x_i + v_{i-1}}{\alpha}\right)^{\beta-1} \times \exp\left[-\frac{(x_i + v_{i-1})^\beta - v_{i-1}^\beta}{\alpha^\beta}\right] \end{aligned} \quad (11.17)$$

$$\begin{aligned} \Lambda = \log L\{\text{data}|\alpha, \cdot\beta, \cdot q\} &= n \times (-\ln \alpha + \ln \beta) \\ &\quad - \left[\left(\frac{T - t_n + v_n}{\alpha}\right)^\beta - \left(\frac{v_n}{\alpha}\right)^\beta \right] \\ &\quad - \sum_{i=1}^n \left[\frac{(x_i + v_{i-1})^\beta - v_{i-1}^\beta}{\alpha^\beta} \right] \\ &\quad + (\beta - 1) \times \sum_{i=1}^n \ln\left(\frac{x_i + v_{i-1}}{\alpha}\right) \end{aligned} \quad (11.18)$$

- Taking logarithms and for a time truncated estimation (up to time T) for failures, considering data from multiple equipment (k) with similar characteristics:

$$\begin{aligned}
 \Lambda = \log L\{\text{data}|\alpha, \cdot\beta, \cdot q\} &= \sum_{l=1}^k n_l \cdot (-\ln \alpha + \ln \beta) \\
 &\quad - \sum_{l=1}^k \left[\left(\frac{T_l - t_{l,n_l} + v_{n_l}}{\alpha} \right)^\beta - \left(\frac{v_{n_l}}{\alpha} \right)^\beta \right] \\
 &\quad - \sum_{l=1}^k \sum_{i=1}^n \left[\frac{(x_{l,i} + v_{l,i-1})^\beta - v_{l,i-1}^\beta}{\alpha^\beta} \right] \\
 &\quad + (\beta - 1) \cdot \sum_{l=1}^k \sum_{i=1}^n \ln \left(\frac{x_{l,i} + v_{l,i-1}}{\alpha} \right)
 \end{aligned} \tag{11.19}$$

- Continuing with the resolution, we equal to 0 partial derivatives with respect to α , β and q .

1. Partial derivative of $\log L$ respect to α :

$$\begin{aligned}
 \frac{\partial \log L}{\partial \alpha} = 0 &= \sum_{l=1}^k -n_l \cdot \frac{1}{\alpha} \\
 &\quad - \sum_{l=1}^k \left(-\frac{\beta}{\alpha} \right) \times \left[\left(\frac{T_l - t_{l,n_l} + v_{n_l}}{\alpha} \right)^\beta - \left(\frac{v_{n_l}}{\alpha} \right)^\beta \right] \\
 &\quad - \sum_{l=1}^k \sum_{i=1}^n \left(-\frac{\beta}{\alpha} \right) \times \left[\frac{(x_{l,i} + v_{l,i-1})^\beta - v_{l,i-1}^\beta}{\alpha^\beta} \right] \\
 &\quad + (\beta - 1) \times \sum_{l=1}^k \sum_{i=1}^n \left(\frac{-1}{\alpha} \right)
 \end{aligned} \tag{11.20}$$

2. Partial derivative of $\log L$ respect to β :

$$\begin{aligned}
 \frac{\partial \log L}{\partial \beta} = 0 &= \sum_{l=1}^k \frac{n_l}{\beta} \\
 &\quad - \sum_{l=1}^k \ln \beta \times \left[\frac{(T_l - t_{l,n_l} + v_{n_l})^\beta - v_{n_l}^\beta}{\alpha^\beta} \right] \\
 &\quad - \sum_{l=1}^k \sum_{i=1}^n \ln \beta \times \left[\frac{(x_{l,i} + v_{l,i-1})^\beta - v_{l,i-1}^\beta}{\alpha^\beta} \right] \\
 &\quad + (\beta) \times \sum_{l=1}^k \sum_{i=1}^n \ln \left(\frac{x_{l,i} + v_{l,i-1}}{\alpha} \right)
 \end{aligned} \tag{11.21}$$

3. Partial derivative of $\log L$ respect to q :

$$\begin{aligned} \frac{\partial \log L}{\partial q} = 0 = & (\beta - 1) \times \sum_{l=1}^k \sum_{i=1}^n \left(\frac{x_{l,i} + \sum_{j=1}^{i-1} q^{i-j} \times x_j}{\alpha} \right)^{-1} \\ & \times \left(\frac{\sum_{j=1}^{i-1} (i-j) \times q^{i-j-1} \times x_j}{\alpha} \right) \end{aligned} \quad (11.22)$$

- Last three equations need to be solved (to find α , β and q), and in order to do so the Newton method is normally used because of its fast convergence. The goodness of results can also be analyzed according to Fisher matrix (see [39]), presented in Eq. 11.23

$$\begin{aligned} \Sigma &= \begin{bmatrix} \text{var}(\alpha) & \text{cov}(\alpha, \beta) & \text{cov}(\alpha, q) \\ \text{cov}(\alpha, \beta) & \text{var}(\beta) & \text{cov}(\beta, q) \\ \text{cov}(\alpha, q) & \text{cov}(\beta, q) & \text{var}(q) \end{bmatrix} \\ &= \begin{bmatrix} -\frac{\partial^2 \log L}{\partial \alpha^2} & -\frac{\partial^2 \log L}{\partial \alpha \partial \beta} & -\frac{\partial^2 \log L}{\partial \alpha \partial q} \\ -\frac{\partial^2 \log L}{\partial \alpha \partial \beta} & -\frac{\partial^2 \log L}{\partial \beta^2} & -\frac{\partial^2 \log L}{\partial \beta \partial q} \\ -\frac{\partial^2 \log L}{\partial \alpha \partial q} & -\frac{\partial^2 \log L}{\partial \beta \partial q} & -\frac{\partial^2 \log L}{\partial q^2} \end{bmatrix}^{-1} \end{aligned} \quad (11.23)$$

For practical resolution of the GRP method the reader is referred to the software provided by ReliaSoft. (Reliasoft Weibull ++ 7—www.reliasoft.com).

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Chapter 12

Control and Knowledge Management System

12.1 Introduction

This section, and the management phase, pursues sustainable development of our network utility maintenance management framework, controlling standardization, knowledge management and proper ICT support over time.

Control systems measure performance and determine the causes of variance, then ensure implementation of corrective actions managing the change.

The information taken as a basis for control has to be practical and reliable in order to increase precision. Internal and external analyses have to be periodical, discovering possible deadlocks jeopardizing continuous improvement.

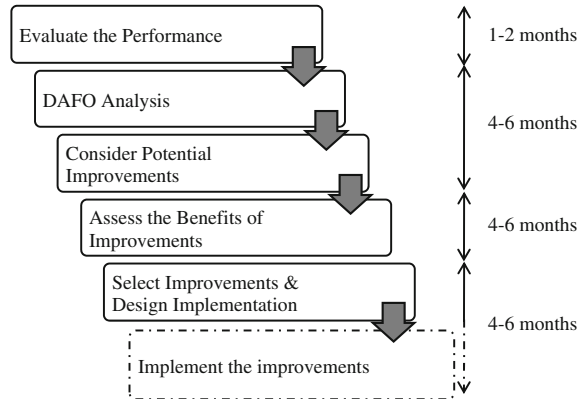
Authors of *change management* books identify a series of steps to follow when evaluating performance and identifying areas of potential change (see Fig. 12.1) [7, 9, 52]. Sizing these steps by time for strategic evaluation is also highly recommended [26], for instance:

1. Evaluate the performance (1–2 months);
2. DAFO analysis, describing the environment and social system (4–6 months);
3. Consider potential improvements (included in previous 4–6 months);
4. Assess the benefits of improvements (4–6 months);
5. Select improvements and design implementation (2–3 months discounting the estimated time for implementation).

Consequently, we will periodically re-evaluate our management framework pillars with our control and knowledge management systems, concentrating on existing:

- Information systems (including advanced “e-maintenance” systems) allowing performance measuring and cause-variance detection;
- Modelling techniques;
- Performance indicators and;

Fig. 12.1 Steps for performance evaluation



- Knowledge maturity assessment (regarding the produced, standardized and exchanged knowledge).

We will deal with the last of these four areas in the last two chapters of the book, and in this chapter we will concentrate on the other three. We will now define, for better control:

- A modular and open architecture for systems;
- A suitable list of techniques and performance indicators per different categories, and the relationships amongst them; and
- The processes, for a better control and overall management.

All these elements will be used, in the final chapters, to identify a suitable path for maintenance improvement, describing maturity steps or levels according to the achievement of certain objectives.

12.2 Setting Up the Required Systems to Control Maintenance

Different systems are required to be in place in order to pursue maintenance objectives through the proper implementation of the maintenance processes previously defined. These systems will ensure quality at a reduced cost and will manage all network knowledge that is constantly generated according to different network scenarios and performances. Maintenance systems should be simple and practical, avoiding information overload and providing analysis tools to make better decisions. According to Higgins et al. [29] the correct implementation of maintenance systems could improve labor utilization by 5–25%, equipment utilization by 1–5% and stores inventory reduction by 10–20%.

Network utilities need a unified vision of service. This unified vision of service should be provided by an integrated management system, because an appropriate network performance is meaningless if it is not translated into quality of delivered services. With the aim of ensuring the quality of services and optimizing their availability, information technologies are the support element.

In the maintenance of networks, we should take the advantage of modern remote management for access to many disparate elements [17], and of advance knowledge generation through artificial intelligence to support fast and suitable decision making. This type of management is called, in the telecommunications sector, *Intelligent Network Management* [5] and network management intelligence and decision making can be distributed or centralized.

The network maintenance management system should include end-to-end and flexible performance monitoring, and should also include central management of the operational functions, administration and procurement (OAM&P). However, the implementation of systems must be in accordance with the organization and processes. That is, prior to system customization, the organization should be clearly defined (including levels of outsourcing), the processes, the activities and the desired management levels demanded by the information system.

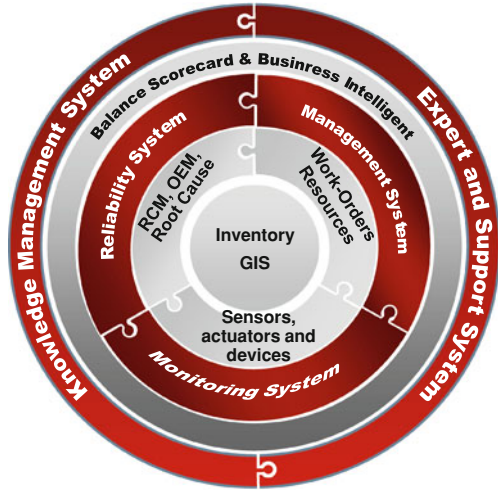
In the case of network utilities, due to the variety and geographical spread of equipment, systems have to be developed considering, in addition to general software principles (flexibility, pragmatism, modularity, scalability, interoperability, confidentiality and integrity), the possibility of a distributed management and intelligence throughout the network.

Distributed computing requires optimization and synchronization of ICT systems for dynamic decision making. Due to the very large volume of data, information and knowledge, some of the more simple processing could be decentralised to a level as low as possible, e.g. to the sensor level [31], supporting this on predictive intelligence (algorithms, software and agents) and mapping relationship between product quality variation and machine and process degradation when required [42], considering machine operational environment throughout its entire life cycle [65].

In practice, ICT management systems integration provides an operations support environment in which management functionality is seamlessly integrated and holistic with the following benefits [14]:

- Simplifying management tasks instead of employing several systems;
- Reducing the need for training and increasing the pool of available personnel, focusing on a few systems;
- Facilitating administration, supervising and intervening to keep network operations running smoothly;
- Reducing the entering of data on multiple times or redundantly in multiple locations;
- Reducing the workload and load of computing for managing the network;
- Facilitating correlation of different data and its availability for other uses;
- Facilitating interoperability with other business systems and its interdependencies.

Fig. 12.2 Basic information systems in network utilities



Therefore, we will divide this section into four parts proposing:

- A set of generic maintenance systems;
- The interconnection among these systems according to modern existing standards;
- The systems processes relationships for network utilities maintenance; and
- A method to evaluate the maintenance systems that can be found in the market according to their functionality;

The maintenance systems could be distinguished according to the maintenance information life cycle [61]: automatic detection, interpretation and execution; communication and store; correlation, analysis and prediction of behaviors and control, evaluation and reporting. Consequently, based on standards and the maintenance models that were presented above, and in order to manage distribution networks maintenance effectively and efficiently, in this document we claim that a total of eight systems should be considered (see Fig. 12.2):

1. Inventory system.
2. Monitoring system.
3. Management system.
4. Geographic information system.
5. Reliability centered system.
6. Balance scorecard and business intelligent system.
7. Knowledge management system.
8. Expert decision and support system.

Through the listed systems it is intended to ensure quality, reduce costs and manage all the knowledge that is constantly being generated in accordance with different scenarios and conditions of the network. But let us now characterize each one of these systems.

12.2.1 Inventory System

A correct network configuration management ensures rigorous bulletins and organization, maintaining in a planned manner the history of elements [23, 37]. So, this is crucial to support the rest of the systems. This will reduce risk of: loss of control, aging or degradation of infrastructure; and variations in quality of service. This system has to detail the network infrastructure elements, their settings and subsystems, their interactions with other systems, in a unique and reliable source for the whole network maintenance. This system implements the management of network assets life cycle and it should contain, at least [25, 74]:

- Inventory of elements, relations and location;
- Logical modelling of networks and services;
- Mapping of services with supporting elements and clients;
- History of activities, incidents, problems, changes and versions;
- Installation, settings and maintenance manuals;
- Support, “on-line”, to be provided to final customer services;
- Occupied resources allocated to each service end-to-end.

12.2.2 Monitoring System

This system is a key factor for maintenance proactivity. It provides information in real time about the infrastructure element status and customer services, facilitating decision making and resources management, in all maintenance activities:

- Corrective, anticipating the customer claims, and minimizing the impact or disinformation that provokes an incident;
- Predictive, improving the release of on-condition activities;
- Preventive, detecting unnecessary maintenance and assets without use.

The main objective of this system is to ensure the maximum network availability [43] with the highest services quality for the user and with a rapid and effective response capability in place for situations of failure. This system should also allow the prevention of potential network problems before they start [32]. Computer systems should be used to make network management more automatic, and specific for each network typology and process [3, 34]. This system should include within its functions:

- The possibility to test and configure the network elements in order to improve network available capacity and services;
- The collection of network elements and services data, easing impact analysis and problem identification;
- The measurement of the network utilization;
- The events correlation between equipment, services and clients;
- The generation of automatic events;
- Mechanisms for notification and escalation.

12.2.3 Geographic Information System

In network utilities—and because of the size of companies, the communications volume and data storages—it is advisable to integrate the inventory and monitoring systems (at least) over a geographic information system (GIS). GIS systems [36] facilitate, with a more intuitive geographical representation, the infrastructure knowledge related to physical location element interactions, environment and customers. This system also integrates cartography with remote sensing technologies, computing spatially related data files, in format raster (cellular-based) or vector (reference-based) data [10] and obtained by computer aided drafting, digitalizing or scanning photography, and image processing [40]. The basic functions of GIS are, according to Aronoff [2]:

- Data input;
- Data management or data storage (DBMS—database management system);
- Data analysis, distinguishing three types:
 1. Storage and retrieval including presentation capabilities.
 2. Constrained queries that allow the user to look at patterns in the data.
 3. Modelling procedures or functions for the prediction of what data might be at a different time and place.
- Data output.

Although from a software point of view of Van Dyke [72], the different components could be the following: Image processing system, Map digitizing system, Database management system distinguishing two database types (spatial and attribute database), Statistical analysis system, Geographic analysis system and Cartographic display system.

Searching interoperability among systems, the open geospatial consortium (OGC) originated in the 1990s as an international industry consortium of 421 companies, government agencies and universities participating in a consensus process to develop publicly available interface standards [55]. Open GIS protocols include standards such as web map service (WMS), web feature service (WFS), web coverage service (WCS); web processing service (WPS); geography markup language (GML), sensor observation service (SOS), sensor planning service (SPS), etc.

The development of this system could be lead by maintenance, network construction or by the engineering department, although maintenance has to be responsible for keeping this system up to date.

12.2.4 Management System

This is a transactional system integrating technology and social variables such as tasks, resources, personnel and organization [21]. Activities have to be automated

by work-flow systems to provide increased levels of availability, reliability, flexibility and speed of services from a technical and economic point of view. The function of this system is managing, planning, documenting and categorizing activities, associated with human resources and infrastructure elements. The main features included in this system could be [16, 49]:

- Scheduling and budgeting jobs, assigning personnel, reserving materials and tools, employing mobile options and recording costs.
- Tracking and recording data concerning equipment and property including specifications, step-by-step instructions, maintenance procedures, warranty information, service contracts, spare parts, asset hierarchy, purchase date, description of the maintenance task, frequency of the task (based on a relevant measure such as elapsed time, operating hours, number of operational cycles or distance), number of personnel, skill level and time required to perform the task, suggestions for further action required, Turnaround and transportation time, etc.
- Maintenance analysis, valuating and reporting with different techniques when a job should be performed, including relevant information such as the cause of the problem, downtime, item reliability and maintainability data (such as the expected lifetime), or the result of the inspection.

Therefore it is characterized by encompassing knowledge management of historical data as a source for managing problems (Symptom-Cause-Solution) and learning. Consequently, it must be practical and simple, and not involve large amounts of information, where all filled information should be objective and truthful to provide an analysis and optimal decision making.

The European standard EN 13460 on Maintenance Documentation [35] is an adequate reference to develop the basic maintenance management systems, in which key issues are defined as follows:

- Thirty basic attributes for maintenance work orders that we could group in relation with the work order and control, implicated resources and their localization, necessary documentation and restrictions and, the symptom-cause and RAMS analysis.
- Seventy procedures for maintenance management, distinguishing between preparatory and operational phases.
- A basic workflow for maintenance work orders, relating the procedures application for each state of the workflow.

12.2.5 Reliability Centered System

Reliability analysis, according to the RCM method, is very complex task. This analysis considers the effects, in the equipment function [69], of the different failure modes. Also this analysis considers that functionality will degrade following a P-F curve (potential-functional failure curve) and that deviations from

standard operating conditions can be corrected to avoid failures that can be produced by a certain set of failure modes [50]. Therefore this type of information has to be interrelated with the intention of predicting failure patterns and mitigating risks, based on real data (as historic events). RCM software is a support tool for the management and optimal planning of equipment maintenance according to each operational context. Reducing multiple or hidden faults, and activities costs according to each operational environment and service requirements, are some of the benefits of an RCM analysis, that is, planning the maintenance efficiently according to a level of risk.

By their own complexity, this system and its associated quantitative tools (representations, simulations and intuitive interfaces) have to be mainly implemented in depth in critical equipment or those equipment in which failure consequences are not admissible (environment, Health and Safety, etc.), facilitating the generation of maintenance knowledge.

Several manufacturers have developed RCM software packages using different strategies or calculation methods, and containing different modules or subpackages such as RAMS analysis (reliability, availability, maintainability and safety), “e-maintenance” technologies based on reliability databases and plant historical data, FMEA/FMECA (analysis of failure modes, effects and criticality), Weibull analysis allowing plots, statistical computation and simulation for risks analysis, hazard and operability (HAZOP) for qualitative valuation of consequences, Markov analysis for prediction, Monte Carlo simulation for estimation in complex problems, and reliability block diagram (RBD) for graphical representation of the relationship among components from the standpoint of reliability.

We have found that existing RCM software can be classified into three different types, according to operational settings, e-maintenance capabilities and integration possibilities:

- Type I has a specific RCM analysis management module which is integrated within the computer maintenance management system (CMMS) interface, or existing database in the organization. In addition, this type defines maintenance strategies and recommendations based on RCM and integrated within the CMMS maintenance plans.
- Type II has a completely modular structure connected to the CMMS database, meaning that they have an operating module with a separate interface for each of the stages proposed by the RCM methodology: FMEA, RCM analysis, RBD and others.
- Type III as a mixture of Types I and II, generally have two modules, a basic module with the main functions to support RCM analysis and an advanced module, more complete, which has all the functionality of the basic module and additional functional features for advanced management of RCM analysis. In addition to the CMMS database interface, sometimes the advanced module has an interface with the monitoring system to update the RCM analysis and maintenance plans in real time.

12.2.6 Balance Scorecard and Business Intelligent System

This is a basic pillar to ensure compliance with the maintenance department objectives. It aligns all activities, processes and resources, with operational objectives and strategy [70]. To this aim, it collects a coherent set of indicators [41] related to: financial aspects, business processes, customer focus and organizational learning. This system implements the data warehousing and data mining based on information distributed in other systems, where the raw information is prepared specifically for each decision making level. It should include key indicators to measure the processes, those activities aligned with business value and service [37], and that reflect the improved profit and, the provided and perceived quality of service. Although, the indicators are not individual, there are relationships among them, which should be described to avoid the worsening of a goal trying to improve another.

Maintenance measurements have to be structured with a controlled access and hierarchically in layers, for each organizational level, with performance indicators and as single and official data source, internally and externally, this is the way to improve decision making in network maintenance by linking the problems' effects to their root causes. BSC has been, however, implemented unsuccessfully in many cases [6]. The reports should also be standardized to facilitate continuous improvement [71], taking special care with the quality, quantity and frequency of the presented information. The maintenance BSC–BI system, as a rule, has to be a component of the Integral BSC–BI of the enterprise.

In this way, it facilitates the management of change in organizations, and decision making due to the effects with the originating causes. It is necessary to implement the balanced scorecard with a lot care to lead it toward the desired benefits, because there are a large number of situations in which the goals have not been achieved [6, 54].

12.2.7 Knowledge Management System

Information in companies should be collected, integrated and stored electronically. Besides this, a good categorization [13] should allow access to information quickly, with geographical independence and at the lowest cost.

Knowledge is one of the intangible assets of enterprises, and as we have explained, maintenance is a key process to improve it [56]. Knowledge management also facilitates the adaptation to changes and this is a source of competitive advantage [60]. However, due to the intangible nature of knowledge, it is not easy to handle [27]. In any case, the maintenance knowledge, fulfilling its lifecycle, has to be created, validated, represented, disseminated and applied to activities through a system, knowledge management system (KMS) [4], which needs to be integrated inside the KMS of the company.

Through a KMS, the access to maintenance information is carried out in a guided and effective form (“tree of knowledge”), making easy to relate information from different sources, as for example suppliers, human resources, standards, best practices, help manuals, contracts, etc. Therefore, the implementation is advisable not only as intranet, but also as a collaborative support system to promote the exchange and generation of knowledge. Alavi and Leidner [1] indicate that KMS seems to require a variety of technological tools in database management, communication and messaging, and browsing and retrieval; distinguishing the following perceived benefits of existing KMS:

- Within process outcomes.
 - *Communication*. Enhanced and faster communication, more visible opinions of staff and increased staff participation.
 - *Efficiency*. Reduced problem solving time customer focus, shortening proposal times targeted marketing, faster results and delivery to market, greater overall efficiency.
- Within organizational outcomes.
 - *Financial*. Increased sales, decreased cost, higher profitability.
 - *Marketing*. Better service, customer focus, targeted and proactive marketing.
 - *General*. Consistent proposals to multinational clients, improved project management, personnel reduction.

12.2.8 Expert and Support System

Maintenance management requires making strategic, tactical and operational decisions [12, 19, 22]. To support these decision-making processes at the three levels of business activities it is recommended a system which integrates:

- A decision support system for decision making (DSS) [67] as an extension of the previous systems and using scientific management models; with
- An expert system (ES) [62] to support decision making in order to emulate, through artificial intelligence, human reasoning such as that of an expert.

This system applies information management and statistical models [11, 15, 48, 57] simulations and human reasoning [8, 33, 39, 45, 59, 63], to facilitate decision making in maintenance [33, 39] which is strongly influenced by conditions of uncertainty, conflicts of interest, emotional factors and high stress. The DSS joins with an ES, that is to say, an expert and support system (ESS) that helps to find the best solutions according to objectives identified by the department. Both systems are usually composed by different software agents [66, 68] programed for each case depending on the variety of the decisions [47]. Thereby, there are two types of agents according to the way in which they could be executed:

- Concurrently with the operation, that is to say, we could run it online, based on monitoring information and predictive simulations, mainly in operational decisions, to automatically warn about levels of risk to optimize maintenance, or to generate corrective and preventive tasks automatically.
- On-demand activation, about medium and long term decisions, mainly tactical and strategic decisions, i.e. the remaining life of the elements to plan the preventive, to predict the power consumption depending on the network growth.

In network utilities, we do consider that the first module type is crucial to facilitate the network maintenance management, reducing time and human resources, materializing acquired know-how about thousands and thousands of network elements.

According to Liao [62] the implementation of this type of systems can be classified into the following types:

- Rule-based systems.
- Knowledge-based systems.
- Systems based on neural nets.
- Systems based on fuzzy logic.
- Object-oriented methodology.
- Case-based reasoning.
- System architecture.
- Systems of intelligent agents.
- Methodology database.
- Modelling.
- Oncology.

12.2.9 Interconnection and Interoperability Among Systems

In this section we are especially interested in exploring interrelation of maintenance with business operations and clients [44]. For this purpose, interconnection must be conducted and controlled among the different existent sources of information in the company [73], facilitating the transparent transference of knowledge [28] between the maintenance systems and, at a minimum, the following enterprise systems [49]:

- Economic system.
- Human resources management system.
- Logistics.
- Customer relationship management (CRM).
- Documentary management system.
- Knowledge management system.

The interconnection and correlation of data among maintenance systems and the rest of the organization must take into account the hierarchical levels of

management. To this end there are standards homogenizing communications and information regardless of the system manufacturer [3]. This is the aim of the federation of interoperability standards Open O&M and the standards ISO 18435, ISO 62264 and MIMOSA (pursuing vertical, horizontal and time integration of systems).

When we look for maximum interoperability, the development of control systems based on the requirements of the open standards Open O&M and TNM, as we have seen in Sect. 2.3, can be considered a good praxis. These standards recommend having:

- A first *central* level, where the inventory system appears as kernel integrated with the geographic information system that is used as a support of network representation and services.
- A second *operative* level divided into four sectors:
 - Two sectors focused on the network and services knowledge based on monitoring and RCM systems. These two systems meet the MIMOSA standard structure with the functions:
 - a. Monitoring system: sensor management, data manipulation, alarm and event detection, evaluation and diagnosis, prediction and prognosis and intelligent generation of warnings.
 - b. RCM system: analysis of spare parts, root cause analysis, RCM analysis and original equipment manufacturers (OEM) reliability model.
 - An organizer sector of activities, processes and resources, adapted to the framework of maintenance management: monitoring, to manage incidents, to prevent incidents, to administrate capacity and to analyze maintenance.
- An *evaluation and reporting* level for the extraction and presentation of all the maintenance data based on the scorecard and business intelligent system and coordinated with the rest of the organization's systems taking into consideration the hierarchical levels of management (operative, tactical and strategic).
- And a third *optimization* level of expert knowledge, based on the above systems and developing the knowledge management systems and expert support systems for decision making. In line with these concepts, nowadays organizations are working internationally in the development of platforms such as Device-to-Business (D2B™) from the Intelligent Maintenance System Centre in USA [44], TELMA from the Nancy University in France [38], and also in projects such as PROMISE from the Intelligent Manufacturing Systems Consortium [44], and DYNAMITE (dynamic decisions in maintenance) from a European joint research and development project within the European Union 6th Framework Programme [30].

From the point of view of information technologies, the logic maintenance system architecture for network utilities could be represented according to the hierarchical levels of organization defined in ISO 18435 (see Fig. 12.3). At the bottom (level 0) the assets and the physical processes are included. Processes are

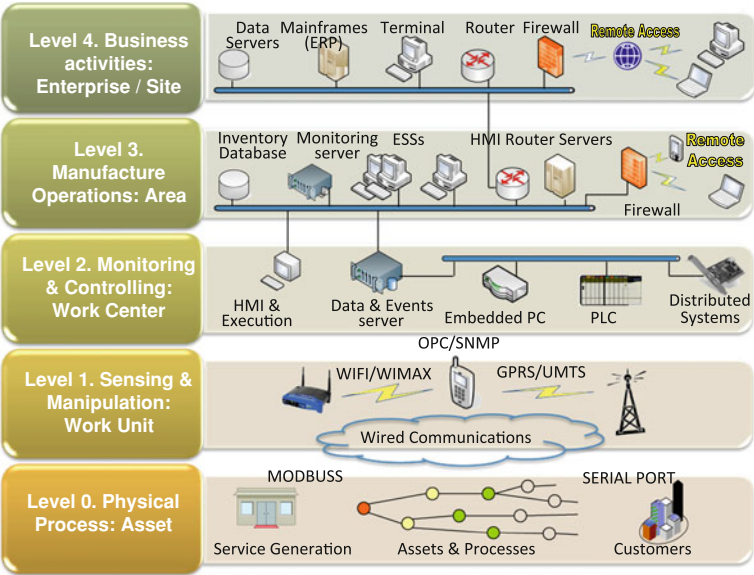


Fig. 12.3 Logic maintenance system architecture

measured and operated (level 1, in the work unit) by sensors, actuators and devices (either by wire or wireless). In level 2 (work center), the data is collected and the control is executed, both mainly on real time. There are different types of data transmissions in sensors and devices depending on the supplier; and different types of connection with them (PLC, fieldbus, and distributed systems or servers), then a server is needed to operate all of them and to communicate data, commands and events to the upper level. Level 3 is characterized by the management of tasks, analysis and reports; it is the kernel of the maintenance systems and the core of the operational processes in maintenance. This level needs a major capacity of computing, store and routing, so specific servers or mainframes are employed to enforce the systems, and also the data warehouse and data mining are developed in this level over all the maintenance information. Centralization is the objective in this level in regional and central data centers, but depending on the quantity of maintenance data generated regionally, distributed software could be maintained in order to minimize the costs of transmission and store while the security is increased based on backup copies and recovery systems on separate storage devices. Therefore, integrated wired and mobile communications are enforced from an operative point of view. Finally, level 4 corresponds to the business activities through central servers in a central data center or two data centers in the case of system replication in order to facilitate the recovery before contingencies and the business continuity. The interoperability and scalability among central and department systems are realized in this level avoiding “technological silos” in software and hardware solutions that could obstruct future developments and

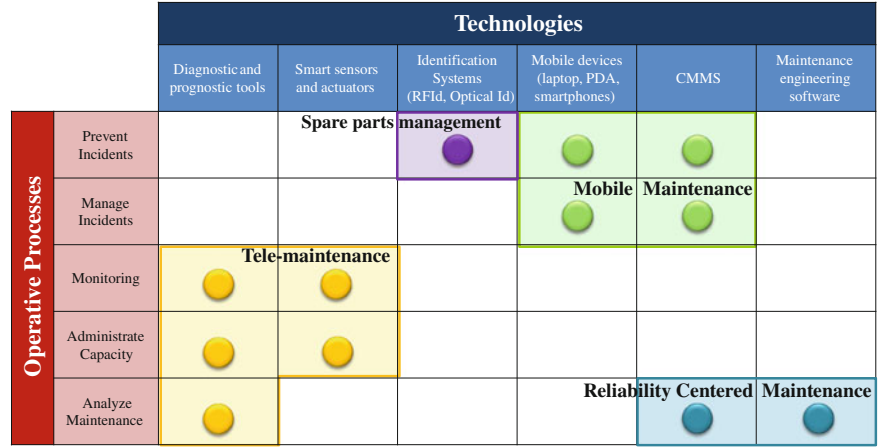


Fig. 12.4 E-maintenance technologies application clusters (modified from [46])

improvements as well as the proper maintenance of the architecture. As a result, integrated wired and mobile communications are administrated in this level from a security point of view, including balanced routing, firewall management and, access rights and protection.

In line with this, the recommendation about e-maintenance technologies application could be more adequate and specific inside a maintenance process than another (see Fig. 12.4), defining homogeneous clusters of technological components [46]:

- Spare parts management, searching real time visibility and traceability of stock levels of the spare parts among different areas of distribution.
- Mobile maintenance, supporting all the activities related to work order management, reporting and execution, in real time and with a large quantity and quality of collected information, thanks to the mobile devices in the hand of maintainers.
- Tele-maintenance, relating to all the activities carried on for equipment continuous online condition monitoring; such activities can be performed also in a remote way, facilitating diagnostic and prognostic analysis and expert decision making.
- Reliability centered maintenance, for analyzing the equipment functions in the operative context in order to make it more productive, reliable and environmental friendly, safer, less risky for the user and other interested parties.

In summary, e-technologies increase the possibilities:

- To utilize data from multiple origins and of different types,
- To process larger volumes of data and to make more advanced reasoning and decision making, and

Table 12.1 Relationship among systems and processes

Process\System	Inventory	Management	KMS	GIS	Monitoring	RCM	BSC	ESS
To monitor	✓	✓	✓	✓	✓	✓		
To administrate capacity	✓	✓	✓	✓	✓	✓		
To manage incidents	✓	✓	✓	✓				
To prevent incidents	✓	✓	✓	✓		✓		
To analyze maintenance	✓	✓	✓	✓	✓	✓	✓	✓
To manage strategy	✓	✓	✓				✓	

- To implement cooperative (or collaborative) activities. The implementation of these e-technologies for the benefit of the maintenance area is the first reason for the emergence of e-maintenance.

12.2.10 Maintenance Processes and Their Supporting Systems

Previously mentioned systems, based on generic recommendations, can be implemented according to standards to support the described processes of the maintenance management framework (see Table 12.1). Notice that these generic systems architecture is characterized by:

- The core of the maintenance systems is the inventory system, being fed from the rest of the systems;
- The management system implements all the processes activities which are carried out by work-flows divided into series of tasks;
- The geographical information system enables the correlation of knowledge, linking data with network and service affection. Then this GIS system is a support environment for the five operational processes.
- Monitoring system is the pillar in the process to monitorize, although also in the process to administrate capacity, in order to supervise the capacity of the network and to avoid saturation. Its produced data is handled by the process to analyze.
- The RCM system implements analysis-based reliability to optimize the strategies of each process through the management system.
- From a strategic point of view, the BSC is used to manage the strategy of the department, but also searching continuous improvement, and could be used in the process to analyze maintenance.
- All the maintenance knowledge is created, validated, represented and available by the KMS, with the intention of being applied to the activities.
- Finally, suitable expert and decision support systems are developed in the process to analyze, whilst once the knowledge is well-established and the rational of the decisions provided from these are accepted, the ESS could be executed in whatever process, operational or strategic, for a specific case of application.

Following with the compliance of standards and once the architecture and hierarchy of systems has been framed; the development of them has to be carried out to implement all the maintenance activities. In order to do this, the activities have to be represented in a data model, based on a generic sequence of status and a defined list of attributes to exchange in each one. The standard ISO 62264 is a generic model for the implementation of operative activities based.

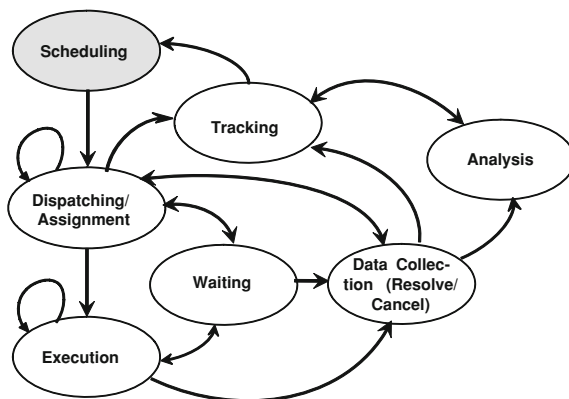
Now we will focus on the definition of the basic system attributes in maintenance systems. In order to take advantages of information processing, the system attributes should be categorized and codified globally for all the architecture. These must be defined considering the following:

- Their application or correspondence within all the systems of the architecture, because one attribute could be employed in different systems or its value could be transferred to another system and;
- Their activation and modification for each state of the activity. One attribute could be activated as a reading or result in a given state and differently in another state.

A wide set of attributes as reference for maintenance systems could be [16, 20, 75]:

- Unique reference number per task;
- Description and type of the maintenance task;
- Frequency of the task;
- Priority and urgency of the work;
- Method of generation (proactive or reactive) and petitioner;
- Status and tracking times about the maintenance task (opened, assigned, accepted, waiting, closed, etc.);
- Number of repair personnel, skill level and time required to perform the task;
- Equipment operational data (running hours, deviations, history, etc.) quantity and location;
- Equipment condition-based data (sensors/actuators/devices, signal, warnings, data manipulation, etc.);
- Feasible repair alternatives;
- Maintenance procedures for disassembly and reassembly;
- Health and safety and other legal procedures or constraints to be followed;
- Special tools, test equipment and support equipment required;
- Spare parts, materials and consumables to be used or replaced;
- Cost factors and centers;
- Turnaround and transportation time to and from repair facilities;
- Item reliability and maintainability data (MTBF, MTTR, symptom, cause, downtime, impact, etc.);
- Diagnostic and prognostic management;
- Decision-making management (horizon, period, event, objectives, parameters, criteria and constraints of the decision);
- Relationship with derived key performance indicators (KPIs);

Fig. 12.5 Generic activity model based on ISO 62264



- Observations and measurements to be made;
- Checkout procedures to verify proper operation and successful completion of the maintenance task.

Based on the standard EN 13460, ISO 62264 and considering the possibility of assignation of activities to external resources/companies, we propose the present diagram of status (see Fig. 12.5).

This generic activity model defines a general request-response cycle of tasks identifying possible data flows. The activity model starts with requests or schedules, generating a detailed schedule whose tasks are dispatched or assigned. The execution management implies that the task could be executed in some parts and considering waiting periods or new assignments. Once the execution ends, the task is resolved collecting data and summarizing reports for the responses. Sometimes the task could be canceled from dispatching, execution or left in a waiting state; in this case the data collection is in charge of documenting the reasons. Inside the tracking state, the maintenance performance and effectiveness information are managed. Finally, the analysis task searches the optimization of resources' operating performance and efficiencies, studying the present-past-future maintenance information to detect problematic areas or areas for improvement.

In Fig. 12.6, the *managing preventive* activity is shown, according to this standard, as an example.

Once the activities are modeled, for their implementation in software, the resulting model is translated into UML.

Considering that the objective is to model with a graphical representation, an analysis of the different options revealed that the best choice is using UML language; which is a standard maintained by the object management group (OMG). UML 2.1 is the version reviewed in depth, formed by thirteen kinds of diagrams that show a specific static or dynamic aspect of a system. In this case, a particular type of diagram: the "class diagram", is used to describe the most important information needed to execute the maintenance designed processes.

For this research the software chosen to model the system is Enterprise Architect 7.1, an UML analysis, design, documentation and project management

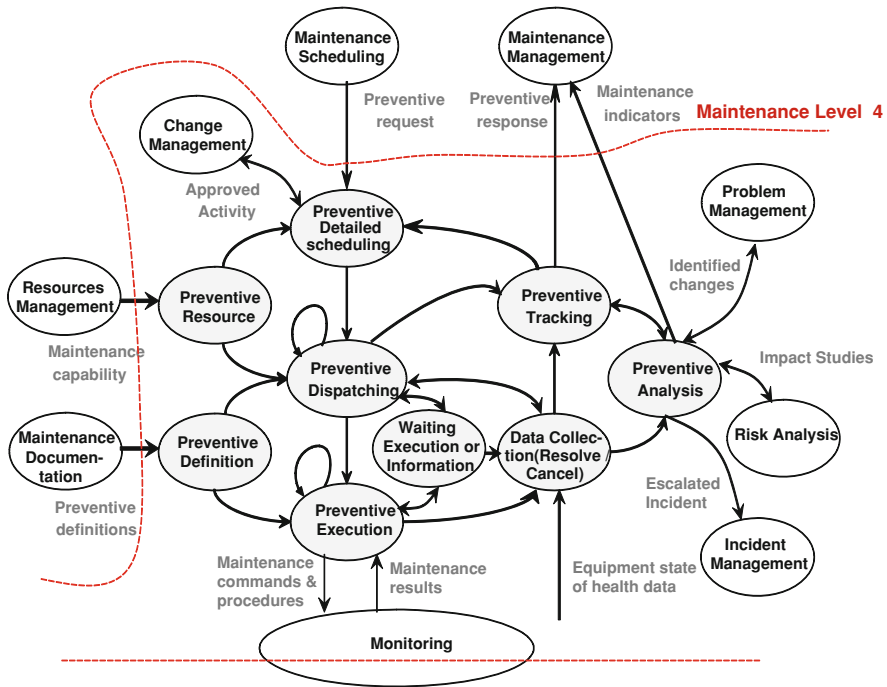


Fig. 12.6 Model of the preventive activity according to ISO 62264

CASE tool, including basic UML models plus testing, metrics, change management, defect tracking and user interface design extensions; developed by Sparx Systems. In Fig. 12.7, a UML diagram for the integration of management, monitoring and reliability systems is presented as an example, in which the relationships among attributes of the three systems are presented (for more detail see [51]).

Thus, taking as starting-point the UML diagram, the implementation of the system in a computer language is easy, translating directly from it. Although we may think about acquiring a commercial maintenance system, instead of developing our own, this gives us the know-how to evaluate the existing choices in the market.

12.2.11 Evaluating a Maintenance Management System

Obviously a maintenance manager does not need to know how to develop a maintenance management system (CMMS), but he should know its fundamental functions. From now on, we will describe a method to evaluate maintenance systems according to the basic functions defined in previous paragraphs. This can be balanced with the main source of costs related to the system implementation.

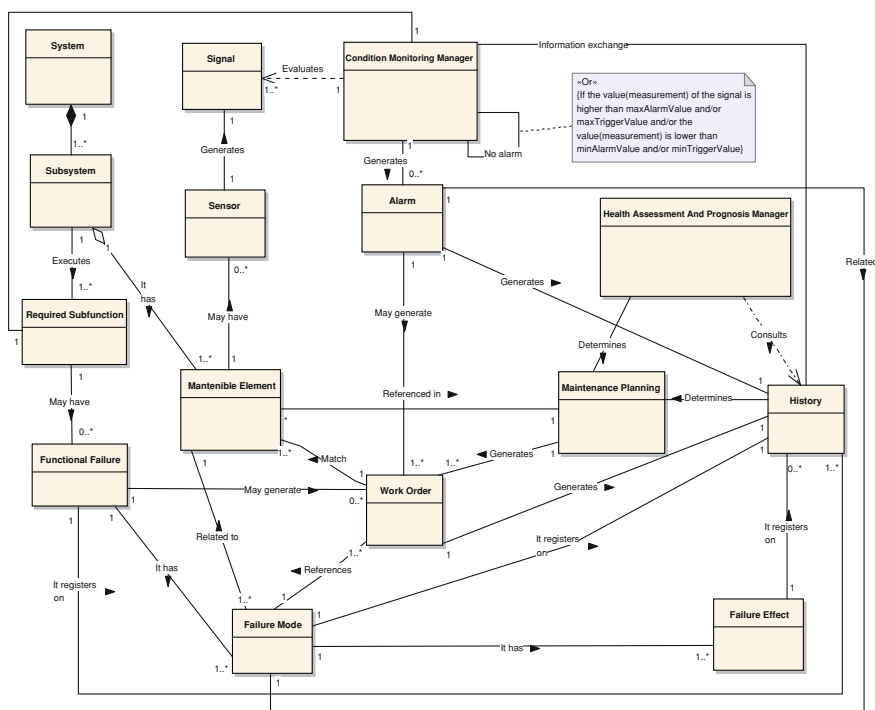


Fig. 12.7 Integrated UML model management, monitoring and reliability systems

The evaluation has been structured in seven sections (adapted from Crespo et al. [18]):

- Functions related to the capture and processing of information:
 - Hierarchical and structural coding of assets and infrastructures, without losing their relation to the functionality and customer.
 - Coding of all maintenance resources and costs.
 - Coding of the symptom-cause-solution, maintenance activities, documents and so on.
 - All coding of data entry must be clear, standard and unified.
 - Facilities to capture “on-line” data, by automated systems, monitoring, etc.
- Functions of knowledge base, providing assistance at the operational level:
 - Management of machinery history from the operative point of view.
 - Development of a “Knowledge Base” from historical information about the symptom-cause-solution coding.
 - Supporting the maintenance operative level by troubleshooting programs and decision-making techniques, toward optimization as expert support system.

- Functions for managing and planning the maintenance department in the operative, tactical and strategic level:
 - Predetermined maintenance management parameters for assets, infrastructures and other resources.
 - ABC classification, such as, on assets, infrastructures, type of costs, type of activities, etc.; related to their maintenance. The evaluation criteria can be simple or multivariable (criticality).
 - Facilitating flexible programing and scheduling, encouraging management by exception and alerting
- Functions for self-control and continuous improvement:
 - Self-assessment procedure by checking the maintenance indicators that determine the system performance.
 - For comparative purposes, employment of standard indicators internally or in relation to similar companies.
 - Implementation of continuous improvement cycle.
- Functions of interoperability and integration with other information subsystems, systems, hardware and applications:
 - Development of a maintenance data base for management, connected with purchasing, human resources, cost accounting and production systems, unifying the relevant encodings.
 - Connection with automated systems of data entry such us from production, inventory or access control.
 - Connecting the document database of maintenance with the general document database.
 - Facilitating the implementation of all stages of the RCM analysis.
 - Online or on-demand risk management and by a qualitative (standardized) or quantitative (in terms of cost) evaluation of impacts/consequences.
 - Complete integration into the overall plant or network organization.
- User facilities:
 - Intuitive and ergonomic interface similar to other applications. Client/server interface or web.
 - Acceptable response time.
 - Graphical facilities for displaying hierarchical and geographical relationships.
 - Management of maintenance documentation in different formats: Word, PDF, Excel, etc.
 - Access control and security.
 - Programmable and automatic protection of backups and restore points.
 - Developed on open source or proprietary.
 - Assistance by telephone, online or in person as a guide in each of the implementation stages and technical service in maintenance incidents.
 - Existence of previous demos for learning and testing.

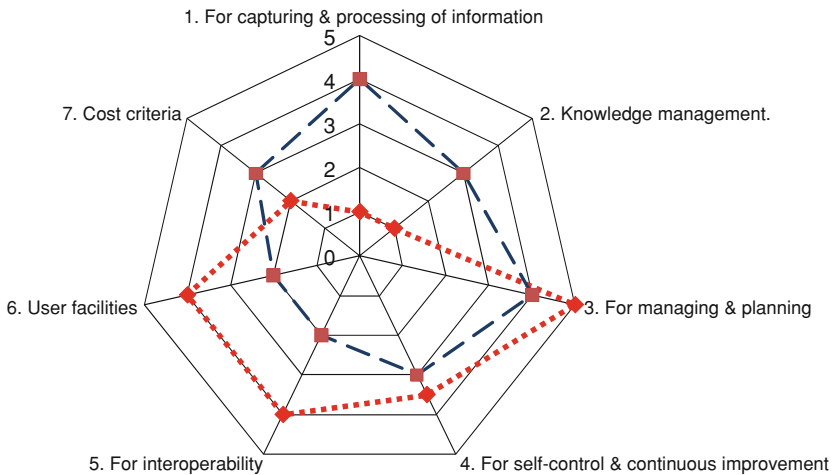


Fig. 12.8 Spider diagram for evaluation of maintenance systems

- The definitions and terms are standardized.
- Inclusion of standardized databases maintenance (ISO 14224, OREDA, etc.).
- Cost criteria:
 - Preparation costs.
 - Cost of software licenses.
 - Implementation costs including internal commitment.
 - Cost of training.
 - Maintenance cost per year including licenses.
 - Cost of updates.
 - Additional consulting costs.
 - Market presence as economy of scale.

Assigning different or equals weights to each one of these factors, according to our preferences, we can determine the fulfilment of each criterion (from 1 to 5), and calculate an average valuation among software alternatives. Results can be presented using a spider diagram as in Fig. 12.8.

12.3 Selecting Proper Performance Indicators

Performance indicators help to control maintenance management and to align maintenance objectives with business objectives. Through indicators control and management experience new knowledge is acquired and improved in the organizations.

There are multiple dashboards or standard indicators that can be used for maintenance management, such as those presented by Kaplan and Norton [41], Nakajima [53], EN15341 [24], SMRP [64] or Parida [58]. In the following lines we have organized some of the referred useful indicators within six categories:

1. *Management and organization*. With the aim of measuring staff skills, performance, leadership, knowledge, motivation and stability;
2. *Finance and economics*. Measuring performance in terms of cost.
3. *Business and production*. Measuring performance parameters of production and equipment.
4. *Quality and customer relations*. Measuring the maintenance influence offered and perceived by the customers.
5. *Safety and environment*. Measuring the maintenance contribution on issues concerning society: health/safety and the environment.
6. *Development and improvement*. Measuring organization's potential to improve and optimize results.

The list that we propose can be used to select convenient indicators for specific network utilities management control.

These groups of indicators are the following:

1. Management and organization.
 - *Average of staff performance indicator (at management and operative level)*. According to the performance management system of the organization.
 - *Average ratio of personnel training (at management and operative level)*. According to the official board of education.
 - *Average ratio of staff competences (at management and operative level)*. According to the competency-based development of the organization.
 - *Average experience of staff (at management and operative level)*. Mean years of experience in the management team.
 - *Ratio of management versus operational personnel*. Resources related to managerial tasks/resources devoted to operational tasks.
 - *Ratio of staff people promoted*. Promoted resources/total number of staff resources.
 - *Ratio of labor turnover*. Rotated resources/total number of staff resources.
 - *Average seniority (of the management and operative team)*. Mean years of seniority in a type of team.
 - *Average of staff use*. Average of the total resources assigned to each activity/average of the total resources necessary in each activity according to standards or procedures.
 - *Overtime ratio*. Total overtime worked/total labor hours.
 - *Absenteeism rate*. Number of absence hours/total number of hours.
 - *Ratio of staff satisfaction*. Ratio of internal staff satisfaction.
 - *Ratio of tools and equipment per worker*. Total cost of tools and equipment (including IT equipment and software licenses)/total number of employees.

- *Ratio of staff with access to information systems.* Human resources with access to ICT/Total number of employees.
- *Ratio of management and documentation based on ICT.* Ratio of activities supported on ICT for management and documentation.

2. Financial—economic.

- *Weight of the department's budget.* Maintenance budget/total budget of the organization.
- *Weight of the maintenance cost in production costs.* Total maintenance costs/total production costs.
- *Weight of the maintenance cost versus replacement net value of assets.* Total maintenance costs/total replacement net value of assets.
- *Compliance with the department's budget.* Total maintenance costs/maintenance budget.
- *Weight of staff cost in the maintenance budget.* Maintenance personnel costs/Maintenance budget.
- *Mean income (at management and operative level).* Total salary cost/total number of staff resources.
- *Ratio of unproductive hours.* Unproductive hours cost/total wage cost.
- *Weight of outsourcing cost.* External service costs referenced to total maintenance costs.
- *Weight of maintenance software systems costs.* Maintenance software systems costs/total maintenance cost.
- *Ratio of operative network assets per worker.* Opening network value/total number of staff resources.
- *Ratio of depreciated network value per worker.* Depreciated network value/total number of staff resources.
- *Ratio of replacement costs versus operative network.* Total cost of employed spares/opening network value.
- *Ratio of network to maintain according to the personnel costs.* Opening network value/total personnel costs.
- *Ratio of maintenance costs per km² of network coverage.* Maintenance budget/km² of network coverage.
- *Depreciation ratio.* Opening network value/depreciated network value.
- *Ratio of maintenance costs in annual sales.* Maintenance costs/annual sales.
- *Ratio of human resources costs per maintenance activity.* Total personnel costs/total number of maintenance activities.
- *Complaints cost.* Cost of guarantees, penalties and bonuses before claims/total number of claims.
- *Customer income ratio per maintenance worker.* Annual customer income/total number of staff resources.
- *Ratio of affected customers by maintenance problems.* Annual customer income of the affected customers by maintenance problems/annual sales.

3. Business and Production.

- *Ratio of each activity type*. Total number of activities per each type (corrective, preventive, etc.)/total number of activities.
- *Dedication per asset*. Total number of hours for each activity type (corrective, preventive, etc.)/total number of operative assets on the network.
- *Ratio of supporting to other departments*. Total costs of supporting activities to other departments/total activity costs.
- *Customer interaction ratio*. Number of activities with customer interaction/total number of maintenance activities. (It is important to measure the influence in dealing with the customer).
- *Generation activity ratio (in a given time)*. Number of activities generated in a given time.
- *Outstanding activity ratio (in a given time)*. Number of outstanding activities in a given time.
- *Average activity life*. Activity life in a given time.
- *Average response time*. Average time to respond to the allocation of an activity.
- *Average resolution time*. Average time in order to resolve completely an activity.
- *Average reestablishment time*. Average time to reestablish the service to its operative condition.
- *Ratio of active assets per worker*. Number of active network assets/total number of staff resources.
- *Ratio of network coverage per worker*. km^2 of network coverage/total number of staff resources.
- *Average of displacement per activity*. Average of HR displacement per activity.
- *Inventory ratio*. Total number of active network assets controlled by software systems/total number of active network assets.
- *Preventive cost ratio*. Total preventive costs/total maintenance costs.
- *Corrective cost ratio*. Total corrective costs/total maintenance costs.
- *Predictive and condition-based maintenance dedication*. Total hours in PDM/CBM/total hours of maintenance.
- *Asset monitoring ratio*. Total number of network assets monitored/total number of active network assets.
- *Ratio of alarms per asset*. Total number of active alarms/total number of network asset monitored.
- *Efficiency of alarms*. Total number of active alarms/total number of dispatched activities.
- *Proactivity ratio*. Total number of proactive corrective activities/total number of corrective activities (reactive+proactive).
- *Frequency of alarms occurrences*. Average of frequency of alarms in all monitored active assets.

- *Ratio of activities without standardized solution internally.* Total number of activities without standardized solution internally/total number of activities.
- *Mean time to fail (MTTF).* The average time operating at full capacity without interruption in a given time.
- *Mean time to repair (MTTR).* Effectiveness in restoring the operating conditions of an asset after a failure.
- *Mean time between failures (MTBF).* Frequency of failures. Total equipment uptime/number of failures during that period.
- *Average of lifetime per removed asset.* Mean lifetime of the removed asset out off the network.
- *Average of lifetime per repaired asset.* Mean lifetime of the repaired asset of the network.
- *Ratio inventory versus depreciated network.* Total value of spare parts/(opening network value—depreciated network value).
- *Weight of inventory out of circulation.* Without moving parts value/total value of spare parts.
- *Efficient use of spare parts.* (Total cost of the installed spare parts—Total cost of residual values of replaced equipments)/total cost of the installed spare parts.
- *Rotating spare parts ratio.* Total cost of installed spare parts/total value of spare parts.
- *Ratio of unplanned activities.* Number of unplanned activities/total number of activities.
- *Ratio of pending activities by spare parts.* Maintenance activities pending of spare parts/total number of maintenance activities.
- *Ratio of rush activities by spare parts.* Rush maintenance activities by spare parts/total number of maintenance activities.
- *Preventive coverage.* Number of asset with preventive activities/total number of active assets.
- *Compliance with the preventive planning.* Percentage of compliance of the preventive planning.
- *Ratio of incidents detected by the preventive.* Total number of incidents detected by the preventive/total number of incidents.
- *Predictive coverage.* Number of asset with predictive activities/total number of active assets.
- *Compliance with the predictive planning.* Percentage of the execution of the predictive planning.
- *RCM coverage.* Number of critical assets with RCM analysis/total number of critical assets.
- *Ratio of changes in the network.* Number of services installed, configured or modified/total number of active assets.
- *Efficiency of network capacity.* Network capacity available/max capacity of the network.
- *Availability.* Average of the percentage of available total time of the asset to perform its function.

- *Performance rate*. Average rate of real production/average rate of planned production.
- *Ratio of unavailability due to corrective*. Total unavailable time of the assets due to remedial activities/total max time of the active assets.
- *Ratio of unavailability due to preventive*. Total unavailable time of the assets due to preventive activities/total max time of the active assets.
- *Dependability*. Average of the probability of assets to perform a specific task under prescribed conditions in a given period.
- *Ratio of risk analysis actualization*. Number of updated risk analysis/total number of risk analysis.
- *Rate of critical assets*. Number of selected assets as critical/total number of active assets.
- *Critical customers ratio*. Number of selected customers as critical/total number of customers.
- *Ratio of contingences implementation to avoid risks*. Number of implemented contingences to avoid risks/total number of analyzed risks.
- *Efficiency of contingencies*. Total cost of implemented contingencies/total cost of estimated risks.
- *Overall network efficiency*. Availability $\times$ Performance Rate $\times$ Quality Rate.
- *Return on asset*. Annual net customer income/total value of operative assets.

4. Quality.

- *Ratio of appraisal costs*. Appraisal costs/total quality costs.
- *Ratio of prevention costs*. Prevention costs/total quality costs.
- *Ratio of failure costs*. Failure costs/total quality costs.
- *Ratio of satisfaction surveys*. Number of satisfaction surveys of customers about maintenance/total number of satisfaction surveys.
- *Quality rate*. (Total sales – Total no quality costs)/total sales.
- *Customer loyalty*. Average of customer life time with the organization.
- *Contract and regulation fulfilment*. As $1 - (\text{Costs of warranties and penalties} / \text{total maintenance costs})$.
- *Brand valuation ratio*. Assessing the degree of brand valuation due to maintenance issues versus the average brand valuation in the sector.
- *Ratio of network penetration*. Number of customers with service in a particular area/total households (with and without service) in a given area. (Market share).
- *Ratio of affected customers by discontinued services*. Number of affected customers by discontinued service/total number of corrective and preventive activities that provoke discontinued service.

5. Safety and environment.

- *Security cost per worker*. Total cost in security/total number of staff resources.
- *Recycling ratio*. Percentage of recyclable parts of the assets.

- *Ratio of risk analysis actualization about health–safety and the environment.* Number of updated risk analysis about health–safety and the environment/total number of risk analysis about health–safety and the environment.
- *Ratio of contingences implementation to avoid risks about health–safety and the environment.* Number of implemented contingences to avoid risks about health–safety and the environment/total number of analyzed risks about health–safety and the environment.
- *Efficiency of contingencies about health–safety and the environment.* Total cost of implemented contingencies about health–safety and the environment/total cost of estimated risks about health–safety and the environment.
- *Ratio of environment nonconformities.* Number of nonconformities about environment/total number of environmental risks.
- *Ratio of security nonconformities.* Number of nonconformities in security/total number of produced personnel access.
- *Ratio of human accidents.* Number of accidents in health–safety/total number of staff resources.
- *Safety accident frequency.* Total number of hours per period/number of safety accidents in the period.
- *Environment accidents frequency.* Total number of hours per period/number of environment accidents in the period.
- *Mean time of waste or debris.* Total number of hours of residues stay/total number of residues sets to be removed.

6. Evolution and improvement.

- *Training ratio.* Number of training hours/total number of hours of staff resources.
- *Planning ratio.* Number of planning hours/total number of hours of staff resources.
- *R&D efficiency.* As $1 - (R\&D \text{ investments} / \text{expected future income})$.
- *Ratio of training hours per worker.* Number of training hours/total number of staff resources.
- *Average cost of training hours.* Total cost in training/total number of trained staff resources.
- *Ratio of internal proposals.* Number of internal proposals/total number of staff resources.
- *Effectiveness of internal proposals.* Number of implemented internal proposals/total number of internal proposals.
- *Continuous management improvement.* Number of modified and created procedures/total number of staff resources.
- *Ratio of investments in automation.* Cost of investments in automation/total investment cost.
- *Perfective improvements per worker.* Number of perfective improvements in maintenance/total number of staff resources.
- *Effectiveness of human reliability.* Real human reliability/human reliability according to standards or procedures.

- *Unavailability due to human error*. Time lost due to human error or lack of knowledge/total hours of active assets.
- *Effectiveness of activities*. Weighted average of resolution time of activities/weighted average of standard resolution time of activities.
- *Ratio of recurrent activities*. Number of repetitive activities/total number of activities.
- *Ratio of lack of data integrity*. Total incomplete and incorrect data/total stored data.
- *Modernization rate*. Number of removed obsolete assets/total number of obsolete active assets.
- *Corrective improvement*. Number of corrective activities with known and standardized solution/total number of corrective activities.
- *Effectiveness of preventive*. Number of corrective activities generated from the preventive/total number of corrective activities.
- *Efficiency of preventive*. Cost of corrective activities generated from the preventive/cost of corrective activities nongenerated from the preventive.
- *Predictive improvement*. Weighted average of cost of corrective activities in assets after the predictive implementation/weighted average of cost of corrective activities in assets previous the predictive implementation.
- *Weight of predictive cost*. Total cost in predictive implementation/total opening network value.
- *Energy efficiency*. Energy consumption cost/estimated energy consumption cost.
- *Weight of call center use due to claims about maintenance*. Total number of claims about maintenance/total number of customer claims.
- *Involved staff resources on improvement forums*. Percentage of involved staff resources on improvement forums.
- *Problems life cycle*. Average time to find solutions to unknown problems.

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Chapter 13

Management of Change

13.1 Introduction

Change in organizations is one of the issues that have been studied from different perspectives, and by different disciplines, as an important aspect in management. Previously we have paid attention and reviewed the physiological aspects of changes, now we will focus on the managerial aspects of changes within organizations.

In order to know how we have to undertake change from a managerial point of view, it is necessary to understand the mechanisms involved, the aspects empowering or obstructing the change, and the implications depending on the different types of organizations and their structures.

The first critical step in designing and conducting a successful large-scale change is to understand the dynamics and performance of the company. It is important to model this performance, identifying relationships among structure levels and organization members. The management of change has to be undertaken according to the organization type as Gareth Morgan [16] indicated. In his study, organizations were classified using metaphors based on assumptions about the organizations' vision and peoples' attitudes before changes:

- **Machines:** the organization is managed in a mechanical manner, consisting of interlocking parts with repetitive and predictable results.
- **Organisms:** the organization is managed as a living system, considering its environmental relations and needs.
- **Brains:** the organization is managed focusing on its ability for learning and training based on parallel information processing and distributed control.
- **Cultures:** the organizations life is managed centring on ideas, beliefs, traditions or rituals.
- **Political systems:** the organization is managed depending on interests, conflicts and power.

- **Psychic prisons:** the organization is managed as a prison where people are trapped, repressed, defended or workaholic, considering conscious and unconscious mind.
- **Flux and transformation:** the organization is managed as a flow with constant change and self-organisation.
- **Instruments of domination:** the organizations are instrument of domination, producing people alienation or exploitation with a strong managerial level.

Although, this generic approach has several critiques, considering it as an epistemologically valid and practical approach instead of using precise scientific languages and procedures [15], it has been used as basis in many studies that provide models of how to manage changes according to the typology or metaphor of the organization. The most frequently used metaphors are those of machine, organism, political and flux [6]:

- Bullock and Batten [5] provide a model for organizations as machines in which the changes are considered as a technical problem, and in which they are viewed as planned changes according to the principles of the project management.
- Nadler and Tushman [17] define a congruence model based on organizations formed by a set of sub-systems interacting with the environment as organisms and subject to power relations as political systems, searching the alignment and congruency amongst themselves in consistency with the vision of the organization. In this model the system is viewed as a box with input (environment, resources and history) to be transformed into outputs (organizational, group and individual) using strategies to guide decisions, where the transformation process is based on four key elements that have to be congruent and aligned (fit) amongst themselves to fulfil the performance effectively:
 - Work which is the basic activity of the organization,
 - People considering abilities and characteristics,
 - Formal organization in order to coordinate the work and people (structure, systems and policies), and
 - Informal organization composed by unwritten rules that influence in global and individual behaviour.
- Stacey et al. [23] present a model as flux where changes emerge naturally and are explained as a way to achieve a new order complex from clean communication, conflict and tension.
- Cameron and Green [6] describe the key assumptions to take into account in the analysis of change management according to applied models or metaphors of organization. For example:
 - Machines rationally structured and unambiguously designed according to the results to be obtained, based on the key assumptions of a unique command chain to each employee with specific roles and managed by objectives summarizing efforts like a team;
 - Political system applying rules and managing power relations and conflicts of interest, based on the key assumptions of recognized power groups with

political dependences among them, where the changes need to be promoted from the leadership through negotiations or competition producing winners and losers;

- Organism compound of several subsystems that interact to maintain proper alignment between themselves with the objectives depending on the changes in the environment; or
- Continuous flux and reflux based on the chaos theory instead of cause-effect relations where the produced changes cannot be managed.

These models and other subsequent models highlight as a crucial pillar, the fit or alignment amongst their elements. One of them is the 7S model of McKinsey [19, 20] in which the organization is considered as a system with seven key correlated and independent factors whose proper fit determines effective performance and change, combining rational factors “hard skills” such as strategy, structure and systems; with emotional factors “soft skills” such as shared values, skills, style, staff. The importance of each factor depends on the change type, varying with the conditions and time.

Following the evolutionary framework of management for network utilities, which implies continuous improvement, and as such is the reason why we have recommend the application on the framework progressively [21], increasing the maturity levels as CMMI indicates, beginning with quick-wins, making incremental and sustainable wins, redirecting the department management towards the optimization of business objectives. That is to say, similar to the defined levels by Nelson and Burns [18] for an effective organization:

- Level I. Reactive organization, acting according to circumstances;
- Level II. Responsive organization, establishing how to react before knowing the circumstances;
- Level III. Proactive organization, defining anticipatory measures to manage circumstances;
- Level IV. High-performing organization, searching for excellence in management.

However, some consequences would appear inside each level of planned change or required performance, because almost 75% of planned changes fail [2], due to, as Kristi and Branch [3] summarize:

- Difficulties in changing the basic ways of thinking within the organization in order to achieve a significant change;
- Absence of a firm leadership in personal and organization issues;
- Disruptive implementation of changes, periods of consolidation are necessary between periods of intensive implementation;
- Complications in business continuity in the interim situation that should be prevented in advance.

Therefore, the changes could be classified in different types depending on their impact on the organization such as the models of:

- Stace and Dunphy [22] describing the following levels:
 - Level I. Fine tuning as quick-wins;
 - Level II. Incremental adjustment as sustainable improvements inside a department;
 - Level III. Modular transformation as transformation of departments; and
 - Level IV. Corporate transformation affecting all the organization.
- Ackerman [1] distinguishes three types of change:
 - Developmental, improving skill or processes of an organization;
 - Transitional, as punctual change from an actual state to a known state;
 - Transformational, as radical change with importance changes in the organization.
- Lorenzi and Riley [13] divide changes into:
 - Micro changes, differences in degree: modifications, enhancements, improvements, and upgrades; and
 - Mega-changes, differences in kind: new system or a very major revision of an existing one.

Consequently, the implementation of a change in an organization has to be undertaken under the above issues in an adequate sequence of phases.

13.2 Implementation of Change Management

As we have previously mentioned, the most frequently used transition models define three involved phases [4, 12]:

- Present or ending phase in which the objective is to accept that the old concepts are ending, presenting the final phase and the enablers or impediments to the change.
- Transition or neutral phase in which the changes occur, implying uncomfortable situations for people but empowering creative solutions.
- Desired or a new beginning phase in which people have to commit themselves to the new concepts.

In order to implement organizational changes, some authors have defined temporal scales from the psychological point of view, but from the managerial point of view we have to plan a correct transition with enough time as Kotter [10] indicates, for instance, years, in the case of mega-changes. Then, we have to minimize potential risks and consequences avoiding a possible regression, undertaking certain actions as Bridges [4] recommends for a proper transition:

- Share the problem;
- Collect information about the problem (the four Is': Information, Interests, Investment, and Influence);

- Audit the aptitude of the organization in order to discover the strengths and weaknesses;
- Educate the leaders;
- Plan for everyone who takes part in the change steps;
- Recognize that people will not (because they cannot) move straight to the new beginning;
- Enable individuals and the organization to step back in the neutral zone for period of assimilation;
- Monitor the progress of individuals and groups using a Transition Monitoring Team;
- Plan how you are going to explain, encourage, and reward the new behaviour and attitudes;
- Keep track of enablers or hinders of the change in every moment in order to guide the transition.

In our framework, for implementing changes we could employ Kotter's eight steps [10]:

- Establish a sense of urgency.
- Create a guiding coalition.
- Develop a vision and a strategy.
- Communicate the change vision.
- Empower broad-based action.
- Generate short-term wins.
- Consolidate gains.
- Anchor new approaches in the culture.

Now, we will focus on two organizational key aspects for implementing changes, previously mentioned, the leadership and change groups.

Change leadership could determine an effective change management, empowering all the organization towards the same vision, through motivation, alignment and integration, overcoming the resistance and avoiding obstacles. The leader has to globally and continuously analyse the change management process, taking corrective or preventive actions, based on the following strategic principles according to Heifetz [9]:

- Identify the adaptive challenge;
- Regulate the level of distress;
- Focus attention, reducing distractions;
- Give the work back to people; and
- Protect voices of leadership without authority.

Consequently, the style of leadership has to adapt according to circumstances, as Goleman [8] indicates; coercive, authoritative, affiliate, democratic, pacesetter, and coaching; recommending each style in certain specific situations because they have a different impact on the organizational aspects: flexibility, responsibility, standards, rewards, clarity, commitment.

In line with this, Bridges [4] recommends the creation of Transition Monitoring Teams (TMTs) which have to be educated to lead and monitor the transition regularly and frequently through meetings and tracking calls. The TMTs are based on individuals from different sections of the organization, no more than ten in number, with implicated people among volunteers and nominees, searching the enthusiasm, experience and specialization, focusing on specific matters inside each group. In TMTs there has to be a facilitator to guide the meetings, decisions and relationships with other teams, following-up the left-over tasks, possible impacts and the produced results, and enhancing communication issues such as feedback, confirmation, dissemination, as well as refreshing the concepts and refuting rumours.

Accordingly, the employment of TMTs are convenient due to the fact that people in organisations work in groups [12], it is a reasonable support to the transition of changes, modelling behaviour from the existing to the new practices and norms. These teams are also maintained after a change project, labelling them as continuous improvement teams [14], which have to overcome the obstacles that weigh down effectiveness [7], taking for granted the team mission, planning, goals, roles, responsibilities, operating processes and, interpersonal and inter-teams relationships.

In major changes, the TMTs are divided by matters or subject, focusing on a clear purpose with the best people that have knowledge about the matter or subject, i.e. according to the managerial dimensions of the Leavitt diamond [11], relating to people, task, structure and technology.

13.3 Change Management Case Study

This case study deals with the project of change within a department of maintenance due to companies merging between two territorial network utilities in the telecommunications sector. Each company has its own processes, procedures and, technologies and suppliers in the network infrastructure. As a result, the initial situation is a complex management environment with different organizations and network structure, which is reinforced with a clear technology-oriented maintenance management, so the aim of the change project is to unify the maintenance management for both companies, searching a final state in which the structure of processes and organization will be integrated, reducing redundant activities and responsibilities and technology dependence.

Therefore, the objective is the implementation of an integrated management framework for maintenance in this multi-supplier and multi-technology environment, in order to facilitate the effectiveness in management and decision making without affecting the rest of the company.

Consequently, the organization has to change to a distributed organization with the decision making authority, regarding budget and technical considerations,

strongly centralized to supervise the operation in both territories, which implies a directive style of management with hierarchical levels of procedures, bureaucracy and authorizations. From an operational point of view, the organization has to transform from functional, depending on technologies and suppliers, to processes-oriented.

Taking into consideration the recommendations of Bridges [4] and Kotter [10], and considering the aspects of project management and software development for service organizations (ITSMF 2010), the implementation phases in the case study were:

- Establish a sense of urgency.
 - Analyse the initial circumstances, collecting information about the problem (information, interests, investment and influence).
 - Audit the aptitude of the organization in order to discover the strengths and weaknesses, analysing the impacts of the changes.
- Create a guiding coalition.
 - Share the problem.
 - Educate the leaders.
 - Plan how you are going to explain, encourage, and reward the new behaviour and attitudes.
- Develop a vision and a strategy.
 - Plan for everyone who takes part in the change steps and support the project in TMT.
- Communicate the change vision.
- Empowering broad-based action.
 - Recognize that people and structures will not (because they cannot) move straight to the new beginning.
- Generate short-term wins or gains.
 - Prioritize the planning and execution including contingencies.
 - Monitor the progress of individuals and groups using Transition Monitoring Teams;
 - Keep track of enablers or hinders of the change in all moment in order to guide the transition.
- Consolidate gains.
 - Enable to individuals and the organization to step back in the neutral zone as period of assimilation;
- Anchor new approaches in the culture.

13.3.1 Establish a Sense of Urgency

This first phase is focused on diagnosing the initial situation, with a clear view of the points that must be tackled as the basis of the definition of the final model. In addition, a TMT was created in order to study the initial situation and compile necessary information about HRM, contracted external companies and services, organization, responsibilities, functions and processes, SLAs, interdepartmental relationships, ICT, techniques and tools, performance, etc. The result is a repository of all procedures, specifications and technical instructions for each company and each network.

The initial situation was characterized by:

- Deviations in maintenance budget;
- Confusion between CAPEX and OPEX;
- Diversity of salaries between both companies;
- Mismanagement of maintenance staff and spares in the territory, using different criteria (overtime ratio 12%, spare parts 60% of maintenance budget);
- Non-unified management of external resources;
- Amalgam of maintenance systems;
- Dispersion of resources in field without unified communication ways;
- Misalignment with business objectives;
- Absence of a unique and updated source of network documentation.

In addition, using the Bridges check-list we obtained that the transition could disrupt the normal operation of maintenance due to providing a $24 \times 7 \times 365$ service. Beyond the personal interests in future responsibilities in the new organization, the necessity of the change was obvious for both staffs, seeing the competitors in the market, and the objectivity of the leadership was supported by an external consulting company.

All these circumstances have to be taken into account in the change management project and then be solved without affecting service level agreements and interdepartmental relationships.

13.3.2 Create a Guiding Coalition

During this phase, the need for change was presented to all the directors and managers, learning and training leaders and facilitator personnel (done by the consulting company). In this managerial level, a monetary reward was funded based on the fulfilments of the change management project.

The middle managers administrate the required personnel for the project restrictively, focusing more on the fulfilment of the annual objectives than on the facilitation of changes, mainly because they were only rewarded by the former.

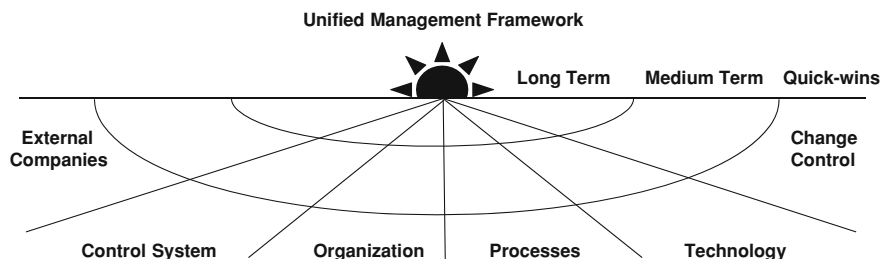


Fig. 13.1 Maintenance strategy and TMT

13.3.3 Develop a Vision and a Strategy

In this phase, different improvement opportunities were valued in order to prioritize those with a high impact and low cost and/or effort. By doing so, the final (considered optimal) model for maintenance management was designed.

Then, the resources and activities required to migrate the current business model to the new situation were assessed and estimated for the implementation. The plan produced was based on the following considerations:

- Dedication of internal resources as principal collaborators.
- Commitment of external resources as support and guide.
- Total involvement of the business management team.

The time table was ambitious, searching in one and half year the transition and implementation of a new management model, without previously ensuring a stable situation (this is due to the fact that sometimes the transition may jeopardize the normal department performance). This was in opposition to Kotters' [10] recommendations for major changes, suggesting that quick advances generate unsatisfying results and negative impacts. The change management project was defined embracing too many changes, integration of organization, information systems, processes, logistic, etc.

According to the preceding paragraphs we sorted out the improvements, sizing and prioritizing a plan considering the impact and benefit of their implementation in the short (quick-wins), medium and long term (see Fig. 13.1):

The initial TMT was divided into subgroups according to functional areas: organization, processes, technology, control system, change control and external contracts. The TMT was determined according to responsibilities, members, objectives and the expected improvements and their benefits:

- Organization.
 - Objective: simplifying the organization in order to reduce costs without quality diminution.
 - Improvements:

1. Definition of the new organization with (central and territorial) responsibilities.
 2. Establishment of the relational model with the rest of the company and external agents (companies, administration, etc.).
 3. Sizing in a high level the resources in line with the number and type of activities.
- Benefits: reduction of resources and the time to perform activities.
- Processes and procedures:
 - Objective: defining the adequate managerial framework with the aim of achieving sustainable improvements.
 - Improvements:
 4. Simplification of activities and systematization in processes.
 5. Definition of the necessary resources per activity.
 6. Implementation of a training plan.
 7. Determination of the logistic sizing.
 8. Development of the maintenance processes.
 9. Design of the contingency and continuity plan.
 - Benefits: reduction of spares and the time to perform activities, increment of efficiency in the use of resources eliminating redundant tasks (cost optimization), and increment of service availability.
 - Technology:
 - Objective: defining the minimum characteristics of equipment according to standards and interoperability, avoiding supplier dependency.
 - Improvements:
 10. Criticality analysis of the network.
 11. Definition of technical specifications and requirements according to standards.
 12. ROI and LCCA of critical equipment and for different suppliers.
 13. Selection of telecommunications technologies.
 - Benefits: reduction of costs, obsolescence, changes and dominant positions of suppliers. Increment of availability, network sustainability and interoperability.
 - Control system and service level management:
 - Objective: implementing mechanism for monitoring the activities and resources.
 - Improvements:
 14. Determination of the objectives of the maintenance department.
 15. Establishment of internal and external service level agreements.
 16. Definition of the control system.

17. Integrated and interoperable governance of the maintenance information: unique repository of information and responsibilities delimitation in its management.
 - Benefits: reduction of cost, cycles and redundant tasks. Facilitation of coordination. Unification of criteria and alignment with the business objectives.
- External contracts:
 - Objective: modifying their management in a unique way without budget increment and quality diminution.
 - Improvements:
 18. Optimization of number of external companies in technologies and activities without critical value and eliminating idle resources, according to real demand and economies of scale.
 19. Analysis of potential suppliers and external services including their agreements.
 20. Redesign of contracts, instead of paying in a flat rate or per activity, searching the payment per performance or fulfilment of common goals.
 - Benefits: reduction of costs. Increment of availability and efficiency.
- Control of change and configuration management:
 - Objective: tracking the entire change project and verifying the configuration of the network.
 - Improvements:
 21. Collection of all the relevant information.
 22. Audition of network configuration, state and reliability of the information.
 23. Detection of competitive advantages and, continuous improvement and weak points.
 24. Systematization of knowledge management.
 25. Prioritization of efforts and improvements.
 - Benefits: reduction of cycles. Increment of effectiveness in the change project. Facilitation of continuous improvement, coordination and knowledge management.

Once the TMT and improvements are described, the global time table of the project could be presented, see Fig. 13.2.

13.3.4 Communicate the Change Vision

The vision of change was communicated in different ways: intranet, conferences, e-mails, direct calls and meetings by the managers, specifically designed for each hierarchical level of organization. The frequency of these communications was

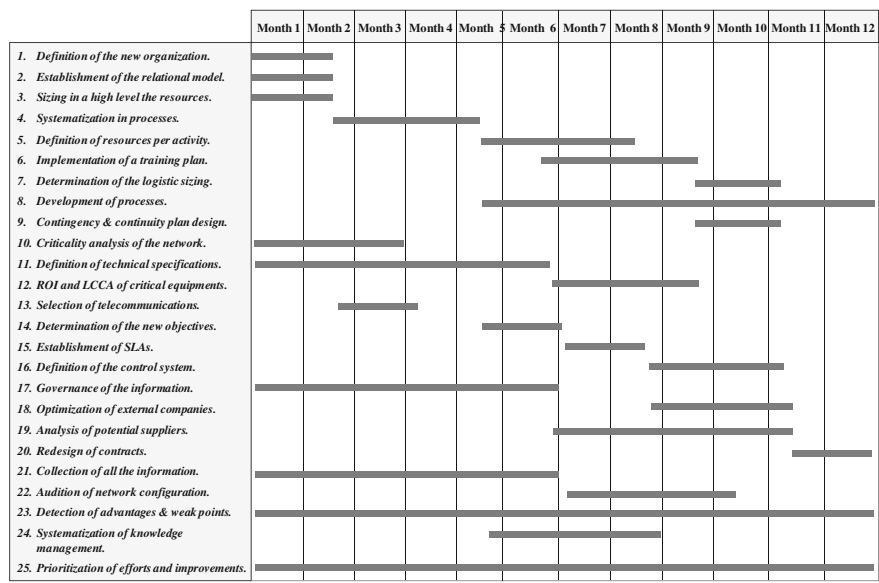


Fig. 13.2 Global planning of the change project

alternated between direct and indirect communications during the life of the project in order to maintain the expectation and motivation, and the feedback was monitored continuously.

At the same time, an external campaign was undertaken in order to align the internal motivation with the external acceptance of the new company, because this transformation also implied affection in the customer relationships.

13.3.5 Empowering Broad-Based Action

There were enough people with adequate technical knowledge, but sometimes inflexible about modifications in the procedures. They were often judging and requiring a higher detail level. The efficiency of the leadership was demonstrated guiding the new model to a level of detail in accordance with the maturity of the organization, that is to say, a level considered “repetitive” and sometimes “reactive” of performance.

Some department managers tried to take advantage, thanks to their participation in TMT, to obtain a promotion at the expense of their colleagues, producing a serious lack of co-operation.

An ignored obstacle from the beginning was the resistance of the suppliers, afraid of losing their dominant position in some aspects, for example in the case of specific critical equipment maintenance. Some equipment had their maintenance information system inside another company. Suppliers obstructed the

interoperability and unification of procedures and logistics during the transition, excusing their participation with their dedication to their normal operations.

13.3.6 Generating Short-Term Wins

At this moment, the prioritized scheduling of quick-wins was defined for each one of twenty five improvements found, based on their weight/importance for the final model of management, the required and available resources and estimated time. In addition to this, during the implementation, the monitoring of the evolution was realized through the TMTs each fortnight, readjusting as necessary.

The impact of changes made in one TMT have to be tested and aligned with each other based on actual circumstances more than on the initial situation, due to difficulties in future prediction (uncertainty) of their consequences. This evaluation was accomplished most of the times on a trial and error basis.

13.3.7 Consolidate Gains

The individual engagement of the staff with the changes and the demand of additional support on ICT were increasing progressively with the success of the quick-wins, but some of them provoked disruptions in others due to unnoticed overlaps among maintenance responsibilities. The trustfulness of the information was tested continuously, rewarding its fulfilment.

The fulfilment of the quick-wins was rewarded, highlighting future potential of the staff.

13.3.8 Anchoring New Approaches in Business Culture

Through knowledge management, norms and rules were defined, managing conflicts searching consensus between territories and hierarchical levels of the organization. The training and learning plan were in concordance with the iterative implementation of improvements.

Periodically, root-causes analysis of performance defects and problems were accomplished. Automation of tasks based on symptom-cause reasoning was accomplished systematically.

In addition, innovations measures and changes in organizational culture were established in order to avoid complacency and regression, encouraging a move towards world class maintenance. The effectiveness of internal proposals and data integrity and the involved staff resources on improvement forums were rewarded. In successive improvements simulations began to be utilized.

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Part III

Management Framework Evaluation

Chapter 14

Qualitative and Maturity Evaluation

14.1 Introduction

In the previous part of this work a reference framework for the maintenance management of network utilities has been defined, and in addition, some considerations about the organization of these types of companies were made. Now we will focus on the development of the framework to improve business effectiveness and efficiency, trying to combine operating criteria with department strategy [8, 9]. Therefore, the purpose of this section is aimed at facilitating the standardization, implementation, monitoring, control, decision making and sustainable continuous improvement supported by the consistency among processes, techniques, resources and systems. In line with this, the benefit of the provided maintenance management framework should be reflected in higher levels of network reliability and risk avoidance.

In many companies, as well as the reference framework that structures the activity, it is important to define the mechanisms that facilitate and guide the improvement and decision making in the right way, limiting the sources of uncertainty in order to minimize error as much as possible due to:

- Uncertain knowledge and data;
- Random and incomplete information; and
- Inaccurate language.

Consequently, this framework will pursue continuous improvement to increase the confidence in its rules under conditions of uncertainty. In order to do so, we suggest the following steps indicated by [10]:

- Problem analysis and communication of reliable information to maintenance staff, providing documented and useful specifications.
- Collection and inclusion of expert knowledge (tacit and explicit), similar to human reasoning and incorporating logic information accepted with consensus.

- Evaluation of consistency, with expert knowledge, improving quality and relevance of decisions.
- Generation of new knowledge and optimization of rules, redesigning the decisions-making process to increase the value of decisions, implementing continuous improvement.
- Decision-making process automation, improving responsiveness, simplicity, investments and accuracy.

Therefore, it is important to define how we must analyze the actual situation, the objectives to reach and the produced deviations. This will facilitate the discrimination among different options, searching higher precision in the choice of strategies. But before this, the following questions arise:

- How can we evaluate the current situation on a permanent basis, and the maturity reached, due to the framework implementation?
- How can we quantify the maintenance contributions due to the framework implementation?

Clearly our maintenance management framework provides a global vision about the dynamic relationships in our organization, facilitating the necessary changes and the alignment between activities and their management. Therefore, through re-engineering we must be able to improve our maintenance management maturity level, adapting our framework conveniently. Of course, higher knowledge and technology levels will be required over time. If we follow this path, the management framework definition and *standardization* is normally continued by a more *quantitative-based* management approach that finally results in a sort of *optimized* management according to circumstances. A mature and optimized process can correlate and consolidate information enabling the effective management of changes focused on the objectives of the organization.

Therefore, the benefit of our current management framework implementation can and should be measured as a maturity level, where this level will reflect the status of the developed mission, objectives, systems, processes, activities and tools, in accordance with the proposed framework. The evolution of the maturity level ensures advances in maintenance management, produces improvements in network reliability and in service quality.

For a proper definition of a maturity model for maintenance management in these customers-oriented companies we have reviewed existing best management practices. We should be able to evaluate the contribution of maintenance in a network utility in a given moment in time, including the existing potential to generate value in the future. We have found that this can be properly done capturing not only over time network reliability, but also business intellectual capital indicators. An important point here is to notice that maintenance may contribute to business excellence, in different areas, due to its internal intellectual capital (see Fig. 14.1): human, structural and relational, as follows:

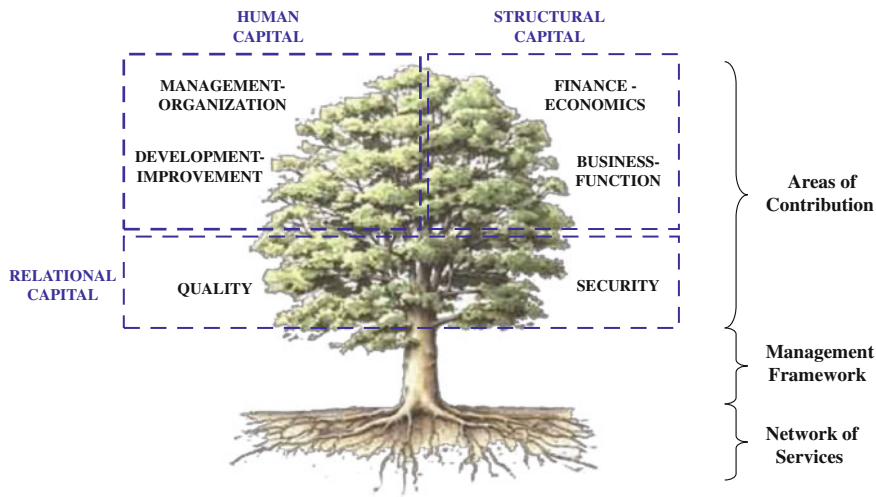


Fig. 14.1 Maintenance contribution areas

- Human capital is reflected in the organization and in its possibilities for improvement;
- Structural capital allows the creation of value in key outcomes of the organization, so it will be reflected in financial features of the business;
- Relational capital essentially reflects its contributions on the environment, customers and society. The affected business features or categories are quality and security.

Thus, in this section, we will try to evaluate the maintenance contribution due to the framework implementation using management maturity levels and also using different performance indicators.

Due to the special characteristics of network utilities, this maintenance management framework evaluation process should be accomplished from two perspectives:

1. Measuring the effectiveness in the management framework standardization. Establishing the maintenance management maturity level according to the implementation level of processes, systems and tools; and
2. Measuring the efficiency in the achievement of the objectives, quantifying key performance indicators of maintenance from the “performability” and “intellectual capital” perspectives.

To develop our maturity model, we will take as starting points the followings examples:

- The CMMI standard’s maturity levels, which will serve as a basis for evaluation of framework, the development through the implementation of best practices; and

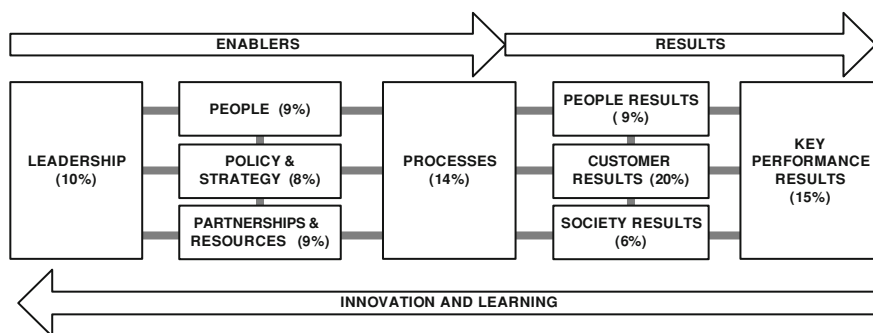


Fig. 14.2 The excellence model of EFQM (Source ww1.efqm.org)

- The EFQM's management excellence evaluation, used for presenting the capabilities of the maintenance department, due to its excellence in management and results.

According to CMMI [2], maturity (for IT infrastructure management in services companies) can be measured in five levels of capacity:

- Level I. Initial, representing unstable and unorganized processes.
- Level II. Repeatable and reactive, characterizing processes with a repeatable performance according to predefined objectives.
- Level III. Defined and proactive, for processes based on well-defined methodologies, procedures, techniques and supporting technologies, established to drive these processes.
- Level IV. Managed, where the processes are qualitatively managed and predictable.
- Level V. Optimized, testing new methodologies, techniques and technologies are tested proactively for continuous improvement aligned with business objectives.

On the other hand, the European Foundation for Quality Management is a non-profit membership foundation founded in 1989 and based in Brussels, and has similar criteria to other international awards such as the Deming Prize managed by the Japanese Union of Scientists and Engineers and established in 1951, or the Malcolm Baldrige National Quality Award managed by the National Institute of Standards and Technology (U.S. Department of Commerce) and created in 1987.

In the EFQM Excellence model [5], nine fundamental principles are defined, seeking business excellence through the ability to achieve planned objectives and to maintain excellent results over time: five enablers (inputs) and four result areas (outputs). The enablers are leadership, people, policy and strategy, partnerships and resources, and processes. The results are oriented to peoples, customer, society, and key performance results. Each of these criteria is subdivided to describe how to evaluate them in further detail (see Fig. 14.2):

- Leadership, enabling the organization to anticipate and react in a timely manner to ensure the ongoing success of the organization.
- Strategy, where policies, plans, objectives and processes are developed and deployed to deliver a stakeholder-focused strategy.
- People, valuing the staff and creating a culture that allows the mutually beneficial achievements of organizational and personal goals.
- Partnerships and resources, managing external partnerships, suppliers and internal resources effectively in order to support strategy and policies and the effective operation of processes.
- Processes, products and services, managed in order to generate value for customers and other stakeholders.
- Customer results, crucial for organizations with respect to their customers.
- People results, key for organizations with respect to their people.
- Society results, decisive for organizations with respect to society.
- Key results, significant for organizations with respect to the key elements of their policy and strategy.

In network utilities, the weight of the information technology in maintenance management processes optimization is very high. We can then use the CMMI's description of the maturity levels to evaluate the performance and excellence in maintenance management of these companies. The EFQM model can also be particularized to our framework, to improve the definition of the maturity levels in maintenance management from different points of view (economics, human resources, systems, etc.).

Now, we will begin presenting the methodology to evaluate qualitatively the effectiveness in framework standardization through management maturity levels, and we will continue describing the methodology for qualitative evaluation of the performance efficiency, to reach the maintenance objectives, from both the “performability” and the “intellectual capital” perspectives.

14.2 Maintenance Maturity Evaluation

In this section we try to answer one of the questions formulated above: how can we evaluate on a permanent basis the current situation, and the maturity reached, as a result of the framework implementation?

The consequent response is evaluating maintenance department maturity according to its management effectiveness, according to existing best management practices as CMMI levels and EFQM Enablers (see Fig. 14.3).

For network utilities, we can measure the maturity level of our framework checking current framework standardization, reviewing the implementation of the defined mission, objectives, responsibilities, strategy, processes, activities, systems and tools.

A precise evaluation methodology can now be defined based on the processes area outlined by CMMI for services [3] and on the enablers of the EFQM model.

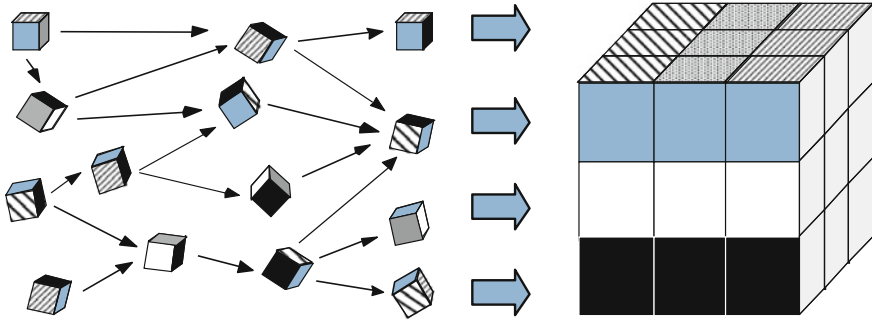


Fig. 14.3 Maintenance maturity evaluation based on best practices

This new release of CMMI considers the concepts of other standards of information technology such as ITIL, ISO2000, COBIT, etc [4, 6, 7].

The best management practices of CMMI-SVC are grouped in the five management maturity levels, and they are also classified into four categories:

- (a) Project management;
- (b) Process management;
- (c) Support and;
- (d) Establishment and delivery of services.

The EFQM enablers are crucial because of the following:

- (a) Policy and strategy, PS (8%). Develops and maintains the mission and vision of the organization via a clear stakeholder-focussed strategy, supported by relevant policies, plans, objectives, targets and processes.
- (b) Leadership, LE (10%). Develops and facilitates the achievement of the mission and vision via appropriate actions and behavior, leading the effective management of the organization and its relationships.
- (c) People, PE (9%). Develops and manages the knowledge and full potential of its people, as an individual as well as team-based, in order to manage its processes effectively according to its policy and strategy.
- (d) Partnerships and resources, PR (9%). Plans and manages its external partnerships and internal resources, in order to manage its processes effectively according to its policy and strategy.
- (e) Processes, PO (14%). Designs, manages and improves its processes in order to generate increasing value for its customers and other stakeholders according to its policy and strategy.

We will now combine these models to obtain a quantitative value, as a percentage, for the effectiveness of our management. To achieve a determined level of maturity, CMMI defines several generic and specific practices, all of which have to be fulfilled in each level [1] in accordance with the defined issues of our framework: mission, objectives, responsibilities, etc. Then, combining with the EFQM

Table 14.1 Check-list in level I of practices to evaluate the maturity and excellence

Process	Practices	PS	LE	PE	PR	PO
For all the processes in all the levels	All the basic functions are implemented as best efforts					✓

model, we could assess the percentage of fulfillment inside each level. To do this, we have to classify the CMMI practices according to the five types of EFQM enablers, dividing proportionally the weight of each enabler into the number of related practices with the same enabler.

- The degree of achievement for each practice will be characterized as:
- 100% when the practice is totally implemented.
 - 66% when the practice is widely implemented.
 - 34% when the practice is partially implemented.
 - 0% when the practice is not implemented.

In summary, multiplying the degree of achievement of each practice by its proportional weight, according to the related enabler, we can deduce the practice contribution to the excellence and the total result is the sum of all the practices contribution. Therefore we will have two values to show the evaluation of our organization: the percentage of excellence and the maturity level.

14.2.1 Maturity Evaluation Implementation

The maturity of a department according to CMMI is indicated by a certain level reached. This level is the highest level in which all indicated best practices, generic and specific, are reached. In the case that, within a level, not all the best practices are in place, then that maturity level is not reached.

From now on, we will present the adaptation of the CMMI-SVC and EFQM models to our reference framework for maintenance management in network utilities, in order to reflect the effectiveness in management.

In Tables 14.1, 14.2, 14.3, 14.4 and 14.5, the compulsory practices per maturity level with their related enablers, and as a check-list, are shown. This is required in order to calculate the weights of excellence for each type of enablers and inside each maturity level.

This maturity evaluation fulfills the principles of simplicity and understanding. The discrete values of CMMI levels show how far the next level of excellence is placed (quantitatively). The weights of EFQM enablers allow the calculation of a percentage value for the maturity within each level and a cumulative value for all of them. We use the following abbreviations in the tables: PS (Policy and strategy), LE (Leadership), PE (People), PR (Partnerships and resources) and PO (Processes).

Table 14.2 Check-list in level II of practices to evaluate the maturity and excellence

Process	Practices	PS	LE	PE	PR	PO
For all the processes of level 2 and uppers	Establish and maintain the mission, policy and strategy to plan and perform the processes	✓				
Requirements management (REQM) of the interest groups (customers, stakeholders, other departments, public administration, etc.)	Identify the requirements	✓				
	Commitment of all involved personnel and groups with the requirements					✓
	Maintain and validate the requirements with bidirectional communication	✓				
Project planning (PP), in high and low level, for the performance	Establish and maintain plans for the performance, assigning responsibilities					✓
	Estimate and assign costs and reasonable efforts					✓
	Prioritize the activities according to the requirements	✓				
	Plan the necessary knowledge, participation and relationships					✓
	Employ techniques more frequently used for planning task such as CPM, PERT, JIT, Criticality analysis, Queuing theory or MRP					✓
	Plan data management and necessary resources in all the life cycle, reconciling plans with real circumstances					✓
Project monitoring and control (PMC) to control the performance according to the plans with appropriate corrective actions	Monitor and control the performance against the plans periodically and in determined milestones, escalating real state, performance and results					✓
	Review in a high level of management the real state, performance and results	✓				
	Monitor the commitment and performance of the personnel, their security and health					✓
	Identify inconsistencies and determine corrective actions					✓
	Record, monitor and maintain the data traceability of the progress, performance and other issues in a management system					✓
Supplier agreement management (SAM) to manage the acquisition of products and services from suppliers	Select and maintain potential suppliers and external services periodically, including their agreements					✓
	Establish, monitor, evaluate and maintain the supplier performance					✓

(continued)

Table 14.2 (continued)

Process	Practices	PS	LE	PE	PR	PO
Process and product quality assurance (PPQA) with objective insight into processes	Establish performance objectives (including quality) according to definitions, procedures or standards	✓				
	Ensure the resolution of the performance, determining corrective actions					✓
	Record, monitor and maintain the quality, performance and other issues, using a monitoring system and techniques more frequently used to evaluate quality such as QLF, QC and QFD				✓	
Configuration management (CM) to ensure the integrity of resources and services Measurement and analysis (MA) capability used to support management information needs	Establish, monitor, record and maintain the configuration and changes management in an inventory system, ensuring the integrity				✓	
	Establish and maintain the measurements and the objectives, specifying the procedures to manage them		✓			
	Obtain and analyze measurement reliable and useful data, and employing techniques more frequently used to assess the results such as check-lists, histograms, TMS or UMS				✓	
	Notify real state, performance and results to different hierarchy levels			✓		
Service delivery (SD) in accordance with service agreements	Establish and maintain service agreements for managing and delivering services	✓				
	Operate and maintain the services					✓

Table 14.3 Check-list in level III of practices to evaluate the maturity and excellence

Process	Practices	PS	LE	PE	PR	PO
Organizational process definition (OPD) to standardize the work environment of processes	Establish, record and maintain the organizational and standardized processes, systems, technologies, procedures and criteria	✓				
	Establish the rules and guidelines to integrate in teamwork and work environment					✓
	Establish and maintain the process improvement opportunities	✓				
Organizational process focus (OPF) to plan, implement and deploy organizational process improvements based on a thorough understanding of current strengths and weaknesses of them	Deploy the processes along with the organization		✓			
	Appraise the processes periodically in order to improve them through corrective actions					✓
	Value personnel performance					✓
	Establish and maintain the repository of data, process information and experiences in a knowledge management system and using techniques such as SWOT or Brainstorming					✓
	Establish, provide and evaluate strategic tools and training needs to perform the roles effectively and efficiently					✓
Organizational training (OT) to develop skills and knowledge of people	Use defined processes and systems to estimate and plan activities, maintaining the global integrity and reviewing according to commitments					✓
	Establish the work environment for improvement	✓				
	Coordinate and collaborate with interest groups and personnel, controlling the access and solving critical issues and defects					✓
	Define a balance score card and the integration between maintenance and the rest systems					✓

(continued)

Table 14.3 (continued)

Process	Practices	PS	LE	PE	PR	PO
Capacity and availability management (CAM) to ensure service agreements using resources effectively	Define the measurements and techniques concerning resources, customers and services, using a geographical information system and techniques such as process capability or casual models					✓
	Monitor the effective use of the resources to ensure the performance of processes and services, taking corrective actions					✓
Risk management (RSKM) to identify potential problems taking risk-handling activities to mitigate adverse impacts on achieving objectives	Determine and categorize the risk sources		✓			
	Evaluate, prioritize and monitor the risks periodically implementing mitigation actions					✓
Service continuity (SCON) to ensure continuity of services during and following any significant disruption of normal operations	Identify and prioritize essential resources and services	✓				
	Establish, validate and analyze the training and service continuity		✓			
Decision analysis and resolution (DAR) to analyze possible decisions using a formal evaluation process that evaluates identified alternatives against established criteria	Establish and maintain guidelines, methods and criteria to take decisions about formal issues, selecting the solutions			✓		
Strategic service management (STSM) in concert with strategic needs and plans	Establish strategic needs and plans for standard services	✓				
Service system development (SSD) to satisfy existing or anticipated service agreements	Translate the requirements in solutions for the service, ensuring integrity and connectivity					✓
Service system transition (SST) to deploy new or significantly changed while managing their effect on ongoing service delivery	Develops the change management taking in count the impacts in the services, making necessary corrective actions				✓	
Incident resolution and prevention (IRP) to ensure timely and effective resolution and prevention of service incidents	Define known and tested solutions to solve or prevent the known incidents					✓

Table 14.4 Check-list in level IV of practices to evaluate the maturity and excellence

Process	Practices	PS	LE	PE	PR	PO
Organizational process performance (OPP) to establish and maintain a quantitative understanding of the performance to achieve quality and process-performance objectives	Encourage quantitative process-performance evaluation through benchmarking	✓				
	Establish and maintain quantitative process-performance models, objectives, measures and techniques and resources, defining baselines			✓		
	Develop, monitor, record and maintain analysis and qualitative prediction of performance, deviations and risks					✓
	Establish, spread and promote techniques for quantitative modeling such as simulation, deterministic, replacement/renewal and Markovian				✓	
Quantitative project management (QPM) to achieve the quality and process-performance objectives	Manage and analyze performance statistically implementing the necessary corrective actions					✓
	Budgetary analysis and prediction including costs of corrective actions and improvements		✓			
	Develop, monitor, record and maintain quantitatively analysis and predictions in a RCM system and knowledge management system and using techniques such as ORA or SPC				✓	
	Stabilize the process-performance to achieve the quantitative objectives					✓

- Level I. Initial or unpredictable. In the first evaluation level I, the maintenance process is designed for the specific purpose, without considering a context of unified management or a reference and without defined records or documentation; this is an unpredictable and unstable process.
- Level II. Managed or reactive. If level II is reached, then a monitoring and planning of the process, projects, products and services exist, through a specific policy. The maintenance process is stable and planned due to repeatable practices and there is documentation concerning resources and incidents, thus the process is defined as managed but without an adequate global integrity.
- Level III. Defined or proactive. Reaching level III implies that a standardization of processes exist, in order to facilitate their understanding through procedures,

Table 14.5 Check-list in level V of practices to evaluate the maturity and excellence

Process	Practices	PS	LE	PE	PR	PO
Organizational innovation and deployment (OID) to deploy incremental and innovative improvements that measurably improve the processes and technologies	Encourage the continuous improvement of processes and technologies toward a world class maintenance	✓				
	Assure the continuous improvement and collect potential improvements and innovations of process and technologies systematically		✓			
	Evaluate the improvement effects (tangible and intangible), including cost analysis		✓			
	Establish, spread and promote techniques for optimization such as Game Theory, RCO, LCCA and OPT				✓	
	Elect, develop and implement the improvements and innovations in the organization, fulfilling the objectives		✓			
Causal analysis and resolution (CAR) of defects and problems to prevent them from occurring in the future	Identify and analyze the root-causes of performance, defects and problems systematically determining improvement actions					✓
	Implement and monitor the improvement actions of the root-causes to mitigate the consequences					✓
	Automate, monitor, record and support actions in an expert support system based in symptom-cause reasoning, and using techniques such as fishbone diagram, FRCA or 5W2H				✓	
	Evaluate the improvement and innovations effects according to expected, including cost analysis		✓			

techniques, methods and tools. The maintenance process is well established and documented, analyzing its performance and tendencies.

- Level IV. Quantitative managed or predictable. Inside this level IV, the numerical analysis of performance, supported by the knowledge management, is implemented on the department objectives or benchmarking in the sector. Therefore, the process is predictable and computerized in a quantitative way,

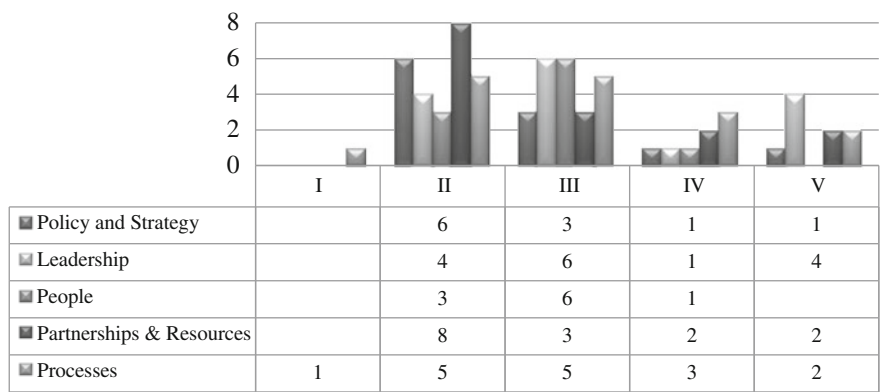


Fig. 14.4 Practices considered per EFQM enablers and CMMI level

- relating direct and probable costs with the performance from an efficient point of view.
- Level V. Optimized or continuously improved. The maximum level of maturity, level V, is reached when a stable, documented and quantitatively known situation of the processes exist and the entire organization is oriented toward the continuous improvement of the department and business objectives, minimizing the causes of variance in the processes and implementing technological improvements. Thus, it is said that the maintenance process is optimized.

Figure 14.4 summarizes the practices that we have considered per EFQM enabler:

- 11 Policy and strategy practices (PS–8%), in CMMI levels 2–5.
- 15 Leadership practices (LE–10%), in CMMI levels 2–5.
- 10 People practices (PE–9%), in CMMI levels 2–4.
- 15 Partnerships and resources practices (PR–9%), in CMMI levels 2–5.
- 16 Processes practices (PO–14%), in CMMI levels 1–5.

Table 14.6 shows the main effects that can be reached with the corresponding enablers within each level of maturity.

To summarize, this evolutionary path, through best practices and levels, improves the deployment of the processes of the reference framework, focusing the effort in different ways for each level:

- Levels II and III of maturity focus on the application of the practices in the definition, planning and monitoring of the processes: managing incidents, preventing incidents, monitoring, administrating capacity and managing the strategy. Thus, maintenance systems to develop are: inventory, work order management, monitoring, geographical information system, balance score card, knowledge management and interconnections with other supporting systems.
- Levels IV and V deal with the improvement of previous processes, focusing on their development based on specialized knowledge of the process analyzing

Table 14.6 Contribution of enablers inside each maturity level

Level	Policy and strategy	Leadership	Personnel	Partnerships and resources	Processes
I	Reactive execution and on purpose	Lack of coordination, without reference nor defined responsibilities	Reactive and variable execution according to personal initiative, disorganization	Requests on demand and without control	Unstable and unpredictable situation
II	Oriented to customer attending to the management of requirements	Commitment of the interest groups and definition of the responsibilities for the projects	Management of the personnel according to the results, monitoring the efficiency and security	Management of partnerships and resources. Specific management of information	Repetitive management and planning according to results
III	Oriented to customer attending to the integrated performance to accomplish with the requirements. Standardization of the organization	Identification of personnel involvement and environment preparation for continuous improvement. Assignment of the responsibility based on processes	Development of knowledge and personal skills, facilitating decision-making activities	Unique and integrated definition of the information, attention to maintain the operation of the resources depend on risks	Unified and coherent management with processes and objectives. Prediction through qualitative techniques
IV	Guide the organization for the statistical analysis with the purpose of improving the objectives and, the with the customers and internal understanding	Defining and implementing the mechanisms for the qualitative analysis	Quantitatively predict and evaluate the need and improvements of the human resources	Quantitatively analyze the operation of the resources, their acquiring and logistics	Estimate future efficiency and possible variances from the actual situation of the processes
V	Innovate in a sustainable way the processes and the technologies for the customers satisfaction and the social perception	Implement the concepts of continuous improvement and pro-activity	Take the potential of the personnel to optimize the efficiency of the organization	Eliminate the causes that produce variations of the operations of the resources	Continuous improvement of the efficiency, through proper adjustment of the processes

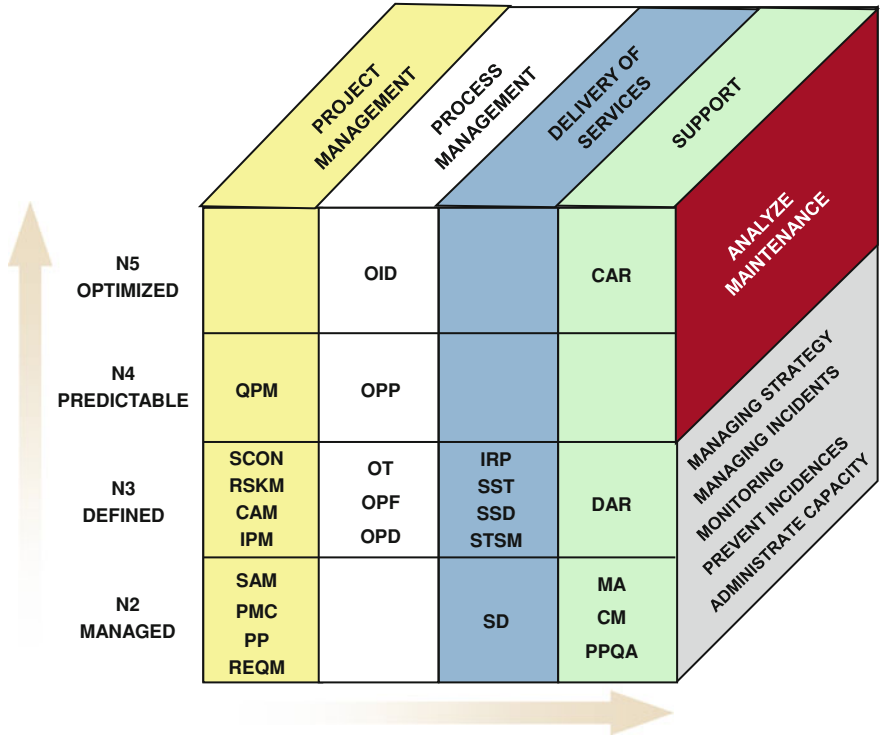


Fig. 14.5 Practices per category of process area and maturity levels

maintenance. Then, the supporting systems are: expert and support decision-making system and reliability system.

In this way, the maintenance department can incrementally improve its maturity level deploying the best practices over time. As depicted in Fig. 14.5, the management evolution may occur according to different sequences of practices implementation, within particular categories or within each level, in order to establish a sustainable improvement. This is because:

- The execution of project management practices allows the planning and controlled repetition of maintenance.
- The implementation of management practices per process stabilizes the maintenance execution.
- The establishment of practices of service providing leads the maintenance deployment toward customer satisfaction.
- The use of supporting practices allows the improvement of the maintenance effectiveness.

If our framework is well standardized, it can provide useful information, through a proper classification of analysis drivers and the proper description of the features of the context that is being analyzed.

We do believe that this approach can be very robust when well formalized. It is based on the CMMI standard and assesses the maintenance maturity of network utilities focusing on the organization, the IT and the maintenance management processes.

We will see how this approach can be easily formalized using a score card and a questionnaire to retrieve information from companies. We will present an example of how this process has been accomplished, during a period of five years, in several network utilities in Spain.

14.2.2 Maturity Evaluation Case Study

This section presents a case study with a brief analysis to test maintenance management maturity evaluation under real business circumstances. This case study is presented for maintenance service provider companies for telecommunications networks, using the defined approach, and considering the 67 practices.

This case study was developed with the collaboration of two companies although we will concentrate mainly on one of them (Company A). For both companies a combination of qualitative information resulting from open questions and from the maintenance maturity valuation was retrieved. The two companies in this case study operate in the telecommunication sector providing maintenance services for telecommunication networks. The common organizational background to both of these network companies was the following:

- They are big companies operating in Spain, where this type of market mainly involves large-scale operators and companies.
- Before full network implementation, the maintenance department was considered as a support department for the network construction, instead of being considered a core department.
- Commissioning of networks elements was a task assigned to the network construction department, and then the responsibility of the inventory system was assigned to the network construction or to the engineering department. For instance, a new network element could be installed without issues related to spare parts being analyzed properly, or with missing documentation and/or knowledge concerning the element.
- No maintenance department would participate, on a regular basis, in decisions concerning new technologies.
- The benefits of maintenance were measured in terms of cost and according to service level agreements established in the sector.
- The ICT applications were implemented without interconnection between the enterprise and the maintenance systems; moreover the monitoring system was composed aggregating several non-standard solutions from the main suppliers.

Table 14.7 Excellence in management obtained through the maturity score card

Excellence according to management levels of companies	A (%)	B (%)
Practices of policy and strategy (PS–8%)	6.5	8
Practices of leadership (LE–10%)	6	10
Practices of people (PE–9%)	9	9
Practices of partnerships and resources (PR–9%)	4.8	9
Practices of processes (PO–14%)	7	14

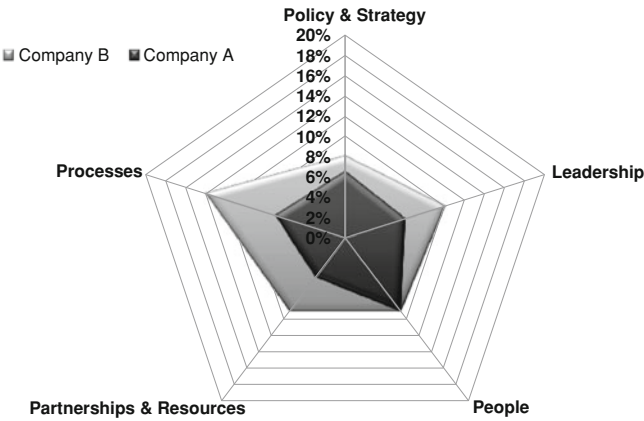


Fig. 14.6 EFQM excellence in spider diagram of company A and B

- Communications with technicians in the field were established by mobile phones and, in the best cases, a copy of the principal documentation was included in laptops without online access to maintenance systems.

Considering that the scores go from I to V, according to CMMI levels, the first company (A) has the level 2.7 of CMMI standard, thus it fulfills all the practices of levels I, II and some of III, while the company (B) has the level 5 CMMI, so it fulfills all the practices of levels I, II, III, IV and V. Therefore, the excellence for the EFQM evaluation would be as presented in Table 14.7, and graphically in Fig. 14.6.

When preparing this case study analysis and through the description provided by the operators, we received some initial (qualitative) impressions regarding the alignment of processes in both companies. The open questions mainly helped in this part of the analysis. During these interviews we could appreciate how the companies had tried to obtain a technical competitive advantage by carrying out different activities, but they had paid less attention to evaluate the impact of each activity for the consecution of the maintenance goals. The scores obtained were then helpful to quantify, by means of standard measurement systems, these “qualitative” impressions.

Table 14.8 Practical implications of maturity evolution in company A

Level	Time period	Net. growth (%)	Objectives			Gross facts		
			Cost reduction (%)	Cycles reduction (%)	Raise quality (%)	Cost reduction (%)	Cycles reduction (%)	Raise quality (%)
II	1–3	100	5	10	5	0	5	0
Real values against the initial situation:			95	90	105	100	95	100
III	3–4	300	5	5	10	5	5	5
Real values against the initial situation:			90	85	115	95	90	105
II	4–5	500	20	5	5	–12	–10	–15
Real values against the initial situation:			60	75	120	97	100	95
II	5	20	0	0	20	–3	5	10
Real values against the initial situation:			60	75	150	100	95	105

It can be appreciated how company A needed to exploit their resources better in order to balance the development of the processes and to align them to the goals the company aimed to reach. Moreover, from interviews and data analysis, it was worth noticing a development trend of company A in the last few years.

In Table 14.8 we can see the expected objectives in several years of maturity improvements, and the real performance achieved. In the first year, the best practices began to be applied, establishing the mission, responsibilities, processes and activities standardization. In addition, the inventory and documentation were updated and stored in servers, and a knowledge management program was initiated. Then, a *defined level* was soon reached in an operational view supported especially in inventory, monitoring and management systems. Results in operational objectives were fundamentally in 5% time reductions. During the second year, the *defined level* was reinforced through adequate resources application, proactive control and monitoring; due to the implementation of GIS and using open standards in a unique monitoring system, network and facilities were monitored and supervised for conditions like temperature, humidity, personnel access control, air conditioning and UPS. Technicians were classified according to the type of equipment they maintained and the level of qualification was considered: basic for repetitive tasks (level 1), and expert to analyze unknown solutions and to optimize the maintenance. In addition to the operational objectives, an important quality improvement was achieved due to reduction of repetitive incidents and the time required for incident identification.

The *managed level* was achieved in the third year, focusing on a tactical as well as strategic view of maintenance, fostering the measurements of process performance, their quantitative understanding and control. During this stage, an

outsourcing process was decided to be applied carefully in repetitive tasks related to corrective activities (only on a first level of attention). This outsourcing was accomplished selecting suitable targets and was considered by the company as one of the main reasons for the network to increase 300% in size very quickly. All past objectives were improved. Up to this point in time, the initial base values were improved 5% in cost reduction, 10% in time reduction and 5% in quality level. The following year expert systems were developed to support maintenance, trying to optimize and automate network maintenance management.

After a short steady state situation, a joint venture with five other companies took place, and the strategic decision was to outsource and to dismiss maintenance-related tasks. The main business plan objective was a cost reduction of 20%. Many activities were outsourced in service contracts and the final number of human resources (internal and external) resulted less than before. The technician motivation decreased and the external company that was selected was in a maturity level II (repeatable and reactive) regardless of the internal procedures and systems imposed in the relationship with the outsourcer. Obviously, this stage implied a regression in the objectives. The most upsetting situation was a 15% loss of service quality.

After this period, the company internalized some technicians to reduce the dependence with the outsourcer and applied the model for two years. The company lost almost 2 years; reputation and the network was seriously degraded (it was impossible to know all possible effects). Cost reduction through the selected process of outsourcing was not a good idea.

The chronology of improvements that we have presented in this case study illustrates clear implications of adopted long-term strategy. This case study also shows the evolutionary nature of the maintenance management framework. During a specific period, knowledge gained over time allowed consistent cost, time and quality improvements.

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Chapter 15

Quantitative Evaluation: An Intellectual Capital Scope

15.1 Introduction

Where does maintenance generate value for the company? How can we measure and compare different maintenance departments' performance? How can we measure the consequences of a poor maintenance management?

In line with Crosby [11] and Dale and Plunkett's [12] ideas, it is important to answer to these questions in a quantitative way, so that managers can acknowledge the actual situation and can make their decisions easier for a more intelligent and competitive maintenance. This section tries to show, using quantitative metrics, how the establishment of our maintenance management reference framework creates value for the network utility organization.

To measure the contribution of our reference framework we will use *performability* metrics besides *Intellectual Capital* (IC) metrics. With the IC metrics we can capture the value generated by our new maintenance organization. Nevertheless human capital has been considered as the most important factor to explain and predict economic growth [5]. We do believe that human capital within our maintenance department can reasonably explain:

- Effectiveness in the processes.
- Quality of activities and services.
- Proper attention to customers.
- Utilization and development of technologies.
- Capacity to improve and innovate.

Then the methodology that will be proposed will allow us to measure the local evolution as well as the comparison with other companies of the sector (see [45], competitors, standards and regulations. Our company's maturity must be measured employing external references as objectives [8, 47] and internal references taking into consideration the relationships of maintenance with other related internal areas.

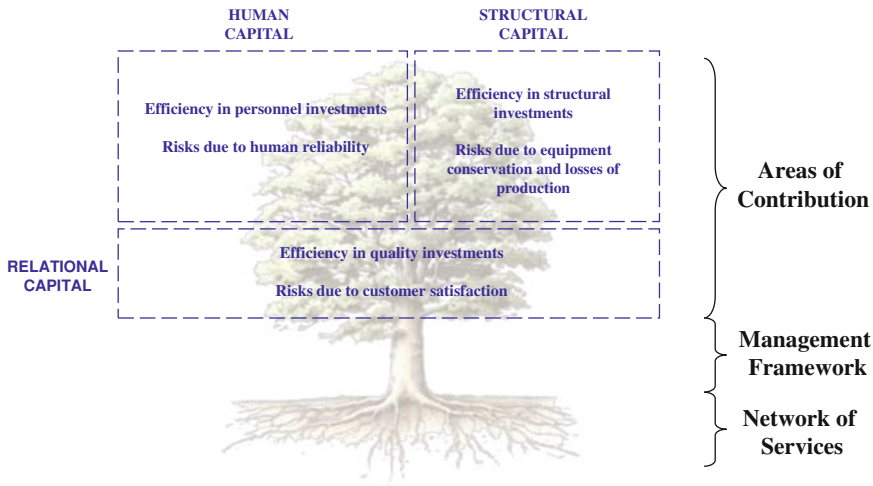


Fig. 15.1 Maintenance evaluation related to intellectual capitals

The EFQM model results' weights will help us to measure the contributions of the company's intellectual capital, in its different categories (human, structural and relational capital). The reader can see Fig. 14.2 to review weights of different results within the EFQM model:

- People Results (EFQM weight: 9%). Perception measurements and performance indicators in relation to people.
- Customer Results (EFQM weight: 20%). Perception measurements and performance indicators in relation to customers.
- Society Results (EFQM weight: 6%). Perception measurements and performance indicators in relation to society.
- Key Performance Results (EFQM weight: 15%). Key performance outcomes and key performance indicators in relation to the planned performance.

Concerning IC indicators, different IC evaluation methods [16, 23, 40, 43, 44] employ diverse indicators. Examples of these indicators are: personnel motivation, customer satisfaction, personnel performance, rate of training, grade of ICT implementation, un-planned activities, efficiency in the use of resources, etc. Our job is to somehow add these categories of indicators to the network performability indicators.

Our evaluation model integrates the intellectual capital and performability indicators in fraction numbers within the four categories of EFQM results. To simplify, we will aggregate the society and customers results of EFQM into a single result category related to quality, linking the society perceptions having influence on the customer perception.

We will measure not only the positive impacts or consequences of a good maintenance, but also the negative impacts or consequences of a poor maintenance, focusing on the most relevant categories of intellectual capital (see Fig. 15.1):

- Human capital or related to people, through efficiency in personnel investments and human reliability.
- Relational capital or related to quality, through efficiency of quality investments and consequences in customer satisfaction.
- Structural capital or related to company structure, through efficiency of structural investments, equipment conservation and production losses.

Thus, we do believe that the scope of the analysis is appropriate and that the evaluation of the company's intangible assets is a must to reduce uncertainties, especially if we can employ suitable references values.

15.1.1 An Intellectual Capital Formulation

This Section's intention is to obtain a generic expression that somehow can capture IC of the organization, and that we can easily apply to different organizations. Our idea is to assess intellectual capital (human, structural and relational) as the sum of two terms or factors, one related to investments in training, education, etc. and another factor taking into account risks:

- Investment values will be multiplied by efficiency indices.
- Risks values will be weighted using indices of excellence.

Thus, for each type of intellectual capital, an assessment will be:

- $CI_H = i_H \cdot CH + q_H \cdot KH$
- $CI_S = i_S \cdot CS + q_S \cdot KS$
- $CI_R = i_R \cdot CR + q_R \cdot KR$

Where:

- CH, CS, CR are the economic values of human, structural and relational capitals respectively.
- KH, KS, KR measure the risks produced by the human, structural and relational capitals respectively.
- i_H, i_S, i_R are the efficiency indices of the human, structural and relational capitals respectively.
- q_H, q_S, q_R are the excellence indices of the human, structural and relational capitals respectively.

In summary, the main formula of intellectual capital would be in the following generic form:

$$CI = i_H \cdot CH + q_H \cdot KH + i_S \cdot CS + q_S \cdot KS + i_R \cdot CR + q_R \cdot KR$$

The economic values of CH, CS and CR are derived from investments accounted in the maintenance budget such as costs of personnel, training, prevention and evaluation, information technology, and customer service, etc. While the KH, KS and KR pretend to embrace the potential savings in costs, lower level of risks (compared to standards or competitors), due to a better organizational efficiency.

Maintenance cost studies show how CH, CS and CR are focused on the real costs of prevention, appraisal and correction of failures, while risks savings deal with failure costs avoidance, requiring the analysis of the probability of damages and/or poor performability due to failures (such as: additional reworks and overtime, additional inventory, damages to staff morale, loss of productivity and opportunity, loss of capacity, risk of equipment obsolescence and waste, loss of income and sales revenue, penalties, loss of customers impacted by failures, loss of customer goodwill, dissatisfaction, damages to reputation, loss of market share, and impacts on health and safety or on the environment).

In utilities [41] the regulator has to define the cost accounting principles (current, historical and future) to be used to determine asset values, but also the depreciation policies for the regulatory asset base and its position toward revaluation of assets and goodwill or intangible assets. Goodwill can be described as “*a global concept capturing a set of company activities related to superior earning power, such as customer loyalty, employees’ expertise, management capabilities, and all the other intangible factors motivating people to do business with the company*”—Rodriguez et al. [41].

Concerning risk, according to IEC 60300-3-9 [21], this can be measured as the product of a frequency or probability of occurrence times the consequences that would occur. We will therefore:

- In terms of dependability, analyse failures within the operational context, but leaving out problems related to confidentiality, integrity and external attacks.
- In terms of probability, estimate the probability of failure events over time.
- In terms of cost, value economic consequences of failures.
- In terms of customer view, include the provided and perceived quality of the services.

In our analysis, we will see equipment from an operational point of view (producing, distributing, assuring, etc.) taking into account particular environmental conditions and context. We will pay special attention to the most severe failures, investigating their causes and estimating the risk of a failure occurrence.

We will work with probabilities of failures as in RAMS Analysis (Reliability, Availability, Maintainability and Safety) [14, 49, 50] to quantify risks. The calculations of the three types of IC (human, structural and relational) will consider that:

- The risks due to people performance will be centred mainly on additional reworks.

- The risks on production will focus on losses due to service interruption (structural capital).
- The risks on dependability will be studied according to the degradation of the equipment that causes aging of the network (structural capital).
- Quality risks measure consequences on customers and company reputation (relational capital).
- Safety risks measure potential consequences of failures on staff health and on the environment. These issues affect customers and society's perception of the company, loss of customers and/or company image (relational capital).

In summary, we have to value the ability of a maintenance department (management structure, staff knowledge, technologies, information, resources, etc.) to effectively and efficiently exploit the full potential of its resources to maintain a competitive advantage [4].

15.1.2 Main Performability Indicators

Most maintenance practitioners would agree that some main measurements of maintenance performance are: reliability, availability, maintainability, resources efficiency or a percentage of un-planned activities. In addition, availability and maintainability measures are usually evaluated according to *standards of services* by national regulators in order to avoid discrimination in the provision of services.

When we consider a sustainable compliance of indicators, in line with the evolution of our management framework within an organization, it could be useful to mark tolerance values inside difference maturity levels, and use these levels as a reference. As a result, we could define ranking levels of performance for each indicator, based on commonly accepted benchmarking studies such as developed by Wireman [51] or Mitchell [34], see Table 15.1. We will use the definitions, in Lowry et al. [31], of the Leveling method as reference: excellent, good, average, fair and poor; in correspondence with the five maturity levels.

Consequently, as the framework implementation progresses, the maintenance objectives should be more demanding to become the best or world class.

15.2 Excellence Evaluation Methodology

Now, we will begin with the development of the evaluation methodology distinguishing for the three categories of intellectual capital (human, relational and structural) and evaluating for them the cost of investments (accounting perspective) and the existing risk (performability perspective).

Table 15.1 Benchmark about maintenance measurements

KPI/Maturity level	II (%)	III (%)	IV (%)	V (%)
Maintenance costs versus replacement costs of assets	15	7	3	1.5
Availability	85	95	97	99
Maintenance cost in training versus maintenance staff cost	1	2	4	8
Ratio of staff satisfaction	70	85	90	95
Personnel performance	60	85	90	95
Productivity of labour	75	85	90	95
Rate of covered staff competences	60	75	85	90
Customer satisfaction	50	65	75	90
Network penetration	25	35	50	65
Ratio of net incomes of unaffected customers	85	95	97	99
Rate of planned activities	60	80	90	95
Efficiency of network capacity	75	85	90	95
Rate of covered staff by ICT	50	75	90	95
Rate of covered critical equipments by predictive tools	40	60	85	90
Rate of covered critical equipments by monitoring	50	75	90	95

15.2.1 Human Capital Evaluation

In this section, we account the investments in maintenance human resources, at the same time we evaluate the payoff of these investments, since they increase staff knowledge and reduce human errors and risks associated to these errors.

15.2.1.1 Accounting Human Capital Investments

From the accounting point of view, the main investments in human capital are two: investment in salaries and investment in learning and training. The implementation of such investments is efficient if it achieves the performance goals and training of staff respectively. Therefore, the efficiency of these investments could be represented by the following indicators:

- Investment in salaries (CH_{salary}).
 - Ratio of staff satisfaction ($i_{H,\text{satisfaction}}$).
 - Global average ratio of staff performance ($i_{H,\text{performance}}$).
 - Productivity of labour (productive hours/total hours = 1 – Absenteeism rate) ($i_{H,\text{productivity}}$).
- Investment in learning and training (CH_{training}).
 - Global average ratio of staff competences to develop the activities ($i_{H,\text{competences}}$).
 - Effectiveness of activities as measurement of the personnel in the resolution of activities according to the standard time ($i_{H,\text{effectiveness}}$).

The first set of indices is more focused on the efficient performance of human resources, while the second is focused on optimizing the realization of activities (reducing the learning curve and costs). In summary, the human capital investments accounting could be as follows:

$$\begin{aligned}
 CI_{CH} &= i_{H_1} \cdot CH_1 + i_{H_2} \cdot CH_2 \\
 &= \frac{(i_{H,\text{satisfaction}} + i_{H,\text{performance}} + i_{H,\text{productivity}})}{3} \cdot CH_{\text{salary}} \\
 &\quad + \frac{(i_{H,\text{competences}} + i_{H,\text{effectiveness}})}{2} \cdot CH_{\text{training}}
 \end{aligned}$$

15.2.1.2 Evaluating Human Capital Risk

It is well known that people working under circumstances of stress, limit times, worries, etc., may perform badly, causing errors and failures with important consequences to the business.

There is always a probability of human errors (Human Error Probability, HEP) [7, 20] that can be quantified in order to know the real personnel potential, determining the likelihood of such failures and their consequences. The HEART method provides human error probabilities according to the complexity of the tasks. Also different authors have researched about work measurement and time study [3, 26, 31, 36], defining levels of performance as standard times with tolerances in proportion to operating conditions such as fatigue, atmospheric conditions, complexity of the task, motivation, etc. The tolerance of performance could change in a percentage depending on the circumstances.

Therefore, our recommendation here is the employment of the so-called Human Reliability Analysis (HRA), which is suitable for studying the reliability of human activity providing standard values of risk as baseline.

In our case, we can compare the standard HEP_S in the sector with the estimated global HEP_H provided by the HEART method.

Consequently, the HEP_H and the Human Reliability ($R_H(t)$) could be described as the weighted sum of the HEP of all the activities of the framework using the frequency of each activity execution δ_g (where M is the number of different types of activities of the framework in the period of evaluation, remember from Chap. 4 that HEP_G is the Human Error Probability of the generic activity, EPC-Error Producing Condition is the Maximum associated affection to the PSF_k, and the APOA-Assessed proportion of Affect is the experts estimate weight of the PSF):

$$HEP = HEP_G \cdot \prod_{k=1}^N ((EPC_k - 1) \cdot APOA_k) + 1$$

$$\text{HEP}_H = \sum_{g=1}^M \text{HEP}_g \cdot \delta_g$$

$$R_H(t) = 1 - \text{HEP}_H$$

- If $\text{HEP}_H \leq \text{HEP}_S$. The organization could be considered excellent, with greater human reliability than the average in the sector.
- If $\text{HEP}_H > \text{HEP}_S$. The organization could not be considered excellent, with reduced human reliability than the average in the sector.

Values for HEP will be more relevant for great number of activities. In this case a great number of errors could be produced and consequently the cost in subsequent activities (additional rework) could be very relevant.

A change in the maturity of maintenance management could reduce or increase the HEP, producing a variation in the same proportion in the in the maintenance budget.

Accordingly, the indicator for human capital excellence will be:

$$q_H = -\text{HEP}_H = -(1 - R_H)$$

Therefore, in order to know the risk due to human capital in terms of costs, we will employ the average cost (C) of one activity in the evaluated period.

$$C = \sum_{g=1}^M C_g \cdot \delta_g$$

Thus, the probability of human-caused failures produces less value for the organization, decreasing the human capital in accordance with it increases. Then, the human capital risk (negative effect depending on HEP) can be calculated as follows (where n_a is the number of total executed activities of the framework in the period of evaluation):

$$\text{CI}_{KH} = q_H \cdot C \cdot n_a = (-\text{HEP}_H) \cdot (C \cdot n_a) = -(1 - R_H) \cdot (C \cdot n_a)$$

In short, a trade off between investments and risk in human capital will be required.

15.2.2 Relational Capital Evaluation

In this section, in order to evaluate relational capital, we account investments in customer service and their payoff in improvements in its perception, and in reduction of service failures and security hazards, contributing to the reputation of the company.

15.2.3 Accounting Relational Capital Investments

From an accounting point of view, companies investments in customer attention ($CR_{customers}$), are seeking to maximize satisfaction, retention and attraction of new customers, so the efficiency of these investments could be represented by the following indicators:

- Investment in customer attention ($CR_{customers}$).
 - Ratio of customer satisfaction ($i_{R.satisfaction}$).
 - Global average ratio of network penetration. ($i_{R.penetration}$).
 - Ratio of net income of unaffected customers ($1 - \text{Ratio of net incomes affected by discontinued services}$) ($i_{R.income}$).

In summary, the relational capital from the accounting point of view can show the efficiency in customer attention as follows:

$$CI_{CR} = i_R \cdot CR = \frac{(i_{R.satisfaction} + i_{R.penetration} + i_{R.income})}{3} \cdot CR_{customers}$$

15.2.3.1 Evaluating Relational Capital Risk

From a performability point of view, the maintenance contributes to the customer perception. Maintenance departments must be focused on pursuing the excellence in those aspects which are important for the customer. We know, for instance, that reliability is one of the most valued aspects in service quality. In addition, maintenance is related to the business activity and contributes with value to the customer from several perspectives [52]:

- Customer relations, increasing service fidelity and reception through a good and quick attention [13, 33].
- Price and quality, thanks to a better availability and cost reduction, the price of the service decreases and the customer satisfaction is improved.
- Internal motivation, image and external reputation [48].

We have also mentioned the difficulty in measuring the quality perceived by customers. The evaluation of critical incidents (significant interaction) is a suitable way, and so it is the number of customer complaints for service failures. Moreover, according to Parasuraman et al. [37], customers evaluate the quality attributes globally; they are not independent and there exists many correlations among them. Therefore, it is difficult to distinguish the weight of each attribute and its influence in the customer's life [10].

A study showing the importance of complaints and the effects of poor quality in the customer relationship was carried out by the Technical Assistance Research Program (TARP), Washington EE.UU [17]. The study is about customer claims in

more than 400 large corporations of services from United States, Japan and Europe. This TARP study determined that:

- Acquiring a new customer costs 5 times more than keeping one.
- Increasing customer retention 5% increases the profits between 25 and 80%.
- A satisfied customer recommends our company to an average of 5 partners, and an unsatisfied customer transmits its bad experience to an average between 9 and 10 partners.
- Not all unsatisfied customers create claims, many drop out without communication, 96% of the unsatisfied customers never complain.
- Only 17% of the unsatisfied customers intend to continue the services.
- 80% of the customer satisfaction is due to deliver services correctly, and the rest 20% is due to solve claims or problems.

Maintenance influences customer perception through service quality, increasing the satisfaction with a high quality and quick resolution and decreasing the satisfaction when failures persist and there is a bad resolution of incidents. We will confine the maintenance effects in service quality through two criteria:

- A poor maintenance decreases reliability and, for each failure, the customers modify their perception about service quality and company image is affected.
- It is crucial to relate the failures to the possible level of affections in customer relationship, and to measure these with the service level agreements (SLA), established with the customers and determined by the market as standard. According to Keaveney [27], service failures are responsible for 44% of customers drop offs.

In addition, the behaviour of the customer could be different; the customer can wait (asking for a monetary refunding) or could not wait, starting legal actions. In both situations, the customer could stop buying our services or simply spread negative propaganda. Sometimes, the customer remains [17, 27].

To study the relationship with the customers, there are qualitative approaches through polls or interviews [1, 2, 18, 37]; or quantitative ones such as “Survival Data Analysis” [9], SERVQUAL [38], SERPERF [10], Model Q [30] or Evaluated Performance [46].

We could estimate service quality (service level agreements, SLA) and relate it to the maintenance indicators, using statistical methods such as the ones in the “Survival Data Analysis” method.

In network utilities sector, where contracts and standards spin around of service level agreements (SLAs), quality measures must mainly capture customer’s perceived quality instead of internal performance. Service quality can therefore be measured according to the fulfilment of the service level agreements like delivery date, response time, recovery time or re-establishment time, precision in the level of service, etc.

The service quality is accepted by the customer when it is maintained within a level of tolerance, but how to characterize this level?

The level of tolerance is frequently defined through the probability of abandoning (P_a) which depends on the service re-establishment time. For instance Bitner et al. [6] show how historical data of failures may help us to find the function representing the risk of losing customers in terms of re-establishment time after a service failure.

If we have a series of failures with affection to (n_f) customers each one, then the computation of this series, using right censored data, is:

- For each failure, all the customers abandoning the service are determined.
- If the real time in which the customer requests the abandonment is greater than the re-establishment time, we consider the re-establishment time, otherwise the real time is used in the computation.
- The survivor customers are estimated in the study, distinguishing this customer status at the re-establishment time.

We will rely on a 2-parameter Weibull distribution and Maximum Likelihood Estimation (MLE) to obtain the Weibull equation that reflects the probability to abandon per customer and per each failure. And estimating the α and β with the MLE [19], for N observations with data for $t_1, t_2, \dots, t_N$, we could present the probability to abandon as:

$$P_a(t_r) = 1 - R(t_r) = 1 - e^{-\left(\frac{t_r}{\alpha}\right)^\beta}$$

With:

$$\begin{aligned} \Lambda &= \log L\{\text{data}|\alpha, \cdot\beta\} = \log L\{t_1, t_2, \dots, t_N|\alpha, \cdot\beta\} \\ &= \log L = \sum_{i=1}^N \log f(t_i|\alpha, \cdot\beta) \\ &= \sum_{i=1}^N \left[(\ln \beta - \beta \cdot \ln \alpha) + (\beta - 1) \cdot \ln(t_i) - \left(\frac{t_i}{\alpha}\right)^\beta \right] \end{aligned}$$

Thus, for a set of n data arranged in pairs $[(t_1, d_1) (t_2, d_2) \dots (t_n, d_n)]$ with $d_i = 1$ if t_i is the a service re-establishment time when customer remained and $d_i = 0$, if t_i is the instant at which the customer abandons. Consequently, we could use this method to obtain, based on the mean service re-establishment time (t_r) in the company, the mean probability to abandon (P_a). According to this idea, our indicator will be:

$$q_{R1} = -P_a(t_r) = -[1 - R(t_r)]$$

The presented indicator affects to the total incomes from customers for the company. The individual income per customer is a known variable in network utilities: “The Customer Present Value” (CPV), which is the updated benefit per customer. Therefore, in a determined period, the risk due to customers drop offs is calculated multiplying q_{R1} by CPV, the mean number of affected customers per

failure (n_c) and the mean number of failures (n_f). Thus, this potential negative impact can be measured as follows:

$$CI_{KR1} = q_{R1} \cdot KR_1 = q_{R1} \cdot CPV \cdot n_c \cdot n_f = -P_a \cdot CPV \cdot n_c \cdot n_f$$

In addition to customer perception, environmental and safety worries could also contribute to a bad network utility reputation. *Green concepts* are nowadays in the mind of the customer mainly in relation to:

- Sustainable consumption of resources. Maintenance can only reduce the energy consumption within the range 6–11% [51].
- Environmental, human health and safety risks. We will concentrate on this issue, whose economic impact is greater.

Maintenance managers should always bear in mind the concern for protecting resources and environment, security of assets, people and environmental safety. Maintenance should ensure the protection of resources and the environment, both internal and external to the company. That is to say, not only for the safety of assets but also for the safety of people (in Spain according to the Law on Occupational Risk Prevention 1995 and Royal Decree 39/1997) and the safety of the environment (according to the principle of European new environmental legislation, those who harm the environment, will pay to clean it [54]). Therefore, we must continually review and update the risks and impacts with which we live according to regulations and legal restrictions. In order to better convey this concern and to decide investments on protective measures, the risks should be quantified in terms of cost implications. This quantification is difficult and subjective to be used as reference, i.e. the valuation of a human life or loss of an animal species is a task of difficult consensus. Consequently, maintenance has to perform a risk assessment for the people health and environmental conservation.

There are simplified methods for assessing security risks [32, 42], expressing the damage based on several factors of calculation (from 1 to 5) such as ABC method which employs one factor and the INSHT method [22] which uses two factors. Although for the present work, searching the characterization to accidents caused by failures we recommend the employment of the William T. Fine (1971) method, which is based on three factors (see Table 15.2):

- Frequency of risk (F_{risk});
- Probability that the accident occurs once the risk has occurred ($P_{accident}$) and,
- The consequences of the accident ($C_{accident}$).

Therefore, the formula to value the security risks of each type of human and environment risk is:

$$R = F_{risk} \cdot P_{accident} \cdot C_{accident}$$

According to this method, all risks must be analyzed and classified (obtaining values between 0 and 10,000).

Table 15.2 William T. Fine method factors (1971)

Factor	Classification	Rating
Frequency	Continuously (or many times daily)	10
	Frequently (approximately once daily)	6
	Occasionally (from once per week to once per month)	3
	Unusually (from once per month to once per year)	2
	Rarely (it has been known to occur)	1
	Remotely possible (not known to have occurred)	0.5
	Impossible	0
Probability	The most likely (and expected result)	10
	Quite possible (has an even 50% chance)	6
	Unusual	3
	Remotely possible	1
	Conceivable (has never happened after many years of exposure)	0.5
	Practically impossible (has never happened)	0.1
Consequences	Catastrophe; numerous fatalities; damage over \$1,000,000	100
	Multiple fatalities; damage \$500,000 to \$1,000,000	50
	Fatality, damage \$100,000 to \$500,000	25
	Extremely serious injury (amputation, permanent disability); damage \$1,000 to \$100,000	15
	Disabling injury, damage up to \$1,000	5
	Minor cuts, bruises, bumps; minor damage	1
Risk score	Immediate (correction required; activity should be discontinued until hazard is reduced)	>200
	Urgent (requires attention as soon as possible)	70–200
	Possible or acceptable (hazard should be eliminated without delay, but situation is not an emergency)	<70

We could utilize this method to evaluate relational capital risk (assuming that the risk value will be 10,000 for the highest economic consequence \$1M, then the cost/risk ratio could be $\$1M/10,000 = \100).

$$q_{R2} = \frac{-1}{K} \cdot \sum_{i=1}^K \frac{R_i}{10.000}$$

where R_i are all the (K) safety and environment risks (we could select those risks over a certain value if we want).

This method also scores the risks, suggesting the prioritization of the corrective or remedial actions, mainly over risk values of 70.

The indicator determining the potential negative impact of this aspect (coherent representation in terms of costs multiplying it by the \$1M) could be as follows:

$$CI_{KR2} = q_{R2} \cdot K \cdot \$1M \cdot z = q_{R2} \cdot KR_2$$

where z is a correction factor that could be necessary in order to adjust the present value of the estimated costs, because the method was defined in 1971 in terms of cost.

Finally, when measuring relational capital, we also have to take into account the impact of both risks (directly over customers or possible over safety and environment) on the company reputation, valuing the economic funds necessary to compensate both effects:

- On customers. We will base on the necessary publicity or marketing campaigns to compensate the bad propaganda, word of mouth, of affected customers. This impact could be important (up to ten times) if we consider the dissatisfied customers bad propaganda transmission ratio indicated by the TARP study (nine or ten partners).
- On safety and environment, going beyond the repair costs and penalties, because it has to consider the impact of catastrophes in market value (8% as average). The economic impact of catastrophes has been measured as almost 15% of the fair market value (FMV) in a year if it is not recovered and 5% during 50 days for recovered catastrophes [29].

Consequently, the correspondent formula to value the impact in company reputation can be estimated as follows:

$$CI_{KR3} = q_{R1} \cdot \frac{TP}{n_s} \cdot n_c \cdot n_f + q'_{R2} \cdot K' \cdot r \cdot FMV = q_{R1} \cdot KR_3 + q'_{R2} \cdot KR_4$$

where TP is the total publicity cost, n_s are the number of spectators for all the TP, K' are, within all the studied risks (K), only those considered as catastrophes; q'_{R2} is a portion of q_{R2} value that can be assigned to catastrophes, and r is the 8% of average impact in the FMV.

15.2.4 Structural Capital Evaluation

In this section, in relation to the intellectual structural capital, we describe two points of view, one dedicated to the accounting perspective and other to the reliability perspective. That is to say, a proper investment in the organization and management of maintenance must improve its key results and dissemination of knowledge, as well as reduce production losses and equipment (network) deterioration.

15.2.4.1 Accounting Structural Capital Investments

From an accounting point of view, the main investments in structural capital are two: investments in management standardization and investments in ICTs.

A measurement of these investments could be represented by the following indicators:

- Investment in standardization, prevention and evaluation of the performance ($CS_{\text{processes}}$).
 - Rate of planned activities ($1 - \text{Ratio of unplanned activities}$) ($i_{S,\text{planning}}$).
 - Efficiency of network capacity. Rate of available capacity against maximum capacity of the network to support services ($i_{S,\text{capacity}}$).
- Investment in ICT (Information and Communication Technologies) and innovation (CS_{ICT}).
 - Ratio of staff with access to ICT ($i_{S,\text{ICT}}$).
 - Predictive coverage in equipments ($i_{S,\text{predictive}}$).
 - Asset monitoring ratio ($i_{S,\text{monitoring}}$).

In summary, we may account for investments in structural capital measuring managerial efficiency as follows:

$$\begin{aligned}
 CI_{CS} &= i_{S_1} \cdot CS_1 + i_{S_2} \cdot CS_2 \\
 &= \frac{(i_{S,\text{planning}} + i_{S,\text{capacity}})}{2} \cdot CS_{\text{processes}} \\
 &\quad + \frac{(i_{S,\text{ICT}} + i_{S,\text{predictive}} + i_{S,\text{monitoring}})}{3} \cdot CS_{\text{ICT}}
 \end{aligned}$$

15.2.4.2 Evaluating Structural Capital Risk

The network's degradation degree is a key factor for the company and influences present maintenance as well as future maintenance. Excessive degradation of the network implies accelerated maintenance cost and unavailability over time.

This section focuses on the value which is generated due to the reliability of the company structure, "the network" in the case of network utilities. With this purpose we will study equipment reliability, considering repairable assets [24, 35], where repairs may impact equipment conditions, either due to the physical conditions modification, or due to the incorporation of latent errors.

The states of the repairable systems after a repair can be [39, 53]:

- As good as new.
- As bad as before the repair.
- Better than before the repair but worse than new.
- Worse than before the repair.
- Better than new.

In the case of distribution networks, the influence of the repair task can be notorious. In the stochastic study of the repairable systems, depending on the type of repair and on the state resulting after the repair, it is common to use different analysis methods. The Generalized Renewal Process (GRP) method [28] is the most flexible technique to cope with all possible states [53] and it includes the

Fig. 15.2 Efficiency of repair according to GRP method

Efficiency of the reapir				
q				
$q < 0$	$q = 0$	$0 < q < 1$	$q = 1$	$q > 1$
Estate	Estate	Estate	Estate	Estate
5	1	3	2	4
Better than new	Equal to new	Better than old and worst than new	Equal to old	Worse than old

Renewal Process (RP) method to describe the first state (as good as new) and the Non-Homogeneous Poisson Process (NHPP) method for describing the second state (as bad as before) [25]. However, such flexibility implies a certain calculation complexity sorted out on computational level. GRP method uses the Weibull distribution to estimate the rate of occurrence of failures.

The GRP method will allow us to improve our maintenance decision making process by comparing different:

- Geographical areas.
- Technical groups.
- Procedures.
- Changes in operations or organization.
- Equipment from different vendors.
- Environmental conditions and operation.
- Etc.

This method calculates the virtual age of the equipment according to the quality of a repair (q_{GRP}) versus its actual age. Properly performed repairs may rejuvenate the equipment reducing its virtual age [15] while bad repairs could make the equipment virtually older (always speaking in terms of equipment reliability, see Fig. 15.2). Of course maintenance will try to make the equipment to be *equal or better than old* after the repair ($q_{GRP} \leq 1$).

$$V_i = q_{GRP} \cdot S_i$$

where V_i and S_i are the virtual and real life of the equipment respectively.

When using the GRP method we obtain the values of α and β (of the Weibull distribution) and the value q_{GRP} that offers information about the repair quality, in each set of equipment with the same characteristics (n_e will be the number of different sets of equipment, $n_{j,e}$ the number of items within each set and n_T the total number of analysed equipments).

Using this method, the repairs can correct the degradation between two consecutive failures or even the degradation of previous intervals. This allows a better fit to all possible repair scenarios. Of course this method assumes that the state (condition) of the equipment is properly captured over time, before and after each repair. Then the values of reliability and q_{GRP} can be useful to show the degree of equipment conservation, allowing a reasonable comparison with values as

specified by the manufacturer or industry standards. In that we do not have a manufacturer or standard value for the comparison, we simply recommend to use of the reliability value for the same equipment but with the value of $q_{GRP} = 1$, equivalent to a repair to the state *as old* (recovering the state prior to the failure).

Therefore, we could deduce indicators of the repair efficiency as a token of equipment conservation.

- If $q_{GRP} < 1$, then the reliability increases, and the equipment deteriorates less than expected.
- If $q_{GRP} > 1$, then the reliability decreases, it is equal or worse than old, and therefore the equipment deteriorates more than expected.

The GRP method (q_{GRP}) gives an idea of how good or bad the repairs are to correct deterioration of the equipment. Then the average value of q_{GRP} represents the mean aging of the network, and it might serve as a key indicator of maintenance excellence.

According to this, we will use q_{S1} as a first indicator of our structural capital risk, based on the GRP reliability analysis, which will be defined as follows:

$$q_{S1} = 1 - \sum_{e=1}^{n_e} q_{GRPe} \cdot \delta_e$$

where we consider n_e sets of critical equipment, using the frequency of each set of critical equipment δ_e , and that the GRP model is applied to each e type of critical equipment to calculate the reliability of the equipment R_e and q_{GRPe} , and then the average sum of all the equipment reliabilities R_T and the average q_{GRP} is obtained as follows:

$$R_T = \sum_{e=1}^{n_e} R_e(t, q_{GRP}) \cdot \delta_e$$

$$q_{GRP} = \sum_{e=1}^{n_e} q_{GRPe} \cdot \delta_e$$

Subsequently, in order to develop q_{S1} our indicator, we will compare the obtained average q_{GRP} with the state equal to old $q_{GRP} = 1$, taking that the values of q_{GRP} will be enclosed between 0 and 2.

$$q_{S1} = 1 - \sum_{e=1}^{n_e} q_{GRPe} \cdot \delta_e$$

- If $q_{S1} < 0$, then the network is preserved worse than old, reliability decreases, and therefore it has been deteriorated more than expected, producing a negative impact in the structural capital. Increasing structural capital risk.

- If $q_{S1} > 0$, then the network is preserved better than old, reliability increases, and therefore it has been deteriorated less than expected, producing a positive impact in the structural capital. Increasing structural capital risk.

Thus the result of 50% corresponds to a normal maintenance which leaves the equipment *as good as old* ($q_{GRP} = 1$), up from 50% improves it with ($q_{GRP} < 1$) and below 50% worsens the state ($q_{GRP} > 1$).

The risk, in terms of cost, can be determined using depreciation calculations, similar to amortization, now using the value of the estimated *virtual life*.

$$CI_{KS1} = q_{S1} \cdot \sum_{e=1}^{n_e} V_{j,e} \cdot n_{j,e} = q_{S1} \cdot KS_1$$

where $V_{j,e}$ is the mean of the book value of each type of equipment and $n_{j,e}$ is the number of equipment inside each type n_e , with $e = 1 \dots n_e$ different types. For more accuracy, the risk calculation could be specified per each type of equipment:

$$CI_{KS1} = \sum_{e=1}^{n_e} q'_{S1} \cdot V_{j,e} \cdot n_{j,e} = \sum_{e=1}^{n_e} (1 - q_{GRPe}) \cdot V_{j,e} \cdot n_{j,e}$$

But from a reliability point of view, we need to evaluate other risk factors of structural capital. In addition to equipment conservation, related production losses may also become an even a more serious risk. But how can we measure this risk in network utilities? The best and regulated (by public administrations) indicator is the *network availability*. We can measure the weighted average of network unavailability per customer (according to the customer's contracted capacity C_t over the total capacity of the network) as follows:

$$q_{S2} = - \sum_{t=1}^{n_u} (1 - A_t) \cdot \left(\frac{C_t}{TC} \right)$$

In network utilities, this indicator measures losses in income from customers as production losses. So if we want to estimate the risk associated to this factor economically, we just need to multiply this indicator by the CPV and by the total number of customers n_u . The value obtained can be compared with the mean value in the sector or with different competitors' values.

$$CI_{KS2} = q_{S2} \cdot CPV \cdot n_u = - \sum_{t=1}^{n_u} (1 - A_t) \cdot \left(\frac{C_t}{TC} \right) \cdot CPV \cdot n_u = q_{S2} \cdot KS_2$$

We could simplify the calculation using customer segmentation and specifying the CPV and unavailability for each segment.

Controlling q_{S1} and q_{S2} we can maximize the lifecycle of the network, avoiding losses in production and bad results of maintenance activities (corrective, preventive and *perfective*).

Table 15.3 Excellence according to results

Excellence According to results		Weight (%)
Related to personnel (9%)	Ratio of staff satisfaction	1.5
	Personnel performance	1.5
	Productivity of labour	1.5
	Rate of covered staff competences	1.5
	Effectiveness of activities according to the standard	1.5
	Excellence of personnel ($1 - q_H = 1 - \text{HEP}_H = R_H$)	1.5
Related to customers (26%)	Customer satisfaction	5
	Network penetration	5
	Ratio of net incomes of unaffected customers	5
	Excellence in customer retention ($1 - q_{R1} = 1 - P_a$)	6
	Excellence in safety and environment conservation ($1 - q_{R2}$)	5
D. Key performance indicators (15%)	Rate of planned activities	2.14
	Efficiency of network capacity	2.14
	Rate of covered staff by ICT	2.14
	Rate of covered equipment by predictive tools	2.14
	Rate of covered equipment by monitoring	2.14
	Excellence in equipment conservation ($1 - R_e$) by GRP method	2.1
	Excellence in service availability ($1 - q_{S2}$)	2.16

15.3 Summarizing the Quantitative Evaluation

In previous sections we have simultaneously developed the quantitative evaluation in terms of costs based on intellectual capital, and in terms of results based on performability indicators and on the EFQM model.

15.3.1 Evaluation of EFQM Results

In Table 15.3, a summary of all the presented indicators in previous sections are included in their appropriate category of results in accordance with the EFQM model. Positive and negative influences in the company due to maintenance performance can be classified according to an existing network utility excellence in four aspects: personnel, customers, society and key performance indicators.

- The achievements thanks to the performance of the staff are reflected mainly in human capital through efficiency ratios related to investments and human reliability.
- The achievements in customer attention are included in the relational capital through efficiency ratios related to investments, the probability of a customer to abandon, the losses due to unavailability and, the bad propaganda and reputation.

Table 15.4 IC accounting and risk factors in network utilities maintenance*Intellectual capital*

$$CI = i_{H_1} \cdot CH_1 + i_{H_2} \cdot CH_2 + q_H \cdot KH + i_R \cdot CR + q_{R1} \cdot KR_1 + q_{R2} \cdot KR_2 \\ + q_{R1} \cdot KR_3 + q'_{R2} \cdot KR_4 + i_{S_1} \cdot CS_1 + i_{S_2} \cdot CS_2 + q_{S_1} \cdot KS_1 + q_{S_2} \cdot KS_2$$

Accounting Factors

$$CI_{CH} = i_{H_1} \cdot CH_1 + i_{H_2} \cdot CH_2 \\ = \frac{(i_{H,\text{satisfaction}} + i_{H,\text{performance}} + i_{H,\text{productivity}})}{3} \cdot CH_{\text{salary}} \\ + \frac{(i_{H,\text{competences}} + i_{H,\text{effectiveness}})}{2} \cdot CH_{\text{training}} \\ CI_{CR} = i_R \cdot CR = \frac{(i_{R,\text{satisfaction}} + i_{R,\text{penetration}} + i_{R,\text{income}})}{3} \cdot CR_{\text{customers}} \\ CI_{CS} = i_{S_1} \cdot CS_1 + i_{S_2} \cdot CS_2 \\ = \frac{(i_{S,\text{planning}} + i_{S,\text{capacity}})}{2} \cdot CS_{\text{processes}} + \frac{(i_{S,\text{ICT}} + i_{S,\text{predictive}} + i_{S,\text{monitoring}})}{3} \cdot CS_{\text{ICT}}$$

Risk Factors

$$CI_{KH} = q_H \cdot C \cdot n_a = (-HEP_H) \cdot (C \cdot n_a) = -(1 - R_H) \cdot (C \cdot n_a) \\ CI_{KR1} = q_{R1} \cdot KR_1 = q_{R1} \cdot CPV \cdot n_c \cdot n_f = -P_a \cdot CPV \cdot n_c \cdot n_f \\ CI_{KR2} = q_{R2} \cdot K \cdot \$1M \cdot z = q_{R2} \cdot KR_2 \\ CI_{KR3} = q_{R1} \cdot \frac{TP}{n_u} \cdot n_c \cdot n_f + q'_{R2} \cdot K' \cdot r \cdot FMV = q_{R1} \cdot KR_3 + q'_{R2} \cdot KR_4 \\ CI_{KS1} = q_{S1} \cdot \sum_{e=1}^{n_e} V_{j,e} \cdot n_{j,e} = q_{S1} \cdot KS_1 \\ CI_{KS2} = q_{S2} \cdot CPV \cdot n_u = - \sum_{t=1}^{n_u} (1 - A_t) \cdot \left(\frac{C}{TC} \right) \cdot CPV \cdot n_u = q_{S2} \cdot KS_2$$

- The achievements in relation to society influence correspond to contributions in the relational capital, synthesized in the worries and risks over safety and environmental issues.
- The key results of the organization according to the planned performance should be in line with the structural capital through efficiency indicators of investments and excellence indicators in production and reliability.

Indicator weights have been distributed proportionality. The category society is considered inside the category customers.

Table 15.3 allows the quantification of the maintenance management maturity and the excellence in its management. It also permits a comparison of the values of different companies or areas of network distribution.

15.3.2 Evaluation of Intellectual Capital

Table 15.4 presents a summary of formula to estimate intellectual capital impacts on maintenance.

These formulas are employed within a determined period of time of evaluation, and their effect could be even more relevant, considering the estimation of current values of a set of future values estimated in a series of periods based on the formulas of [Sect. 3.2.4](#).

As a result, we can evaluate maintenance from different points of view, as a necessary expense to ensure service levels, but also as a contributor to higher availability minimizing safety and environmental risks.

15.4 Quantitative Evaluation Case Study

We now try to compare maintenance management maturity in two water network utilities in the same area of distribution and using the annuity as the evaluation period.

In [Table 15.5](#) the investments and main KPIs are presented for both companies.

As we can see, the investments in both companies are the same, 900,000 €/year; although the worth of intellectual capital is less than the investments in both companies, 682,917 in company A and 748,750 in company B. In addition, we can appreciate that:

- The structural capital is higher in company B than in A due to the investments in ICT.
- The relational capital is higher in company B than in A due to customers' satisfaction. The number of affected customers by failures is less.
- The human capital is similar for both. A seems to be more efficient due to higher employee satisfaction.

Now, we will continue measuring risk in these two companies.

(a) Risk associated to human reliability.

We will deal with a simple example to estimate the HEP according to the HEART method. We suppose an activity of type (C), a complex task requiring high level of comprehension and skills. In circumstances for the company A and B: operator inexperience, unreliable instrumentation; and additionally for company B: low workforce morale.

Then the estimations for both companies are indicated in [Table 15.6](#) considering the proposed remedial measures per each factor.

For these estimations, the expert technicians decided to weigh each FSP with $APOA_A = 0.2$, $APOA_B = 0.2$, y $APOA_C = 0.6$. Then, the HEP is higher in company B than in A, and supposing that the mean cost of an activity and the number of activities is the same for both, then the risk associated to human reliability is greater in the company B (see [Table 15.7](#)):

(b) Risk associated to customer retention.

To simplify we will apply the same values for both companies and we will show results of the survival method.

Table 15.5 Intellectual capital in case study

	Company A	Company B
<i>Investments</i>		
Salaries	300,000	250,000
Investment in learning and training	50,000	100,000
Investment in customer attention	100,000	100,000
Investment in quality and processes	300,000	250,000
Investment in ICT and Innovation	150,000	200,000
Total	900,000	900,000
<i>KPI</i>		
	%	%
Personnel satisfaction	90	80
Personnel performance	75	85
Productivity of labour	90	90
Fulfilment of personnel competences	90	90
Fulfilment of effectiveness versus. standards	85	80
Customer satisfaction	75	85
Network penetration	45	55
Customer incomes unaffected by failures	80	95
Planning	60	80
Capacity	75	85
Personnel with ICT	75	95
Equipments with predictive	80	80
Equipments with monitoring	75	75
<i>Accounting capital factors</i>		
$i_{H_1} \cdot CH_1 = \frac{(i_{H.satisfaction} + i_{H.performance} + i_{H.productivity})}{3} \cdot CH_{salary}$	255,000	212,500
$i_{H_2} \cdot CH_2 = \frac{(i_{H.competences} + i_{H.effectiveness})}{2} \cdot CH_{training}$	43,750	85,000
$i_R \cdot CR = \frac{(i_{R.satisfaction} + i_{R.penetration} + i_{R.income})}{3} \cdot CR_{customers}$	66,667	78,333
$i_{S_1} \cdot CS_1 = \frac{(i_{S.planning} + i_{S.capacity})}{2} \cdot CS_{processes}$	202,500	206,250
$i_{S_2} \cdot CS_2 = \frac{(i_{S.ICT} + i_{S.predictive} + i_{S.monitoring})}{3} \cdot CS_{ICT}$	115,000	166,667
$CI_{human} = CI_{CH} = i_{H_1} \cdot CH_1 + i_{H_2} \cdot CH_2$	298,750	297,500
$CI_{relational} = CI_{CR} = i_R \cdot CR$	66,667	78,333
$CI_{structural} = CI_{CS} = i_{S_1} \cdot CS_1 + i_{S_2} \cdot CS_2$	317,500	372,917
Total	682,917	748,750

Assume that we suffer 30 incidents in equipment supporting around 200 customers each. In Table 15.8, the lost and maintained customers per incident are presented besides the time when the incident took place. Customers leaving the service (one or more) are marked as F in second column and those remaining as S. We then deduce the probability of a customer abandoning using the Weibull analysis (we used Weibull++ from Reliasoft—www.reliasoft.com).

Table 15.6 HEP according HEART

Generic task (GT)			
(C) Complex task requiring high level of comprehension and skill			0.16 (0.12–0.28)
	EPC	APOA	$[(EPC-1)*APOA]+1$
a.— Operator inexperience (15)	X3	0.2	$[(3-1)*0.2]+1=1.4$
b.— Unreliable instrumentation (23)	X1.6	0.2	$[(1.6-1)*0.2]+1 = 1.12$
c.— Low workforce morale (31)	X1.2	0.6	$[(1.2-1)*0.6]+1 = 1.12$
In company A $HEP1 = 0.16*1.4*1.12 = 0.25$			
In company B $HEP2 = 0.16*1.4*1.12*1.12 = 0.28$			

Table 15.7 Impact due to human reliability

	Company A	Company B
–HEP	–0.25	–0.28
Mean cost of activity (€), C	150	150
Number of activities, n_a	589	589
Risk due to human reliability (€), CI_{KH}	–22,088	–24,738

Then, the Weibull function parameters are $\beta = 2.24$ and $\text{Eta} = 945.89$, and the following Figs. (15.3,15.4, 15.5, 15.6, 15.7) represent the customer chronology, probability, reliability and maximum likelihood of the estimation (Fig. 15.7).

In this last figure, you can see the Gauss form of the ML surface estimating the best Weibull parameters.

Considering values in Table 15.9, if the re-establishment time is different in the two companies and the CPV is the same, the risk in the company A is higher because the probability to abandon is higher. In addition, for Company A, due to the large easement of this equipment (around 200 customers each) any decision about them has major implications. The effects of bad propaganda and the potential customers to loose within a given area may be determinant. On the other hand, for Company B, if repair times are shorter than the market average, attracting and retaining customers could be an easier task than for the competitors. Company A must pay special attention to critical equipment, deserving prioritized maintenance strategy to minimize the impact on customers.

These estimates, connected to relational capital, should be checked a posteriori, and periodically, based on the customer retention and acquisition marketing department data.

(c) Safety and environmental related risks.

In order to simplify, we will consider a unique risk factor with possible safety affections, that we will analyze by the W.T. Fine Method, presented in Sect. 15.2.2.2. With the aim of characterizing possible accidents caused by failures, this method has a base on three factors:

- Frequency of risk (F_{risk});
- Probability that the accident occurs once the risk has occurred (P_{accident}) and,
- The consequences of the accident (C_{accident}).

Table 15.8 Estimation of the probability of abandoning

Number of customers	Status F o S	Time	ID	Date	06/09/2010
220	S	1	1	Weibull-2P	
165	S	7	2	Analysis	EMV
187	S	8	3	Method for LC	LLR
131	S	10	4	Rank	MED
119	S	12	5	Beta	2.241209263
1	F	22	6	Eta	945.8907389
164	S	22	6	LK	−807.7817704
61	S	24	7	Failures\Susp.	89\4033
2	S	25	8		
47	S	31	9		
160	S	35	10		
80	S	42	11		
120	S	45	12		
1	F	52	13		
175	S	52	13		
1	F	54	14		
100	S	54	14		
160	S	54	15		
2	F	73	16		
138	S	73	16		
1	F	50	17		
2	F	80	17		
160	S	80	17		
1	F	68	18		
1	F	87	18		
142	S	87	18		
1	F	83	19		
13	S	83	19		
3	F	95	20		
136	S	95	20		
3	F	110	21		
139	S	110	21		
1	F	84	22		
4	F	132	22		
181	S	132	22		
1	F	86	23		
5	F	140	23		
132	S	140	23		
1	F	92	24		
1	F	144	24		
5	F	162	24		

(continued)

Table 15.8 (continued)

Number of customers	Status F o S	Time	ID	Date	06/09/2010
137	S	162	24		
1	F	126	25		
1	F	139	25		
7	F	186	25		
230	S	186	25		
1	F	201	26		
55	S	201	26		
1	F	104	27		
1	F	132	27		
3	F	228	27		
123	S	228	27		
174	S	231	28		
1	F	97	29		
1	F	135	29		
1	F	159	29		
1	F	242	29		
1	F	283	29		
1	F	314	29		
1	F	357	29		
15	F	411	29		
190	S	411	29		
1	F	122	30		
1	F	208	30		
1	F	242	30		
1	F	290	30		
1	F	320	30		
12	F	367	30		
192	S	367	30		

Now, it is a simple example about working on voltage, where there is a high probability that the worker suffers a shock with: frequency values, would occasionally, with probability remotely possible, and consequently different for each company of one dead worker in the company (A) and several deaths of workers in the company (B). So the risk is (according to the Table 15.2 about the possible values of the three factors):

$$R = F_{\text{risk}} \cdot P_{\text{accident}} \cdot C_{\text{accident}}$$

- $R = 3 * 1 * 25 = 75$ in the company (A)
- $R = 3 * 1 * 50 = 150$ in the company (B)

According to this method, all risks must be evaluated depending on their consequences in terms of cost (assuming that the cost value of risk will be \$ 1M for the highest economic consequence that corresponds to catastrophe, numerous fatalities or damage about its quantity).

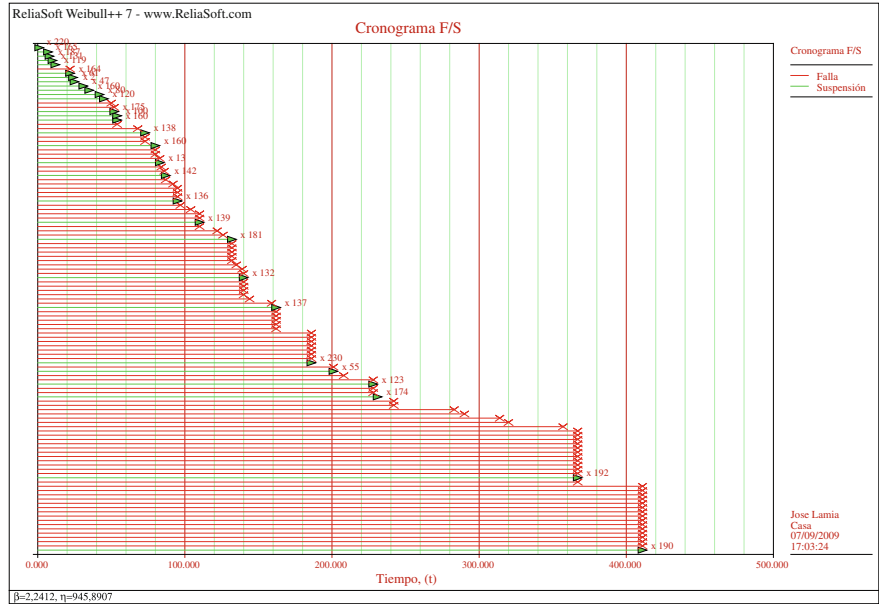


Fig. 15.3 Customer chronology

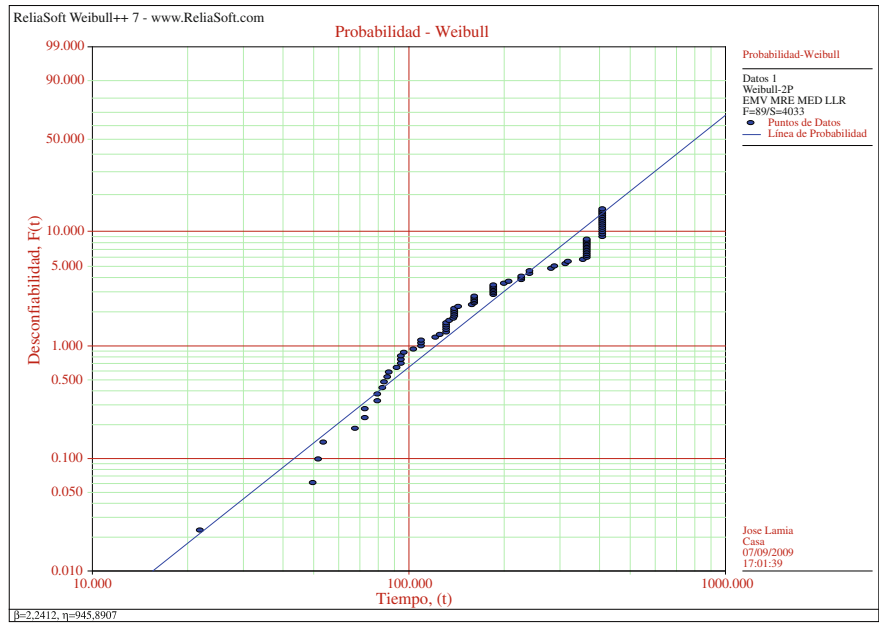


Fig. 15.4 Probability curve

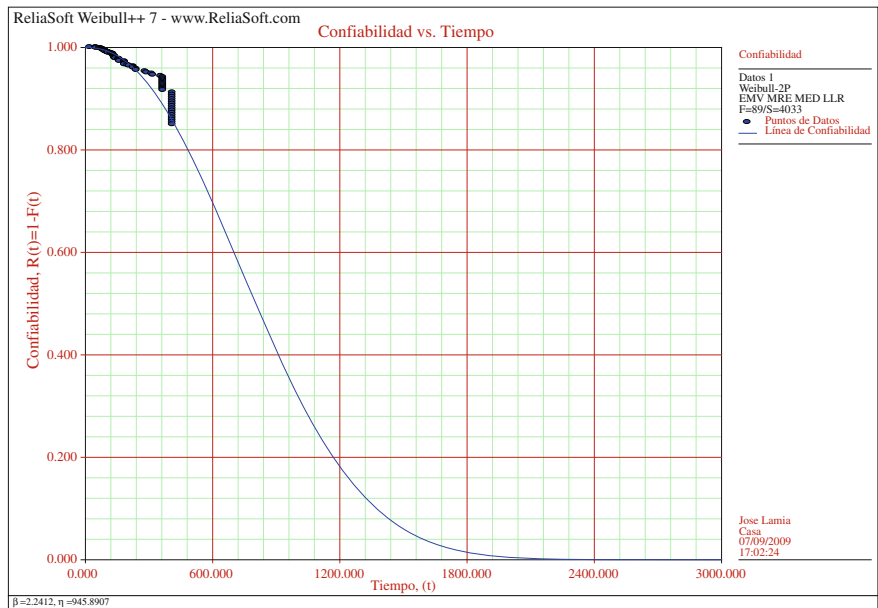


Fig. 15.5 Reliability curve

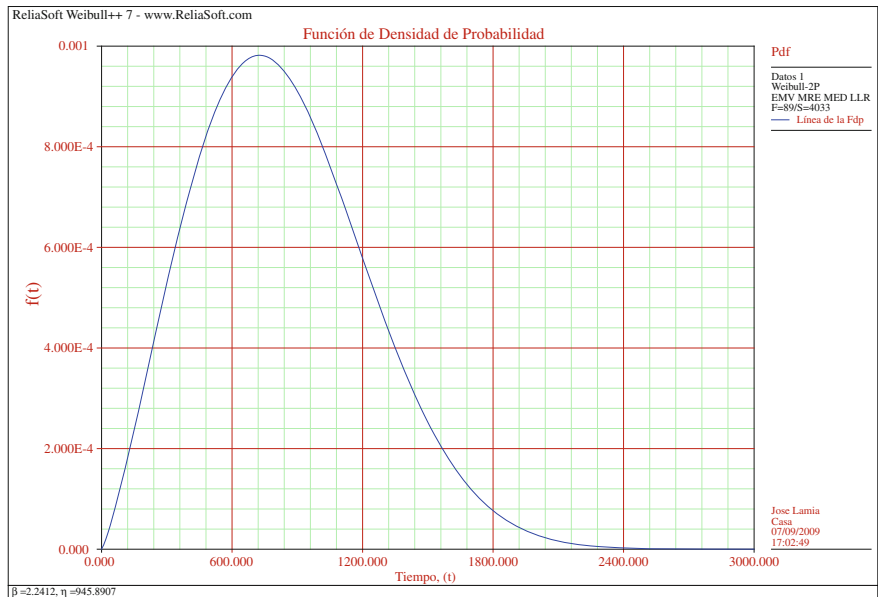


Fig. 15.6 fdp curve

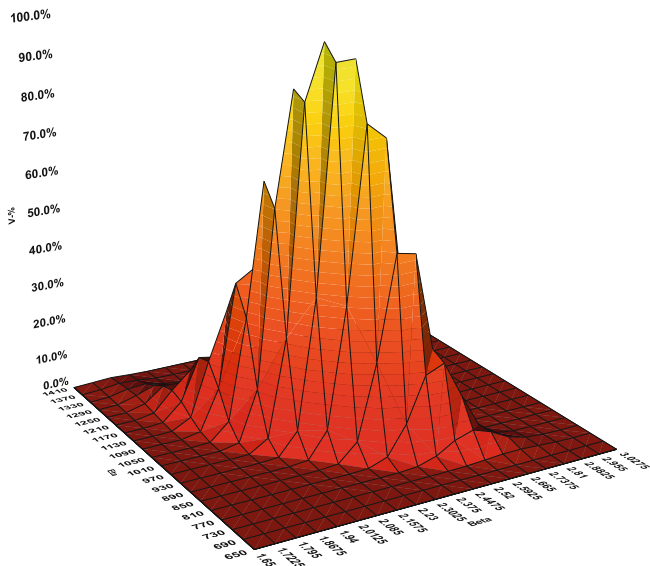


Fig. 15.7 Maximum Likelihood surface

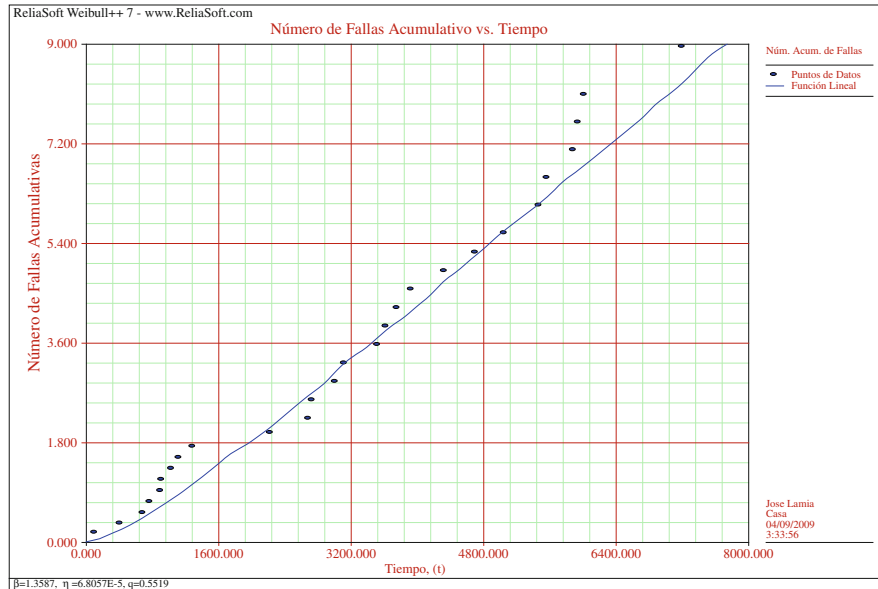


Fig. 15.8 Cumulative number of failures by GRP

Considering the risk score of the W.T. Fine method, the risk that needs corrective or remedial actions are those whose risk value is over 70, and so we have

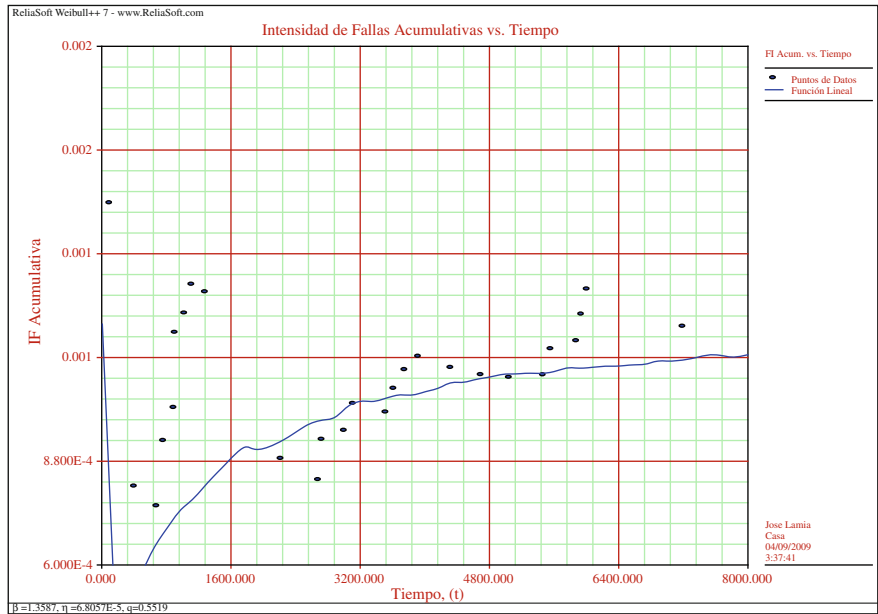


Fig. 15.9 Failure rate by GRP

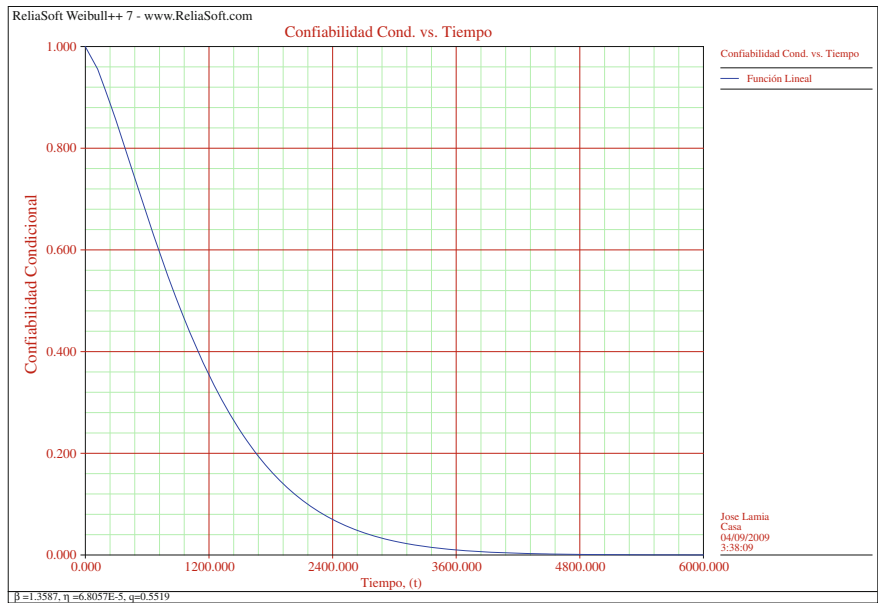


Fig. 15.10 Reliability rate by GRP

Table 15.9 Estimation of risk of customers abandoning

	Company A	Company B
Re-establishment time	48	32
Pa	-0.0013	-0.0005
CPV	3,000	3,000
Number of affected customers, n_c	200	200
Number of failures, n_f	30	30
Customer retention/abandon risk, CI_{KR1}	-23,400	-9,000

Table 15.10 Impact on safety

	Company A	Company B
q_{R2}	-75/10,000	-150/10,000
Higher consequence	1,000,000 \$	1,000,000 \$
Corrective factor (z)	1.5	1.5
Impact on safety (€), CI_{KR2}	-22,500 €	-11,250 €

Table 15.11 Impact on company reputation

	Company A	Company B
Re-establishment time	48	32
Pa	-0.0013	-0.0005
TP/n_s	300	300
Number of affected customers, n_c	200	200
Number of failures, n_f	30	30
Risk affecting reputation (€), CI_{KR3}	-2,340	-900

an unacceptable significant risk level for both companies, 75 and 150, for which urgent correction is recommended. In order to translate this risk in terms of cost and impact, we will employ a correction factor (z) equal to 1.5 in order to adjust the past consequences costs of the method (defined in 1970) to the present value in euros (see Table 15.10):

We have a high value of the risk, due to the death as affection, which is detracted from our intellectual capital value. Although the most important factor is to know its amount in order to decide whether to take any action to reduce it or simply to live with it.

(d) Impact on company reputation.

Now, we will now suppose that the publicity cost per spectator for both companies is the same, and no presence of catastrophic disaster in none of them Table 15.11.

Thus, the impact on company reputation is worst in the Company A than in Company B, mainly because of the proportion of customers with a high probability of generating negative propaganda.

Table 15.12 GRP model calculations for reliability data of six systems

System ID	F or E	Time	FI	Power Law
1	F	2227,08	Beta	1,358744637
1	F	2733,229	Lambda	6,81E-05
1	F	3524,214	q	0,551859894
1	F	5568,634	LK	-209,9571106
1	F	5886,165		
1	F	5946,301		
1	F	6018,219		
1	F	7202,724		
1	E	8760		
2	F	772,9542		
2	F	1034,458		
2	F	3011,114		
2	F	3121,458		
2	F	3624,158		
2	F	3758,296		
2	E	5000		
3	F	900,9855		
3	F	1289,95		
3	F	2689,878		
3	F	3928,824		
3	F	4328,317		
3	F	4704,24		
3	F	5052,586		
3	F	5473,171		
3	E	6200		
4	F	411,407		
4	F	1122,74		
4	E	1300		
5	F	688,897		
5	F	915,101		
5	E	2650		
6	F	105,824		
6	E	500		

e) Impact in relation to equipment reliability.

In this example, we will apply the GRP method to failures (F) of six systems (1–6) with a known end-of-life time (E), obtaining a Weibull parameters estimation (using again Weibull++—www.reliasoft.com), the cumulative number of failures, the rate of failures and the reliability (see Table 15.12) (Figs. 15.8, 15.9, 15.10).

According to the results obtained, the repair efficiency (q_{GRP}) is 0.55, and then the virtual equipment life is 45% (less than the real life). To calculate the corresponding associated risk we will suppose that this is the q_{GRP} value for the company A and that the q_{GRP} value for the Company B is 0.60. The reliability at

Table 15.13 Risk due to equipment increased reliability

	Company A	Company B
q_{GRP}	0.45	0.40
Purchase value, $V_{j,e}$	30,000	30,000
Number of equipments, $n_{j,e}$	6	6
Risk due to higher reliability (€), CI_{KS1}	81,800	72,000

Table 15.14 Risk associated to service interruption

	Company A	Company B
Availability, A_i	85%	95%
Unavailability	15%	5%
CPV	3,000	3,000
Number of affected customers, n_u	200	200
Ris— Impact on production, CI_{KS2}	−90,000	−30,000

Table 15.15 Overall maintenance contribution to Intellectual Capital

	Company (A) (€)	Company (B) (€)
Accounting capital	900,000	900,000
Human capital	299,167	299,167
Relational capital	73,333	71,667
Structural capital	327,500	378,333€
Total accounting capitals	700,000	749,167
Human reliability associated risk	−22.088,00	−24.738,00
Customer retention associated risk	−23.400	−9.000,00
Safety and environment associated risk	−22.500,00	−11.250,00
Company reputation associated risk	−2.340	−900,00
Equipment reliability associated risk	81.800	72.000,00
Service interruption associated risk	−90.000	−30.000,00
Total risk	−78.528	−3.888
Overall maintenance contribution to IC	604.389	744.862

the time of evaluation for (A) is 0.36 and for (B) we will suppose that is 0.3. Thus, the risk for both companies is presented in Table 15.13.

In this example the impact is positive because the maintenance reduces the probability of equipment failure against the supplier estimation.

(f) Service interruption risk.

To estimate this risk, we will consider a unique segment of customers with the same capacity of service; that the type of equipment supports 200 customers (n_u) for both companies, with the same capacity; and that the CPV is 3,000 €/year (as mentioned above). With these assumptions Table 15.14 contains risk calculations for both companies.

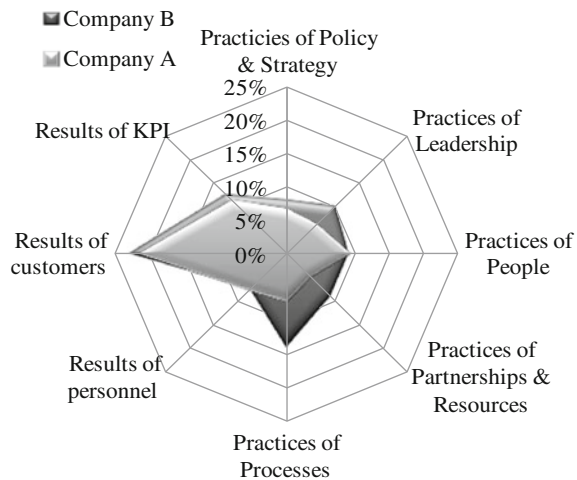
Table 15.16 Overall maintenance contribution to EFQM results

Excellence according to results		(A)	(B)	Weight (%)
Related to personnel (9%)	Ratio of staff satisfaction	0.90	0.80	1.50
	Personnel performance	0.75	0.85	1.50
	Productivity of labour	0.90	0.90	1.50
	Rate of covered staff competences	0.90	0.90	1.50
	Effectiveness of activities	0.85	0.80	1.50
	Excellence of personnel ($1 - q_H = 1 - \text{HEP}_H$)	0.75	0.72	1.50
	<i>Total results related to personnel</i>	7.58%	7.46%	
Related to customers (26%)	Customer satisfaction	0.75	0.85	5
	Network penetration	0.45	0.55	5
	Ratio of net incomes of unaffected customers	0.80	0.95	5
	Excellence in customer retention ($1 - q_{R1} = 1 - Pa$)	0.9987	0.9995	6
	Excellence in safety and environment conservation ($1 - q_{R2} $)	0.9925	0.9850	5
	<i>Total results related to customers</i>	20.95%	22.67%	
Key performance indicators (15%)	Rate of planned activities	0.60	0.80	2.14
	Efficiency of network capacity	0.75	0.85	2.14
	Rate of covered staff by ICT	0.75	0.95	2.14
	Rate of covered equipment by predictive tools	0.80	0.80	2.14
	Rate of covered equipment by monitoring	0.75	0.75	2.14
	Excellence in equipment conservation ($1 - R_T$) by GRP method	0.64	0.70	2.14
	Excellence in service availability ($1 - q_{S2} $)	0.85	0.95	2.16
	<i>Total results related to KPIs</i>	11.02%	12.43%	

Table 15.17 Excellence according to EFQM model enablers and results

Excellence	Company A (%)	Company B (%)
Policy & Strategy	7	8
Leadership	6	10
People	9	9
Partnerships & Resources	5	9
Processes	7	14
Results related to personnel	7.58	7.46
Results related to customers	20.95	22.67
Results related to KPIs	11.02	12.43

Fig. 15.11 EFQM excellence in spider diagram of company A and B



Lack of service availability in Company A may generate more significant customers' complaints, higher customer probability of abandoning, and of course the loss of the estimated CPV portion which is not provided.

To conclude, Table 15.15 summarizes all previous calculations related to intellectual capital and Table 15.16 calculates the EFQM results for these two companies.

We can appreciate that the efficiency levels of investments are below 100%, and therefore the accounting value of the intellectual capital is lower than investments in both companies. Besides this, management should pay attention to substantial risks related to maintenance activities, because some of them could generate serious problems in the customers and society relationships.

Table 15.17 and Fig. 15.11 summarize the EFQM excellence evaluations in spider diagrams for both companies.

15.5 Evaluation Remarks

While searching for the excellence through this holistic framework, companies can also pay attention to risks. This can be a clear maintenance management self-diagnosis, controlling the fulfilment of objectives in a culture of continuous improvement.

This framework measures the overall performance, identifying weaknesses, strengths and also improvement opportunities. Consequently our valuation will look for obtaining these savings or expenditures, in terms of costs, whose combination will reflect the generated value by the maintenance management in a service provider company, considering the effects of reversing or reducing budgets due to current and future consequences. Moreover, these issues can also be used to assess the criticality or priority of maintenance policies. Given this, there is a methodology for assessing the present situation, the planned objectives, the

deviations and the contribution of maintenance in terms of cost, so as to facilitate the choice of strategies.

Consequently, we can evaluate the maintenance:

- According to maturity of management offering a value between 1 and 5.
- According to excellence of management with values between 0 and 100%.

As a conclusion, any organization can use one model or another to assess the intellectual capital, but what is important is that the model:

- Offers information for the organization maturity measurement and control, in a quantitative way and including intangible costs.
- Facilitates the decision-making and its comparison with external agents, either with competitors or with regulated and internationally accepted values or indices.
- Aligns the maintenance with the company's strategy and with other departments.
- Detects potential capabilities for the organization to improve the current situation and to adapt for new challenges, as well as detecting the weak points in order to make decisions on how to fix them.
- Brings management and management decisions closer to the client as an improvement in offered and perceived quality.
- Cares about personnel and monitors the way the work is done (processes) impacting on safety.
- Standardizes the communication concerning maintenance resources evolution.

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